

Figure 1

A

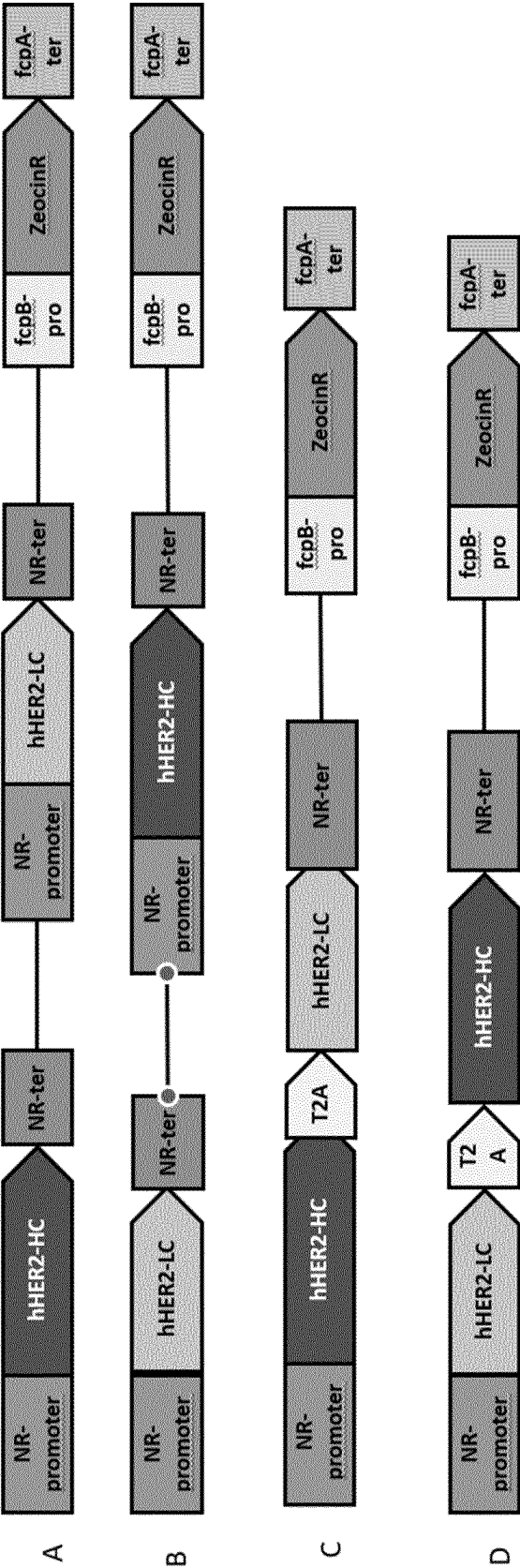


Figure 1 (continuation)

B

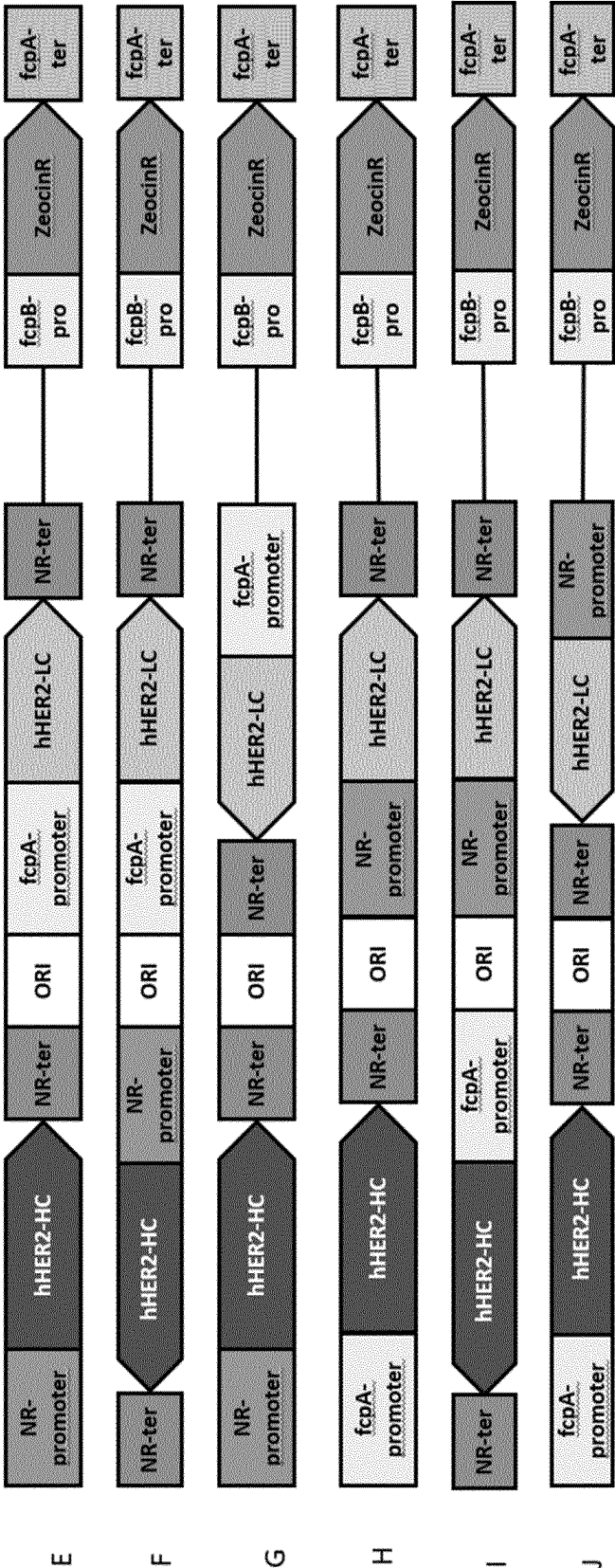


Figure 2

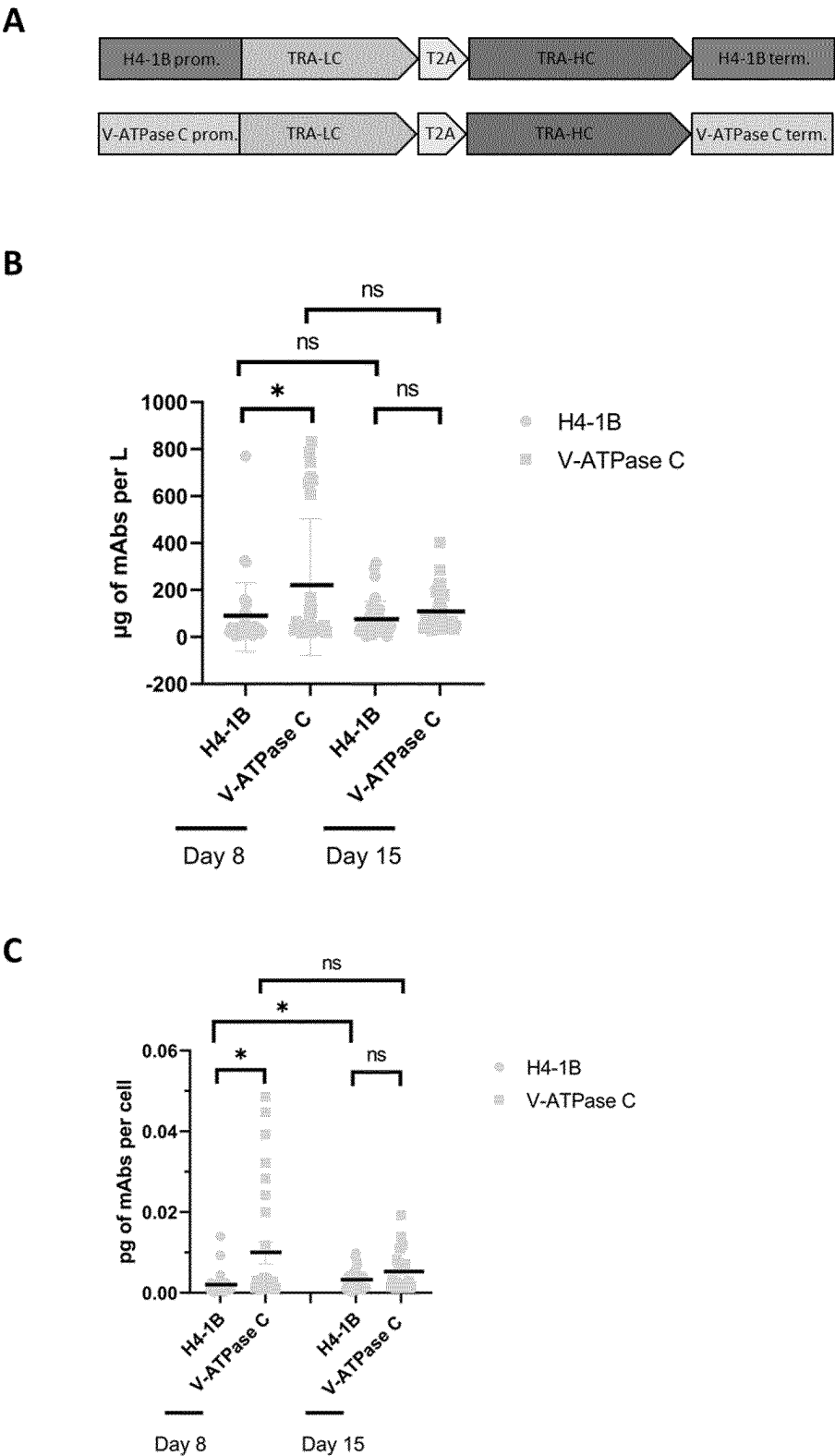


Figure 3

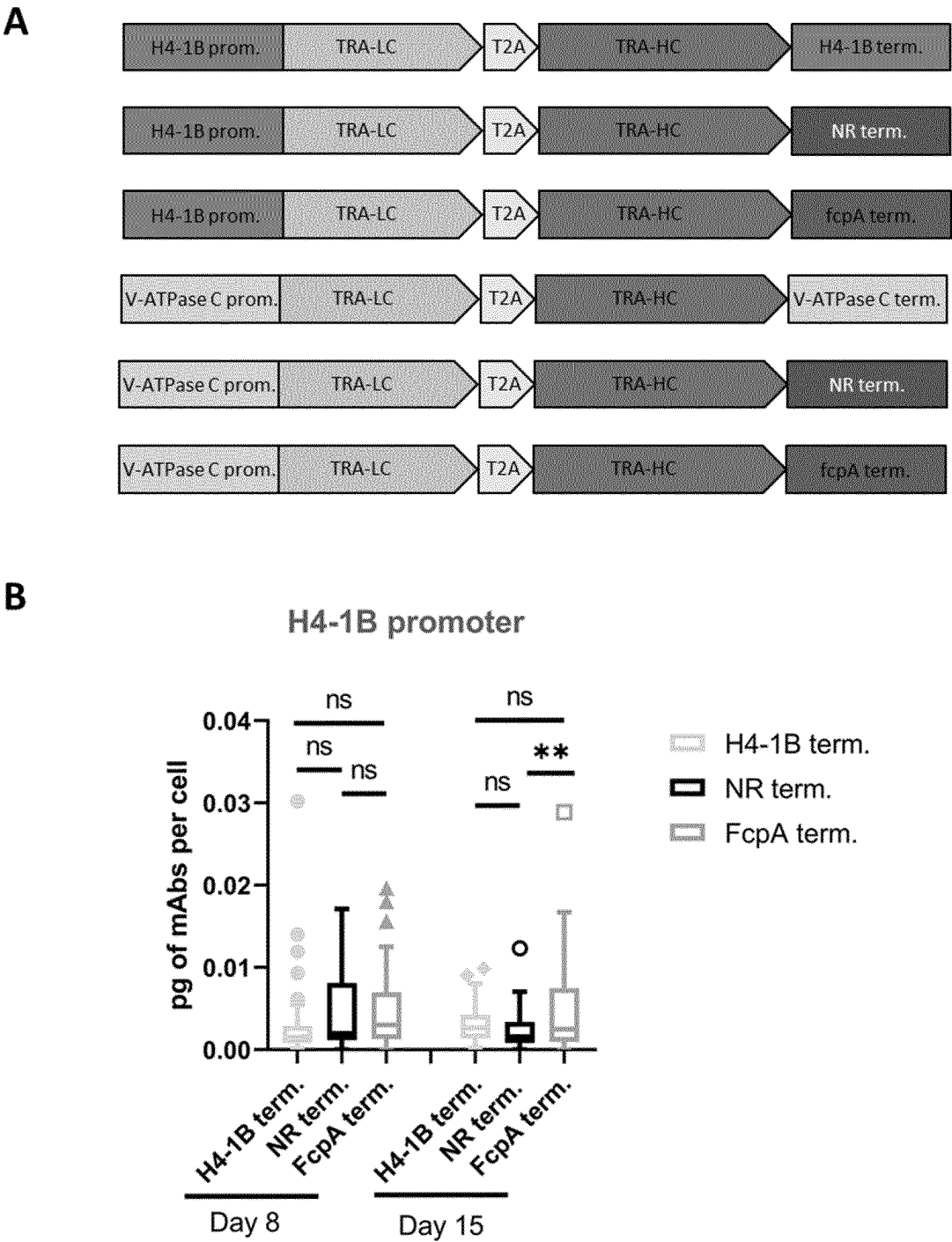


Figure 3 (continuation)

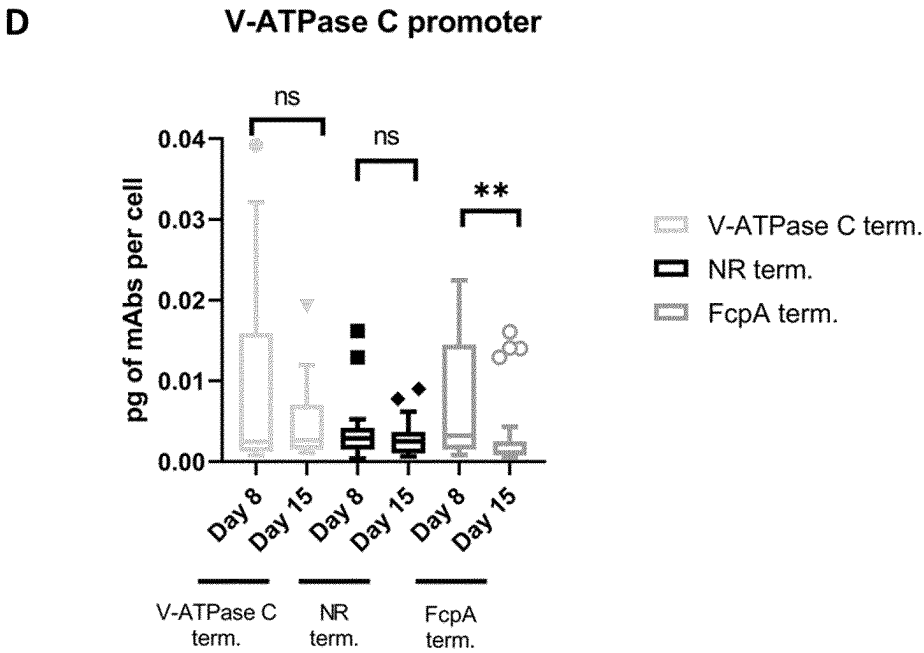
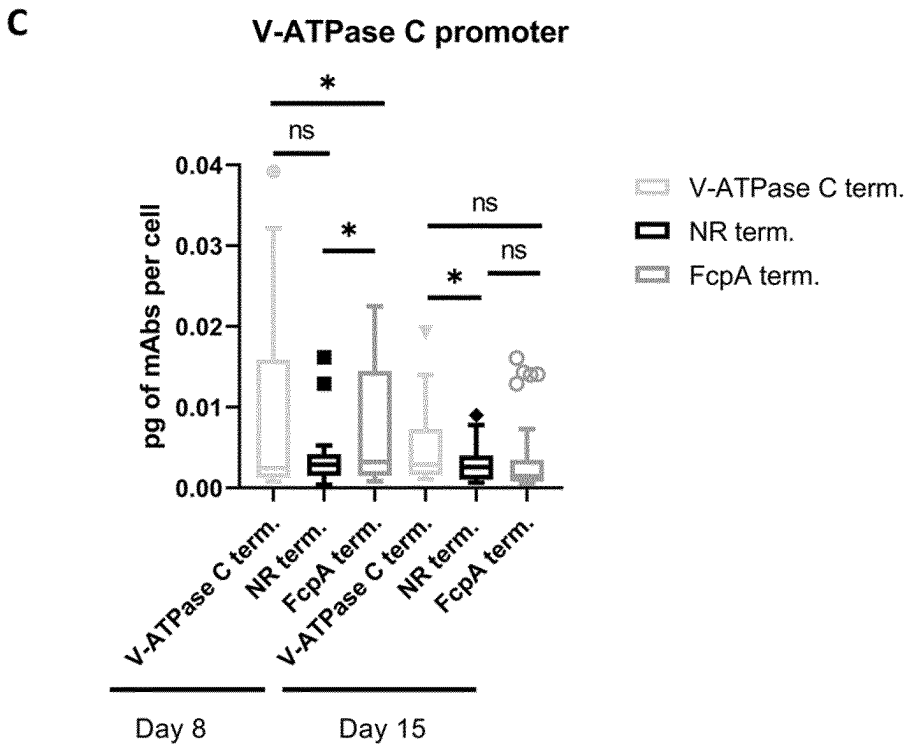


Figure 4

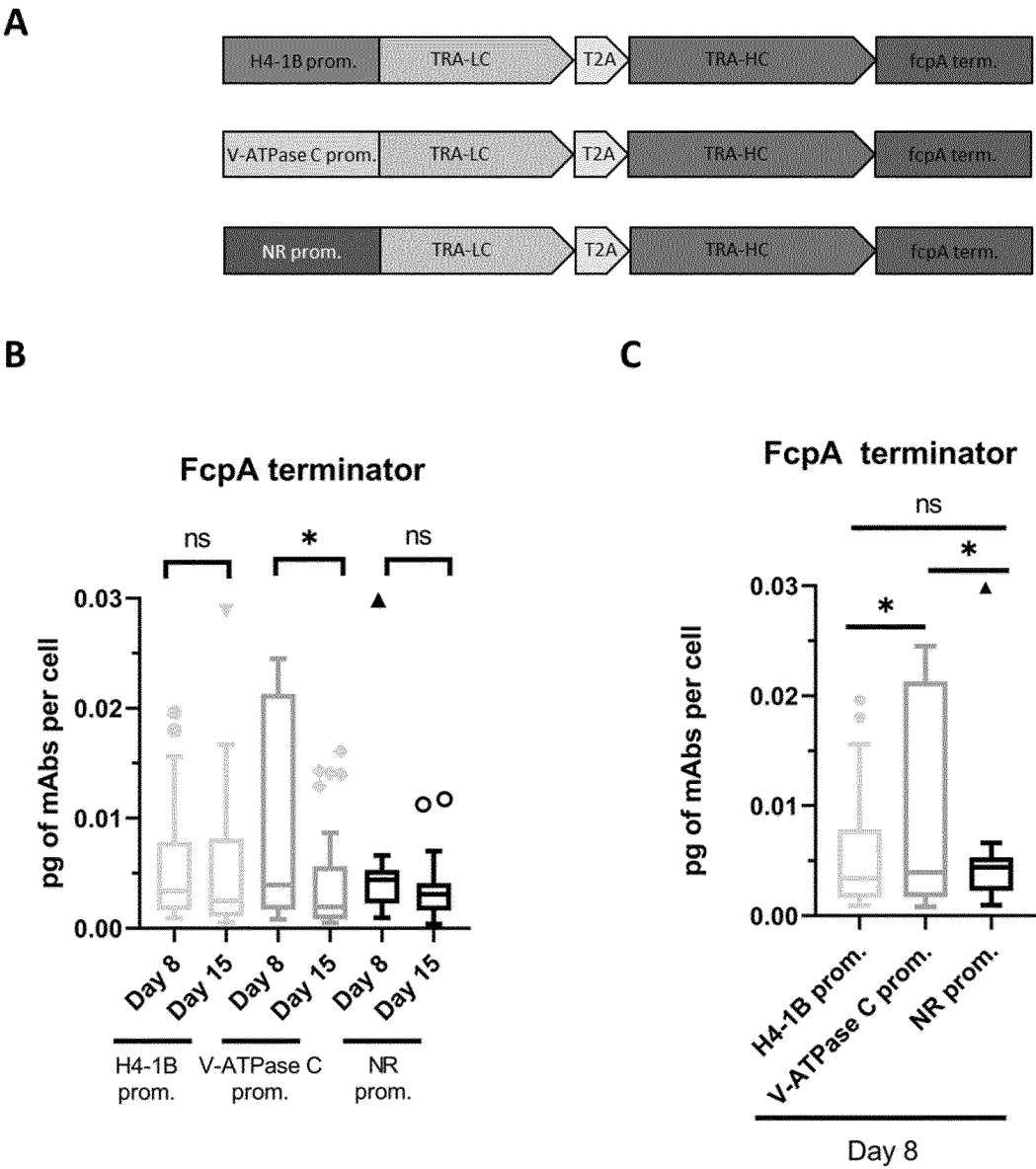
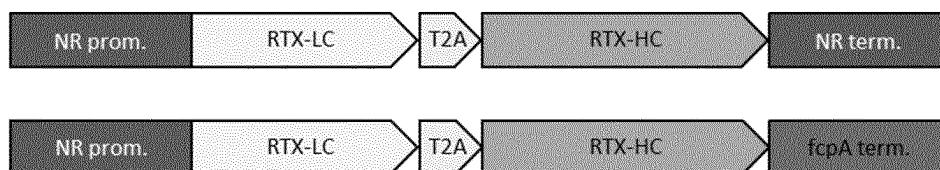
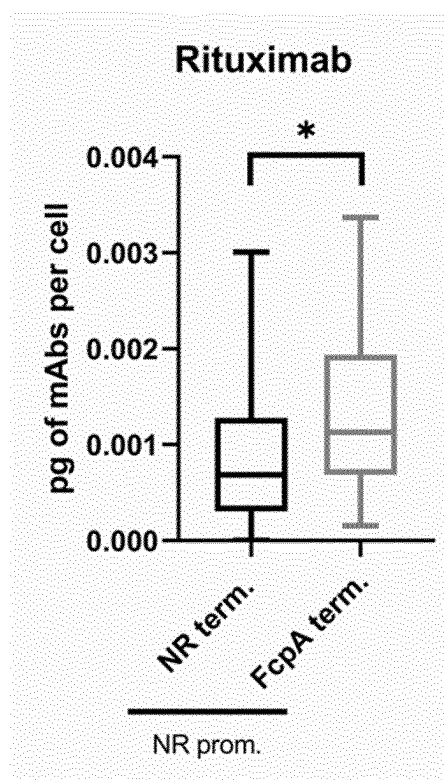


Figure 5

A**B**

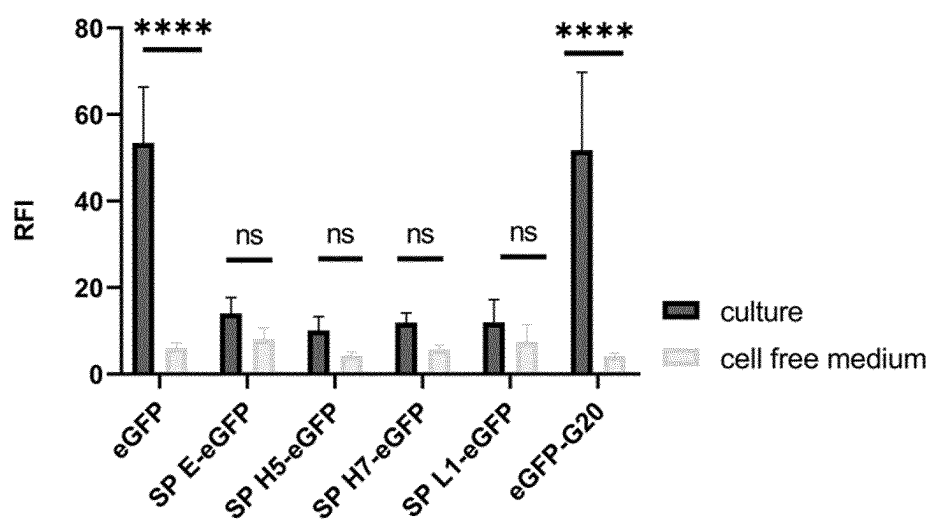


Figure 7

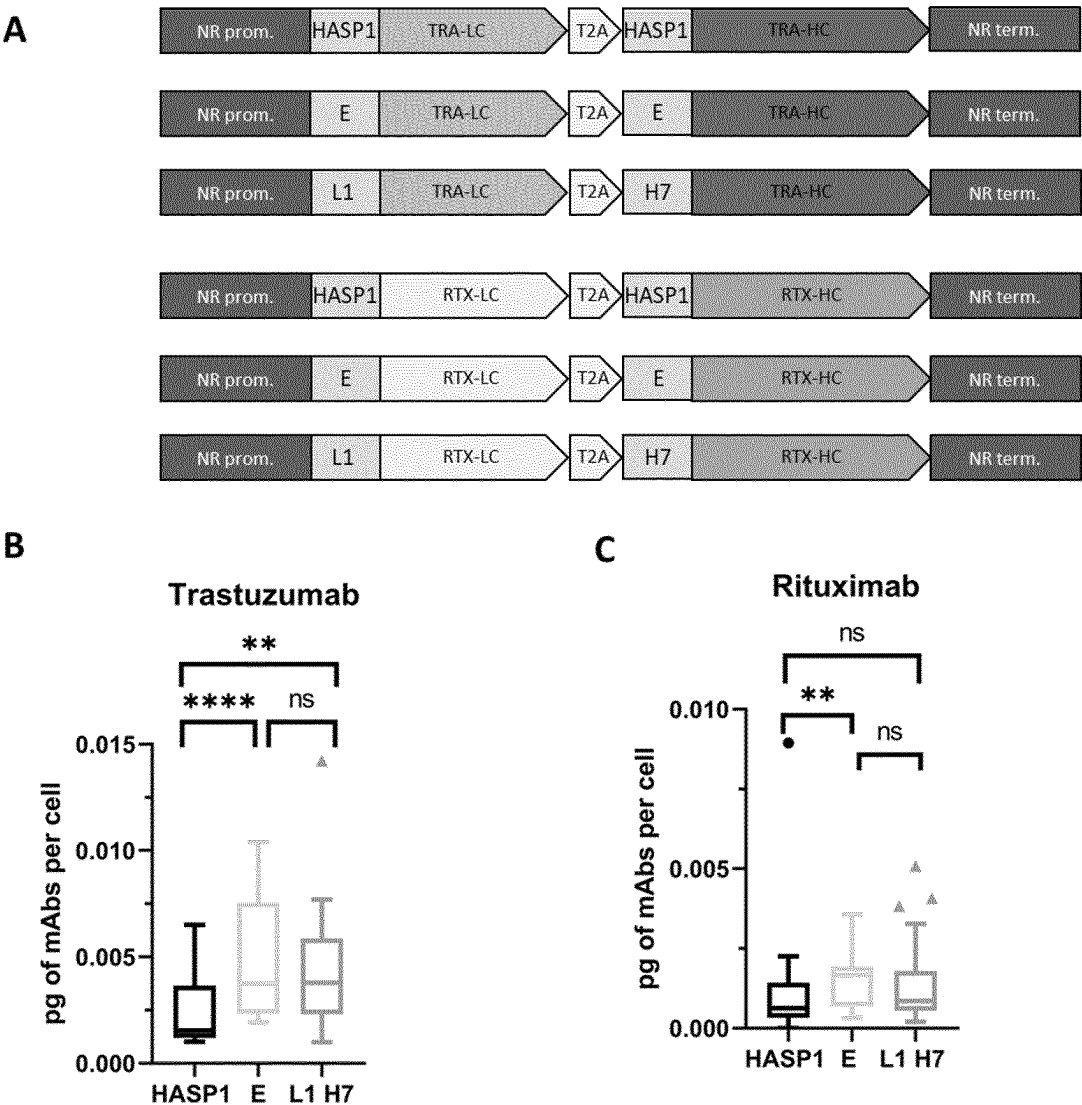


Figure 8

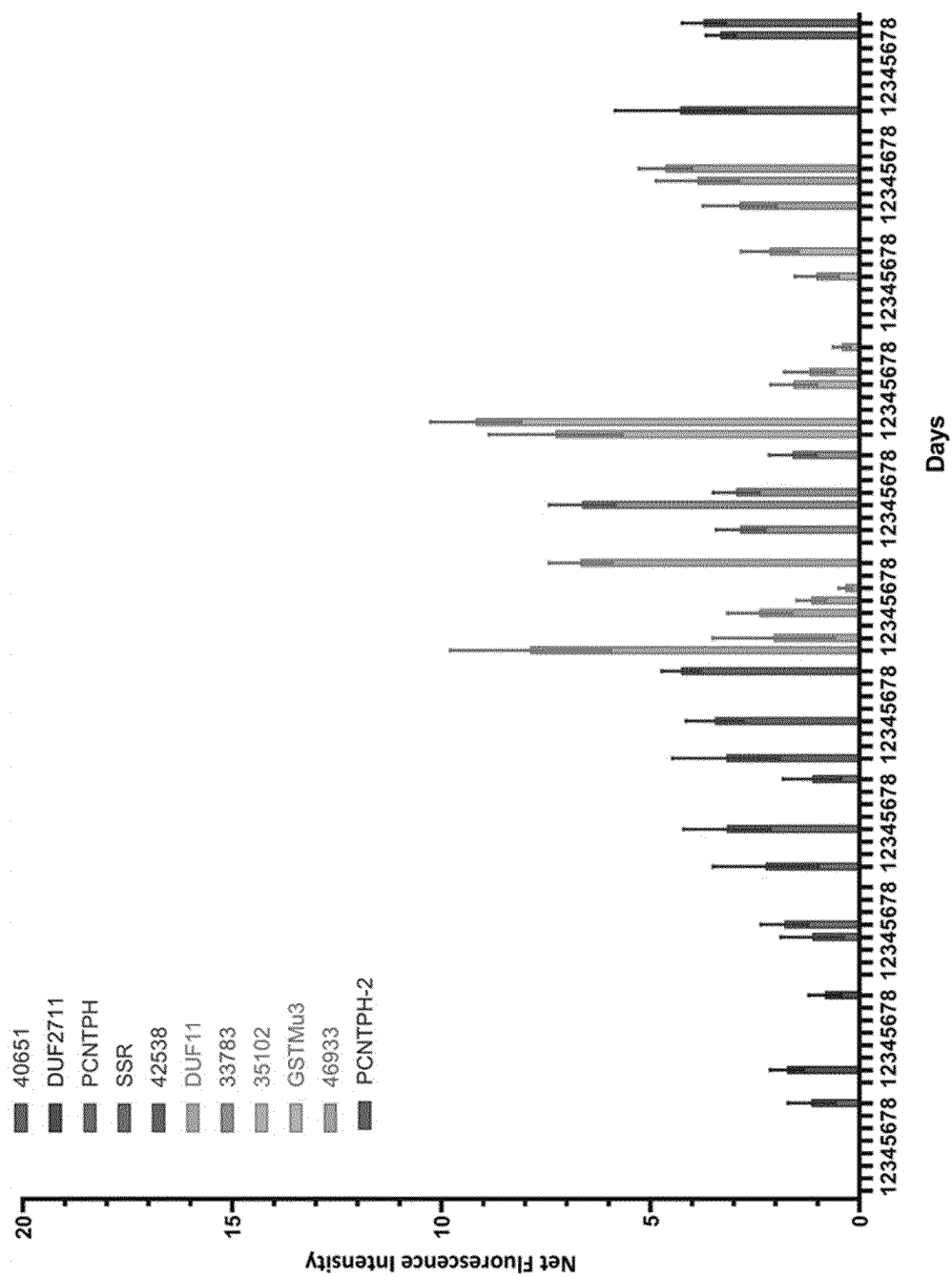
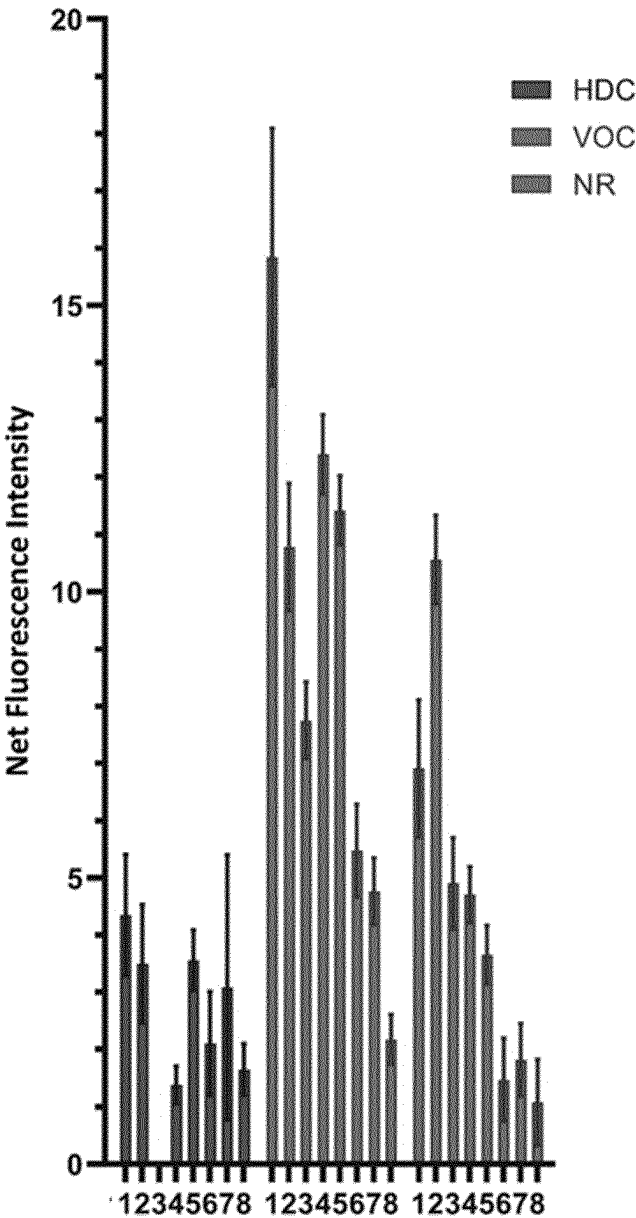


Figure 8 (continuation)

A2



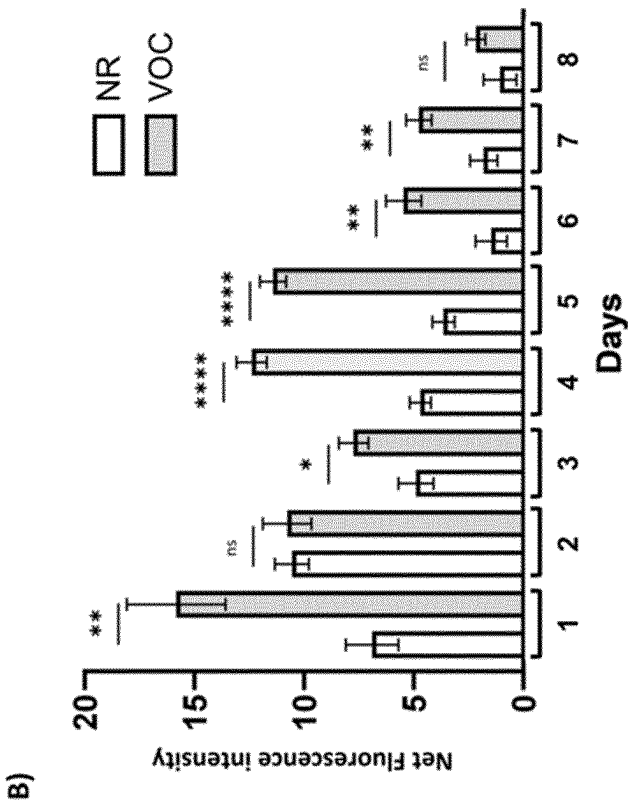
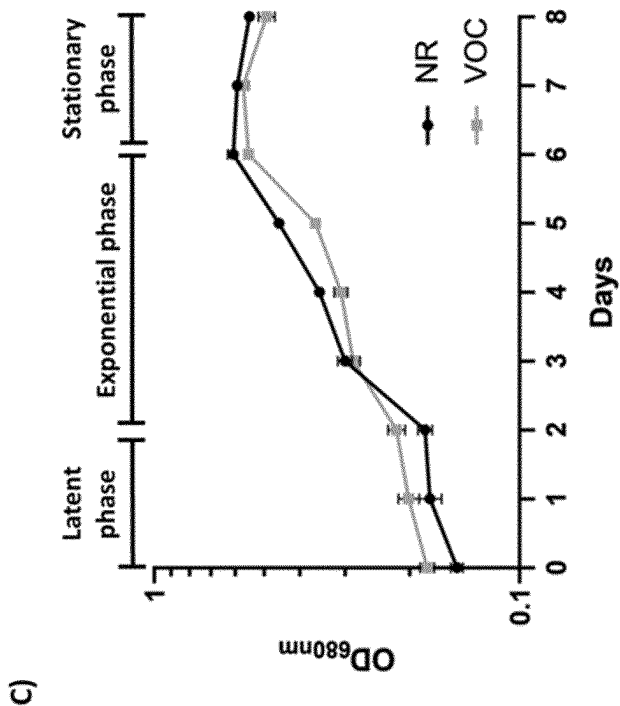


Figure 9

A

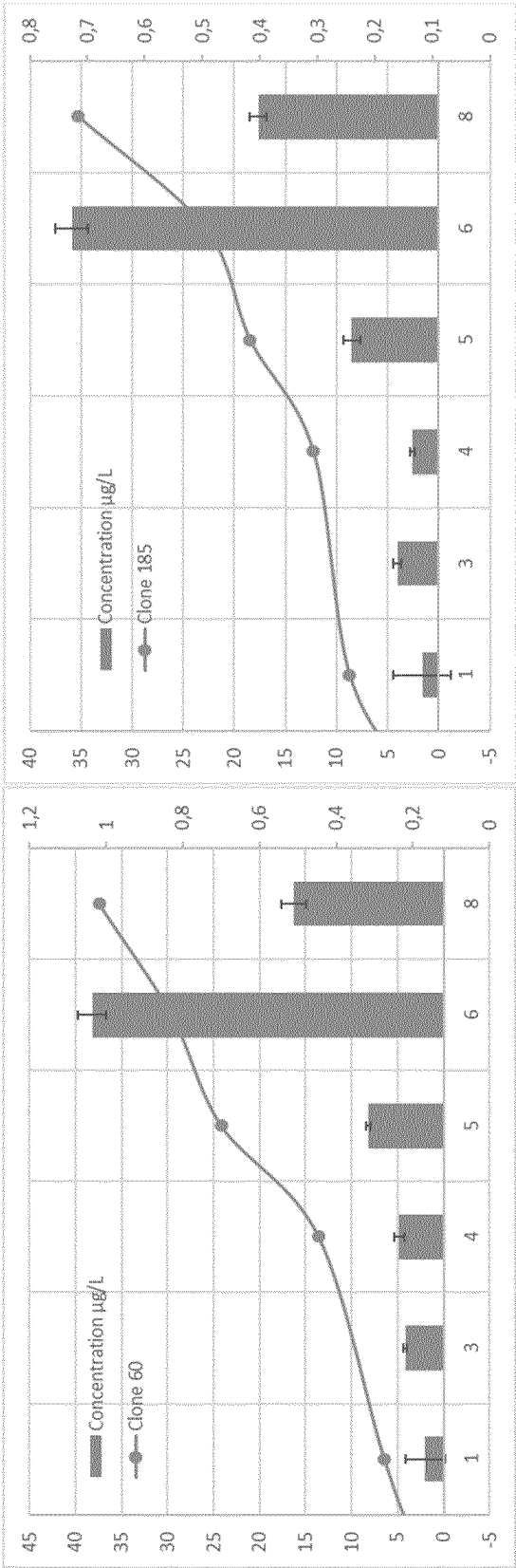
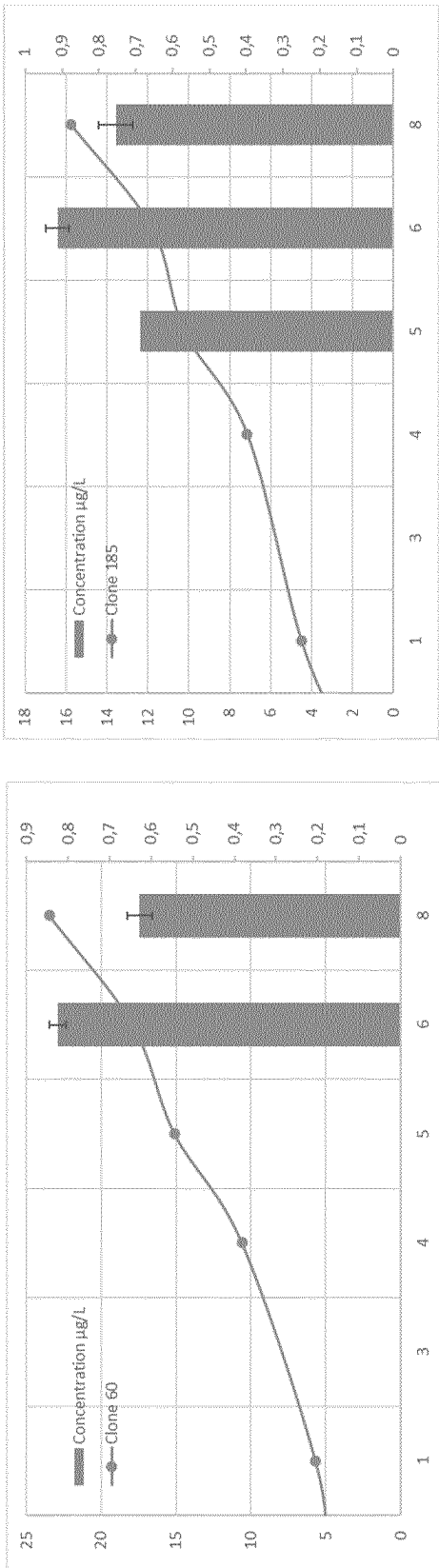


Figure 9 (continuation)



B

PRODUCTION OF THERAPEUTIC ANTIBODIES BY THE MICROALGAE PHAEODACTYLUM TRICORNUTUM

DESCRIPTION OF THE PRIOR ART

[0001] Monoclonal antibodies (mAbs) represent the most rapidly growing category of the recombinant therapeutic protein pipeline, with more than 85% of the therapeutic indications. In 2018, the overall sales of mAbs represented above US\$115 billion and is expected to increase up to US\$300 billion towards 2025. As of December 2019, 79 mAbs were granted approval by the US FDA and several hundred are still in developments or clinical trials worldwide.

[0002] mAbs are glycosylated and heterotetrameric proteins folded as Y shaped containing two Heavy Chains (HC) and two Light Chains (LC). mAbs are currently produced through a bioprocess that involved recombinant DNA technology, cell transfection, selection of clones and finally expansion of the best-producer clone within large-scale bioreactor to accumulate the mAbs secreted in the culture medium. The post-translational modifications like glycosylation are necessary for proper folding, oligomerization, stability, tissue targeting and biological function of the mAbs. Currently, the main manufacturers (Abbott/Abbvie, Sanofi, Merck, Johnson and Johnson, Bristol-Myers Squibb, Amgen, Regeneron Pharmaceuticals Inc., Novartis, Genentech/Roche, Pfizer, AstraZeneca . . .) are producing mAbs using mammalian cell lines as bio-factories, like for example Chinese Hamster Ovary (CHO) cell line. This is because of its growth behavior, its ability to excrete proteins in significant amounts and its human-like glycosylation.

[0003] However, after more than 40 years of optimization, production in CHO cells remains imperfect. For example, mAb produced in CHO cells are heterogeneous. Indeed, they do not have a homogenous glycosylation pattern but rather a mixture of glycoforms, which is not easily controlled and can vary between batches. Thus, metabolic engineering strategies have still to be implemented in CHO cells. Moreover, mAbs produced in CHO cells are subjected to proteolytic degradations, thus increasing the heterogeneity of the product. In addition, CHO cell expression system present other disadvantages like contamination risk with viruses and prions that could be harmful to humans, thus requiring numerous steps of purification and quality control. In routine manufacturing at a 10,000 L scale, culture media, complex nutritional requirement and fermentation as well as downstream processing steps contribute to the high manufacturing costs of mAbs using mammalian cells as bio-factories. Finally, mAbs produced in such conditions are very expensive.

[0004] Due to these limitations, there is interest to develop alternative expression systems for the production of mAbs in order to decrease their production costs, making them more accessible.

[0005] Alternative expression systems like plants have emerged. However, the production of biopharmaceuticals in tobacco plants require specific hands-on to transform and grow plants efficiently and require to build entirely new infrastructures and production unit including greenhouses.

[0006] Microalgae are photosynthetic organisms like plants while being unicellular microorganisms that can be grown as mammalian cells in self-contained systems like Petri dishes, flasks or bioreactors requiring cell culture

expertise. Culture media is estimated to be much lower (0.002 US\$ per liter) that is especially true for photosynthetic microalgae for which sunlight drives the production of biomass (Rosales-Mendoza et al., 2020). Furthermore, microalgae are less susceptible to virus harmful to humans (Yolken et al., 2014). In addition, significant improvement in genome-editing tools led recently to consider microalgae as alternative cell bio-factory for the production of recombinant proteins (Rosales-Mendoza et al., 2020).

[0007] Diatoms like *Phaeodactylum tricornutum* (later abbreviated as *P. tricornutum* or PHAEO) have been proposed to be used for the production of recombinant proteins. They are unicellular eukaryotic cells responsible for about 20% of the global carbon dioxide fixation and for biogeochemical cycling of important nutrients. *P. tricornutum* is believed to have arisen via a serial secondary endosymbiotic event in which a green alga and subsequently a red one were engulfed by a heterotrophic eukaryote, thus generating specific genomic characteristics. Indeed, a recent investigation of *P. tricornutum* genome revealed that 3,170 genes (26%) are unique and specific to this organism (Rastogi et al., 2018). *P. tricornutum* also harbors a combination of metabolic pathways that belongs to either the plant or animal kingdoms (Butler et al., 2020). This represents unique biological features, responsible for the great adaptability of *P. tricornutum* within the environment (Tirichine and Bowler, 2011) and its capacity to grow in different culture conditions such as hyposaline conditions, low temperature or low light (Gutenbrunner et al., 1994; Bartual et al., 2008; De Martino et al., 2007; De Martino et al., 2011). Thus, the flexible and unique nature of *P. tricornutum* offers immense possibilities to produce sustainable recombinant proteins.

[0008] *P. tricornutum* naturally synthesizes numerous compounds of interest like pigments, carotenoids, EPA and omega-3 fatty acids (Butler et al., 2020). Transformation methods and development of molecular tools, especially genetic engineering such as gene silencing, TALEN and Crispr/cas9, pave the way for biotechnological applications like the production of recombinant therapeutic proteins in *P. tricornutum* (Rosales-Mendoza et al., 2020). Indeed, *P. tricornutum* has been used to produce recombinant human anti-Marburg virus mAbs (Hempel et al., 2017) and functionally glycosylated human anti-hepatitis B mAbs (Hempel et al., 2011; Hempel and Maier, 2012; Vanier et al., 2015 and 2018). The quality of these anti-Hepatitis B mAb was confirmed using a Multi-Attribute Monitoring (MAM) approach, thus demonstrating that the production of mAbs in *P. tricornutum* was homogeneous and of good-quality. It was shown that the C-terminal ends of the heavy chains of the *P. tricornutum*-made mAbs remain intact in contrast to what is usually observed in mAbs produced in CHO cells. In addition, cleavage of the human signal peptide from the anti-Hepatitis B mAb produced in *P. tricornutum* was efficient, thus confirming that the protein secretion in *P. tricornutum* occurs as reported in mammals (Vanier et al., 2015). With regards to its biological activity, it was demonstrated that the *P. tricornutum*-made mAb is able to bind human Fcγ receptors including FcγRI and FcγRIIIa, which suggests that it could be used efficiently in human immunotherapy to induce phagocytosis and antibody-dependent cell mediated cytotoxicity (ADCC) response (Vanier et al., 2018). In addition, it was demonstrated that *P. tricornutum* is able to glycosylate the Fc parts of mAbs with oligomannosides bearing 5 to 9 mannose residues (Vanier et al., 2015). These oligomanno-

sides are structurally identical to those of mammals (Dumontier et al., 2021), which contrast to other microalgae like the green microalgae *Chlamydomonas reinhardtii* and *Chlorella vulgaris* (Toustou et al., 2022). Thus, *P. tricornutum* is able to produce and secrete mAbs exhibiting a human-like glycosylation.

[0009] Even successful, industrial exploitation is still limited due to the low amount of mAbs secreted within the culture medium. Thus, increasing the production yield is the bottleneck to unlock before any industrialization of *P. tricornutum* as a green eco-friendly expression system for the production of mAbs can be envisioned.

[0010] There is therefore a need to identify innovative strategies to increase the secretion and production yields of monoclonal antibodies in *P. tricornutum*.

[0011] The present inventors solve this need by proposing a unique improved *P. tricornutum* nucleotide construct encoding therapeutic monoclonal antibodies (for example Trastuzumab and Rituximab), and recombinant *P. tricornutum* strains containing said construct, said cells exhibiting an higher secretion and mAbs production yield while being robust at the industrial scale.

[0012] The example studies below have been carried out on two anti-cancer mAbs namely Trastuzumab and Rituximab. Both mAbs belong to the 21st WHO Model List of Essential Medicines (World Health Organization, 2019) and are used to treat cancers that in 2018 caused 9.6 million deaths with cancer worldwide incidence increasing to 18.1 million cases, representing multibillion dollar industries.

DESCRIPTION OF THE INVENTION

[0013] In order to increase the production yield and secretion of these mAbs in *P. tricornutum*'s culture medium, different promoter/terminator couples were tested. It is herein demonstrated that the V-ATPase C promoter/FcpA terminator pair can increase the yield of mAbs production in *P. tricornutum* in an 8-day bioprocess. It has also been highlighted that the use of the FcpA terminator is preferential to the endogenous terminator associated with the promoter for an increase in production yield. In parallel, the influence of different heterologous and endogenous signal peptides was tested on the secretion of antibodies in the culture medium. Surprisingly, the heterologous signal peptide "E" was shown to be more effective than the endogenous one of the HASP1 protein that is the most abundant secreted protein in *P. tricornutum*'s culture medium. These results contribute to the characterisation of various key elements to be considered in the vector design in order to increase the production yield of recombinant proteins in *P. tricornutum*.

[0014] Moreover, the selection of a strong promoter that drives high yields of mAbs is a prerequisite for developing a cost-effective production platform. To date, only few endogenous promoters have been characterized in *P. tricornutum*. In the present work, transcriptomic data from *P. tricornutum* Pt3 strain were analysed and thirty-three potential "strong" endogenous promoters were identified. First, these putative promoter sequences were cloned and fused to the coding sequence of the enhanced green fluorescent protein (eGFP). Their strengths were assessed by measuring eGFP fluorescence, thus reflecting the level of eGFP expression. Among the thirty-three promoters, thirteen were able to drive eGFP expression. Best results were obtained with the VOC promoter (Phatr3_J34085, SEQ ID NO:33) which

allowed a significant increase expression of eGFP by comparison to the one obtained with the inducible NR promoter. These findings contribute to the discovery of new genetic tools that could be used to improve recombinant protein production in *P. tricornutum*.

[0015] In the present studies, two antibodies were chosen as proof of concepts. The first one is a therapeutic mAb used to target human epidermal growth factor receptor 2 involved in HER2-positive breast and gastric cancers, named Trastuzumab. The second one is the Rituximab, an anti-CD20 recombinant mAb that has been commercialized to treat lymphoma. Its clinical use has now been extended to treat autoimmune diseases.

[0016] Altogether, the present invention provides a new system for producing high amounts of therapeutic antibodies, functional fragments or derivatives thereof, in microalgae. Said therapeutic antibodies, functional fragments or derivatives thereof are secreted in the extracellular medium of said microalgae, from which they can easily be purified.

[0017] In a first aspect, the present invention relates to a single polynucleotide vector expressing a therapeutic monoclonal antibody or a functional fragment or derivative thereof, said vector sequence comprising:

[0018] a) nucleotide sequences encoding, in the same or in two different Open Reading Frame(s):

[0019] the light chain of said therapeutic antibody, fragment or derivative thereof, preceded at its N-terminal end by a first heterologous secretion signal peptide,

[0020] the heavy chain of said therapeutic antibody, fragment or derivative thereof, preceded at its N-terminal end by a second heterologous secretion signal peptide,

[0021] said light and heavy chain being separated by a nucleotide sequence encoding a linker,

[0022] b) at least one promoter operatively linked with the sequences encoding said light and heavy chains, said promoter being able to drive the expression of said light and heavy chains in the microalgae *Phaeodactylum tricornutum*,

[0023] c) at least one translation termination sequence, and

[0024] d) optionally, at least one detectable marker.

[0025] All the important features of the vector of the invention will now be described one by one.

Secretion Signal Peptides Useful in the Vector of the Invention

[0026] The polynucleotide vector of the invention encodes at least two heterologous secretion signal peptides, said heterologous signal peptides enabling the secretion of the two different polypeptide chains of the therapeutic antibody, of the functional fragment or of the derivative thereof, into the extracellular environment, herein being the culture medium.

[0027] Preferably, these heterologous secretion signal peptides are located at the N-terminal end of each polypeptide chains, whose secretion is therefore independently regulated. These secretion signal peptides are typically 15-30 amino acids long.

[0028] In particular, the polynucleotide of the invention encodes:

[0029] the light chain of said therapeutic antibody, fragment or derivative thereof, preceded at its N-terminal end by a first heterologous secretion signal peptide,

[0030] the heavy chain of said therapeutic antibody, fragment or derivative thereof, preceded at its N-terminal end by a second heterologous secretion signal peptide.

[0031] In one embodiment, these first and second heterologous secretion signal peptides are identical.

[0032] In another embodiment, these first and second heterologous secretion signal peptides are different.

[0033] In the context of the invention, the secretion signal peptides can be chosen in the group consisting of:

[0034] Signal peptide E of azurocidine disclosed in Kober et al 2013 (SEQ ID NO:1)

[0035] Signal peptide B disclosed in Kober et al 2013 (SEQ ID NO:5).

[0036] Signal peptide B from Human Serum Albumin modified as described in Attalah et al., 2017 (SEQ ID NO:85)

[0037] Signal peptides disclosed in Table 1 of Kober et al 2013

[0038] Signal peptide H5 disclosed in Haryadi 2015 (SEQ ID NO:2)

[0039] Signal peptide H7 disclosed in Haryadi 2015 (SEQ ID NO:3)

[0040] Signal peptide L1 disclosed in Haryadi 2015 (SEQ ID NO:4)

[0041] Signal peptide H1, H2, H3, H4, H6, H8 and L2 by reference to Table 1 of Haryadi et al. 2015

[0042] Signal peptide of the human BIP protein (SEQ ID NO:6).

[0043] Signal peptide of the BIP protein from *Phaeodactylum tricornutum* (SEQ ID NO:7 or SEQ ID NO:8)

[0044] Signal peptide of the human PDI protein (SEQ ID NO:9)

[0045] Signal peptide of the PDI protein from *Phaeodactylum tricornutum* (SEQ ID NO:82)

[0046] Signal peptide used in Hempel and Maier 2012 which is incorporated by reference

[0047] Signal peptide used in Hempel et al., 2017 which is incorporated by reference

[0048] Signal peptide of gametolysin, disclosed in Ramos-Martinez et al., 2017.

[0049] The signal peptides found in proteins that are naturally secreted by *Phaeodactylum tricornutum*, identified recently in Chuberre et al, 2022, which is incorporated by reference.

[0050] In a preferred embodiment, the polynucleotide vector of the invention contains at least one secretion signal peptide chosen in the group consisting of: E of SEQ ID NO:1, H5 of SEQ ID NO:2, H7 of SEQ ID NO:3 or L1 of SEQ ID NO:4.

[0051] In a preferred embodiment, said first and second heterologous secretion signal peptides are L1 of SEQ ID NO:4 (preceding the sequence of the light chain) and H7 of SEQ ID NO:3 (preceding the sequence of the heavy chain) respectively.

[0052] In a preferred embodiment, said first and second heterologous secretion signal peptides are identical, and are

more preferably the E signal peptide of SEQ ID NO:1 (in other terms, both chains are regulated by the same signal peptide, which is E).

[0053] In a preferred embodiment, said first and second heterologous secretion signal peptides are followed by or include a sequence encoding a cleavage site, so that the signal peptides can be removed in the endoplasmic reticulum once their function has been fulfilled.

[0054] These signal peptides can be encoded by codon optimized sequences that are easily identifiable by the skilled person. For example, it is possible to use in the vectors of the invention any of:

[0055] SEQ ID NO:86 (Codon-optimized DNA sequence coding for signal peptide "E")

[0056] SEQ ID NO:87 (Codon-optimized DNA sequence coding for signal peptide "H5")

[0057] SEQ ID NO:88 (Codon-optimized DNA sequence coding for signal peptide "H7")

[0058] SEQ ID NO:89 (Codon-optimized DNA sequence coding for signal peptide "L1")

[0059] SEQ ID NO:90 (Codon-optimized DNA sequence coding for signal peptide of the human BIP)

[0060] SEQ ID NO:91 (Codon-optimized DNA sequence coding for signal peptide of the human PDI).

Promoters Useful in the Vector of the Invention

[0061] Promoters can be incorporated into a vector using standard techniques known in the art. In one embodiment, the promoter can be positioned at about the same distance from the transcription start site as it is from the transcription start site in its natural genetic environment. Some variation in this distance is permitted without substantial decrease in promoter activity. A transcription start site is typically included in the vector.

[0062] Exemplary promoters that can be used in the vector of the invention in order to drive the expression of the encoded polypeptides include, but are not restricted to, promoters such as:

[0063] The promoter of the H4-1B gene disclosed in Slattery et al, 2018 (SEQ ID NO:24).

[0064] The promoter of the V-ATPase C gene disclosed in Watanabe et al, 2018 (SEQ ID NO:25).

[0065] The endogenous promoter of the NR gene of *Phaeodactylum tricornutum* disclosed in Hempel et al., 2009 and in Stork et al., 2012 (SEQ ID NO:26).

[0066] The endogenous promoter of the fcpA/lhcf1 gene of *Phaeodactylum tricornutum* (disclosed in Bhaya and Grossman, 1993; Apt et al., 1995; Apt, Kroth-Pancic and Grossman, 1996; Falcioratore et al., 1999) (SEQ ID NO:27)

[0067] The endogenous promoter of the fopb/lhcf2 gene of *Phaeodactylum tricornutum* (disclosed in Bhaya and Grossman, 1993; Apt et al., 1995; Apt, Kroth-Pancic and Grossman, 1996; Falcioratore et al., 1999) (SEQ ID NO:28)

[0068] The promoter HASP1 as described in Erdene-Ochir et al., 2019 (SEQ ID NO:83).

[0069] The promoter 40SRPS8 as described in Slattery et al, 2018 (SEQ ID NO:84)

[0070] It is also possible to use any of the following promoters, which have been identified as they are associated to genes that are upregulated in *Phaeodactylum tricornutum* being subjected to saline stress (culture in 10% sea water):

[0071] Promoter of the Phatr3_J37038 gene (SEQ ID NO:29)

[0072] Promoter of the Phatr3_Jdraft1443 gene (SEQ ID NO:30)

[0073] Promoter of the Phatr3_J35102 gene (SEQ ID NO:31)

[0074] Promoter of the Phatr3_J50252 gene (SEQ ID NO:32)

[0075] Promoter of the Phatr3_J34085 gene (SEQ ID NO:33)

[0076] Promoter of the Phatr3_J48164 gene (SEQ ID NO:34)

[0077] Promoter of the Phatr3_EG02422 gene (SEQ ID NO:35)

[0078] Promoter of the Phatr3_J42538 gene (SEQ ID NO:36)

[0079] Promoter of the Phatr3_J48356 gene (SEQ ID NO:37)

[0080] Multiple copies of promoters or multiple promoters can be used in the vectors of the invention. For example, it is possible to associate parts of the above-disclosed promoters to construct chimeric promoter sequences that are as efficient as (or more efficient than) the original promoters.

[0081] In particular, it is possible to use vectors in which the coding sequence of each chain of the therapeutic antibody is operatively linked to a different promoter (FIG. 1B). In a particularly preferred case, the vector of the invention can contain an origin of replication (ORI) between the two sequences encoding each antibody chain (see for example the schemes of the vectors E) to J) on FIG. 1B).

[0082] It is possible to use vectors in which the two coding sequences are positioned in the same direction for translation (in this case, all the coding sequences can be located within the same Open Reading Frame, see vectors E) and H) of FIG. 1B), or in opposite direction for translation (in this case, the vector shall contain two Open Reading Frames). The vectors F, G, I and J proposed on FIG. 1B typically display the two coding sequences in opposite directions.

[0083] The two promoters are preferably chosen in the group consisting of: the promoter of the Nitrate Reductase gene from *Phaeodactylum tricornutum* (SEQ ID NO:26), the promoter of the FcpA gene from *Phaeodactylum tricornutum* (SEQ ID NO: 27), the promoter of the FcpB gene from *Phaeodactylum tricornutum* (SEQ ID NO:28), the promoter of the V-ATPase-C gene (SEQ ID NO:25), the promoter of the histone H4-1B gene (SEQ ID NO:24), the promoter of the Phatr3_J37038 gene (SEQ ID NO:29), the promoter of the Phatr3_Jdraft1443 gene (SEQ ID NO:30), the promoter of the Phatr3_J35102 gene (SEQ ID NO:31), the promoter of the Phatr3_J50252 gene (SEQ ID NO:32), the promoter of the Phatr3_J34085 gene (SEQ ID NO:33), the promoter of the Phatr3_J48164 gene (SEQ ID NO:34), the promoter of the Phatr3_EG02422 gene (SEQ ID NO:35), the promoter of the Phatr3_J42538 gene (SEQ ID NO:36), and the promoter of the Phatr3_J48356 gene (SEQ ID NO:37).

Self-Cleaving Linkers Used in the Vector of the Invention

[0084] When the two polypeptide chains are within the same ORF, they should be separated by a self-cleaving

“linker” sequence ensuring that the two chains of the same ORF are appropriately cleaved after their translation in the microalgae cells. This self-cleaving peptide is for example a 2A linker peptide, that is, a 18-22 amino acid long peptide which can induce ribosomal skipping during translation of the antibody chains in the microalgae cells. This linker peptide and others are described for example in Szymczak et al., 2004.

[0085] In a preferred embodiment, said self-cleaving linker sequence is chosen in the group consisting of:

[0086] Self-cleaving linker peptide E2A (SEQ ID NO:10)

[0087] Self-cleaving linker peptide F2A (SEQ ID NO:11)

[0088] Self-cleaving linker peptide variant of F2A (SEQ ID NO:12) from Zou et al 2018 or from Ho et al, 2013 (SEQ ID NO:13)

[0089] Self-cleaving linker peptide P2A (SEQ ID NO:14)

[0090] Self-cleaving linker peptide T2A (SEQ ID NO:15)

[0091] It is also possible to use any known means to induce appropriate cleavage, and in particular the Internal Ribosome Entry Site (IRES) elements that have successfully been used to induce monoclonal antibody expression in CHO cells (Ho et al., 2012, 2013 and Chng et al., 2015). Useful IRES linkers have for example the SEQ ID NO:16 and SEQ ID NO:17.

[0092] In a particular embodiment, it is advantageous to add a furin cleavage sequence upstream of the 2A linker peptide to eliminate two amino acids that would otherwise remain attached to the upstream antibody chain after cleavage. This furin-cleavage site has for example the sequence RRKR (SEQ ID NO:18). As disclosed in Chng et al., 2015, it would be also advantageous to add a GSG linker between the furin-cleavage site and the 2A linker, in order to enhance the cleavage efficiency and the expression of the antibodies. Repeated GSG sequences can be also used as preferred linkers.

Translation Termination Sequences Useful in the Vector of the Invention

[0093] The polynucleotide vector of the invention may contain other regulatory components that are functional in the microalgae cells in which the vector is to be expressed. Regulatory elements include, for example, 5' promoters, transcription factor binding sites, Ubiquitous Chromatin-Opening Elements, InRNAs, Kozak, transcription termination sequences, translation termination sequences, enhancers, and polyadenylation elements.

[0094] Translation termination are typically positioned downstream of a coding sequence to provide for efficient termination of the translation into the encoded polypeptide.

[0095] In the vector of the invention, it is possible to use, in particular, the following translation termination sequences:

[0096] Termination sequence of the FcpA gene as disclosed in Apt et al 1996 (SEQ ID NO:19)

[0097] Termination sequence of the NR gene as disclosed in Hempel et al., 2009 and Stork et al., 2012 (SEQ ID NO:20)

[0098] Termination sequence of the H4-1B gene as disclosed in Slattery et al 2018 (SEQ ID NO:21)

[0099] Termination sequence of the V-ATPase C gene (SEQ ID NO:22).

[0100] Termination sequence of the γ -tubulin gene as disclosed in Slattery et al 2018 (SEQ ID NO:23)

[0101] It is also possible to use other termination sequences, such as those cited in Slattery et al., 2018, notably the termination sequences of the EF1- α gene, of the 40SRPS8 gene, of the RBCMT gene, of the fcpB, fcpC, fcpD, or fcpF genes.

[0102] It is also possible to use other termination sequences, such as those cited in Windhagauer et al., 2021, notably the termination sequences of the NADH: Ubiquinone oxidoreductase gene, of the Synaptobrevin/VAMP-like protein gene, of the Phatr3_J45582 gene, or of the Prohoitin gene.

[0103] Finally, it is also possible to use the termination sequences of the genes whose promoters are recited in SEQ ID NO:29-37, namely the termination sequence of Phatr3_J37038 (SEQ ID NO:59), of Phatr3_Jdraft1443 (SEQ ID NO:60), of Phatr3_J35102 (SEQ ID NO:61), of Phatr3_J50252 (SEQ ID NO:62), of Phatr3_J34085 (SEQ ID NO:63), of Phatr3_J48164 (SEQ ID NO:64), of Phatr3_EG02422 (SEQ ID NO:65), of Phatr3_J42538 (SEQ ID NO:66), or of Phatr3_J48356 (SEQ ID NO:67).

[0104] In a preferred embodiment, the vector of the invention contains a translation termination sequence chosen among: the termination sequence of the Nitrate Reductase gene (SEQ ID NO:20) or the termination sequence of the FcpA gene of *Phaeodactylum tricornutum* (SEQ ID NO:19), the termination sequence of the V-ATPase-C gene (SEQ ID NO:22), the termination sequence of the γ -tubulin gene (SEQ ID NO:23), the termination sequence of the histone H4 gene (SEQ ID NO:21) and the termination sequence of the of Phatr3_J34085 gene (SEQ ID NO:63).

Coding Sequences for Therapeutic Molecules

[0105] The vector of the invention can encode any monoclonal therapeutic antibody.

[0106] In particular, it can encode any chimeric, humanized, or human monoclonal antibody, said antibody having anti-viral, anti-auto-immune or anti-cancer therapeutic action on human or on animal subjects.

[0107] In a particular embodiment, said antibody can be chosen in the group consisting of: rituximab, trastuzumab, adalimumab, bevacizumab, infliximab, cetuximab, motavizumab, palivizumab, alemtuzumab, dinutuximab, naxitamab but also comprising for instance, benralizumab, catumaxomab, daratumumab, elotuzumab, epratuzumab, farletuzumab, galiximab, gemtuzumab ozogamicin, ibritumomab tiuxetan, lumiliximab, necitumumab, nimotuzumab, ocrelizumab, ofatumumab, oregovomab, pertuzumab, tositumomab, zalutumumab, and zanolimumab. Any other useful antibodies can be used (see Walsh et al, 2018).

[0108] The polypeptide sequences of the constant and variable chains of all these antibodies are well known, as they have been published and well described. They do not need to be detailed here.

[0109] In a preferred embodiment, the vector of the invention encodes either the Trastuzumab (Herceptin) antibody or the Rituximab antibody or the dinutuximab antibody or the naxitamab antibody or any functional variant thereof.

[0110] For producing the antibody TrastuzumAb (Herceptin), the vector of the invention can contain the light chain of SEQ ID NO:38 and the heavy chain of SEQ ID NO:39 (chains without signal peptides).

[0111] For producing the antibody RituximAb, the vector of the invention can contain the light chain of SEQ ID NO:40 and the heavy chain of SEQ ID NO:41 (chains without signal peptides).

Particular Vectors of the Invention

[0112] In a preferred embodiment, the vector of the invention contains:

[0113] said promoter is the promoter of the Nitrate Reductase gene of SEQ ID NO:26 or the promoter of the V-ATPase-C gene of SEQ ID NO:25 or the promoter of the Phatr3_J34085 gene of SEQ ID NO:33 and

[0114] the termination sequence of the FcpA gene of *Phaeodactylum tricornutum* of SEQ ID NO:19.

[0115] In a more preferred embodiment, the vector of the invention contains, in this order:

[0116] the promoter of V-ATPase C gene of SEQ ID NO:25 or the promoter Phatr3_J34085 gene (SEQ ID NO:33),

[0117] the sequence encoding the secretion peptide signal E of SEQ ID NO:1,

[0118] the sequence encoding the light chain of said therapeutic antibody, fragment or derivative thereof,

[0119] the sequence encoding the linker T2A of SEQ ID NO:15 or others variants,

[0120] the sequence encoding the secretion peptide signal E of SEQ ID NO:1,

[0121] the sequence encoding the heavy chain of said therapeutic antibody, fragment or derivative thereof,

[0122] the termination sequence of the FcpA gene of *Phaeodactylum tricornutum* of SEQ ID NO:19.

[0123] The following vectors have been successfully generated by the inventors for producing Herceptin (TrastuzumAb). As shown in the examples below (and FIGS. 2-4), they are able to produce high yields of said monoclonal antibody in *Phaeodactylum tricornutum* cells of different strains.

Vector NVB5	promoter of NR- LC of herceptin-T2A-HC of herceptin -fcpA terminator	SEQ ID NO: 42
Vector NVB6	promoter of H4-1B- LC of herceptin -T2A-HC of herceptin - pH4-1b terminator	SEQ ID NO: 43
Vector NVB7	promoter of V-ATPase-C- LC of herceptin -T2A-HC of herceptin - V-ATPase-C terminator	SEQ ID NO: 44
Vector C	promoter NR-HC of herceptin -T2A-LC of herceptin - NR terminator	SEQ ID NO: 45
Vector D	promoter NR-LC of herceptin -T2A-HC of herceptin - NR terminator	SEQ ID NO: 46

-continued

Vector Pro1D	Promoter 37038 + LC of herceptin -T2A-HC of herceptin - NR terminator	SEQ ID NO: 47
Vector Pro2D	Promoter D1443 + LC of herceptin -T2A-HC of herceptin - NR terminator	SEQ ID NO: 48
Vector Pro3D	Promoter 35102 + LC of herceptin -T2A-HC of herceptin - NR terminator	SEQ ID NO: 49
Vector Pro4D	Promoter 50252 + LC of herceptin -T2A-HC of herceptin - NR terminator	SEQ ID NO: 50
Vector Pro5D	Promoter 34085 + LC of herceptin -T2A-HC of herceptin - NR terminator	SEQ ID NO: 51
Vector Pro6D	Promoter 48164 + LC of herceptin -T2A-HC of herceptin - NR terminator	SEQ ID NO: 52
Vector Pro7D	Promoter EG02422 + LC of herceptin -T2A-HC of herceptin - NR terminator	SEQ ID NO: 53
Vector Pro7D + Kozak	Promoter EG02422 + Kozak sequence + LC herceptin -T2A-HC herceptin - NR terminator	SEQ ID NO: 54

[0124] The following vectors have been successfully generated by the inventors for producing RituximAb. As shown in the examples below (NVB12, NVB13 and NVB16 on FIGS. 5 and 7), they are able to produce high yields of said monoclonal antibody in *Phaeodactylum tricornutum* cells of different strains (Pt1 to 10, Rastogi et al, 2019).

present invention via a plasmid, virus sequences, double or simple strand DNA, circular or linear DNA.

[0132] It is generally desirable to include into each nucleotide sequences or vectors at least one selectable marker to allow the selection of the microalgae cells that have been stably transformed. Examples of such markers are antibiotic

Vector NVB12	promoter of NR- LC of RituximAb -T2A-HC of RituximAb -NR terminator	SEQ ID NO: 55
Vector NVB13	promoter of NR- LC of RituximAb -T2A-HC of RituximAb - fcpA terminator	SEQ ID NO: 56
Vector NVB14	promoter of NR- L1-LC of RituximAb -T2A-H7-HC of RituximAb - NR terminator	SEQ ID NO: 57
Vector NVB16	promoter of NR- E-LC of RituximAb -T2A-E-HC of RituximAb - NR terminator	SEQ ID NO: 58

Recombinant Cells of the Invention

[0125] A second aspect of the invention concerns transformed or recombinant microalgae cells containing the polynucleotide vector of the invention, as disclosed above.

[0126] Said microalgae cells according to the invention are selected from the genus *Tetraselmis*, *Porphyridium*, *Symbiodinium*, *Thalassiosira*, *Nannochloropsis*, *Emiliania*, *Pavlova*, *Isochrysis*, *Eutreptiella*, *Euglena*, *Phaeodactylum*, *Dunaliella*, *Chlorella*, *Haematococcus*, and *Scenedesmus*.

[0127] In a preferred embodiment, said microalgae cells according to the invention are selected among the species *Phaeodactylum tricornutum*, *Nannochloropsis oculata*, *Isochrysis galbana*, *Tetraselmis suecica*, *Dunaliella salina*, *Chlorella*, *Scenedesmus*, and *Hematococcus*.

[0128] In an even more preferred embodiment, said microalgae cells correspond to *Phaeodactylum tricornutum*, and are from the strains Pt1, Pt2, Pt3, Pt4, Pt5, Pt6, Pt7, Pt8, Pt9 or Pt10.

[0129] In a most preferred embodiment, said microalgae cells are transformed cells of *Phaeodactylum tricornutum* from the strains Pt1, Pt3, Pt4, Pt5 and Pt7. The inventors indeed demonstrated that the vectors of the invention produce high levels of monoclonal antibodies from Pt1 and Pt3 cells (example 2).

[0130] Transformation of the microalgae cells of the invention can be carried out by conventional methods such as microparticles bombardment (Biolistic), electroporation, bacterial conjugation leading to episomal expression.

[0131] In an embodiment of the invention, nucleotide sequences may be introduced into the microalgae cells of the

resistant genes such as sh ble gene enabling resistance to zeocin, nat or sat-1 genes enabling resistance to nourseothricin, aph8 enabling resistance to paromomycin, nptII enabling resistance to G418. The use of auxotrophic strains of *P. tricornutum* can also be used for the selection of microalgae that have been transformed (both stably or transiently). Auxotrophic strains for uracil, tryptophan and histidine have been generated as described in literature, but auxotrophy to leucin or methionine can also be considered. The generation of strains with particular genes inactivated can also allow positive selection of transformed strains. For example, inactivation of the umps and apt genes allows resistance to 5-FOA (5-Fluoroorotic acid) and 2-FA (2-fluoroadenine) respectively.

[0133] After transformation of microalgae of the present invention, transformants producing the desired therapeutic antibodies, functional fragments or derivatives thereof secreted in the culture media are selected. Selection can be carried out by one or more conventional methods comprising: enzyme-linked immunosorbent assay (ELISA), mass spectroscopy such as MALDI-TOF-MS, ESI-MS chromatography, spectrophotometer, fluorimeter, immunocytochemistry by exposing cells to an antibody having a specific affinity for the desired therapeutic antibodies, functional fragments or derivatives thereof.

Optimized Methods to Produce Therapeutic Molecules Using the Vectors of the Invention

[0134] In another aspect, the invention relates to a method for producing a therapeutic antibody, a functional fragment or a derivative thereof which is secreted in the extracellular medium, said method comprising the steps of:

[0135] (i) culturing the transformed microalgae cells of the present invention as described here above;

[0136] (ii) harvesting the extracellular medium of said culture; and

[0137] (iii) purifying the therapeutic antibody, a functional fragment or a derivative thereof, which is secreted in said extracellular medium.

[0138] The method of claim 13, wherein said appropriate culture conditions involve the use of a culture medium containing a nitrate concentration comprised between 0.1 g/L and 1 g/L and low salt concentration comprised between 5 to 30%.

[0139] In a particular example, the strain of *P. tricornutum* can be cultivated between 15° C. and 25° C., preferably at 19° C. It can be cultivated for example on a photoperiod of 16/8 h light/dark cycle with light intensity of 30 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$. It can be cultivated either on agar plates or in liquid cultures under agitation at 150 rpm. The cells can be grown in 100% artificial seawater (33.3 g·L⁻¹ of sea salt (Instant Ocean®, Aquarium System, Sarrebourg, France)) filtered through 0.45 μm filters and sterilized by autoclave. The culture can last 15 days, preferentially between 3 and 10 days, more preferentially 8 days.

[0140] They are preferably cultivated in a sterilized culture medium containing between 5% to 30% sea water. This medium should be supplemented with 1 mL·L⁻¹ of Conway NaNO₃ nutritive medium (said nutritive medium containing Na₂EDTA·2H₂O:45 g·L⁻¹; NaNO₃:100 g·L⁻¹; H₃BO₃:33.6 g·L⁻¹; NaH₂PO₄:20 g·L⁻¹; FeCl₃:0.768 g·L⁻¹; ZnCl₂:21 mg·L⁻¹; CoCl₂, 6H₂O:20 mg·L⁻¹; (NH₄)₆Mo₇O₂₄, 4H₂O:9 mg·L⁻¹; CuSO₄, 5H₂O:20 mg·L⁻¹; MnCl₂, 4H₂O:360 mg·L⁻¹; B₁ vitamin:200 mg·L⁻¹; B₁₂ vitamin:10 mg·L⁻¹). Optionally 80 mg·L⁻¹ of sodium metasilicate (Na₂SiO₃) can be added.

[0141] In a preferred embodiment, the Man and Meyers medium (M&M medium) is used to cultivate the strains of *P. tricornutum*.

[0142] The composition of the M & M medium is given below:

Mann and Myers' medium		
	Component	Concentration
Salinity	NaCl	5 g/L
	MgSO ₄ ·7H ₂ O	1.2 g/L
	KCl	0.6 g/L
	CaCl ₂	0.3 g/L
Nitrate	NaNO ₃	11.76 mM
Phosphate	K ₂ HPO ₄	0.574 mM
Tris	Tris	1.0 g/L
Trace element	Na ₂ EDTA	8.92 × 10 ⁻⁵ M
	H ₃ BO ₃	9.70 × 10 ⁻⁵ M
	FeSO ₄ ·7H ₂ O	7.19 × 10 ⁻⁶ M
	MnCl ₂	1.11 × 10 ⁻⁵ M
	ZnSO ₄ ·7H ₂ O	1.15 × 10 ⁻⁶ M
	Co(NO ₃) ₂ ·6H ₂ O	2.40 × 10 ⁻⁸ M
	CuSO ₄ ·5H ₂ O	8.01 × 10 ⁻⁹ M

Definitions

[0143] The term “polynucleotide” used herein refers to DNA molecules (e.g., cDNA or genomic or synthetic DNA) and RNA molecules (e.g., mRNA or synthetic RNA), as well as analogs of DNA or RNA containing non-natural nucleotide analogs, non-native internucleoside bonds, or both. Preferably, said polynucleotide is a DNA molecule. The polynucleotide can be in any topological conformation, like linear or circular.

[0144] “Operatively linked” promoter refers to a linkage in which the promoter is contiguous with the gene of interest to control the expression of said gene.

[0145] The term “signal peptide” as used herein refers to an amino acid sequence which is generally located at the amino terminal end of the amino acid sequence of a therapeutic antibody or functional fragment thereof. The signal peptide mediates the translocation of said therapeutic antibody, functional fragment or derivative thereof through the secretion pathway and leads to the secretion of said therapeutic antibody, functional fragment or derivative thereof in the extracellular medium.

[0146] As used herein, the term “secretion pathway” refers to the process used by a cell to secrete proteins out of the intracellular compartment. Such pathway comprises a step of translocation of a polypeptide across the endoplasmic reticulum membrane, followed by the transport of the polypeptide in the Golgi apparatus, said polypeptide being subsequently released in the extracellular medium of the cell by secretory vesicles. Post-translational modifications necessary to obtain mature proteins, such as glycosylation or disulfide bonds formation, are operated on proteins during said secretion pathway.

[0147] The term “heterologous”, with reference to a signal peptide according to the invention, means an amino acid sequence which does not exist in the corresponding microalga before its transformation, and preferably originate from another organism (not a microalga). It is intended that the term encompasses proteins that are encoded by wild-type genes, mutated genes, and/or synthetic genes.

[0148] The term “antibody” is used herein in the broadest sense and specifically covers monoclonal antibodies (including full length monoclonal antibodies) of any isotype such as IgG, IgM, IgA, IgD, and IgE, polyclonal antibodies, multi-specific antibodies, chimeric antibodies, and antigen-binding fragments. An antibody reactive with a specific antigen can be generated by recombinant methods such as selection of libraries of recombinant antibodies in phage or similar vectors, or by immunizing an animal with the antigen or an antigen-encoding nucleic acid.

[0149] In a preferred embodiment, the antibodies of the invention are monoclonal antibodies. As used herein, the term “monoclonal antibody” refers to an antibody arising from a nearly homogeneous antibody population. More particularly, the individual antibodies of a population are identical except for a few possible naturally-occurring mutations which can be found in minimal proportions. In other words, a monoclonal antibody consists of a homogeneous antibody population arising from the growth of a single cell clone (for example a hybridoma, a eukaryotic host cell transfected with a DNA molecule coding for the homogeneous antibody, a prokaryotic host cell transfected with a DNA molecule coding for the homogeneous antibody, etc.) and is generally characterized by heavy chains of one and only one class and subclass, and light chains of only one

type. Monoclonal antibodies are highly specific and are directed against a single antigen. An “antigen” is a pre-determined molecule to which an antibody can selectively bind. The target antigen may be a polypeptide, a carbohydrate, a nucleic acid, a lipid, a hapten or any other naturally occurring or synthetic compound. Preferably, the target antigen is a polypeptide.

[0150] A typical antibody is comprised of two light chains and two heavy chains that are joined by disulfide bonds. As used in the invention, the term “light chain” refers to mammalian immunoglobulin light chain, lambda (λ) or kappa (κ), having two successive domains: one constant domain and one variable domain. As used in the invention, the term “heavy chain” refers to chain of mammalian immunoglobulin denoted by: alpha (α), delta (δ), epsilon (ϵ), gamma (γ), and mu (μ). Each heavy chain has two regions, the constant region and the variable region. The constant region is identical in all antibodies of the same isotype. The variable region of each heavy chain is composed of a single Ig domain. The “variable region” or “variable domain” of an antibody refers to the amino-terminal domains of the heavy or light chain of the antibody. The variable domain of the heavy chain may be referred to as “ V_H ”. The variable domain of the light chain may be referred to as “ V_L ”. These domains are generally the most variable parts of an antibody and contain the antigen-binding sites. Each variable region contains three segments called “complementarity-determining regions” (“CDRs”) or “hypervariable regions” (“HVRs”), which are primarily responsible for binding an epitope of an antigen and are interspersed with regions that are more conserved, designated “Framework Regions” (FR). The CDRs thus direct the specificity of the binding of the antibody. They are usually referred to as CDR1, CDR2, and CDR3, numbered sequentially from the N-terminus.

[0151] Each V_H and V_L is composed of three CDRs and four FRs, arranged from amino-terminus to carboxy-terminus in the following order: FR1, CDR1, FR2, CDR2, FR3, CDR3, FR4. The assignment of amino acid sequences to each domain is in accordance with well-known conventions (for example, the IMGT unique numbering convention as disclosed by Lefranc, M.-P., et al., *Dev. Comp. Immunol.*, 27, 55-77 (2003)). The functional ability of the antibody to bind a particular antigen depends on the variable regions of each light/heavy chain pair, and is largely determined by the CDRs. The variable region of the heavy chain differs in antibodies produced by different B cells, but is the same for all antibodies produced by a single B cell or B cell clone (or hybridoma). By contrast, the constant regions of the antibodies mediate the binding of the immunoglobulin to host tissues or factors, including various cells of the immune system (e.g. effector cells) and the first component (C1q) of the classical complement system.

[0152] The term “chimeric antibody” as used herein refers to an antibody containing a natural variable region (light chain and heavy chain) derived from an antibody of a given species in combination with constant regions of the light chain and of the heavy chain of an antibody of a species heterologous to said given species. Thus, a “chimeric antibody”, as used herein, is an antibody in which the constant region, or a portion thereof, is altered, replaced, or exchanged, so that the variable region is linked to a constant region of a different species, or belonging to another antibody class or subclass. “Chimeric antibody” also refers to an

antibody in which the variable region, or a portion thereof, is altered, replaced, or exchanged, so that the constant region is linked to a variable region of a different species, or belonging to another antibody class or subclass. Such chimeric antibodies, or fragments of same, can be prepared by recombinant engineering. For example, the chimeric antibody could be produced by cloning recombinant DNA containing a promoter and a sequence coding for the variable region of a non-human monoclonal antibody of the invention, notably murine, and a sequence coding for the human antibody constant region. A chimeric antibody according to the invention coded by one such recombinant gene could be, for example, a mouse-human chimera, the specificity of this antibody being determined by the variable region derived from the murine DNA and its isotype determined by the constant region derived from human DNA.

[0153] As used herein, the term “humanized antibody” refers to a chimeric antibody which contain minimal sequence derived from non-human immunoglobulin. In one embodiment, a humanized antibody is a human immunoglobulin (recipient antibody) in which residues from a CDR of the recipient are replaced by residues from a CDR of a non-human species (donor antibody) such as mouse, rat, rabbit or non-human primate having the desired specificity, affinity, and/or capacity. In some instances, framework (“FR”) residues of the human immunoglobulin are replaced by corresponding non-human residues. Furthermore, humanized antibodies may comprise residues that are not found in the recipient antibody or in the donor antibody. These modifications may be made to further refine antibody performance, such as binding affinity. In general, a humanized antibody will comprise substantially all of at least one, and typically two, variable domains, in which all or substantially all of the hypervariable loops correspond to those of a non-human immunoglobulin sequence, and all or substantially all of the FR regions are those of a human immunoglobulin sequence, although the FR regions may include one or more individual FR residue substitutions that improve antibody performance, such as binding affinity, isomerization, immunogenicity, etc. The number of these amino acid substitutions in the FR are typically no more than 6 in the Heavy (H) chain, and in the Light (L) chain, no more than 3. The humanized antibody optionally will also comprise at least a portion of an immunoglobulin constant region (Fc), typically that of a human immunoglobulin.

[0154] An “antibody fragment” comprises a portion of an intact antibody, preferably the antigen binding and/or the variable region of the intact antibody. Examples of antibody fragments include Fab, Fab', F(ab')₂ and Fv fragments; diabodies; linear antibodies (see U.S. Pat. No. 5,641,870); single-chain antibody molecules and multispecific antibodies formed from antibody fragments. Papain digestion of antibodies produced two identical antigen-binding fragments, called “Fab” fragments, and a residual “Fc” fragment, a designation reflecting the ability to crystallize readily. The Fab fragment consists of an entire L chain along with the variable region domain of the H chain (V_H), and the first constant domain of one heavy chain (C_{H1}). Each Fab fragment is monovalent with respect to antigen binding, i.e., it has a single antigen-binding site. Pepsin treatment of an antibody yields a single large F(ab')₂ fragment which roughly corresponds to two disulfide linked Fab fragments having different antigen-binding activity and is still capable of cross-linking antigen. Fab' fragments differ from Fab

fragments by having a few additional residues at the carboxy terminus of the C_H1 domain including one or more cysteines from the antibody hinge region. Fab'-SH is the designation herein for Fab' in which the cysteine residue(s) of the constant domains bear a free thiol group. F(ab')₂ antibody fragments originally were produced as pairs of Fab' fragments which have hinge cysteines between them. Other chemical couplings of antibody fragments are also known. The term "Fv" as used herein refers to the minimum antibody fragment which contains a complete antigen-recognition and -binding site. This fragment consists of a dimer of one heavy- and one light-chain variable region domain in tight, non-covalent association. From the folding of these two domains emanate six hypervariable loops (3 loops each from the H and L chain) that contribute the amino acid residues for antigen binding and confer antigen binding specificity to the antibody. However, even a single variable domain (or half of an Fv comprising only three hypervariable regions specific for an antigen) has the ability to recognize and bind antigen, although at a lower affinity than the entire binding site. "Single-chain Fv" also abbreviated as "sFv" or "scFv" are antibody fragments that comprise the V_H and V_L antibody domains connected into a single polypeptide chain. Preferably, the sFv polypeptide further comprises a polypeptide linker between the V_H and V_L domains which enables the sFv to form the desired structure for antigen binding.

[0155] In a preferred embodiment, the antibody or fragment of the invention is multispecific, and in particular bispecific. As such, it can be chosen in the group consisting of: bispecific IgGs, IgG-scFv₂, (scFv)₄-IgG, (Fab')₂, (scFv)₂, (dsFv)₂, Fab-scFv fusion proteins, (Fab-scFv)₂, (scFv)₂-Fab, (scFv- C_H2 - C_H3 -scFv)₂, bibody, tribody, bispecific diabody, disulfide-stabilized (ds) diabody, 'knob-into whole' diabody, single-chain diabody (scDb), tandem diabody (TandAb), flexibody, DiBi miniantibody, [(scFv)₂-Fc]₂, (scDb- C_H3)₂, (scDb-Fc)₂, Di-diabody, Tandemab., etc. or any combination.

[0156] "Functional fragments" of the antibodies of the invention comprise a portion of an intact antibody, generally including the antigen binding or variable region of the intact antibody or the Fc region of an antibody which retains or has modified FcR binding capability. Examples of antibody fragments include linear antibody, single-chain antibody molecules and multispecific antibodies formed from antibody fragments.

[0157] As used herein, the term "derivative" refers to a polypeptide having a percentage of identity of at least 75% with the amino acid sequence of the therapeutic antibody or functional fragment thereof as disclosed previously and having the same activity. Preferably, a derivative has a percentage of identity of at least 75%, at least 80%, at least 85%, at least 90% and more preferably of at least 95% with said amino acid sequence, and preferably of at least 99% with said amino acid sequence.

[0158] The identity percentage between said two derivative sequences can be identified by a global alignment of the sequences in their entirety (e.g., when the sequences are of about the same size). This alignment can be performed by means of an algorithm that is well known by the skilled person, such as the one disclosed in Needleman and Wunsch (1970). Accordingly, sequence comparisons between two amino acid sequences or two nucleotide sequences can be performed for example by using any software known by the

skilled person, such as the "needle" software using the "Gap open" parameter of 10, the "Gap extend" parameter of 0.5 and the "Blosum 62" matrix.

[0159] When local alignment of the sequences is to be considered (e.g., in case of derivatives that have a smaller size than the sequences of the invention), then said alignment can be performed by means of a conventional algorithm such as the one disclosed in Smith and Waterman (J. Mol. Evol. 1981; 18 (1) 38-46).

[0160] The invention also provides vectors encoding homologous amino acid sequences that are "similar" to the ones recited herein. "Similarity" of two targeted amino acid sequences can be determined by calculating a similarity score for the two amino acid sequences. As used herein, the "similarity score" refers to the score generated for the two sequences using the BLOSUM62 amino acid substitution matrix, a gap existence penalty of 11, and a gap extension penalty of 1, when the two sequences are optimally aligned. Two sequences are "optimally aligned" when they are aligned so as to produce the maximum possible score for that pair of sequences, which might require the introduction of gaps in one or both of the sequences to achieve that maximum score. Two amino acid sequences are substantially similar if their similarity score exceeds a certain threshold value. The threshold value can be any integer ranging from at least 1190 to the highest possible score for a particular reference sequence. For example, the threshold similarity score can be 1190, 1200, 1210, 1220, 1230, 1240, 1250, 1260, 1270, 1280, 1290, 1300, 1310, 1320, 1330, 1340, 1350, 1360, 1370, 1380, 1390, 1400, 1410, 1420, 1430, 1440, 1450, 1460, 1470, 1480, 1490, 1500, or higher. Amino acid substitution matrices and their use in quantifying the similarity between two sequences are well-known in the art and described, e.g., in Dayhoff et al. (1978), "A model of evolutionary change in proteins", "Atlas of Protein Sequence and Structure," Vol. 5, Suppl. 3 (ed. M. O. Dayhoff), pp. 345-352. Natl. Biomed. Res. Found., Washington, D.C. and in Henikoff et al. (1992) Proc. Natl. Acad. Sci. USA 89:10915-10919. While optimal alignment and scoring can be accomplished manually, the process is facilitated by the use of a computer-implemented alignment algorithm, e.g., gapped BLAST 2.0, described in Altschul et al., (1997) Nucleic Acids Res. 25:3389-3402, and made available to the public at the National Center for Biotechnology Information website. To generate accurate similarity scores using NCBI BLAST, it is important to turn off any filtering, e.g., low complexity filtering, and to disable the use of composition-based statistics. One should also confirm that the correct substitution matrix and gap penalties are used. Optimal alignments, including multiple alignments, can be prepared using, e.g., PSI-BLAST, available through the NCBI internet site and described by Altschul et al., (1997) Nucleic Acids Res. 25:3389-3402.

[0161] The vector of the invention can contain a detectable marker. This marker confers resistance to the selection and allows cells to stably integrate the plasmid into a chromosome and be expanded into a cell line. Usually this detectable marker is an antibiotic resistance gene, such as ZeocinR or nourseothricin omeycin resistance gene. The use of auxotrophic strains of *P. tricornutum* can also be used for the selection of microalgae that have been transformed (both stably or transiently). Auxotrophic strains for uracil, tryptophan and histidine have been generated as described in literature, but auxotrophy to leucine or methionine can also be

considered. The generation of strains with particular genes inactivated can also allow positive selection of transformed strains. For example, inactivation of the *umps* and *apt* genes allows resistance to 5-FOA (5-Fluoroorotic acid) and 2-FA (2-fluoroadenine) respectively.

FIGURE LEGENDS

[0162] FIG. 1—Structure of different vectors according to the invention

[0163] FIGS. 2—H4-1B and V-ATPase C promoters are able to drive expression of Trastuzumab in the diatom *P. tricornutum*. A. Design of the constructs used to produce Trastuzumab (TRA) in *P. tricornutum* under the control of the H4-1B (Slattery et al., 2018) and the V-ATPase C (Watanabe et al., 2018) promoters. B. Expression of trastuzumab under the control of each promoter/terminator couple was assessed by ELISA after 8 and 15 days of culture on at least 29 independent clones. The results are expressed in $\mu\text{g/L}$. C. Output is represented as pg of mAbs produced per cell. Statistical differences between each construction were analysed by a two-way ANOVA using the Tukey's multiple comparisons test. ns corresponds to no statistical difference observed here ($p\text{-value}>0.05$), one asterisk (*) corresponds to a $p\text{-value}<0.05$.

[0164] FIG. 3—Evaluation of the influence of different terminators on Trastuzumab expression driven by the H4-1B and V-ATPase C promoters in *P. tricornutum*. A. Three different terminators were tested in combination with the H4-1B (Slattery et al. 2018) and V-ATPase C promoters (Watanabe et al., 2018): the native H4-1B terminator, the native V-ATPase C terminator, the NR terminator and the *fcfA* terminator. B. Expression of trastuzumab under the control of each promoter/terminator couple was assessed by ELISA after 8 and 15 days of culture, respectively, on at least 31 independent clones and is represented as pg of mAbs produced per cell. Statistical differences between each construction were analysed by a two-way ANOVA using the Tukey's multiple comparison test: ns corresponds to a $p\text{-value}>0.05$; one asterisk (*) corresponds to a $p\text{-value}<0.05$; two asterisks (**) corresponds to a $p\text{-value}<0.005$.

[0165] FIG. 4—Comparison of HA-1B, V-ATPase C and NR promoter strengths on Trastuzumab expression in *P. tricornutum*. A. Three different vectors were used to compare the strengths of each promoter in combination with the *fcfA* terminator. B. Expression of Trastuzumab under the control of each promoter was assessed by ELISA after 8 and 15 days of culture, respectively, on 31 independent clones. Output is represented as pg of mAbs produced per cell. Statistical differences between each construction were analysed by a Mann-Whitney-Wilcoxon test: ns corresponds to a $p\text{-value}>0.05$; one asterisk (*) corresponds to a $p\text{-value}<0.05$. C. Comparison of the expression of Trastuzumab under the control of each promoter at day 8. ELISA output is represented as pg of mAbs produced per cell. Statistical differences between each construction were analysed by a two-way ANOVA using the Tukey's multiple comparison test: ns corresponds to a $p\text{-value}>0.05$; one asterisk (*) corresponds to a $p\text{-value}<0.05$.

[0166] FIG. 5—Confirmation of the better efficiency of the NR promoter/*fcfA* terminator couple on mAb production yield using the Rituximab. A. The production yield of Rituximab under the control of NR promoter was evaluated in combination with either the NR or *fcfA* terminators. B. Expression of rituximab under the control of each different

promoter was assessed by ELISA after 8 days of culture on 36 independent clones. Output is represented as pg of mAbs produced per cell. Statistical difference was analysed by a Mann-Whitney-Wilcoxon test: one asterisk (*) corresponds to a $p\text{-value}<0.05$.

[0167] FIG. 6—Evaluation of the influence of different heterologous signal peptides (SP) or glycomodule on the secretion of eGFP produced under the control of the NR promoter in *P. tricornutum*. A. Schematic diagrams of the expression cassettes used. Six different vectors were constructed to evaluate the influence of the heterologous SP E (Kober et al., 2013), H5, H7, L1 (Haryadi et al., 2015) and G20 glycomodule (Ramos-Martinez et al., 2017). The peptide sequences of the different heterologous SP E, H5, H7 and L1 as well as the one from the SP20 C-terminal tail are indicated. A SP/G20-less vector was used as a control. B. Relative fluorescence intensity of the eGFP measured on an 8-days culture, either on the total culture or on the cell free medium (CFM) after removing the cells. Error bars represent the standard deviation (SD, $n=9$). Statistical differences were analysed by a two-way ANOVA using Sidak's multiple comparison's test: ns corresponds to a $p\text{-value}>0.05$; three asterisks (***) correspond to $p\text{-value}<0.0001$.

[0168] FIG. 7—Evaluation of the influence of the different heterologous and endogenous signal peptides (SP) on the production of Trastuzumab and Rituximab under the control of the NR promoter in *P. tricornutum*. A. Six different vectors were constructed to evaluate the influence of the endogenous SP HASP1 (Erdene-Ochir et al., 2019) and heterologous SP E (Kober et al., 2013), H7, L1 (Haryadi et al., 2015). The SP HASP1 and E were added on both HC and LC of Trastuzumab and Rituximab. The SPL1 was added to LC and H7 to HC also on the two different mAbs. B. Secretion of Trastuzumab (T) under the control of each SP was assessed by ELISA after 8 days of culture. Cell free culture media of 36 independent clones were analysed. Output of the ELISA is represented as pg of mAbs produced per cell. Statistical differences were determined using a Mann-Whitney test: ns $p\text{-value}>0.05$; ** $p\text{-value}<0.005$; **** $p\text{-value}<0.0001$. C. Secretion of Rituximab® under the control of each SP was assessed by ELISA after 8 days of culture. Cell free culture media of 36 independent clones were analysed. Output of the ELISA is represented as pg of mAbs produced per cell. Statistical differences between each construction were analysed using a Mann-Whitney test: ns corresponds to a $p\text{-value}>0.05$; two asterisks (**) corresponds to a $p\text{-value}<0.005$.

[0169] FIG. 8—Levels of eGFP expression driven by novel promoters in *P. tricornutum* cells grown under low salinity conditions. A) Mean of net eGFP fluorescence in transformed *P. tricornutum* cell lines grown over 8 days under low salinity conditions. Only promoters for which eGFP expression could be detected are shown. A1) Promoters allowing a cyclic expression of eGFP; A2) Promoters allowing eGFP expression throughout the culture. B) Comparison of eGFP expression driven by the Phatr3_J34085 promoter (VOC) and NR promoter over an 8-day culture in low salinity conditions. C) Growth curves of cultures containing the 16 clones of the Phatr3_J34085 (VOC) and NR cell lines over an 8-day period. Error bars represent the standard error of means (SEM, $n=9$ for every day except on day 5 where $n=8$). Statistical differences were analyzed by a Welsh's t test comparison and by a Mann-Whitney test for data that are following a normal distribution or not (accord-

ing to Shapiro-Walk test). ns: p-value>0.05, *p-value<0.05, **: p-value<0.01, ***: p-value<0.001.

[0170] FIG. 9—Production of recombinant mAb in two independent clones of *Phaeodactylum tricornutum* cultured in different media: A) the Conway medium containing NO₃ as a nitrogen source and B) the MM medium.

EXAMPLES

Example 1: Increased Production of Monoclonal Antibodies in *P. tricornutum* Strains

[0171] In order to increase the production yield and secretion of these mAbs in *P. tricornutum*'s culture medium, different promoter/terminator couples were tested. We demonstrated that the V-ATPase C promoter/FcpA terminator pair could increase the yield of mAbs production in *P. tricornutum* in an 8-day bioprocess. We also highlighted that the use of the FcpA terminator was preferential to the endogenous terminator associated with the promoter for an increase in production yield. In parallel, we tested the influence of different heterologous and endogenous signal peptides on the secretion of antibodies in the culture medium. Surprisingly, the heterologous signal peptide "E" was shown to be more effective than the endogenous one of the HASP1 protein that is the most abundant secreted protein in *P. tricornutum*'s culture medium. These results contribute to the discovery and characterisation of various key elements to be considered in the vector design in order to increase the production yield of recombinant proteins in *P. tricornutum*.

1. Materials and Methods

1.1. Culture Conditions

[0172] The strain of *P. tricornutum* Pt1.8.6 (CAP 1055/1) were acquired from the Culture Collection of Algae and Protozoa (CCAP, UK). *P. tricornutum* was cultivated at 19° C. on a photoperiod of 16/8 h light/dark cycle with light intensity of 30 $\mu\text{mol photons}\cdot\text{m}^{-2}\cdot\text{s}^{-1}$ either on 1.5-2% agar plates or in liquid cultures under agitation at 150 rpm. Cells were grown in 100% artificial seawater (33.3 $\text{g}\cdot\text{L}^{-1}$ of sea salt (Instant Ocean®, Aquarium System, Sarrebourg, France)) filtered through 0.45 μm filters and sterilized by autoclave. The sterilized culture medium was supplemented with 1 $\text{mL}\cdot\text{L}^{-1}$ of Conway NaNO₃ nutritive medium (Na₂EDTA·2H₂O:45 $\text{g}\cdot\text{L}^{-1}$; NaNO₃:100 $\text{g}\cdot\text{L}^{-1}$; H₃BO₃:33.6 $\text{g}\cdot\text{L}^{-1}$; NaH₂PO₄:20 $\text{g}\cdot\text{L}^{-1}$; FeCl₃:0.768 $\text{g}\cdot\text{L}^{-1}$; ZnCl₂:21 $\text{mg}\cdot\text{L}^{-1}$; CoCl₂, 6H₂O:20 $\text{mg}\cdot\text{L}^{-1}$; (NH₄)₆Mo₇O₂₄, 4H₂O:9 $\text{mg}\cdot\text{L}^{-1}$; CuSO₄, 5H₂O:20 $\text{mg}\cdot\text{L}^{-1}$; MnCl₂, 4H₂O:360 $\text{mg}\cdot\text{L}^{-1}$; B₁ vitamin:200 $\text{mg}\cdot\text{L}^{-1}$; B₁₂ vitamin:10 $\text{mg}\cdot\text{L}^{-1}$) and 80 $\text{mg}\cdot\text{L}^{-1}$ of sodium metasilicate (Na₂SiO₃). The medium supplemented in Conway and sodium metasilicate was named SAS (Supplemented Artificial Seawater) medium. Transformants were grown in a SAS medium where NaNO₃ was replaced by 80.25 $\text{g}\cdot\text{L}^{-1}$ of NH₄Cl (SAS-NH₄ medium) as sole nitrogen source. For the transformation steps, 50% artificial seawater was used, but Conway NH₄Cl and sodium metasilicate were used at the same concentration. This medium was named 50% SAS-NH₄ medium. For ELISA experiments, the positive transformants were grown in the Mann and Myers' medium described recently (Song, Lye and Parker, 2020). Indeed, the Mann and

Myers' medium contained NO₃. Therefore, in this culture condition, the NR promoter is expressing continuously the recombinant mAbs.

1.2. Plasmid Construction

[0173] Various human signal peptides ("H5", "H7", "L1" from (Haryadi et al., 2015) and "E" from (Kober et al., 2013)) shown in the literature to be effective for the secretion of recombinant mAbs in CHO cells were selected. Their capacity to lead the secretion of recombinant proteins in *P. tricornutum* was first assessed using eGFP as reporter. The addition of a C-terminal 20-unit tandem repeat Serine-Proline glycomodule was also assessed as described for *Chlamydomonas reinhardtii* (Ramos-Martinez, et al., 2017). The eGFP DNA sequence was recovered from the pPhaT1-eGFP vector (Zaslavskaja et al., 2000) and was cloned in the pPha-NR shuttle vector (JN180663.1) optimized by e-Zyvec (Loos, France). This "pPhaNR-eGFP" vector was used as a reference to assess the intracellular fluorescence of eGFP.

[0174] DNA sequences from the different signal peptides, glycomodule as well as sequences encoding the Heavy Chain (HC) and Light Chain (LC) of Trastuzumab (Li et al., 2011) and of Rituximab (<https://go.drugbank.com/drugs/DB00073>) were synthesized and codon optimized for expression in *P. tricornutum*. Signal Peptides and glycomodule nucleotide sequences were cloned respectively either at the N- or C-terminal extremity of eGFP in the pPhaNR-eGFP vector by e-Zyvec. Codon optimized sequences encoding for HC and LC of Trastuzumab and Rituximab were cloned in fusion with the T2A peptide (Szymczak et al., 2004). Different constructs containing either the Nitrate Reductase (NR), the V-ATPase C and H4-1B promoters (Slattery et al., 2018; Watanabe et al., 2018) or different terminators (NR, FcpA, V-ATPase, H4-1B) were tested for their ability to increase expression level of recombinant mAbs like Trastuzumab in *P. tricornutum*. The most promising human signal peptides resulting from the screening step with eGFP, namely L1, H7 and E were evaluated for their capacity to secrete recombinant mAbs in *P. tricornutum*. The secretion of both Trastuzumab and Rituximab under these heterologous signal peptides was analysed. In parallel, the use of the endogenous signal peptide corresponding to the HASP1 protein, the most abundant secreted protein in *P. tricornutum* that was previously used to successfully drive the secretion of eGFP (Erdene-Ochir et al., 2019) was also evaluated for its capacity to secrete Rituximab in *P. tricornutum*'s culture medium.

1.3. *P. tricornutum* Transformation and Screening of Positive Clones

[0175] The different plasmids were introduced into diatom cells in exponential growth phase (10⁸ cell per ml) by using the Biolistic Particle Delivery System (PDS)-1000/He (Bio-Rad Laboratories, CA) and a protocol adapted from (Apt et al., 1996). Approximately 2×10⁹ cells of *P. tricornutum* were taken from culture in exponential growth phase and centrifuged at 3.000×g, during 5 minutes at 19° C. Cell pellets were gently resuspended with 150 μL of fresh 50% SAS-NH₄ medium and were spreading on 2% agar plates containing 50% SAS-NH₄ medium. Cells were stably transformed 24 h later by biolistic transformation using 1350 psi rupture disks and M-17 tungsten particles (Biorad, Marnes la Coquette, France) coated with 5 μg of plasmid DNA with Biolistic PDS-1000/He. After transformation, cells were incubated in the dark at 19° C. After 48 h, cells were plated on 1.5% agar

plates containing 50% SAS-NH₄ medium and 75 µg·mL⁻¹ of Zeocin (Invitrogen). Plates were then incubated at 19 °C during at least two weeks, on a photoperiod of 16:8 h light:dark cycle with light intensity of 30 µmol photons·m⁻²·s⁻¹. Isolated Colonies were streaked on agar SAS-NH₄ medium containing 75 µg·mL⁻¹ of Zeocin. This selection step was repeated twice. To confirm the insertion of the transgenes within *P. tricornutum* cells, 100 clones of each transformation events were tested by PCR using specific primers (Table 1). Cells were resuspended in 30 µL of the following extraction buffer (10 mM Tris HCl, 1 mM EDTA, 1M NaCl and 1% Nonidet P40 (NP40), pH8) and were heated at 95 °C for 10 min. After centrifugation at 3.000×g for 5 min, supernatants were diluted 5-fold with DNase-free water and 1 µL was used to perform the PCR. Transgenes were amplified with the DreamTaq DNA polymerase (Promega) and specific primers designed on both sides of the HC and LC chains of the antibodies were used. Specific primers named Tra-F/Tra-R and RTX-F/RTX-R respectively were used to screen transformants that have potentially received constructs containing trastuzumab or rituximab as gene of interest, respectively (Table 1). PCR products were analysed by electrophoresis using 1% agarose gel electrophoresis.

TABLE 1

Table summarizing the specific primers used for the PCR screening of the <i>P. tricornutum</i> clones.		
Primers	5'-3' Nucleotides Sequences	Hybridization Temperature
TRA-F	CGATAATGCGTTGCAGTCGG (SEQ ID NO: 68)	60° C.
TRA-R	TCCGACCCTTAGTACTCGCA (SEQ ID NO: 69)	60° C.
RTX-F	ATCTCACGCGTTGAAGCAGA (SEQ ID NO: 70)	61° C.
RTX-R	CCCTGGCTTCACTAGTTCGG (SEQ ID NO: 71)	61° C.

1.4. eGFP Screening of Transformants

[0176] Colonies growing on selective media after biolistic transformation were grown in 96-well plates in 150 µL of SAS-NH₄ medium under agitation (150 rpm) at 19° C. on a photoperiod of 16/8 h light:dark cycle with light intensity of 30 µmol photons m⁻²·s⁻¹. Six to seven days after the first inoculation, 50 µL of cultures were recovered and subcultured in 100 µL of fresh SAS-NH₄ medium under the same conditions. In order to allow the activation of the NR promoter and thus the expression of eGFP, NaNO₃ was added to the cells at a final concentration of 1 mM on day six. Two days after induction of the NR promoter, cultures were transferred into a black, clear-bottom 96-well plates (Costar®, Sigma Aldrich, Saint Quentin Fallavier, France). Screening of transformants was performed by exciting eGFP at 485 nm/20 nm and emission was measured at 538/20 nm using a fluorometer (FlexStation 3 MultiMode Microplate Reader—Molecular Devices, Tecan Group Ltd, Männedorf, Switzerland). A cut-off filter of 530 nm was applied.

1.5. Evaluation of the Secretory Efficiency of Signal Peptides

[0177] The culture volumes of the top three expressing eGFP clones for each construct were increased to a volume

of 50 ml in SAS-NH₄ medium with an initial cell density adjusted to 5×10⁶ cells·mL⁻¹. Cells were grown and induced as described above and harvested on day 8. 150 µL of total culture were recovered and the rest of the culture was centrifuged at 3.000×g for 5 minutes. Overall culture and supernatant were loaded in a black, clear-bottom 96-well plates (Costar®) and eGFP fluorescence was measured as described above. All measurements were made with three technical replicates on three independent biological replicates. The normality of the data was tested using GraphPad Prism 8, version 8.0.2 prior to an ANOVA analysis using Sidak's multiple comparison test.

1.6. Western Blot Analyses

[0178] For Western Blot analyses, cell cultures of 100 mL were performed and harvested as described above. After centrifugation, the cell pellets were resuspended in 500 µL of lysis buffer (0.1M Tris HCL, pH 7.5 containing 1 tablet of SIGMAFAST™ Protease Inhibitor Cocktail (SIGMA)). Total proteins were extracted using lysing beads (E-matrix lysing tubes, MP Biomedicals®, Fisher Scientific, Illkirch, France) and grounded for 3 cycles of 30 seconds at 6.5 m·s⁻¹ in a FastPrep-24™ homogenizer (MP Biomedicals®). Lysing tubes were centrifuged at 5,000×g for 5 minutes and supernatant was collected, then 500 µL of lysis buffer was added to the lysing tubes and a second homogenisation step was performed. Secreted eGFP was recovered by precipitating total proteins from the supernatant using a Trichloroacetic acid (TCA)/NP40 mixture. Briefly, 75 UL of NP40 was added to the supernatant, samples were vortexed and incubated at room temperature for 30 minutes. Then, 2.5 mL of TCA was added and samples were homogenised by vortex and incubated on ice during 1 h 30. Samples were centrifuged at 10,000 g for 30 minutes at 4° C. Pellets were washed with 500 µL of cold acetic acid three times with a centrifugation step (10,000×g for 30 min) between each wash. Pellets were left at room temperature (RT) until complete evaporation of acetone.

[0179] The presence of eGFP in supernatant or in cell pellets was analysed by Western Blot analysis under reducing conditions (Leamml Buffer containing β-mercaptoethanol heated at 100° C. during 10 min). Succinctly, the same amount of each sample was loaded on Tris-Glycine gel (12%), and the migration of proteins was carried out in a Tris-Glycine migration Buffer at 150 V. Purified eGFP produced in plants was used as positive control and Bovine Serum Albumin was used as negative control. One µg of each control was loaded on the SDS-PAGE gel. After migration, proteins were transferred to a nitrocellulose membrane using a wet tank transfer system (Bio-Rad) either during 2 hours at 60 V or overnight at 10 V. After transfer, the membrane was saturated in Tris buffer salt (TBS) complemented with Tween 20 at 0.1% (TBS-T) buffer and 5% skim milk for at least 2 hours at RT under agitation at 150 rpm. Detection of eGFP was performed by incubating the membrane with a rabbit anti-eGFP polyclonal antibody (Invitrogen, ThermoFisher Scientific, Illkirch, France) diluted at 1:2,000 in TBS-T buffer with 1% skim milk for 3 hours at RT. After washing, the membrane was then incubated with Horseradish peroxidase (HRP)-conjugated goat anti-rabbit antibody (Invitrogen) diluted at 1:30,000 in TBS-T buffer for 1 h 30 at RT. Finally, revelation of the

membrane was performed using the ECL Pico Plus kit (ThermoFisher Scientific) according to the manufacturer's instructions.

1.7. Quantification of the Level of Expression of Rituximab and Trastuzumab in *P. tricornutum* Culture Supernatant

[0180] *P. tricornutum* selected positive clones were grown in Mann and Myers' medium (Song, Lye and Parker, 2020). For each set of transformants containing a different construct, at least 30 independent clones were grown in 5 mL of fresh medium under an agitation of 200 rpm at 19 °C during either 8 or 15 days. At the end of the culture, an aliquot of 1 mL was collected and centrifuged at 14,000×g for 5 minutes in order to remove the diatom cells and cell debris. The amount of mAbs present in the *P. tricornutum* culture medium was finally quantified by ELISA using the MabTech Human IgG ELISA BASIC kit (ALP) using the instructions from the manufacturer (MabTech, Sweden). The quantities of mAbs produced in the different conditions were then determined as compared to the standard curve using GraphPad Prism 8. Statistical analyses were performed with GraphPad Prism 8, version 8.0.2. The normality of the data was first tested prior to either an ANOVA analysis using the Tukey's multiparametric comparisons test or a Mann-Whitney test. Details of the tests that have been used are described in the legend of each figure.

2. Results

2.1. Search for a Promotor/Terminator Couple that Allows Increasing the Production Yield of Recombinant mAbs in the Diatom *P. tricornutum*

[0181] Among the various possibilities of promoter/terminator couples to be tested, we decided to evaluate the capacity of the promoter and terminator of the H4-1B and V-ATPase genes to drive the production of Trastuzumab. These promoters correspond to the promoter of the histone H4 (Slattery et al., 2018) and the one of the vacuolar ATP synthase 16-kDa proteolipid subunit clumping factor A (V-ATPase C), respectively (Watanabe et al., 2018). So far, they have never been used to produce recombinant mAbs. The H4-1B promoter has been described previously to improve by 30% the mRNA relative expression of a reporter gene as compared to other promoters such as 40SRPS8, γ -Tubulin, FcpC and FcpF (Slattery et al., 2018). The activity of the V-ATPase C gene promoter has been demonstrated to increase by approximately 2.73 times the expression of Sh ble mRNA transcripts under both light and dark conditions at the stationary phase as compared to the fopA gene promoter (Watanabe et al., 2018). The constructs allowing expression of the Trastuzumab as a single cistron (promoter-TRA-LC-T2A-TRA-HC-terminator) were built in combination with either the H4-1B promoter or the V-ATPase C promoter using their respective terminator sequences (FIG. 2A). These constructs were used to transform via biolistic the diatom cells. After PCR screening of the clones on genomic DNA, expression of the mAbs was chosen as an output for the assay. Therefore, we investigated the strength of the promoters by following the amount of protein produced under their control. In agreement, after PCR screening using specific primers, at least twenty-nine independent positive clones containing the gene of interest were analysed by ELISA to evaluate the amounts of recombinant mAbs accumulated in the cell-free culture medium after 8 and 15 days of culture, respectively. These time points were chosen to maximize productivity of the recom-

binant mAbs as it was reported previously that proteins are mainly expressed at the stationary phase of cell culture (Erdene-Ochir et al., 2016). Moreover, the efficiency of some promoter such as the V-ATPase C one was assessed in the late stationary phase (Watanabe et al., 2018).

[0182] Data reported in FIG. 2B demonstrated that both H4-1B and V-ATPase C promoters allow the production of Trastuzumab in *P. tricornutum*. The quantity of mAb produced under the control of the H4-1B promoter reached an average of 84 $\mu\text{g}\cdot\text{L}^{-1}$ at day 8 and 78 $\mu\text{g}\cdot\text{L}^{-1}$ at day 15 (FIG. 2B). In contrast, the quantity of Trastuzumab produced under the control of the V-ATPase C promoter achieved an average of 212 $\mu\text{g}\cdot\text{L}^{-1}$ at day 8 and 112 $\mu\text{g}\cdot\text{L}^{-1}$ at day 15. The difference observed between the two promoters is significant at day 8 as demonstrated by the use of a non-parametric Mann-Whitney test (p-value=0.0171) (FIG. 2B). In order to perform a fair comparison and avoid differences due to the variation in biomass between the clones, the amount of mAbs produced was normalised for all the experiments relatively to the number of cells per mL. This gave a result expressed in pg of mAbs produced per cell, representing the productivity per diatom cell unit (FIG. 2C). Such a treatment of the data provided stronger results and confirmed at day 8 the significant difference of mAbs produced under the control of the H4-1B promoter (mean of 2.2×10^{-3} pg per cells) and under the control of the V-ATPase C (mean of 9.9×10^{-3} pg per cells), the latest representing a better promoter to drive the expression of mAb at day 8 (4.5 times increase as compared to the H4-1B).

[0183] In order to go further and evaluate the influence of the terminators on the activities of the H4-1B and V-ATPase C promoters, we built 6 vectors containing the LC and HC of the Trastuzumab driven either by the H4-1B promoter in combination with the H4-1B, NR or FcpA terminators or by the V-ATPase C promoter tested in combination with the same terminator sequences (FIG. 3A). As shown in FIG. 3B, the use of different terminators in combination with the H4-1B promoter did not significantly improve the production yield of Trastuzumab in *P. tricornutum* at day 8. Even not significant, the terminator FcpA seemed to rise the quantity of mAb at day 8. In contrast, the amount of mAb produced per cell unit increased significantly at day 15 using the FcpA terminator as compared to the NR terminator (p value=0.0039, two-way ANOVA Turkey's multiple comparisons test).

[0184] When using the V-ATPase C promoter in combination with its endogenous, the NR or the FcpA terminator sequences (FIGS. 3C and 3D), the better efficacy of the FcpA terminator was confirmed as it allowed a significative improvement of the mAb quantities secreted by the diatom as compared to the V-ATPase C and NR terminator sequences (above 2-fold increase). Moreover, it gave a better production yield of mAbs at day 8.

[0185] Then, we compared the activities of the H4-1B and V-ATPase C promoters to the NR promoter that was previously used for the production of recombinant anti-viral antibodies in *P. tricornutum* (Hempel et al., 2011, 2017; Hempel and Maier, 2012). All these promoters were used in combination with the FcpA terminator (FIG. 4A) to drive the expression of the Trastuzumab. At least thirty-one independent clones were analysed for each construct. Their respective strength was first evaluated at day 8 and day 15. Even not always significant using a non-parametric Mann-Whitney test, the three promoters presented higher activities for

the production of mAb at day 8 (FIG. 4B). Therefore, we decided to compare their respective performance at day 8 (FIG. 4C). It resulted from this comparison that, at day 8, the H4-1B is the weakest promoter for the production of mAb (Mean=0.0054 pg of mAbs per cell) as compared to the V-ATPase C and NR promoters (Mean=0.0094 pg of mAbs per cell), the best one being, in the conditions of culture used being the V-ATPase

[0186] C promoter leading to a mean of 0.1348 pg of mAb per cell unit (p value=0.0325 as compared to the H4-1B promoter and p value=0.0395 as compared to the NR one using a two-way ANOVA multiparametric Turkey's test). Therefore, usage of the V-ATPase C promoter allowed 14- and 25-fold increases as compared the NR and H4-1B promoters, respectively.

[0187] In order to validate the observation made for the production of Trastuzumab at day 8 with the FcpA terminator and demonstrate that this is not specific for this mAb and can be extended to other mAb candidates, we expressed the Rituximab under the control of the NR promoter in combination with either the NR or FcpA terminator sequences (FIG. 5A). The expression of Rituximab was analysed by ELISA in thirty-six independent clones at day 8. The median of the mAbs quantity produced per diatom cell was $0.6 \cdot 10^3$ pg for the vector containing the NR terminator and $1.1 \cdot 10^3$ pg for the one containing the FcpA terminator, thus confirming the positive effect of using FcpA terminator. This result was demonstrated to be significant with a p-value of 0.0131 using the non-parametric Mann Whitney test (FIG. 5B).

[0188] Altogether, these results demonstrated that the use of the V-ATPase C promoter in combination to the FcpA terminator is the best option to produce recombinant mAbs in *P. tricornutum* for an 8-day bioprocess.

2.2. Heterologous Signal Peptides Increase the Secretion of eGFP in *P. tricornutum*

[0189] We then evaluated the impact of signal peptides on the secretion of mAbs in the culture medium of *P. tricornutum*. Different signal peptides have been previously described in the literature to increase the secretion of mAbs in CHO cell cultures. Thus, we decided to evaluate the efficiency of such signal peptides for the production and secretion of mAbs in *P. tricornutum* cells. Therefore, signal peptides named "H5", "H7", "L1" (Haryadi et al., 2015) and "E" (Kober et al., 2013) were chosen. First, we tested the capacity of the selected heterologous signal peptides to improve the secretion of eGFP in the culture medium of *P. tricornutum*. Therefore, different constructs containing eGFP under the control of the NR promoter and terminator were designed with or without the different signal peptides. These constructs are represented in FIG. 6A. Another construct containing a glycomodule C-terminal tail rich in Serine-Proline repeating unit (G20) at the end of the eGFP was also evaluated. Such a strategy was previously demonstrated to increase the secretion of Venus in the green microalgae *C. reinhardtii* (Ramos-Martinez et al., 2017). The different constructs were used to transform *P. tricornutum* cells. Positive transformants were then screened based on their fluorescence intensity that is reflecting the level of eGFP expression. Transformants presenting the highest expression levels were selected for further analysis. Then, the overall fluorescence intensity of a 50 mL culture of the three best expressing clones for each construct was investigated. Finally, the cultures were centrifugated and the

fluorescence of the cell-free medium reflecting the eGFP secreted in the culture medium was measured and compared to the fluorescence of the total culture (FIG. 6B and Table 2).

TABLE 2

Relative percentage expressed as percent of eGFP fluorescence measured in the cell-free culture medium of <i>P. tricornutum</i> transformants in comparison to the intracellular fluorescence. The given value represents the mean of three technical replicates for three independent clones for each construct.	
Constructs	Relative percentage of eGFP fluorescence in the cell free culture medium (%)
eGFP	11.3
SP E-eGFP	56.5
SP H5-eGFP	43.7
SP H7-eGFP	48.2
SP L1-eGFP	60
eGFP-G20	8.5

SP: signal peptide; G20 corresponds to the Glycomodule C-terminal tail containing 20 repeating units of Serine and Proline

[0190] When comparing the intensity of the fluorescence in the culture and in the cell-free medium for the clones expressing the different constructs, it appears that between 43% to 60% of eGFP fluorescence was observed in the culture media. In contrast, as expected, only 11% of the fluorescence appears to be secreted in the cell-free medium in the control experiment of eGFP clones without any signal peptide. The clones expressing eGFP-G20 construct showed predominantly fluorescence inside the cells as observed for the cytosolic eGFP used as a control (FIG. 6B and Table 2). These results were confirmed by Western Blot using a specific anti-eGFP antibody (data not shown). Altogether, these data demonstrate the ability of heterologous signal peptides to drive the secretion of eGFP in *P. tricornutum* cells. Therefore, the best signal peptides namely H7, L1 and E retrieved from this screening step were then used either alone or in combination to express both Trastuzumab and Rituximab in *P. tricornutum*.

2.3. Heterologous Signal Peptides Increase the Secretion of mAbs in *P. tricornutum*

[0191] In order to demonstrate the efficacy of the H7, L1 and E heterologous signal peptides to drive the secretion of mAbs in *P. tricornutum* cells, constructs allowing the expression of Trastuzumab and Rituximab were built (FIG. 7A) in parallel to a construct that contained the endogenous signal peptide of the HASP1 protein (Uniprot: B7G4A0). This protein has been described to be the most abundant protein naturally secreted in *P. tricornutum* culture medium (Erdene-Ochir et al., 2019). The different constructs were used to stably transformed *P. tricornutum* cells. The positive clones were selected by PCR and ELISA. Finally, the amounts of mAbs produced in thirty-two independent clones expressing the Trastuzumab were analysed for the different constructs after 8 days of culture. FIG. 6B reports on the amount of Trastuzumab produced in the culture medium at day 8 normalised per diatom cell (pg per cell). These results demonstrated that the amount of Trastuzumab secreted in *P. tricornutum*'s culture media with the endogenous HASP1 signal peptide is the lowest. The heterologous signal peptide E and the L1/H7 combination used allowed a 1.9-fold improvement for E and a 2.7-fold improvement for L1/H7, respectively by comparison to HASP1. These differences were significant according to a non-parametric Mann-Whit-

ney test (p-value<0.0001 for the SP E and p-value=0.0012 for the L1/H7 combination). To confirm these results, similar studies were performed for the Rituximab (FIGS. 7A and 7C). Thirty-six independent clones were analysed by ELISA. Again, the highest production yield of secreted Rituximab was obtained with the heterologous signal peptide E (FIG. 7C) which results in a 2.6-fold increase.

Example 2: Production of Trastuzumab in Different *P. Tricornutum* Strains

1. Material and Methods

1.1. Culture Conditions

[0192] Prior to the transformation, the ten *P. tricornutum* wild type strain were cultivated at 19° C. under a photoperiod of 16 h light/8 h dark with light intensity of 30 $\mu\text{mol photons m}^{-2}\text{s}^{-1}$ either on 1.5% agar plates or in liquid cultures (glass Erlenmeyer) under agitation at 150 rpm. Cells were grown in 100% artificial seawater (33.3 g/L of sea salt (Instant Ocean®)) filtered through 0.45 μm filters and sterilized by autoclave. The sterilized culture medium was supplemented with 1 mL·L⁻¹ of Conway NaNO₃ nutritive medium (Na₂EDTA·2H₂O:45 g/L; NaNO₃:100 g/L; H₃BO₃:33.6 g/L; NaH₂PO₄:20 g/L; FeCl₃:0.768 g/L; ZnCl₂:21 mg/L; CoCl₂·6H₂O:20 mg/L; (NH₄)₆Mo₇O₂₄·4H₂O:9 mg/L; CuSO₄·5H₂O:20 mg/L; MnCl₂·4H₂O:360 mg/L; B1 vitamin:200 mg/L; B12 vitamin:10 mg/L) and 80 mg/L of sodium metasilicate (Na₂SiO₃). The medium supplemented in Conway and sodium metasilicate was named SAS (Supplemented Artificial Seawater) medium. Transformants were grown in a SAS medium where NaNO₃ was replaced by 80.25 g·L⁻¹ de NH₄Cl (SAS-NH₄ medium).

[0193] Expression of eGFP was carried out on 2 ml cultures in 12 well plates containing 10% artificial seawater supplemented with the same proportions of Conway medium and sodium metasilicate (SAS medium: supplemented artificial seawater). Cultures were grown at 19° C. under continuous agitation at 150 rpm on a 16 h/8 h light/dark cycle.

[0194] For antibody expression, *P. tricornutum* cells containing the heavy and Light chains transgenes were grown in 5-1000 mL culture inoculated at a density of 0.1-0.6 (OD_{680 nm}) in SAS-NH₄ medium at 19° C. on a photoperiod of 16 h light/8 h dark (30 $\mu\text{mol photons m}^{-2}\text{s}^{-1}$) for four days. The fifth day, cultures were transferred to 23° C. under continuous illumination (37 $\mu\text{mol photons m}^{-2}\text{s}^{-1}$). On day six, trastuzumab production was induced by the addition of 1 mM NaNO₃. Cells were harvested two days after induction.

1.2. Plasmid Construction

[0195] DNA sequences encoding for the fully-humanized heavy chain (HC) and light chain (LC) of the monoclonal antibody trastuzumab was retrieved from the patent WO 2011/103700 (Li, Tong and Kan, 2011) and the T2A sequence—EGRGSLITCGDVEENPGP (SEQ ID NO:9)—from (Szymczak et al., 2004). The specific codon bias of *P. tricornutum* was used to synthesize the different genes. Polycistronic vectors containing HC-T2A-LC (vector C), or LC-T2A-HC (vector D) were constructed by e-Zyvec (Loos,

France). Briefly, codon-optimized DNA sequences of HC and LC of trastuzumab and T2A were cloned in a modified version of pPha-NR shuttle vector (JN180663.1).

1.3. *P. tricornutum* Transformation

Transformation of *P. tricornutum* by Conjugation

[0196] First, an *E. coli* strain containing the pTA-Mob (mobilisation helper) vector was transformed with the SLIC eGFP vector containing the NR promoter. Transformation of *P. tricornutum* was adapted from (Karas et al., 2015). Exponentially growing cultures of transformed *E. coli* (37° C. in LB medium) and *P. tricornutum* (19° C. in SAS medium) were harvested via centrifugation (10 min at 3 000 g). Cell pellets were resuspended in 500 μL of SOC medium and 50% SAS medium for bacteria and diatoms respectively. 200 μL of both prokaryotic and eukaryotic cells were mixed and plated on 1% agar plates containing 50% SAS medium without silica and 5% LB. Agar plates were incubated at 30° C. during 1 h30 in dark, then at 19° C. on a 16 h/8 h light/dark cycle for 72 h. Then, 1 mL of 50% SAS medium was added to the plates and cells were scratched and resuspended. 200 μL of resuspended cells were plated on selective agar plates containing 50% SAS medium and 20 $\mu\text{g}\cdot\text{mL}^{-1}$ of phleomycin. Plates were incubated at 19° C. on a 16 h/8 h light/dark cycle and colonies of transformed *P. tricornutum* appeared approximately 10 and 15 days after the transformation.

[0197] Biolistic transformation was adapted and modified from (Apt et al., 1996). Approximately 1×10^9 cells of *P. tricornutum* Pt1.8.6, Pt2, Pt3, Pt4, Pt5, Pt6, Pt7, Pt8, Pt9 and Pt10 strains were taken from culture in exponential growth phase and centrifuged at 3,000×g, during 5 minutes at 19° C. Cell pellets were gently resuspended with 150 μL of fresh 50% artificial seawater (16.6 g/L Sea Salt Medium) containing 1 mL·L⁻¹ of Conway NH₄⁺ medium and were spreading on 2% agar plates containing 50% artificial seawater medium supplemented by 1 mL·L⁻¹ of Conway NH₄⁺. Cells were stably transformed 24 h later by biolistic transformation using 1350 psi rupture disks and M-17 tungsten particles coated with 5 μg of plasmid DNA using the Particle Delivery System-1000/He (Bio-Rad). After transformation, cells were incubated in the dark at 19° C. After 48 h, cells were plated on 1.5% agar plates containing 50% artificial seawater medium supplemented with 1 mL·L⁻¹ of Conway NH₄⁺ medium, 40 mg/L of sodium metasilicate and 75 $\mu\text{g}\cdot\text{mL}^{-1}$ of Zeocin (Invitrogen).

1.4. PCR Screening of Potential Transformants

[0198] 15 days after biolistic transformation, colonies growing on selective medium were subcultured on fresh 1.5% agar plates containing SAS medium with 75 $\mu\text{g}/\text{ml}$ of Zeocin. Prior to PCR screening, a rapid DNA extraction was performed on approximately 200-250 Zeocin resistant clones. For this purpose, a small amount of biomass from each putative transformant was scraped directly from the agar plates and placed in 30 μL of lysis buffer (1% NP-40, 10 mM Tris HCl, pH 8; 1 mM anhydrous EDTA; 1 M NaCl). Cell lysis was achieved by heating the cells at 95° C. for 10 min. Then, lysed cells were vortexed briefly and 120 μL of DNase-free water were added. DNA extractions were maintained at 4° C. until performing PCR. PCR amplification was performed using GoTaq® G2 DNA Polymerase (Promega). 1 μL of each DNA extraction was used and manufacturer's recommendations were adapted to a final volume of 20 μL . The presence of both HC and LC in transformants that

potentially received vector 1 was confirmed using primers 5'-CTGTCATTATAAGTCGGGGGG-3' (SEQ ID NO:72) and 5'-TCGCGACTAATCGTGAAACGG-3' (SEQ ID NO:73) for HC and primers 5'-AAACTCCTCTATCCTCACCTG-3' (SEQ ID NO:74) and 5'-CAACTTAGTCCCTTGACCG-3' (SEQ ID NO:75) for LC. For transformants that potentially received vector 2, the presence of both chains was confirmed using a single pair primers 5'-CAGCAAGGACTCCACATACAGC-3' (SEQ ID NO:76) and 5'-TCGCGACTAATCGTGAAACGG-3' (SEQ ID NO:77). Amplicons were analyzed on 1.5% agarose gels.

1.5. Samples Preparation and ELISA Assay

[0199] For the ELISA assay, cell cultures were adjusted to the same density ($OD_{680\text{ nm}}=0.6$) and samples were harvested 48 hours after induction. Cells were centrifuged (3 000×g, 5 min) and resuspended in 500 µl of lysis buffer (0.1M Tris HCL, pH 7.5 containing 1 tablet of SIGMA-FAST™ Protease Inhibitor Cocktail (SIGMA)) and the supernatant was cleared of residual cells by a second centrifugation step (10 000×g, 5 min) and collected. Total proteins were extracted using lysing beads (E-matrix lysing tubes, MP Biomedicals®) and grounded for 3 cycles of 30 seconds at 6.5 m/s in a FastPrep-24™ homogenizer (MP Biomedicals®). Lysing tubes were centrifuged at 5 000×g for 5 and supernatant was collected, then 500 µl of lysis buffer was added to lysing tubes and a second homogenization step was performed.

[0200] Samples were analyzed by ELISA on the day of harvest using the Human IgG ELISABASIC kit (MabTech). Briefly, Nunc® MaxiSorp™ plates were coated with 100 µl per wells of MT145 antibody diluted at 2 µg/ml in PBS (pH 7.4) and incubated overnight at 4° C. The following day, plates were washed twice with 200 µl per wells of PBS (pH 7.4), then saturated with 200 µl per wells of PBST (PBS (pH 7.4) with 0.05% Tween 20) and 0.1% BSA and then incubated for 1 hour at room temperature (RT). Plates were washed five times with 200 µl per wells of PBST, then 100 µl of samples or standard dilution (ranging from 500 to 2.5 ng/ml) were added and plates were incubated for 2 hours at RT. Plates were washed as described above and 100 µl per wells of MT78-ALP antibody diluted at 1:1000 in PBST containing 0.1% BSA were added, and plates were incubated for 1 hour at RT. Plates were washed as previously described and 100 µl per wells of pNPP solution at 5 mM, and plates were incubated 45 min at 37° C. in dark and plates were read at 405 nm using a plate reader. Medium and wild-type cells pellets served as a negative control and all measurements were carried out on 3 biological replicates and technical triplicates for each sample.

1.6. Purification of the Trastuzumab Biosimilar

[0201] For further functional analysis, the trastuzumab biosimilar was purified using protein-A sepharose. The culture medium was collected as described above, filtered through a 0.22 µm filter as described in (Hempel et al., 2012) then diluted with ultrapure water (v/v) and finally purified

through protein-A sepharose (GE HealthCare). After washing with 10 column volumes (CV) of Tris 0.1 M (pH 8). The antibody was eluted with 5 CV of 0.1 M of Glycine-HCL buffer, pH 2.5 into a glass test-tube containing 0.25 CV of Tris buffer 1M, pH 9. The entire purification step was performed at 4° C.

1.7. Western BLOT

[0202] Correct antibody assembly and purification validation were performed by Western Blot analysis under native (Leamml Buffer without β-mercaptoethanol) or reducing conditions (Leamml Buffer with β-mercaptoethanol at 100° C. during 10 min). Succinctly, the same amount of each sample was loaded on Bis-Tris (4-12%) or Tris-Glycine gel (12%), and the migration of proteins was carried out in a MOPS-Buffer or a Tris-Glycine migration Buffer respectively at 130 V. Commercial trastuzumab from Roche was used as positive control (under natives or reducing conditions) and BSA was used for negative control. 100 ng of each control was loaded on gel. After migration, proteins were transferred to a nitrocellulose membrane using a wet tank transfer systems (Bio-rad) either during 2 hours at 60 V or overnight at 10 V. After transfer membrane was saturated in TBS-T (TBS with 0.1% Tween 20) buffer with 5% skim milk for at least 2 hours at RT. Detection of the full-length antibody or heavy and light chains was realized by incubating the membrane with a biotin-conjugated goat anti-human IgG (H+L) (Invitrogen) diluted at 1/3 000 in TBS-T buffer with 1% skim milk for 3 hours at RT. The membrane was then incubated with HRP-conjugated streptavidin (Sigma) diluted at 1/30 000 in TBS-T buffer for 1 h30 at RT. Finally, revelation was performed using the ECL Pico Plus kit (ThermoFisher) according to the manufacturer's instructions.

[0203] A glycosylation analysis was also performed using eastern blots revealed by specific lectins prior to mass spectrometry analyses. After antibody loading under reducing conditions, followed by migration and transfer as described above, nitrocellulose membranes were saturated in TBS-T buffer and then incubated in TBS-T buffer containing either 5 µg/ml of Concanavalin A (Vector Laboratories) containing 10 mM CaCl₂, a rabbit anti-xylose antibody (Agrisera) or a rabbit anti-fucose antibody (Agrisera) for 3 hours at RT. Membranes were subsequently incubated with HRP-conjugated streptavidin (Sigma) or HRP-conjugated goat anti-rabbit antibody (Invitrogen) respectively, diluted at 1/30 000 in TBS-T buffer for 1 h30 at RT. Revelation was performed as described above.

2. Results

[0204] Different expression vectors were designed to drive the expression of TrastuzumAb under the control of the NR promoter with or without the T2A linker between the genes encoding for both heavy (H) and light (L) chains of TrastuzumAb (FIG. 1A showing the structure of the tested vectors A, B, C and D). After the clone selection by PCR, the amounts of mAbs produced in the culture media and within the *P. tricornutum* cells were determined using an ELISA assay. It is to note that for such mAbs quantification, no purification or desalting step was introduced. Ten independent clones of *P. tricornutum* expressing the same construct were studied. Two different PHAEO strains were studied (Pt1.8.6 and Pt3).

TABLE 3

Quantitative results in Pt1.8.6					
Construct	Number of transformants (after double subculturing)	Positive clones by PCR on gDNA	Percentage of clones receiving construct	Protein-expressing clones	Percentage of clones expressing the protein
HerA	15	0	0.0	0	0.0
HerB	3	0	0.0	0	0.0
HerC	185	3	1.6	3	100.0
HerD	205	42	20.5	10	23.8

TABLE 4

Quantitative results in Pt3					
Construct	Number of transformants (after double subculturing)	Positive clones by PCR on gDNA	Percentage of clones receiving construct	Protein-expressing clones	Percentage of clones expressing the protein
HerD	176	87	49.4	84	96.6%

[0205] In average for clones expressing the vector D, the mAb production yield is above 20 µg/L for strain Pt1.8.6 using 100 mL of an 8-day culture and 4 g/L in average for Pt3 strain. It is to note that for the latest, the production yield was only measured for the screening stage (5 ml cultures where the culture conditions are not optimal).

Example 3: Identification of New Endogenous Promoters of *P. Tricornutum*

[0206] The selection of a strong promoter that drives high yields of mAbs is a prerequisite for developing a cost-effective expression system. To date, only few endogenous promoters have been characterized in *P. tricornutum*. In the present work, we analysed our previously published transcriptomic data from *P. tricornutum* Pt3 strain and identified thirty-three potential “strong” endogenous promoters. First, these putative promoter sequences were cloned and fused to the coding sequence of the enhanced green fluorescent protein (eGFP). Their strengths were assessed by measuring eGFP fluorescence, thus reflecting the level of eGFP expression. Among the thirty-three promoters, thirteen were able to drive eGFP expression. Best results were obtained with the Phatr3_J34085 (VOC) promoter which allowed a significant increase expression of eGFP by comparison to the one induced by the NR promoter. These findings contribute to the discovery of new genetic tools that could be used in future studies to improve recombinant protein production in *P. tricornutum*.

1. Material and Methods

1.1. Cell Culture and Growth Conditions

[0207] The *Phaeodactylum tricornutum* strain Pt3 (CCAP 1052/1B) was used for this study. Cells were grown at 19° C. on a 16 h/8 h light/dark cycle. Screened transformants were maintained on 1.5% agar plates containing 100% artificial seawater (33.3 g·L⁻¹ of sea salt (Instant Ocean®)) filtered on 0.45 µm filters and sterilized by autoclave. The sterilized culture medium was supplemented by 1 ml·L⁻¹ of Conway medium (Na₂EDTA·2H₂O:45 g·L⁻¹; NaNO₃:100

g·L⁻¹; H₃BO₃:33.6 g·L⁻¹; NaH₂PO₄:20 g·L⁻¹; FeCl₃:0.768 g·L⁻¹; ZnCl₂:21 mg·L⁻¹; CoCl₂, 6H₂O:20 mg·L⁻¹; (NH₄)₆Mo₇O₂₄, 4H₂O:9 mg·L⁻¹; CuSO₄, 5H₂O:20 mg·L⁻¹; MnCl₂, 4H₂O:360 mg·L⁻¹; B1 vitamin:200 mg·L⁻¹; B12 vitamin:10 mg·L⁻¹) and 80 mg·L⁻¹ of sodium metasilicate.

[0208] Expression of eGFP was carried out on 2 ml cultures in 12 well plates containing 10% artificial seawater supplemented with the same proportions of Conway medium and sodium metasilicate (SAS medium: supplemented artificial seawater). Cultures were grown at 19/16° C. under continuous agitation at 150 rpm on a 16 h/8 h light/dark cycle.

1.2. Regulatory Sequences Selection

[0209] In order to identify putative promoter sequences, we used the results of a RNA-seq transcriptomic analysis of the three main morphotypes of *P. tricornutum* Pt3 strain (Ovide et al., 2018). A selection of the 55 most overexpressed genes (Log 2 Fold Change values>5) in the oval morphotype compared to the other two morphotypes was performed. In silico analyses of these 55 genes and their upstream regulatory sequences were performed using the Integrative Genomic Viewer (IGV v2.11.4) and Blast2GO software in order to retain only the sequences that present higher probability to be promoter regions. Transcriptomic data were visualised using IGV to identify and eliminate mispredicted sequences. Blast2GO software was used to verify the sequence description, and sequences for which no annotation could be made, not even “predicted proteins” were also removed from the study.

1.3. Isolation of Putative Promoter Regions

[0210] DNA extraction from Pt3 cell line was carried out using the Nucleospin Plant II kit (Macherey-Nagel) following the manufacturer’s instructions. Putative promoter sequences were amplified by PCR from Pt3 genomic DNA using the Phusion Green Hot Start II High Fidelity DNA Polymerase (ThermoFisher Scientific). Amplicons were purified using “Wizard SV Gel and PCR Clean-Up System” (Promega) and cloned into either the pJET1.2/blunt cloning

vector (ThermoFisher Scientific) or in the SmaI linearised pUC19 vector (ThermoFisher Scientific). Cloning into the pJET1.2/blunt cloning vector was realised following the instructions provided by the ‘CloneJET PCR Cloning Kit’ (ThermoFisher Scientific) and cloning into the pUC19 was realised using T4 DNA ligase (Promega). Cloning vectors were used to transform DH5a chemically competent *Escherichia coli* (Invitrogen) and colony PCR was performed using the GoTaq G2 polymerase (Promega) and primers as described above. Plasmids were purified from *E. coli* cells using the Nucleospin Plasmid Kit (Macherey-Nagel). Finally, the putative promoter sequences were verified by Sanger sequencing prior to further experiments (Eurofins, France).

1.4. Transformation Vectors Construction

[0211] The pPtpuC3 vector backbone (Karas et al., 2015) required for *P. tricornutum* transformation by conjugation was modified by insertion of a “SLIC-eGFP-NRt” transcriptional unit (TU), in order to allow a rapid cloning of the different promoters to be tested in this study. Based on prior work, the TU was amplified from the vector “pPhaNR-eGFP” (Mekhalfi et al., 2022) using the primers 5'-GCT-CTA-GAT-AGT-TGG-AAT-GGT-ACG-TAC-CAA-CTC-CAT-AAG-GAT-CCA-TGG-TGA-GCA-AGG-GCG-AGG-AGC-3' (SEQ ID NO:78) and 5'-ACA-TGC-ATG-CTC-CGG-ATG-CGT-TCA-TTT-TAG-ATC-CTG-ATC-CG-3' (SEQ ID NO:79). After digestion of the pPtpUC3 vector backbone and of the cassette “SLIC-eGFP-NRt” TU with the restriction enzymes SphI and XbaI, the cassette was ligated into the pPtpUC3 vector using T4 DNA polymerase (Promega) thus allowing the creation of the pPtpUC3_SLIC-eGFP-NRt vector.

[0212] For the construction of final vectors, the pPtpUC3_SLIC-eGFP-NRt vector was digested with the restriction enzyme SnaBI (Promega), and the putative promoter regions were amplified from the previously sequencing-validated vectors. The putative upstream regulatory sequences were cloned into the pPtpUC3_SLIC-eGFP-NRt vector using the sequence- and ligation-independent cloning method (SLIC) by adapting the previously described method (Jeong et al., 2012). The insert:vector ratio was increased to 10:1 and 2 units of T4 DNA polymerase (Promega) were used. All the resulting SLIC vectors (empty or promoter containing) constructed were also checked by sequencing as described above for validation prior to *P. tricornutum* transformation (Eurofins, France).

1.5. Transformation of *P. tricornutum*

[0213] First, an *E. coli* strain containing the pTA-Mob (mobilisation helper) vector were transformed with the SLIC vectors (empty or promoter containing). Transformation of *P. tricornutum* was adapted from (Karas et al., 2015). Exponentially growing cultures of transformed *E. coli* (37° C. in LB medium) and *P. tricornutum* (19° C. in SAS medium) were harvested via centrifugation (10 min at 3 000 g). Cell pellets were resuspended in 500 µl of SOC medium and 50% SAS medium for bacteria and diatoms respectively. 200 µl of both prokaryotic and eukaryotic cells were mixed and plated on 1% agar plates containing 50% SAS medium without silica and 5% LB. Agar plates were incubated at 30° C. during 1 h30 in dark, then at 19° C. on a 16 h/8 h light/dark cycle for 72 h. Then, 1 mL of 50% SAS medium was added to the plates and cells were scratched and resuspended. 200 µl of resuspended cells were plated on

selective agar plates containing 50% SAS medium and 20 µg·mL⁻¹ of phleomycin. Plates were incubated at 19° C. on a 16 h/8 h light/dark cycle and colonies of transformed *P. tricornutum* appeared approximately 10 and 15 days after the transformation.

1.6. Confirmation of the Presence of Transgene by PCR

[0214] After transformation, colonies growing on selective medium were subcultured on fresh 1.5% agar plates containing SAS medium with 75 µg/ml of Zeocin. Prior to PCR screening, a rapid DNA extraction was performed on approximately 20 Zeocin resistant clones. For this purpose, a small amount of biomass from each individual clone was scraped directly from the agar plates and placed in 30 µl of lysis buffer (1% NP-40, 10 mM Tris HCl, pH 8; 1 mM anhydrous EDTA; 1 M NaCl). Lysis was achieved by heating cells at 95° C. for 10 min. Then, lysed cells were vortexed briefly and 120 µl of DNase-free water were added. DNA extractions were maintained at 4° C. until further use. PCR amplification was performed using GoTaq® G2 DNA Polymerase (Promega). 1 µl of each DNA extraction was used and manufacturer’s recommendations were adapted to a final volume of 20 µl. The presence of the gene encoding eGFP in transformants was confirmed using primers 5'-AT-CATGGCCGACAAGCAGAA-3' (SEQ ID NO:80) and 5'-GACTGGGTGCTCAGGTAGTG-3' (SEQ ID NO:81). Amplicons were analysed on 1.5% agarose gels.

1.7. Assessment of the Activity of Potential Promoter Sequences by eGFP Fluorescence Intensity Measure.

[0215] For the same construct, between 7 to 16 independent clones were grown together in pools in 6-well plates containing 5 mL of culture. 150 µl of each culture was sampled every day to measure cell density (OD680 nm) and eGFP fluorescence for 8 days. Measurements were carried out in black, clear-bottom 96-well plates (Costar®). Screening of transformants was performed by exciting eGFP at 485 nm/20 nm and emission was measured at 538/20 nm using a fluorometer (FlexStation 3 MultiMode Microplate Reader—Molecular Devices). A cut-off filter of 530 nm was applied. The eGFP fluorescence emitted under the control of each promoter was normalized to a one OD unit (considering the OD680 nm linearity region). The normalized fluorescence of the empty vector control (pPtpUC3_SLIC-eGFP-NRt) was subtracted from each condition in order to ignore the chlorophyll background signal. All measurements were made with three technical replicates on three independent biological replicates.

1.8. Statistical Analyses

[0216] Mean values of the net eGFP fluorescence per OD680 nm unit, mean values of the growth curve, standard error of the means (SEM) and statistical analyses were calculated and performed using Prism software. The normalised net eGFP fluorescence intensity of the VOC promoter was compared to the values of the reference NR promoter, over the 8 days period of culture. The Shapiro-Walk test was used to assess whether the data were normally distributed or not. The difference between the values obtained at different days for the NR and VOC promoters were evaluated using a Welch’s t-test comparison for values following a normal distribution and by a Mann-Whitney test for values not following a normal distribution.

1.9. In Silico Analyses of Validated Promoter Sequences

[0217] Sequences that drove eGFP expression were further analysed in silico. The prediction of cis-regulatory elements in these sequences was performed using the PlantCARE database. Additional analysis was performed on promoter that allowed a higher eGFP expression than that

were retrieved for further analysis. The Blast2GO software was used to attribute a functional annotation to the 55 selected genes. Eight out of the 55 genes (14.5%) still do not have any functional annotation. Altogether, these in silico analyses reduced the number of potential promoters to analyse to thirty-three (Table 5).

TABLE 5

Summary of the information regarding the 33 potential promoter sequences selected from our previous RNA-Seq transcriptomic data analyses of the three *P. tricornutum* morphotypes (Ovide et al., 2018).

Gene ID	Protein annotation	Chromosomal position	Promoter predicted length ¹	Promoter name	Promoter final length ²
Phatr3_J39391	predicted protein	19: 379218-380555	896 pb	39391	896 pb
Phatr3_J33266	solute carrier family 34	3: 462819-464704	1741 pb	SOCF34	1701 pb
Phatr3_J37038	syringomycin synthesis regulator	12: 20667-21773	1890 pb	SSR	1246 pb
Phatr3_J46468	predicted protein	10: 427467-428684	817 pb	46468	828 pb
Phatr3_Jdraft1668	taurine catabolismfamily	bd_31x35: 110713-112101	560 pb	TCF	559 pb
Phatr3_J40651	predicted protein	25: 73019-73327	756 pb	40651	756 pb
Phatr3_J42538	major intrinsic	1: 361483-362842	1089 pb	42538	1092 pb
Phatr3_J48356	DUF11 domain-containing	17: 217814-220228	986 pb	DUF11	985 pb
Phatr3_J33783	predicted protein	4: 340826-341837	747 pb	33783	744 pb
Phatr3_J41599	predicted protein	32: 49730-50980	1087 pb	41599	1076 pb
Phatr3_J36794	predicted protein	11: 381859-382908	1491 pb	36794	³
Phatr3_J50361	predicted protein	29: 82531-83523	755 pb	50361	754 pb
Phatr3_J43621	DUF2711 family	2: 1005210-1005785	478 pb	DUF2711	477 pb
Phatr3_J34976	alkaline phosphatase	6: 788784-789464	681 pb	AP	681 pb
Phatr3_J43494	predicted protein	2: 626125-627632	1669 pb	43494	³
Phatr3_J47869	alkaline phosphatase	15: 296103-298532	1972 pb	AP	³
Phatr3_EG02507	RING-H2 finger ATL74-like	14: 535986-536588	830 pb	RHFA	830 pb
Phatr3_J8683	peptide methionine sulfoxide reductase	1: 241098-242192	597 pb	PMSR	596 pb
Phatr3_Jdraft1443	P-loop containing nucleoside triphosphate hydrolase	bd_32x35: 212637-213710	1138 pb	PCNTPH	1125 pb
Phatr3_J40433	solute carrier family 34	23: 381101-383068	735 pb	SOCF34-2	727 pb
Phatr3_J35771	predicted protein	8: 716893-717834	1215 pb	35771	³
Phatr3_J35102	predicted protein	7: 63434-63919	535 pb	35102	535 pb
Phatr3_J8537	type VI secretion system tip	5: 930561-935372	1542 pb	SST5	1542 pb
Phatr3_J36570	serine hydrolase	10: 784031-785449	717 pb	SH	725 pb
Phatr3_J50252	glutathione S-transferase Mu 3	28: 107611-109050	1095 pb	GSTMu3	1098 pb
Phatr3_J49693	glycerophosphodiester phosphodiesterase	23: 448289-450075	860 pb	GPPD	³
Phatr3_J34085	VOC family	4: 1061396-1062934	957 pb	Voc	957 pb
Phatr3_J40539	predicted protein	24: 193701-194375	687 pb	40539	687 pb
Phatr3_J46933	predicted protein	11: 854712-855482	1767 pb	46933	1772 pb
Phatr3_J36444	predicted protein	10: 464796-465641	851 pb	36444	851 pb
Phatr3_J48164	hemerythrin domain-containing	16: 393379-394224	905 pb	HDC	872 pb
Phatr3_EG01906	predicted protein	26: 436989-438052	479 pb	E1906	472 pb
Phatr3_EG02422	P-loop containing nucleoside triphosphate hydrolase	bd_32x35: 126016-129603	953 pb	PCNTPH-2	953 pb

¹Theoretical lengths of upstream intergenic sequences of genes overexpressed in the oval morphotype cultured under salt stress conditions.

²Experimental lengths of upstream intergenic sequences of genes overexpressed under salt stress conditions, obtained after cloning and sequencing.

³ Potential promoters for which no PCR amplification from genomic DNA have been generated or for which the DNA sequencing results were inconclusive.

obtained with the NR promoter to characterise the binding sites of TFs whose genes have already been identified in *P. tricornutum*39 using PlantPAN2.0.

2. Results

2.1. Identification and Cloning of Potential Promoter Sequences

[0218] In silico analyses of the most overexpressed fifty-five genes under salt stress conditions (10% sea water) in the oval morphotype of *P. tricornutum* were performed using the Integrative Genomic Viewer (IGV) and Blast2GO software. This allows identifying the potential promoter sequences possessing a good probability to drive transgene expression. A total of 22 sequences were identified as mispredicted because of the lack of consistency between transcriptomic data and the gene prediction. These mispredicted sequences

[0219] In this study, we considered the entire intergenic region between the STOP codon of the upstream gene and the ATG of the gene of interest as being the putative promoter. The sequences of the thirty-three putative promoters were retrieved from the ASM15095v2 version of the *P. tricornutum* CCAP 1055/1 genome available on the EnsemblProtists website (<https://protists.ensembl.org/index.html>). PCR amplification of these putative promoters was only possible for twenty-eight of them. Finally, the 28 potential promoter sequences that could be amplified and validated by sequencing exhibited lengths ranging from 477 bp to 1701 bp (Table 5).

2.2. Construction of Transformation Vectors

[0220] In order to create a versatile vector that would allow a fair comparison of the 28 different potential promoters, we prepared a plasmid vector compatible with the

method called (Sequence and) Ligation Independent Cloning ((S)LIC). For this purpose, the pPtpuC3 transformation vector used for episome expression in *P. tricornutum* (Karas et al., 2015) was modified to introduce a ‘SLIC-eGFP-NRt’ transcriptional unit (TU). This TU contains the eGFP coding sequence, the NR terminator and a “SLIC” cassette allowing the promoters to be cloned easily. As this cloning technique is based on the 3' exonuclease activity of T4 DNA polymerase, complementary ends must be created in the transformation vector and the fragment to be inserted. The twenty-eight cloned potential promoters validated by sequencing were thus amplified from the cloning vector by polymerase chain reaction (PCR) with specific primers. The inducible NR promoter previously used for the production of mAbs in *P. tricornutum* was also amplified from the pPha-NR vector and cloned in the SLIC vector. In this study, the NR promoter was used as a reference in order to select promoters presenting a strength higher than the one of the NR. To overcome the biolistic transformation “position effect” and the variability in copy number of the transgene inserted into the genome, we used episomal expression as recently reported for the characterisation of new promoters (Slattery et al., 2018; Windhagauer et al., 2021). Indeed, compared to biolistic transformation, episomal expression limits the variability of gene expression between clonal lines that received the same construct. After the transformation, *P. tricornutum* colonies started to appear on selective medium between 10 to 15 days after the transformation. After two subcultures on fresh agar medium with selective antibiotic, about twenty transformants were screened by PCR with primers specific to the eGFP reporter gene. For almost all transformed cell lines, 16 positive clones in PCR were used to assess the ability of the promoter to express eGFP. For cell lines that received the 42538 and DUF11 promoters, only 12 and 7 clones were used respectively due to a lower number of colonies grown on selective medium. Since the promoters tested in this study were isolated from genes overexpressed under salt stress conditions, *P. tricornutum* cells were cultivated in hyposaline medium (10% sea water) from colonies isolated on agar plates. In order to induce the NR promoter under comparable conditions, the cell line that received this reference promoter were placed in hyposaline medium containing NaNO₃ to activate this nitrate inducible promoter from the beginning of the culture.

2.3. Levels of eGFP Fluorescence

[0221] The fluorescence of eGFP emitted in the different transgenic lines was measured by fluorometry. For a fair assessment of the promoter strength, up to 16 clones that received the same transgene were pooled and grown together for fluorometry analysis. Indeed, even if conjugation is supposed to limit the variability of gene expression between clones, difference in the fluorescence level of a reporter protein reaching up to 30-fold has been reported between two independent clones from the same conjugation event (Windhagauer et al., 2021). Promoter-less vector (pPtpuC3_SLIC-eGFP-NRt) was used to ensure that the observed fluorescence results specifically from the expression of eGFP driven by one of the tested promoter and not from residual background due to chlorophyll. To this purpose, fluorescence emitted by the pool of cell line which was transformed with promoter less vector was subtracted from the fluorescence emitted by each cell line pool. The mean net fluorescence of each transformed cell line pool is presented as normalized fluorescence net per OD_{680 nm} unit (FIG. 8A).

Among the 28 potential promoters tested, these results showed that promoters SSR (Phatr3_J37038, SEQ ID NO:29), 40651 (Phatr3_J40651), 42538 (Phatr3_J42538, SEQ ID NO:36), DUF11 (Phatr3_J48356, SEQ ID NO:37), 33783 (Phatr3_J33783), DUF2711 (Phatr3_J43621), PCNTPH (Phatr3_Jdraft1443, SEQ ID NO:30), 35102 (Phatr3_J35102, SEQ ID NO:31), GSTMu3 (Phatr3_J50252, SEQ ID NO:32), VOC (Phatr3_J34085, SEQ ID NO:33), 46933 (Phatr3_J46933), HDC (Phatr3_J48164, SEQ ID NO:34) and PCNTPH-2 (Phatr3_EG02422, SEQ ID NO:35) (Table 5) are able to drive eGFP expression in addition to the NR promoter.

[0222] In order to compare the efficiency of each promoter during *P. tricornutum* growth curve, measurement of the fluorescence intensity of each cell line pool was performed over a period of 8 days. As shown in FIGS. 8A and 8B, the expression of eGFP driven by the NR promoter, our experimental reference, increased until it reaches its maximum on day-2 after culturing the diatom cells in the presence of NaNO₃. This step is followed by a decrease in eGFP fluorescence from day-3 onwards. The expression of eGFP then continued to reduce until it stabilised at its lower level in the stationary phase.

[0223] In contrast, some of the promoter candidates only allowed to drive a low-level expression of eGFP compared to the NR promoter and moreover this expression is observed during a single day. This is the case for promoters: 40651, DUF2711 and PCNTH for which fluorescence intensity is low and only detectable at day-8, day-2, and day-5 respectively. The SSR, 42538, DUF11, 33 783, 35102, 46933, HDC and PCNTH-2 promoters allowed a more cyclic and variable expression of eGFP but mostly at an intensity lower than the eGFP fluorescence under the control of the NR promoter. Finally, the VOC promoter (Phatr3_J34085, SEQ ID NO:33) seemed to stand out from the others and exhibited a better promoter activity than the NR one on the basis of eGFP expression (FIG. 8A).

2.4. Comparison of eGFP Expression Under the Control of the VOC and NR Promoters

[0224] To further compare the activity of the VOC promoter versus the NR promoter over time, statistical analyses were performed using a Wesh's t-test comparison for data following a normal distribution and a Mann-Whitney test for samples with data not following a normal distribution. The NR promoter seemed to have a preferential activity in the latent phase with a maximum intensity at day-2. From day-3 onwards, eGFP expression decreased until it stabilised in the stationary phase from day-6 (FIGS. 8B and 8C). In contrast, eGFP expression under the control of the VOC promoter did not correlate with the different phases of the growth curve (FIGS. 8B and 8C). Indeed, as soon as cells are cultured in a salt-stress condition, the VOC promoter is active. After a maximum expression on day-1, the fluorescence intensity decreased until day-3 before reaching again a maximum value on day-4. From day-6 onwards, eGFP expression slightly decreased again until reaching the same intensity as of the one induced by the NR promoter at day-8. These data are in agreement with previous transcriptomic results reported in the DiatOmicBase (<https://www.diatomicsbase.bio.ens.psl.eu/>) showing that the VOC gene (Phatr3_J34085 gene; Table 5) is largely under-expressed (Log 2Fold-Change: -4.55) in the stationary phase compared to the early growth phase. Interestingly, according to the DiatOmicBase,

nitrate and phosphate starvation or exposition to dark does not seem to influence the expression of the gene VOC.

2.5. In Silico Analyses of Promoters Driving eGFP Expression

[0225] Transcription factors (TFs) have been shown to be key elements for the regulation of gene expression. Therefore, an in silico analysis was performed to identify the potential cis-acting regulatory elements present on the 13 promoters that allowed eGFP expression. The PlantCARE tool was used to identify numerous CAAT-box or TATA-box motifs. These motifs belonging to the minimal promoter to drive expression were found in all promoter sequences allowing to drive eGFP expression. Numerous motifs involved in light response were also identified in all promoters. In contrast, low temperature response motifs were only identified for the SSR, 33783, GSTMu3, HDC and PCNTPH-2 promoters. Surprisingly, putative stress response motifs were not found in all promoters that drove eGFP expression.

[0226] Finally, we have deeply investigated the VOC promoter sequence and the binding sites of TFs. The analysis was performed using the “Promoter Analysis” tools available on PlantPAN2.0. Numerous binding sites for four of the five most reported TFs in *P. tricornutum* were identified (data not shown).

Thus, the VOC promoter exhibited binding sites for the Myb, bZIP (basic leucine zipper), Zinc-finger (C2H2 type) and bHLH (basic Helix-loop-helix) TFs which represent respectively an average of 17.9%, 11.8%, 5.7% and 3.7% of the TFs identified in *P. tricornutum* (Rayko et al., 2010; Thiriet-Rupert et al., 2016). Surprisingly, no binding sites for heat shock factors (HSFs) were identified, although they represent major TF family (30-35%) in *P. tricornutum* (Rayko et al., 2010; Thiriet-Rupert et al., 2016). Several cis-acting elements involved in a stress response have been identified in the sequence of the VOC promoter (data not shown).

Example 4: Optimization of the Growth Parameters of *P. Tricornutum*

1. Material and Methods

[0227] Cell cultures were inoculated at approximately 3×10^6 cells/ml and grown at 19° C. under constant agitation (150 rpm) and under a photoperiod of 16 h light/8 h dark during 8 days. The cultures were performed either in sea water supplemented with Conway or Conway supplemented or F/2. For comparison, the MM medium was also used for the production of the mAbs (Table 6).

TABLE 6

detailed composition of the culture medium used to produce mAbs in <i>P. tricornutum</i> .					
Compounds	Concentration in final solution (SAS)				
	Conway		Conway supplemented		f/2 - SI
	Mass concentration (g · L ⁻¹)	Molar concentration (M)	Mass concentration (g · L ⁻¹)	Molar concentration (M)	Mass concentration (g · L ⁻¹)
Instant Ocean Salt	33				
Sodium chloride					
Magnesium sulfate heptahydrate					
Potassium Chloride					
Calcium chloride					
Ethylenediaminetetra acetic acid disodium salt dihydrate	0.045	1.21E-04	0.045	1.21E-04	0.00436
Sodium nitrate or Ammonium chloride	0.1	1.18E-03	0.47	5.53E-3	0.075
Boric acid	0.08025			8.79E-3	
Sodium phosphate monobasic	0.0336	5.43E-04	0.0336	5.43E-04	
Tris hydrochloride	0.02	1.67E-04	0.3	2.50E-03	0.005
Iron(III) chloride					
Zinc chloride	0.000768	4.74E-06	0.000768	2.84E-06	0.00315 g/L
Cobalt(II) chloride hexahydrate	0.021	1.54E-07	0.021	1.54E-07	
Ammonium molybdate tetrahydrate	0.020	8.41E-08	0.020	8.41E-08	0.01 mg/L
Sodium molybdate dihydrate	0.009	7.28E-09	0.009 mg/L	7.28E-09	
Copper(II) sulfate pentahydrate					0.0063
Zinc sulfate heptahydrate	0.020	8.01E-08	0.66	2.64E-06	0.098
Manganese(II) chloride tetrahydrate					0.022
Thiamine Hydrochloride (Vitamin B1)	0.36	1.82E-06	0.36	1.82E-06	0.18
Cyanocobalamin (Vitamin B12)	0.2	5.93E-07	0.2	5.93E-06	0.1
Biotin (Vitamin B7, H)	0.01	7.38E-09	0.01	7.38E-09	0.0005
					0.0005
Compounds	Concentration in final solution (SAS)				
	f/2 - SI		M&M		
	Molar concentration (M)		Mass concentration (g · L ⁻¹)	Molar concentration (M)	
Instant Ocean Salt					
Sodium chloride			5		
Magnesium sulfate heptahydrate			1.2		
Potassium Chloride			0.6		
Calcium chloride			0.3		

[02228] Both the Conway medium containing NO₃ as a nitrogen source and the MM medium allowed the production of the recombinant mAb in PHAEO in two independent clones (FIG. 9).

Sequence total quantity: 91		
SEQ ID NO: 1	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Signal peptide E of azurocidine	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 1		
MTRLTVLALL AGLASSRA		19
SEQ ID NO: 2	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Signal peptide H5	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 2		
MDWTWRFLFV VAAATGVQS		19
SEQ ID NO: 3	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Signal peptide H7	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 3		
MEFGLSWVFL VALFRGVQC		19
SEQ ID NO: 4	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
REGION	1..22	
	note = Signal peptide L1	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 4		
MDMRVPAQLL GLLLLWLSGA RC		22
SEQ ID NO: 5	moltype = AA length = 18	

-continued

FEATURE	Location/Qualifiers	
REGION	1..18	
	note = Signal peptide B	
source	1..18	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 5		
MKWVTFISLL FLFSAYS		18
SEQ ID NO: 6	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Signal peptide of the human BIP protein	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 6		
MKLSLVAAML LLLSAAR		17
SEQ ID NO: 7	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Signal peptide of the BIP protein from Phaeodactylum	
	tricornutum	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 7		
MMFMRIAVAA LALLAAPSIR		20
SEQ ID NO: 8	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Signal peptide of the BIP protein from Phaeodactylum	
	tricornutum	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 8		
MIFMRIAVAA LALLAAPSIR		20
SEQ ID NO: 9	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
REGION	1..16	
	note = Signal peptide of the human PDI protein	
source	1..16	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 9		
MLRRALLCLA VAALVR		16
SEQ ID NO: 10	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Self-cleaving linker peptide E2A	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 10		
QCTNYALLKL AGDVESNPGP		20
SEQ ID NO: 11	moltype = AA length = 22	
FEATURE	Location/Qualifiers	
REGION	1..22	
	note = Self-cleaving linker peptide F2A	
source	1..22	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 11		
VKQTLNFDLL KLAGDVESNP GP		22
SEQ ID NO: 12	moltype = AA length = 27	
FEATURE	Location/Qualifiers	
REGION	1..27	
	note = Self-cleaving linker peptide variant of F2A	
source	1..27	

-continued

	mol_type = protein organism = synthetic construct	
SEQUENCE: 12		
KIVAPVKQTL NFDLLKLAGD VESNPGP		27
SEQ ID NO: 13	moltype = AA length = 24	
FEATURE	Location/Qualifiers	
REGION	1..24	
	note = Self-cleaving linker peptide variant of F2A	
source	1..24	
	mol_type = protein organism = synthetic construct	
SEQUENCE: 13		
APVKQTLNFD LLKLAGDVES NPGP		24
SEQ ID NO: 14	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Self-cleaving linker peptide P2A	
source	1..19	
	mol_type = protein organism = synthetic construct	
SEQUENCE: 14		
ATNFSLLKQA GDVEENPGP		19
SEQ ID NO: 15	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
REGION	1..18	
	note = Self-cleaving linker peptide T2A	
source	1..18	
	mol_type = protein organism = synthetic construct	
SEQUENCE: 15		
EGRGSLLTCG DVEENPGP		18
SEQ ID NO: 16	moltype = DNA length = 23	
FEATURE	Location/Qualifiers	
misc_feature	1..23	
	note = wild type encephalomyocarditis virus (EMCV) internal ribosomal entry site (IRES)	
source	1..23	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 16		
atgataatat ggccacaacc atg		23
SEQ ID NO: 17	moltype = DNA length = 46	
FEATURE	Location/Qualifiers	
misc_feature	1..46	
	note = mutated EMCV IRES with attenuated translation efficiency (IRESatt)	
source	1..46	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 17		
atgataagct tgccacaacc cggggagatg aggatcgttt cgcgatg		46
SEQ ID NO: 18	moltype = AA length = 4	
FEATURE	Location/Qualifiers	
REGION	1..4	
	note = furin-cleavage site	
source	1..4	
	mol_type = protein organism = synthetic construct	
SEQUENCE: 18		
RRKR		4
SEQ ID NO: 19	moltype = DNA length = 241	
FEATURE	Location/Qualifiers	
misc_feature	1..241	
	note = Termination sequence of the FcpA gene	
source	1..241	
	mol_type = other DNA organism = synthetic construct	
SEQUENCE: 19		
accttcctta aaaatttaaat ttccattagt tgcagtcact ccgctttggt ttcacagtca		60

-continued

```

ggaataacac tagctcgtct tcaccatgga tgccaatctc gcctattcat ggtgtataaa 120
agttcaacat ccaaagctag aacttttga aagagaaaaga atatccgaat agggcacggc 180
gtgccgtatt gttggagtgg actagcagaa agtgagggaag gcacaggatg agttttctcg 240
a                                                    241

```

```

SEQ ID NO: 20      moltype = DNA length = 273
FEATURE           Location/Qualifiers
misc_feature       1..273
                   note = Termination sequence of the NR gene
source            1..273
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 20
caatttactt taagactagc gcaatttttt tcttacatta gatgaacata aaacgacgat 60
gaggatcact gttagggaaa cgatgaacta cgtcaggac ttcctattat tcgtcctttg 120
atggcggtgt tggtctttgg ttgcgggcgg agcgcaacga ccaaggcctg taccaaaaac 180
gaagttgata atcagatatt tgagttgtaa caagaaaacta aagcattatt ttagcagtag 240
cacgaatcag gatctaaaat gaacgcatcc gga                                                    273

```

```

SEQ ID NO: 21      moltype = DNA length = 501
FEATURE           Location/Qualifiers
misc_feature       1..501
                   note = Termination sequence of the H4-1B gene
source            1..501
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 21
gcttgcgctt gatcctcgac tttgttgctc gtctttaaaa ccttggaaacg attatataca 60
atcaccaact caaaaacggg atttttctaaa atccacccaa acaaccaaag aaaatacttc 120
tcattgcatt tatgaatcac agcagacctg cgtcctttta aaacttagat cctgttttct 180
tataaaaaac agatcaaatt ttctgggagt tcattgactc tgccagtcag aatcaatcct 240
gcagtaattc tttattttaca ggtgaaagta aaagagaatc ccaatttttt gcttgtacat 300
taaggtcctc cttacattac agctaatttc aaaataagat gaagttggat tcgtgtcctt 360
tcatggtgat gtgatatttc gcactatacc aaacaccgtg aaatgtcagc tagagcttgt 420
catgaggcag ttgtgtccaa tcactacata tagatccttc acggagaaaaa gttggcttca 480
ttctctgttg ctaatcggtc g                                                    501

```

```

SEQ ID NO: 22      moltype = DNA length = 501
FEATURE           Location/Qualifiers
misc_feature       1..501
                   note = Termination sequence of the V-ATPase C gene
source            1..501
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 22
acggacgctt gttggagatt ccgcttttac ctgatttcca gttcctgtaa tttgtcgtt 60
ccttatgttt ccacacaaac gactgcaagc aaaatcaact aatttgttta tcacatttca 120
atagccagtt tgaatgtttt tcttacgata ctgtaggcag tggaagcgga attggttttt 180
gtgggcaacc gaaaaaggct tgcaattttc aattagggta acagtagtcg gtaaacaaatt 240
gctcatctag ctctaggaag tctattaatc cacagggtgg aaacgcaatg aatgtgccat 300
gtatgacggc gccgggtgtag gcaggagaat ttctctaatt taatctttgt tccacgactc 360
ttacagaggg acccaaaacca tcgcttcgat cagttgacga gaggtttctg aaaacgatct 420
tatcgaggcg ccgttggtgg tcctttcgaa agaatacatt ccaacatgta gctttgggat 480
gcaaaacttt cattggatgt g                                                    501

```

```

SEQ ID NO: 23      moltype = DNA length = 501
FEATURE           Location/Qualifiers
misc_feature       1..501
                   note = Termination sequence of the gamma-tubulin gene
source            1..501
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 23
agcaaaactc ttatgatgca tgggagtgcg accgagtttc gaacgatgct aacgaacatt 60
attagtggaa ggcagtcatt gttgtatgcy tcaaagtata taatcagacg gacagtagtt 120
cattttaaac ttttgctcgg aaagcgttga tcattcatcg gggaaatcgg ctaacgagca 180
gtaattggag ttgtaactgc aggtcacaac cgtttctggc tcccggtggg taccataatg 240
ctaactatca ttctttattt gcagtatcac cagtccaaga attattcgtg gtcattttctg 300
atgctactgt ggagaagtga gagtaatttc gccgactttg aagtgaagac gcgtctccga 360
acgtagattg ttgttattgt accattgaag gagagttttt gaactcaggt tgttgatcgt 420
aatgtccgcy accctctggc gcactacacg ccaatgacca taacatgtct tggtcgccct 480
cctcgtaagt catcgccatt g                                                    501

```

```

SEQ ID NO: 24      moltype = DNA length = 598
FEATURE           Location/Qualifiers
misc_feature       1..598

```

-continued

```

note = promoter of the H4-1B gene
source      1..598
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 24
gtctctttcc cgggaaacgg gtacactcct cgcgcgccaac aacatattac tactactacc 60
aagaacgtcc acggccttgt cgtgcgttac gctctcccaa cgcgtgcggg gtaaattacg 120
tctcggtttg ctaagtagcg cacagctaaa tagatgaccg ttattgtatt taagatcatt 180
caatattgat tgcattgtac tttgcgtcaa actgaaatcc cctcgtacta acggttaacc 240
cgtcaacctc aagcgttcgc ccaaagtagt caaccgggac acgcgaaccg acattgggca 300
gatctttcac agacagaaaa ccatttccaa tccaaataag catgactatt acacacccat 360
tcgtagcgcg aggacaaact gatagctcca acaaaatgcg ccaacatcgt acattgtaag 420
aagcttacgg aacactatgt atgtagaacc atacgaacag caactagtac tggccatcga 480
gcagcgggtg ctcccggttc tcgtagcgct gtgaagggtt cactctcaca attcgtcttc 540
ggctacaacc gacaaaagtc ttactcacag tcaataccga aaacaaacaa cagccaac 598

SEQ ID NO: 25      moltype = DNA length = 500
FEATURE            Location/Qualifiers
misc_feature        1..500
                    note = promoter of the V-ATPase C gene
source              1..500
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 25
cgattggaat ggttttggag aggaaaccgt ttttatggcg gttggatggg aaaggaaggg 60
atgggtgttc cactcagatt tgggagctca cccaccgtat gttgacagta caaacatcc 120
tgcaatcctg cgaaccaact gtttctcgtc tttttgggtc gtacgtggta ggatagagag 180
tctgccggga taccatttcg tgcgcagacg tggaaagatc attattcagc ttgcacgaca 240
cagattttat acagacaaac gtacaatacg caattgctcc gacaaaaatt gccgaccatt 300
cctacaggca tatttttgca tataattagg ggaacgcgat agaatgcaat cttcagttac 360
atacagacaa cattctgtcg cagctcgcca cagctaccaa ctccagattc cgagcttcgc 420
cgcgcccttt tggaaagatc gaaaaggcgc cgccgcggct gtcgttcccg cagaaaagtc 480
aagcagccac tagcgcaaaa 500

SEQ ID NO: 26      moltype = DNA length = 422
FEATURE            Location/Qualifiers
misc_feature        1..422
                    note = promoter of the NR gene of Phaeodactylum tricornutum
source              1..422
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 26
catatgcgga agtgactgta aacgagaagt gcacgaagcc ttttctgttg acgtcacaaa 60
ccgaacagcc ctacgtggcg tgcgactttc cggcactttg gatatcggtg ccctatatgt 120
attgggtgta tacatgtcgt attccaacta cgacagaaaa cagtcagtta cgggtattcg 180
aaacttcggc acctgcagcg aggagttttg tgtccgtcgc acatcctccg tgctttcggc 240
acctacgtag gcggcaaaact tcctgccttc cgcccgcttc gcggcatttc ggctccgggc 300
cagaattgcc cgggtgttca caattttgcc tcctcacgaa aaaacgttct actttgtatt 360
ttggtcgggt ttcggatcct tccagcaacc attctcattc aaagtaccca cttgtgcgaa 420
cg 422

SEQ ID NO: 27      moltype = DNA length = 444
FEATURE            Location/Qualifiers
misc_feature        1..444
                    note = promoter of the fcpA/lhcf1 gene of Phaeodactylum
                    tricornutum
source              1..444
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 27
gggctgcagg acgcaatgga ggattatcac cgcaaaaatg aacttcgaaa aaaactttcg 60
agcgaccatg gaaaaggagg atcagattca gattacaaca gtggattgct ctggtagcaa 120
atatcttctg ctagattggc tcatggtcgg ttttggacgt tcgaagctca ccgtcaaaag 180
aaacaaaaga gaagaatgac gtcttcgtga cgtagaatct acgactgtac tcggatctgg 240
gaaatgaatt gactcacggt ctctctcgag tcctgttaca ggcccttggt ccgaaccccc 300
acacgatttt tgcaccaaag atttgcttca atttgctgga tgttttgact gcaagatcag 360
ctggcctagc aagagtgttc gtgttgcttc gtcgggaatc cctacgaatt tcagttctgc 420
acaaatttgt ctgcggtttc gaga 444

SEQ ID NO: 28      moltype = DNA length = 187
FEATURE            Location/Qualifiers
misc_feature        1..187
                    note = promoter of the fcpb /lhcf2 gene of Phaeodactylum
                    tricornutum
source              1..187
                    mol_type = other DNA

```

-continued

```

organism = synthetic construct

SEQUENCE: 28
gagattcaaa attcaatctg tttggacctg gataagacac aagagcgaca tcttgacatg 60
aacgccgtaa acagcaaatc ctggttgaac acgtatcctt ttgggggcct cccgctacga 120
cgctcgctcc agctggggct tccttactat acacagcgcg catatttcac ggttgccaga 180
tgtcaag 187

SEQ ID NO: 29      moltype = DNA length = 1246
FEATURE            Location/Qualifiers
misc_feature        1..1246
                    note = Promoter of the Phatr3_J37038 gene
source              1..1246
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 29
tgtcgctact tcaccacagc tttattgaat agggacgagc ttttattaat catgtgctgc 60
tgtccttttt gcatcaggat tgtcacgtag ccacttttga ctgtaattaa tcttgacttc 120
ttatctctaa aaagcggcga gatcaaggtc cttctcagtc aaacctcgca gcttgtgtgt 180
ccaaattttt acctogatat ctcggttaact agtaagatgg aagtcggggg ggtgactttc 240
gcgttcagct atggccccc tctcgttgat gcatttttaa gcagactgaa aattgctgtc 300
cgtaaattta cgagatatat gcaaagggtc tgattcactc gagcaaatg tccacagcga 360
tccccttata ccaagtttgt ccactgcact ttcaaatgtc aatagaaaag atcggtccag 420
agtagagcaa ggagcgcaat gtttcgggtc ttgcggggag cctgaagcca tcactaatac 480
tgtgaaaatg agagccggca atttctgaag actatgtaag tctccgaaca ggaagggaagc 540
gcaagcaaac cacaattcga actccagata aacttgctgt tagcaccaat aacatgacca 600
tgtacaagcg tcgtctgctt gtgagccctt caatatgatt gattgagggc agatccttag 660
cactttaacc ctgtaaacat tttgtttttg gctctcatca acggcagctg tagcgaatct 720
atagcctgcc gggcagttat acgttcatga tcaaaaatga agtgtatata agtcgataaa 780
tgtaacctct tttcttgeta tataattttt ttggaacgat gaacactaaa cacacatagc 840
tatctgacat cgaacataaa tgtaagagtc attccaaata agtagagatg caatgtggac 900
atcatgggat gatgtgtgga ttacgatcc cctcaactgt aaaactattc ttaggcacaa 960
tttaccgagc gaacaaaaat actaaagaaa cttagcatat ttctgaaagc attgtaattt 1020
tcaacctttg tgaatttttt tctatatatt ccgaagtgtg acagtgaatt ccaaatcttg 1080
agctgtgctt gtgtactttg acatgacaaac ttccggacac gctcactttt tgacgaccac 1140
acgttcaatc gttgctgcgc tccggcacag actaaagttc aggaattgct tgtcacctac 1200
gcagcatcac cactctctcg tctctaccgt gattcaaggc tgaatc 1246

SEQ ID NO: 30      moltype = DNA length = 1125
FEATURE            Location/Qualifiers
misc_feature        1..1125
                    note = Promoter of the Phatr3_Jdraft1443 gene
source              1..1125
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 30
tcgggttgga gtactgctag gagcaccata agccaaagac tatgtgctgc tggatgaactc 60
taaaagcgga cggcaattaa cggtttgtgt ttcttaacca ttgtcatgat ggtggctgc 120
gaggttggtt ctgctgcttg tttagcaagat tcaacctcgc atctcggtat aaagtgcgca 180
atgctgtctac agcaactcac catcaatctg cgatcttgtt cggtcgcaag accacgaaat 240
ttgctggcct tcaggaaagg aaaatactgt tggctctatat ccgagctctg gtagtagggca 300
acacgacaat gctgtgctgt atatgaagtc agacaacaca ccacaaaccc atactcccaa 360
ataaatgtat agcgatttat gaaaagagct cgggttcaaa aatttgtcca aacatggctg 420
gaaatattcc aaaagcgaca aaggctaaagt aagctaaagtc ctcttggtgc ttgaaaggaa 480
agctgcagag gggatgattt aaaattacgc gaaagggaag agcttctaag atggagagag 540
agagagaccg cagtcattgt tacaaagtga agctacaaaa tggccagaac aaatgatgta 600
ctattttggc tacgtgtagg tggaccggtc ctattggcca aatgagccca aacagaaccc 660
tagccacaat ttttttacta attaaagctgt ggtctttctt atgctgcagg tgggtttcat 720
ttctcgaaga acgttttcgc tccccgaggc ggtcttgctc cattagtcaa tgatgtcacg 780
cataccatat tctgcgact accgtctcaa ataaagccgg taacgcaaca atatgtaggc 840
tgatgttatg gtgtgattta acaaggccca aacagttttt taacaggtta ctattggcga 900
ttcaagacat cacaataatg accatcttaa acgattcaga gtgtgactca tccgggaaag 960
tagggtcgca tgtttcctga gagaatcgct tcatgagatc tcatcggttg caactggctc 1020
cacaagtga cgttagataa gcaataacag caaacctttg tgatttcagc taactgtgac 1080
cagatcgctt ctagactgga aacaactctc atccattcca cgact 1125

SEQ ID NO: 31      moltype = DNA length = 535
FEATURE            Location/Qualifiers
misc_feature        1..535
                    note = Promoter of the Phatr3_J35102 gene
source              1..535
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 31
taattatcat tgtatgttgc ttctctagga atttatagtc aagcttcctt ttgctactac 60
cctgctctcc aatcagctgc agggaagcaa gatcctctcc ggctagggtc ttagcatggt 120
tcttggccta cgctagctag gaaggtacca taaaaacgag agctattaat ggctcagcat 180

```

-continued

```

ttatccgata catgtttaac agaagacag aaaacacgaa aaaagtgtgt gacagtgaat 240
gctccagatg tcacaagaaa ggcgttgggg cgaatcgatt gggcctttgg aggcattagt 300
aaaaaggaga aagcttttta atctttgtcg acatgtactt ctatacaata ttgctataga 360
attgaaggac agcttgaat gtctatcgac atagtgactc cgacttaagg tcattcacta 420
tttttacgcg tgaccgcgga cccaaaaccg gaccggcagt tcctgcccc aacataatac 480
aaattatagc tccaaaattg actctctttg aaccgacgac ccaattggtt ctacc 535

```

```

SEQ ID NO: 32      moltype = DNA length = 1098
FEATURE           Location/Qualifiers
misc_feature       1..1098
                   note = Promoter of the Phatr3_J50252 gene
source            1..1098
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 32
aaattgtata ctccagcagt cgtttaatac cggagacggt agacgcaaga gcagctcaca 60
ctctttggcc acgactgtaa atgcctctag tgttgacgag tttgcgtgca cgagagaaat 120
tttcgagaga acacaggaag gaccccagta cttttgggtc cttttgagct tgcaacgctt 180
ttcgtttcga aaaccaaattc caaatccctt tcgacacagg ggctttttct cgcgcagtgt 240
ataaaaaagca aaaaaaaccc ggagatcctc caaggtcgtc actgtcagaa tacggtgact 300
ctagaaggca atgcggaact cccagccggc cgcatactat tgacgatgaa tcatcttgct 360
tttcattact ataggttggt ggcgcctctc cgatttattt cgagtgtatc tgtttaaaat 420
gttctctcatt cgggtgaattt tcttgacgac catgtcccaa gtctacact accacagcag 480
atgatggaat taaggtcacc gtcaaaaggt agacagagtc gcgttagaaa ccatgttaga 540
taaagggtca aacggaatgt tgcggattcc ccagggtat agacaggact accaaagaga 600
tctgcgatca aaattttggt gcacagcaca tataccgaaa gataacttga gttatcctta 660
cattgggagt cccaaaaatc aatctttcga cgtctctcgt gccgcaacaa ttatagtgcc 720
agttggaaca aaaaaagtat caaatgggtc aaaaaccaca agagaaacgc tcactaatag 780
aaattctgtc gcctcttata tttctttgaa catgagcgtt attagcaggt aaacagttgt 840
ttcgtaaatc tgacattaat agtggagtgt ggtcataatg ccgcgtcctc ctgatgtcga 900
aaactatggat tcattgaaaa cgttcacctc tcatagcaaa ggaacttcgc tttctagtaa 960
ataaccacac aaacaaatct cctcacctct ttcgttgaat ctggcatttc catcaacacc 1020
ttttcatagc tttgtggatg gattgagttg cgctcgctgt caccatttgc gccgaattcc 1080
ttgcccttta cagtaatg 1098

```

```

SEQ ID NO: 33      moltype = DNA length = 957
FEATURE           Location/Qualifiers
misc_feature       1..957
                   note = Promoter of the Phatr3_J34085 gene
source            1..957
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 33
aatcataggc aatgtttggt acacaataaa tttgacgtag ttcattatgt gctgcaatac 60
tgaatacttg ccaggagtga caggatttcta gctcctgagg agcgcttaca gtggaattac 120
acacatgact tgtaccgaca gccaaactaac tataataaga agcaatgcgg acttggtagg 180
atcctttcat caattcaaac attcactatc aatcaaaaat tacaaaattt acttgcttgc 240
agtcagcata tatttgtcaa cgcattgggt tcttctcatg tcaaatagaa atgtcacgag 300
aatagctccc gggcccgtga agttcttaat tctgtaaact gtccttgcca ctccatttgc 360
acttgacgta tcacttgtcg ctctcaacac agcattacac aatggcactc tacagagcat 420
tgggttaagt aaattattga aaggcaaaac atagcccttc ttgactgtga gtcccaccat 480
cgctacgagt cacttacatg taaaatatct gtccttccat cgagggcacg gaattctgaa 540
acgaaaacaa tcttgttggg ctccggtagtc cgttacgcaa aaacgcttct atcaattcaa 600
ttaggattgg tcttggtttg gagctgcccg acgcgctcaa aaattgactg ggtatgggta 660
cactttctct ctgtcgaaat attttgtaca gtcattttgg ttcttgggta tgcttatagg 720
agatgatcag ttgatgcgag acggattgag cccaaacaat ttacatcata gagatatgca 780
ctttggaaaa gaccctctgc ggtcatctgc aaatcccaca agctttttgc tctcgaaact 840
tcacaactct acccgagttg ctatcggtgc tagcagcctt tcccattgat tgccaacatt 900
ttgatcctga caaccatag attcgctctg ttaaaccaac agcaacagca agcaaat 957

```

```

SEQ ID NO: 34      moltype = DNA length = 872
FEATURE           Location/Qualifiers
misc_feature       1..872
                   note = Promoter of the Phatr3_J48164 gene
source            1..872
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 34
gttgaacta cattgtgtgc ggatgactcg ccgactgcga tacgatagat tggatgatcg 60
gcgatcgcca atgaagaca gcccgctgag attgcagctt ggcgtgctct ccgtgtggta 120
gactagtga gcaacgagga agcggccaaa gagcttgtga aagtggggca tcgaatcaaa 180
ggaatagcta attgtatttc cacttcaaga agaaaagata caactgagcg attgtcacta 240
tacttttcaa gttggagagc agttgtcact ggattttgtc tccctactag aaccgacggt 300
cgaagatctg actggtgcca tgtgcacgag aaaaatatca taccggttgg tagtcttttg 360
aaatgtctct ttgattttga aagcgaatcg agatttttat cacagtcaat tttagtttgt 420
ttcgggttaga tagaactgta aacttgaaca attgtaaata atcattgaga cataagttcc 480

```

-continued

```

gtcggaaata ctttaagcag tttgtgtgca aacatgtaca aacatttact tacttatatc 540
tactgtcgca tggagatttt gcgttcttgt cttttagttt tgettccgc ttatgcttct 600
tagttcaagt tgtgaaacag tgcctactct cttcagcagg atactaacat aaatcgattt 660
tctttccccc ttacagttta taggtcaaat ggaagtagtc atcattttca tggcctcgcg 720
ctggctaccg agacatttca ggaatcggtt gaagtaataa aacttgaaaa cccgaataaa 780
ctgctctggc atcttggcga gtttcgcaag cacttacatc tcacaacagc aatactcggt 840
tcctccgttg tgtagaaaca agcctagaga at 872

```

```

SEQ ID NO: 35      moltype = DNA length = 953
FEATURE           Location/Qualifiers
misc_feature       1..953
                   note = Promoter of the Phatr3_EG02422 gene
source            1..953
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 35
ttaccaccaa gcgctcttct atctaccagt caacactaat gataatgcac gtatatgaat 60
gacatatttg tttttggact aaagattgac ttgatctgta tcattaaaaac ttgtaggcgg 120
cgtggcattg agacacaggc tgtatttcgc cgcaaaaagc atgaagtctc gttttgaagg 180
ggcgggagcg tccctacacct gtaactccgtc attgttgtgc acactaggat gtgaaagcgt 240
gtcgaaagga gtcattgtcag gactggacag agaaacaagc aaacctaacg tgacgatgtc 300
tgactgtttg gattctgaag ggtgattacg ttgctggct tacattaaat catgttttta 360
tttactgtca gtatagataa atagcgattt ttcggtgaac taattttttt caaagtgttt 420
cgttgaaagc agtaggatcg gacgcagagc atttacgggt tcacatgagg ttgaaagga 480
cgagcggttc cttcgtctcg acacaaatca tagatgttc acctggcgcg acgatcttcc 540
tgacggaaag ctaaatatgg ccaactagatt gccaaacctc atgtatttca atagtggaaa 600
agcccgctca gacctaggcc caagtatata ttcggaagga catgtccgat atatctctac 660
aaaggatgag cggagtcaca ctcggtcgta gcgtctttca aacacttgct ctcttggttt 720
catttttacc ggtcatttgg tcattgagca aaaaggccct ctggtagcgt ttgtaggcat 780
cgctacggca aatgcctgtt cagcacggtc acgagccatt cgctctgtgt cgttgggcca 840
cttttctcgg attgtgaag agtttcagca tcagtcgtca aaatatcaaa ttatattcag 900
aggtaaacag cgcttctgcc aaattgcaac tctatcgaaa aaacagaacg aac 953

```

```

SEQ ID NO: 36      moltype = DNA length = 1092
FEATURE           Location/Qualifiers
misc_feature       1..1092
                   note = Promoter of the Phatr3_J42538 gene
source            1..1092
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 36
gaagatgagc agtttccgca tcaccggaag ggacacatca ttacacgaga ttaatctaag 60
caaaactcag ctcaagaaag gcgatgttcc aacaagcggg tggatgggct cttgttgcaa 120
tgcaaaaaag ctcagaaaaa actagctttc cgacctttta cgtatttctt ggtaattcgt 180
agaagacgcg gcgacgactc acttttaacc ttttgtgact gtctcgccca tcgtctacaa 240
ttagttgttg ctgactgtga ccaacaaacg gagactcttt gccatgtct ttctgtttac 300
gtcgcaaaat gacagtcgtg tgettctcgg gattgactgt gagtcggtgt gacagactct 360
cgcaatcgaa aatgcttcat aataccgaat agagatacgg ttgtactaag tcatcactct 420
aactctgtcg cggtaaatgt gaagatgtat ttcgaatgta gaattgacag tgaatgccat 480
ggattgtgcg ggctaaatat ttgtgttata tgcaagagtt ttgcgatgta attatctaag 540
caaaactttt cattgtaaca agagaaaaag ctatagatgg agctatgggc cgattgtgtt 600
ttgaaatcac gtacaaatcc atactgggtc tacgtgttac gtcctcaacc ccacctaaac 660
tggtagtgac tgtgaattcg cgcggaaata tatacgtgta gttgtgggca ccaacctatc 720
tgaatcactc ctatgtatcc ctatcagaat agtctatggt gaatttcatt gaagtgaacc 780
cgagtcgata cagcaaaacc tatctagaaa acggggatga taaagcaaaa agaaaaggct 840
tatttgtaaa tcgtgtagat ccataaatcg gtactggtag aatcgatcca ggaatgctat 900
tcaatatact atcctgtttg tcgggattgt tccatcagaa atcgagtaga actggtagac 960
gctcgaccgc cagctgtttc tgttcgagaa aatcttttca tcaaacgtgt tcctcactct 1020
gaaaaagcct tggtttcatt ccggccacaa atctgaaaag acatctttca tcgatcccg 1080
atctcaacca cc 1092

```

```

SEQ ID NO: 37      moltype = DNA length = 985
FEATURE           Location/Qualifiers
misc_feature       1..985
                   note = Promoter of the Phatr3_J48356 gene
source            1..985
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 37
cctttaggag aacaagctgc ggaaggacag cgaataatcg aagaatgtat ggggagggg 60
gcgatccag tataaagtga aatatcgata gggatggaag cagcagaaag gaaagtttaa 120
ttcgtgatc cagttctgga tgagtccttg tctttgcggg cttcgtgtaa gttacttact 180
gttagtggtta ctgttgttca gaggcaaaact tgcgtctgtc cttcggagtt tttttgttcg 240
cagtcgctgg ctggtcgtgt ccgtgttgct gtctatgtac cgcgttttga ccactgctcg 300
taaatgcacg cctaccagat gaggaacatt tccatatagt gtctcacagt cgtgttttga 360
atcattcact agtatcccat cagattgca agcaccatg cttttcattc gtagttagt 420

```

-continued

```

ggctgattct ctacacttat tgttggttagc agccgaggaa gacattttct ctgtgacact 480
ttcaaagggtg gtttcgcgaa acgactccga ttgttacaat taggtcaaga ttaagagggtt 540
cgtacctctt atacgcgcaa agaacaagat tcagaagaaa aaaatgattc ttagctcgta 600
ggttcgggca ttggtctctc aagagggtcta tagttattgc gtttttaata agcttctatt 660
gttcgcctta gtcttttagct cgtaaatgac tacgtttcac attgttgcaa aaatggaaaa 720
caacgggtgta taacctagaa tgtacaaatg ctggatgtct ccttagccgc gccactgta 780
tcgatcgatc ctttcgggca gagccaaata tgttcaattc gtctatacta gaattaagat 840
ggctctttgat gccttcgagc tcgaatgacc aatcaaaaat cttgccggtc cctcatgaa 900
ttagaacaca aatcgaggga gctctcattt tctctcacat atttcagtct agtaggcata 960
attttaccaa aaggctgcag cgaag 985

```

```

SEQ ID NO: 38      moltype = AA length = 221
FEATURE           Location/Qualifiers
REGION            1..221
                  note = light chain of the antibody TrastuzumAb (Herceptin)
source            1..221
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 38
DIVMTQTPLS SPVTLGQPAS ISCRSSQSLV HSDGNTYLSW LQQRPGQPPR LLIYKISNRF 60
SGVPRFRFSGS GAGTDFTLKI SRVEAEDGVV YYCMQATQFP HQWTFGQGTK VEIKRTVAAP 120
SVFIFFPSDE QLKSGTASVV CLLNNFYPRE AKVQWKVDNA LQSGNSQESV TEQDSKSDSTY 180
SLSSTLTLSK ADYEKHKVYA CEVTHQGLSS PVTKSFNRGE C 221

```

```

SEQ ID NO: 39      moltype = AA length = 294
FEATURE           Location/Qualifiers
REGION            1..294
                  note = Heavy chain of the antibody TrastuzumAb (Herceptin)
source            1..294
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 39
EVHLVESGGG LVQPGGALRL SCAASGFTFS NYDMHWVRQA TGKGLEWVSA NGTAGDTYYP 60
GSVKGRFTIS RENAKNSLYL QMNSLRAGDT AVYYCARDVD IVATITDYFD YWGQGLVTV 120
SSASTKGPSV FPLAPSSKST SGGTAALGCL VKDYFPEPVT VSWNSGALTS GVHTFPAVLQ 180
SSGLYSLSV VTPSSSLGT QTYICNVNHNK PSNTKVDKRV EPKSCDKTHT CPPCPAPELL 240
GGPSVFLPPP KPKDTLMISR TPEVTCVVVD VSHEDPEVKF NWYVDGVEVH NAKT 294

```

```

SEQ ID NO: 40      moltype = AA length = 213
FEATURE           Location/Qualifiers
REGION            1..213
                  note = light chain of the antibody RituximAb
source            1..213
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 40
QIVLSQSPAI LSASPGKVT MTCRASSSVS YIHWFOQKPG SSPKPWIYAT SNLASGVPVR 60
FSGSGSGTSY SLTISRVEAE DAATYYCQQW TSNPPTFGGG TKLEIKRTVA APSVFIFPPS 120
DEQLKSGTAS VVCLLNNEYP REAKVQWKVD NALQSGNSQE SVTEQDSKDS TYSLSSTLT 180
SKADYEKHKV YACEVTHQGL SSPVTKSFNR GEC 213

```

```

SEQ ID NO: 41      moltype = AA length = 451
FEATURE           Location/Qualifiers
REGION            1..451
                  note = heavy chain of the antibody RituximAb
source            1..451
                  mol_type = protein
                  organism = synthetic construct

```

```

SEQUENCE: 41
QVQLQQPGAE LVKPGASVKM SCKASGYTFT SYNMHWVKQT PGRGLEWIGA IYPGNGDTSY 60
NQKFKGKATL TADKSSSTAY MQLSSLTSED SAVYYCARST YYGDDWYFNV WGAGTTVTVS 120
AASTKGPSVF PLAPSSKSTS GGTAAALGCLV KDYFPEPVTV SWNSGALTSV VHTFPAVLQS 180
SGLYSLSVVV TVPSSSLGTQ TYICNVNHNK SNTKVDKKEV PKSCDKTHTC PPCPAPELLG 240
GPSVFLFPPK PKDTLMISR PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY 300
NSTYRVVSVL TVLHQDWLNG KEYCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRD 360
ELTKNQVSLT CLVKGFYPSD IAEWEESNGQ PENNYKTPPP VLDSDGSEFFL YSKLTVDKSR 420
WQQGNVFPSCS VMHEALHNHY TQKSLSLSPG K 451

```

```

SEQ ID NO: 42      moltype = DNA length = 5722
FEATURE           Location/Qualifiers
misc_feature       1..5722
                  note = Nucleotide sequence of NVB5
source            1..5722
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 42

```

-continued

ttcatatg	gaagtgc	taaacgaga	gtgcacga	cctttttt	tgacgtaca	60
aaccgaac	ccctacgt	cgtgcgact	tccggcact	tggatcgt	tgccctata	120
gtattgggt	tatacatg	gtattcca	tacgacaga	aacagtcag	tacgggtatt	180
cgaacttc	gcacctgc	cgaggagtt	tgtgtccgt	gcacatcct	cgtgcttcg	240
gcacctac	aggcgga	cttctgcct	tccgccgcc	ttggcgcat	cgggtccgg	300
gccagaatt	cccggtgt	cacaatttt	cctcctcac	aaaaacgtt	ctactttgt	360
ttttggtcg	gtttcggtc	cttccagca	ccattctcat	tcaaaagtc	cacttgtgc	420
aacgatct	ggatttcca	gtacaaat	tcagttttc	acttatctc	gccagtgtg	480
taattagta	ggcgacatt	gtcatgac	aaactcctc	atcctcacc	gtgaccttg	540
gacaaccgc	ctctatcag	tgcggttcc	cccaatcgt	ggtccattc	gatggaaca	600
cgtacttgt	ttggtccaa	cagcgccct	ggcagcccc	gcggtcttt	atttataaaa	660
taagtaac	atttagcgc	gttccggac	ggttcagtg	ttcgggcga	ggtaccgatt	720
tcaccctaa	gatctccgc	gtagaggcg	aggatgtag	agtgtatt	tgtatgcaa	780
caactcaat	tccccacca	tggagtttc	gtcaaggga	taaagttga	attaagcgt	840
cgggtgctg	gccatcgtc	tttattttt	cgccatctg	cgaacagct	aaaagcgga	900
cgggttctg	cgtctgtct	ctgaacaat	tttaccgcg	agaggcta	gttcagtga	960
aggtcgata	tgcgttcag	tcgggcaac	cgcaggaa	tgtgacgga	caggacagc	1020
aggactcc	atacagcta	tcgtcgac	tgacactgt	aaaagcgga	tacgagaa	1080
acaaggtga	tgcctgcga	gttaccatc	agggtttgt	atccacagt	accaaatca	1140
ttaatcgag	tgaatcgag	ggcggaggt	cgtctttga	gtgcggtga	gtcgaagaa	1200
acccgggac	catggattt	caggtccaa	tattttctt	tcttctgta	tcagcctct	1260
tcattata	tcgggggga	gtccacctc	tggaaagcg	gggggggtt	gtacagccg	1320
gtggcgctt	gagactctc	tgtgcagct	cgggattca	cttttcaaa	tacgatatg	1380
attgggtga	gcaggccac	ggcaagggt	tggagtggg	gagtgtaat	ggcaccgag	1440
gtgatacta	ttatccagt	agcgtcaag	gccgtttca	gattagtgc	gaaaacgta	1500
agaattctc	ctatttcag	atgaattca	tacgagcag	tgacacagc	gtctactat	1560
gtgcgagga	cgttgatatt	gttgcgaca	tcaccgatta	cttcgatt	tggggtcaa	1620
gcactttgt	aacagtttc	agtgcgag	ctaagggtc	gagtgtatt	cctttagcc	1680
cttctctaa	aagcaattc	ggcggaac	cagcccttg	ctgtttggg	aaagactat	1740
tcccggaac	tgtaatctg	agctggaat	ctggcgctc	cactagtga	gttcacact	1800
ttccagcgt	gttacagtc	atccctctt	ctccgtctg	accgtcccg		1860
cgagttcact	aggtacgc	acatacatt	gtaacgtca	tcacaagcc	tcgaacaca	1920
aggtgggaa	gcgtgtgaa	cccgaagct	gcgataaag	ccacacctg	cctccctgc	1980
cagctccga	gcttctcgg	ggaccttcg	ttttctctt	tccgcgaag	cccaaagata	2040
ctttgatga	ttcacggag	ccagaagtga	catgcgtgt	tgttgacgt	agccatgag	2100
atccggga	gaaattta	tggtacgtc	atggagtga	ggtccata	gccaaagca	2160
agccccgca	ggaacaat	aattccac	atcgtgtgt	gtcggtact	acggtcttc	2220
atcaggact	gctaaatga	aaagaata	aatgcaagg	gtccaata	gcgctcccc	2280
ctccatcga	aaagacgat	agcaaggcg	aggggcaac	acgtgaacc	caagataca	2340
cgtcccgcc	ctcccgag	gaaatgaca	aaaaccaag	ctccctgac	tgtttgggt	2400
aaggatttt	cccttcgg	atcgcggt	agtgggagt	caacggaca	cctgagaat	2460
actacaaa	gactcctca	gtcctggac	cggacggat	cttctcttg	tactccaat	2520
tgacggtga	caaatcccg	tggcaaca	gcaacgttt	tagctgctc	gtcatgcat	2580
aagctttga	caaccact	acacaaaa	cgttgtcgt	gtcgccag	aaataaaa	2640
acctctcta	aaaattta	ttcattagt	tgcagtcac	cgcgtttgt	ttcacagt	2700
ggaataaac	tagctcgt	tcaccatga	tgccaatct	gcctattca	ggtgtata	2760
agttcaac	ccaaagct	aacttttga	aagagaa	atatccga	agggcacgg	2820
gtgcggtat	gttggagt	actagcaga	agtgaagg	gcacaggat	agttttctc	2880
agagccagg	catcaata	aacgaaagg	tcagtcgaa	gactgggct	ttcgtttat	2940
ctgtgtgtt	tcggtgaac	ctctctact	gagtcacac	ggctcacc	cgggtgggc	3000
ttctgcgtt	tatatagag	gaaatgtgc	cggaaaccc	atttgttat	ttttctaat	3060
acattcaaa	atgtatccg	tcatgagac	ataacccgt	taaatgctt	aataatatt	3120
aaaaagga	agtatgag	ttcaacatt	cgtgtgcgc	cttattccc	tttttgcgc	3180
attttgctt	cctgttttt	ctcaccaga	aacgctggt	aaagtaaaa	atgctgaag	3240
tcagttggg	gcacgagt	gttacatca	actggatct	aacagcggt	agatccttg	3300
gagtttttc	ccgaagaac	gttttcca	gatgagcac	tttaaagt	tgctatgtg	3360
cgcggtata	tcccgatt	acgccccga	agagcaact	ggtcgccga	tacactatt	3420
tcagaatga	ttggttgag	actcaccag	cacagaaa	catcttacg	atggcatga	3480
agtaagaga	ttatgcagt	ctgccata	catgagtga	aacactgcg	ccaacttact	3540
ctcgacaac	atcggagg	cgaaggag	aaccgcttt	ttgcacaac	tgggggatca	3600
tgtaaactc	cttgatcgt	gggaaccga	gctgaatga	gccatacca	acgacgagc	3660
tgacaccag	atgcctgt	caatggca	aacgttgcg	aaactatt	ctggcgaa	3720
acttactct	gcttccgcg	aacaattat	agactggat	gaggcgata	aagttgcag	3780
accacttct	cgtcggccc	ttccggctg	ctggtttat	gtgataaat	ctggagccg	3840
tgacggtgg	tcacgcggt	tcattgcag	actggggca	gatggtaag	cctcccgat	3900
cgtagtatt	tacacgag	ggagttag	aactatgat	gaacgaaat	gacagatcg	3960
tgagatag	gctcactga	ttaagcatt	gtaactgtc	gaccaagtt	actcatatat	4020
actttagatt	gatttaaa	ttcattttt	atttaaaag	atctaggtg	agatccttt	4080
tgataatct	atgacaaaa	ttccctaac	tgagttttc	ttccactga	cgtcagacc	4140
cgtagaaa	atcaaaagt	cttcttgag	tcctttttt	ctgcgctga	tctgctgct	4200
cgaaacaaa	aaaccacgc	taccagcgt	ggtttgttg	cggatcaag	agctacca	4260
tttttttcc	aaggttaact	gcttcagca	agcgcagat	ccaaatact	ttcttctagt	4320
gtagccgt	ttaggccac	acttcaaga	ctctgtagc	cgcctacat	acctcgtct	4380
gctaactct	ttaccagt	ctgctgcc	tggcgata	tcgtgtctt	cgggttgga	4440
ctcaagac	tagttacc	ataaggcga	cgggtcggc	tgaacgggg	gttcgtgc	4500
acagccagc	ttggagcga	cgacctac	cgaactgag	tacctacag	gtgagctat	4560

-continued

```

agaaagcgcc acgtctcccg aaggagagaaa ggccgacagg tatccggtaa gcggcagggt 4620
cggaacacgga gagcgacga gggagcttcc agggggaac gcctgggtatc tttatagtcc 4680
tgctcggttt cgccacctct gacttgagcg tcgatttttg tgatgctcgt caggggggcg 4740
gagcctatgg aaaaacgcca gcaacgcggc ctttttacgg tcctggcct tttgctggcc 4800
ttttgctcac atgttcttcc ctgcgttatc ccctgattct gtggataacc gtattaccgc 4860
cagagaatat aaaaagccag attattaatc cggtcttttt attatttgac gcgttttgag 4920
attcaaaatt caatctgttt ggacctggat aagacacaag agcgacatcc tgacatgaac 4980
gccgtaaaaca gcaaatcctg gttgaacacg tatccttttg ggggcctccc gctacgacgc 5040
tcgctccagc tggggcttcc ttactataca cagcgcgcat atttcacggt tgcacagatgt 5100
caagatggcc aagttgacca gtgcggttcc ggtgctcacc gcgcgcgacg tcgccggagc 5160
ggtcgagttc tggaccgacc ggctcggttt ctcccgggac ttcgtggagg acgacttcgc 5220
cggtgtggtc cgggacgacg tgacctgtt catcagcgcg gtccaggacc aggtggtgcc 5280
ggacaacacc ctggcctggg tgtgggtgcg cggcctggac gagctgtacg ccgagtgggc 5340
ggaggtcgtg tcacgaactc tcgggacgc gccatgaccg agatcggcga 5400
gcagccgtgg gggcggaagt tcgccctgcg cgaccggccc ggcaactgcg tgcaattcgt 5460
ggccgaggag caggactgaa ccttccttaa aaatttaatt ttcattagtt gcagtcactc 5520
cgctttggtt tcacagtcag gaataacact agctcgtctt caccatggat gccaatctcg 5580
cctattcatg gtgtataaaa gttcaacatc caaagctaga acttttgtaa agagaagaa 5640
tatccgaata gggcacggcg tgcggtattg ttggagtggc ctacgagaaa gtgaggaagg 5700
cacaggatga gttttctcga tt 5720

```

```

SEQ ID NO: 43      moltype = DNA length = 6157
FEATURE            Location/Qualifiers
misc_feature       1..6157
                    note = Vector NVB6
source             1..6157
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 43
ttgtctcttt cccgggaaac ggggtacactc ctccgcgcga acaacatatt actactacta 60
ccaagaacgt ccacggcctt gtcgtgcgtt acgctctccc aacgcgtgcg gggtaaatga 120
cgtctcggtt tgctcaagtag gcacagccta aatagatgac cgttattgta ttaagatca 180
ttcaatattg attgattgt actttgcgtc aaactgaaat tccctcgtag taacggttaa 240
ccggtcaacc ctaagcgctt gcccaagta gtcaaccggg acacgcgaac cgacattggg 300
cagatcttcc acagacagaa aaccatttcc aatccaaata agcatgacta ttacacaccc 360
attcgtagcg cgaggacaaa ctgatagctc caacaaaatg cgccaacatc gtacattgta 420
agaagcttac ggaacactat gtatgtagaa ccatacgaac agcaactagt actggccatc 480
gagcagcggt gactcccgcc tttcgtagcg ctgtgaaggt tacactctca caattcgctc 540
tcgggtacaa ccgacaaaag tcttactcac agtcaatacc gaaaacaaac aacagccaac 600
atctatggat ttccaagtac aaatcttcag ttttctactt atctcgccca gtgtaataat 660
tagtaggggc gacattgtca tgacacaaac tctctctatc tcacctgtga cccttgga 720
accgcctctc atcagctgcc gttcctccca atcgctggtc cattcgatg gaaacacgta 780
cttgtcttgg ctccaacagc gccctgggca gcccccgcg ctcttgattt ataaaataag 840
taacagattt agcggcgctc gggaccgggt cagtgggtcg ggccgaggta ccgatttcac 900
ccttaagatc tcccgcgtag aggcggagga ttagtaggtg tattactgta tgcaagcaac 960
tcaatttccc caccaatgga cgttcggtca agggactaaa gttgaaatta agcgtacggg 1020
tgctgcgcca tccgtcttta tttttccgcc atctgacgaa cagctcaaaa gcggaacggc 1080
ttctgtcgtc tgtctctcga acaattttta cccgcgagag gctaagggtc agtggaaggt 1140
cgataatgcg ttgcagtcg gcgaactcga ggaatctgtg acggaacagg acagcaagga 1200
ctccacatac agcttatcgt gcacattgac actgtcaaaa gcggactacg agaaacacaa 1260
ggtgtatgcc tgcgaagtta cccatcaggg attgtcatca ccagtgaaca aatcatttaa 1320
tcgaggtgaa tgcgagggcg gagggctcgt cttgacgtgc ggtgacgtcg aagaaaaacc 1380
gggacccatg gattttcagg tccaaatatt tttctttctt ctgatatcag cctctgtcat 1440
tataagtcgg ggggaagtcc accctgtgga aagcgggggg ggttttagtac agccgggtgg 1500
gccttgaga ctctcttggc cagcctcggg attcaccttt tcaaaactacg atatgcattg 1560
ggtgaggcag gccaccggca aggggtctgga gtgggtgagt gctaattggca ccgcagggtg 1620
tacttattat ccaggtagcg tcaaaaggccg tttcacgatt agtcgcgaaa acgctaagaa 1680
ttctctctat ttgcagatga attcactacg agcagggtgac acagcagttc actattgtgc 1740
gagggacgtt gatattgttg cgaccatcac cgattacttc gattactggg gtcaaggcac 1800
tttgtaaca gtttctagtg cgagtactaa gggtcgaggt gtatttctct tagccccctc 1860
ctctaaaagc atttcagtcg gaacagcagc ccttggtgtg ttggtgaaag actacttccc 1920
ggaacctgta actgtgagct ggaattctgg cgcgctcact agtggagttc acacttttcc 1980
agccgtgtta cagtcactcg gtctttattc cctgtcctcc tgcgtgacgg tcccgtcgag 2040
ttcactaggt acgcagacat acatttgtaa cgtcaatcac aagccatcga acacaaagg 2100
ggacaagcgt gttgaaccca agagctgcga taagaccac acctgccctc cctgcccagc 2160
tcccagagct ctccggcgac cttcggtttt cctctttccg ccgaagccca aagatacttt 2220
gatgatttca cggacgccag aagtgcacat cgtgggtgtt gacgtaagcc atgaggatcc 2280
ggaagtgaat ttttaactgtt acgtcgatgg agttgagtc cataacgcga agaccaagcc 2340
ccgcgaggaa caatataatt ccacatatcg tgtggtgtcg gtacttacgg tctttcatca 2400
ggactggcta aatggaaaag aatacaaatg caaggtgtcc aataaagcgc tcccgcctcc 2460
catgaaaag acgatcagca aggcgaaggg gcaaccacgt gaaccgcga tatacgcgt 2520
cccgccctcc cgagaagaaa tgacaaaaaa ccaagtctcc ctgacctgtt tgggtaaagg 2580
attttaccct tcggacatcg cggtagagtg ggagtcacac ggacaacctg agaataacta 2640
caaaacgact cctccagtc tggactcgga cggatccttc tcttctgact ccaaattgac 2700
ggttgacaaa tcccgatggc aacaaggcaa cgttttttag tgcctggtca tgcagtgaag 2760
tttgacaaac cactacacac aaaaatcggt gtcgctgtcg ccaggaaaaa aaaaccgctt 2820

```

-continued

```

gcgcttgatc ctcgactttg ttgctcgtct ttaaaacctt ggaacgatta tatacaatca 2880
ccaactcaaa aaccggatgt tctaaaaatcc acccaaaacaa ccaaagaaaa tactttctcat 2940
tgcatttatg aatcacagca gacctgcgtc cttttaaaaac ttagatcctg ttttcttata 3000
aaaaacagat caaattttct gggagttcat tgactctgcc agtcagaatc aatcctgcag 3060
taattcttta tttacaggtg aaagtaaaaag agaatoocaa ttttttgctt gtacattaag 3120
gtcctcctta cattacagct aatttcaaaa taagatgaag ttggattcgt gtcctttcat 3180
ggatgatgtg tattctgcac tataccaaac accgtgaaat gtcagctaga gcttgctatg 3240
aggcagttgc tgccaatcac tacatataga tccctcacgg agaaaagtgt gcttcattct 3300
ctggttgctaa tcgctgagc ccaggcatca aataaaacga aaggctcagt cgaagactg 3360
ggcctttcgt tttatctgtt gttgtcgggt gaacgctctc tactagagtc aactggctc 3420
accttcgggt gggcctttct gcgtttatat agagggaat gtgcgcggaa cccctatttg 3480
tttatttttc taaatacatt caaatatgta tccgctcatg agacaataac cctgataaat 3540
gcttcaataa tatgaaaaa ggaagagtat gagtattcaa catttccgtg tcgcccttat 3600
tccctttttt cggcattttt gcttccctgt ttttgctcac ccagaaacgc tggtgaaagt 3660
aaaagatgct gaagatcagt tgggtgcacg agtgggttac atcgaaactg atctcaacag 3720
cggtaagatc cttgagagtt ttgcgccgga agaactgttt ccaatgatga gcacttttaa 3780
agttctgcta tgtggcggg tattatcccg tattgacgcc gggcaagagc aactcggctg 3840
ccgcatacac tattctcaga atgacttggt tgagtactca ccagtcacag aaaagcatct 3900
tacggatggc atgacagtaa gagaattatg cagtgcctgc ataaccatga gtgataacac 3960
tcgggccaac ttacttttga caacgatcgg aggacggaag gagctaaccg cttttttgca 4020
caacatgggg gatcatgtta ctgccttga tcgttgggaa ccggagctga atgaagccat 4080
accaaacgac gagcgtgaca ccacgatgcc tgtagcaatg gcaacaacgt tgcgcaaac 4140
attaactggc gaactactta ctctagcttc ccggcaacaa ttaatagact ggatggaggc 4200
ggataaagtt gcaggaccac ttctgcgtc ggcccttccg gctggctggt ttattgctga 4260
taaatctgga gccggtgagc gtgggtcacg cggatcatt gcagcactgg ggccagatgg 4320
taagccctcc cgatcgttag ttatctacac gacggggagt caggcaacta tggatgaacg 4380
aaatagacag atcgtgaga taggtgcctc actgattaag cattggtaac tgcagacca 4440
agtttactca tatatacttt agattgattt aaaacttcat ttttaattta aaaggatcta 4500
ggtgaagatc ttttttgata atctcatgac caaaatccct taacgtgagt tttcgttcca 4560
ctgagcgtca gaccccgtag aaaagatcaa aggatcttct tgagatcctt tttttctgcy 4620
cgtaatctgc tgccttgcaaa caaaaaaac accgctacca gcggtggttt gtttgccgga 4680
tcaagagcta ccaactcttt ttccgaaggt aactggcttc agcagagcgc agataccaaa 4740
tactgttctt ctagtgtagc cgtagttagg ccaccacttc aagaactctg tagcacccgc 4800
tacatacctc gctctgctaa tctgtttacc agtggctgct gccagtggcg ataagtctg 4860
tcttaccggg ttggactcaa gacgatagtt accggataag gcgcagcggg cgggctgaac 4920
gggggggttc tgcacacagc ccagcttgga gcgaacgacc tacaccgaac tgagatacct 4980
acagcgtgag ctatgagaaa gcgccacgct tcccgaaagg agaaaggcgg acagggtatcc 5040
ggtaagcggc agggctcgaa caggagagcg cacgagggaag cttccagggg gaaacgcctg 5100
gtatctttat agtcctgtcg ggtttcgcca cctctgactt gagcgtcgat tttgtgatg 5160
ctcgtcaggg gggcggagcc tatggaaaaa cgccagcaac gcgcctttt tacgggttct 5220
ggccttttgc tggccttttg ctacatggtt ctttctctgc ttatccctg attctgtgga 5280
taaccgtatt accgccagag aatataaaaa gccagattat taatccggct tttttattat 5340
ttgacgcgtt ttgagattca aaattcaatc tgtttggacc tggataagac acaagagcga 5400
catctgaca tgaacgcgtt aaacagcaaa tcttggttga acacgtatcc ttttgggggc 5460
ctcccgctac gacgctcgct ccagctgggg cttccttact atacacagcg cgcataattc 5520
acggttgcca gatgtcaaga tggccaaagt gaccagtgc gttccgggtg tcaccgcgcg 5580
cgacgtcgcc ggagcggctg agttctggac cgaccggctc ggggttctcc gggacttcgt 5640
ggaggacgac ttgcgcgggt tggtcgggga cgacgtgacc ctgttcatca gcgcgggtcca 5700
ggaccaggtg gtgcaggaca acaccctggc ctgggtgtgg gtgcgcggcc tggacgagct 5760
gtacgccgag ttgctggagg tctgttccac gaacttccgg gacgcctccg ggccggccat 5820
gaccgagatc ggcgagcagc cgtgggggcg ggagttcgcc ctgcgcgacc cggccgggcaa 5880
ctgcgtgcac tctgtggcag aggagcagga ctgaaccttc cttaaaaatt taattttcat 5940
tagttgcagt cactccgctt tggtttcaca gtcaggaata acactagctc gtcttcacca 6000
tggatgccaa tctcgcttat tcatggtgta taaaagttca acatccaaag ctagaacttt 6060
tgaaaagaga aagaatatcc gaatagggca cggcgtgccc tattgttgga gtggactagc 6120
agaaagttag gaaggcacag gatgagtttt ctcgatt 6157

```

```

SEQ ID NO: 44      moltype = DNA length = 6059
FEATURE            Location/Qualifiers
misc_feature       1..6059
                    note = Vector NVB7
source             1..6059
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 44
ttcgattgga atggttttgg agaggaaacc gtttttatgg cggttggatg ggaaggaag 60
ggatggtggt cacactcaga tttgggagct caccaccggt atgttgacag tacaacaat 120
cctgcaatcc tgcgaaccaa ctgtttctcg tctttttggt ctgtacgtgg taggatagag 180
agtctgccgg gataccattt cgtgcgcaga cgtggaagag tcattattca gcttgcacga 240
cacagatttt atacagacaa acgtacaata cgcaattgct ccgacaaaaa ttgccgacca 300
tctctacagg catatttttg catataatta ggggaaacgc atagaatgca atcttcagtt 360
acatacagac aacattctgt cgcagctcgc cacacgtacc aactccagat tccgagcttc 420
gccgcggcct tttggaaaga tcgaaaaggc gccgcgcggc ctgtcgttcc cgcagaaagt 480
gcaagcagcc actagcga aaatctatgg atttccaagt acaaatcttc agttttctac 540
ttatctcgcc cagtgtata attagtaggg cgcacattgt catgacacaa actcctctat 600
ctcacctgtg gacccttgga caaccgcct ctatcagctg ccgttctccc caatcgtg 660

```

-continued

tccattccga	tggaaacacg	tacttgtctt	ggctccaaca	gcgcctggg	cagcccccgc	720
ggctcttgat	ttataaaaata	agtaacagat	ttagcggcgt	tccggacgg	ttcagtggtt	780
cgggcgcagg	taccgatttc	acccttaaga	tctcccgcgt	agaggccgag	gatgtaggag	840
tgtattactg	tatgcaagca	actcaatttc	cccaccaatg	gacgttcggt	caagggacta	900
aagttgaaat	taagcgtacg	gttgcgtcgc	catccgtctt	tatttttccg	ccatctgacg	960
aacagctcaa	aagcggaaacg	gctctctgctg	tctgtctcct	gaacaatttt	taccgcgcag	1020
aggctaaggt	tcagtggaaag	gtcgataatg	cgttgcagtc	gggcaactcg	caggaatctg	1080
tgacggaaaca	ggacagcaag	gactccacat	acagcttata	gtcgacattg	acactgtcaa	1140
aagcggacta	cgagaaacac	aaggtgtatg	cctgcgaagt	taccocatcag	ggattgtcat	1200
caccagtac	caaatcattt	aatcgagggtg	aatgcgaggg	cagagggtcg	ctcttgacgt	1260
gcgggtgacgt	cgaagaaaac	ccgggaccca	tggattttca	ggccaaaata	ttttcttttc	1320
ttctgataatc	agcctctgtc	attataagtc	ggggggaagt	ccacctcgtg	gaaagcgggg	1380
gggggttagt	acagccgggt	ggcgccctga	gactctcttg	tgacgcctcg	ggattcacct	1440
tttcaaaata	cgatatgcac	tgggtgaggc	agggccacgg	caaggggtcg	gagtggtgga	1500
gtgctaattg	caccgcagggt	gatacttatt	atccaggtag	cgtcaaaagg	cgtttcacga	1560
ttagtgcgga	aaacgctaag	aattctctct	atttgcagat	gaattcacta	cgagcagggtg	1620
acacagcagt	ctactattgt	gcgagggagc	ttgatatgtg	tgcgaccatc	accgattact	1680
tcgattactg	gggtcaaggc	actttggtaa	cagtttctag	tgcgagtagt	aagggtcoga	1740
gtgtatttcc	tttagccctc	tctctaaaaa	gcacttcagg	cggaaacagca	gccctggcct	1800
gtttgggtgaa	agactacttc	ccggaacctg	taactgtgag	ctggaattct	ggcgcgctca	1860
ctagtggagt	tcacactttt	ccagccgtgt	tacagtcac	gggtctttat	tcctctgctc	1920
ccgtcgtgac	cgtcccgtcg	agttcactag	gtacgcagac	atacatttgt	aacgtcaatc	1980
acaagccatc	gacacaaaag	gtggacaagc	gtgttgaaac	caagagctgc	gataagagacc	2040
acacctgccc	tccttgccca	gctcccagag	ttctcggcgg	accctcgggt	ttctcttttc	2100
cgcgaagcc	caaagatact	ttgatgattt	cacggacgcc	agaagtgcga	tgctggtgtg	2160
ttgacgtgaa	ccatgaggat	ccggaagtga	aatttcaactg	gtacgtcgat	ggagtgtgag	2220
tcataaacgc	caagaccaag	ccccgcgagg	aacaatataa	ttccacatat	cgtgtggtgt	2280
cggtaactac	ggtcctctat	caggactggc	taaatggaaa	agaatacaaa	tgcaaggtgt	2340
ccaataaagc	gtcccccgtc	cccatcgaaa	agacgatcag	caagggcgaag	gggcaaccac	2400
gtgaaccgca	agtatacacg	ctccccccct	cccagagaag	aatgacaaaa	aaccaagtct	2460
ccctgacctg	tttggttaaa	ggattttacc	cttcggacat	cgcggtagag	tgggagttca	2520
acggacaacc	tgagaataac	tacaaaaacga	ctcctccagt	cctggactcg	gacggatcct	2580
ttctcttgta	ctccaaattg	acggttgaca	aatcccagatg	gcaacaaggc	aacgttttta	2640
gctgctcggt	ctatgcatac	gctttgcaca	accactacac	acaaaaatcg	ttgtcgtgtg	2700
gcccaggaaa	ataaaaccac	ggacgtttgt	tggagattcc	gcttttacct	gatttccagt	2760
tcctgtaat	ttgtcgttcc	ttatgtttcc	acacaaaacga	ctgcaagcaa	aatcaactaa	2820
tttgtttatc	acattttcaat	agccagtttg	aatgtttttc	ttacgatact	gtaggcagtg	2880
gaagcggaa	tggtttttgt	gggcaaccga	aaaaggcttg	caattttcaa	ttagggtaac	2940
agtagtcggt	aaacaattgc	tcacttagct	ctaggaagtc	tattaatcca	caggttgga	3000
acgcaatgaa	tgtgcctgac	atgacggcgc	cgggttaggc	aggagaattt	ctctaagtga	3060
atctttgttc	cagactctt	acagaggcac	ccaaaccatc	gcttcgatca	gttgacgaga	3120
ggtttctgaa	aacgatctta	tcgaggcgcc	gttggtgtgc	ctttcgaaa	aatcaacttc	3180
aaacgtgtag	cttgggtagc	aaaaactttca	ttggatgtga	gccagggcat	caataaaaac	3240
gaaaggctca	gtcgaaagac	tgggcctttc	gttttatctg	ttgtttgtcg	gtgaacgctc	3300
ctactagag	tcacactggc	tcaccttcgg	gtgggccttt	ctcggtttat	atagagggaa	3360
atgtgcgcgg	aaacctat	tgtttatttt	tctaataaca	ttcaaatatg	tatccgctca	3420
tgagacaata	accctgataa	atgcttcaat	aatattgaaa	aagggaagagt	atgagtattc	3480
aacattttcgt	tgtgcgcctt	atcccttttt	ttgcggcatt	ttgccttcc	gtttttgtct	3540
accagaaaac	gtgggtgaaa	gtaaaagatg	ctgaagatca	gttgggtgca	caggtgggtt	3600
acatcgaact	ggatctcaac	agcggtaaga	tccttgagag	ttttcgcccc	gaagaacggt	3660
ttccaatgat	gagcactttt	aaagttctgc	tatgtggcgc	ggattattcc	cgtattgacg	3720
ccgggcaaga	gcaactcggg	cgcgcatac	actattctca	gaatgacttg	gttgagtact	3780
caccagtcac	gaaaaagcat	cttacggatg	gcattgacag	aagagaatta	tgacgtgctg	3840
ccataaccat	gagtataaac	actgcggcca	acttacttct	gacaaacgat	ggaggaccga	3900
aggagctaac	cgtctttttg	cacaaacatg	gggatcatgt	aactgccttc	gatcgtttgg	3960
aaccggagct	gaatgaagcc	ataccaaaacg	acgagcgtga	caccacgatg	cctgtagcaa	4020
tggcaacaac	gttgccgaaa	ctattaactg	gcgaactact	tactctagct	tcggcgcaac	4080
aattaataga	ctggatggag	cgggataaag	ttgcaggacc	actctgcgc	tcggcccttc	4140
cggctggctg	gtttattgct	gataaatctg	gagccgggtga	gcgtgggtca	cgcgggtatca	4200
ttgcagcact	ggggccagat	ggtaagccct	cccgtatcgt	agttatctac	acgacgggga	4260
gtcagggaac	tatggatgaa	cgaaaatagac	agatcgctga	gatagggtgc	tcactgatta	4320
agcattggta	actgtcagac	caagtttact	catatatact	ttagattgat	ttaaaacttc	4380
atttttaatt	taaaaggatc	taggtgaaga	tcctttttga	taactctcatg	accaaaatcc	4440
cttaacgtga	gttttcgttc	cactgagcgt	cagaccccg	agaaaagatc	aaaggatctt	4500
cttgagatcc	ttttttctg	cgcgtaatct	gctgcttgca	aacaaaaaaa	ccaccgctac	4560
cagcgggtgt	ttgtttgcg	gatcaagagc	taccaactct	ttttcgaag	gtaactggct	4620
tcagcagagc	gcagatacca	aatactgttc	ttctagtgtg	gccgtagtta	ggccaccact	4680
tcaagaactc	tgtagcaccg	cctacatacc	tcgctctgct	aatcctgtta	ccagtggctg	4740
ctgccagtgg	cgataagtcg	tgtcttaccg	ggttggactc	aagacgatag	ttaccggata	4800
aggcgacagc	gtcgggtcga	acgggggggt	cgtgcacaca	gcccagcttg	gagcgaaacga	4860
cctacaccga	actgagatag	ctacagcgtg	agctatgaga	aagcgccacg	cttcccgagg	4920
ggagaaaaggc	ggacaggtat	ccggtaagcg	gcagggctcg	aacaggagag	cgcacagagg	4980
agcttccagg	gggaaacgcc	tggtatcttt	atagtcctgt	cgggtttcgc	caactctgac	5040
ttgagcgtcg	atttttgtga	tgctcgtcag	ggggcgcgag	cctatggaaa	aacgccagca	5100
acggggcctt	tttacgggtc	ctggcccttt	gctggccttt	tgctccatag	ttctttcctg	5160
cgttatcccc	tgattctgtg	gataaacgta	ttaccgccag	agaatataaa	aagccagatt	5220

-continued

```

attaatccgg cttttttatt atttgacgcy ttttgagatt caaaattcaa tctgtttgga 5280
cctggataag acacaagagc gacatcctga catgaacgcc gtaaacagca aatcctgggt 5340
gaacacgtat ccttttgggg gccctccgct acgacgctcg ctccagctgg ggcttcctta 5400
ctatacacag cgcgcataat tcacgggttc cagatgtcaa gatggccaag ttgaccagt 5460
ecgttcgggt gctcaccgcy cgcgacgtcg ccggagcggg cgagttctgg accgacgggc 5520
tcgggttctc ccgggacttc gtggaggagc acttcgcccg tgtggtccgg gacgacgtga 5580
ccctgttcat cagcgcggtc caggaccagg tgggtccgga caacaccctg gcctgggtgt 5640
gggtgcggcy cctggagcag ctgtacgccc agtggtcgga ggtcgtgtcc acgaacttcc 5700
gggacgccto cgggccggcc atgaccgaga tcggcgagca gccgtggggg cgggagttcg 5760
ccctgcgcga cccggccggc aactgcgtgc acttcgtggc cgaggagcag gactgaacct 5820
tccttaaaaa ttttaatttc attagtgtca gtccactccg tttggtttca cagtacggaa 5880
taacactagc tcgtcttcac catggatgcc aatctcgcc attcatgggt tataaaagt 5940
caacatccaa agctagaact tttggaaga gaaagaatat ccgaataggg cacggcggtgc 6000
cgtattgttg gagtggaacta gcagaagtg aggaaggcac aggatgagtt ttctcgatt 6059

```

```

SEQ ID NO: 45      moltype = DNA length = 5748
FEATURE           Location/Qualifiers
misc_feature       1..5748
                  note = Vector C
source            1..5748
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 45
catatgcgga agtgactgta aacgagaagt gcacgaagcc ttttctgttg acgtcacaaa 60
ccgaacagcc ctacgtggcg tgcgactttc cggcactttg gatcctgttg ccctatatgt 120
attgggtgta tacatgtcgt attccaaacta cgacagaaaa cagtcagtta cgggtattcg 180
aacttcggcg acctgcagcg agtagttttg tgcctcgctc acatcctccg tgctttcggc 240
acctacgtag cgggcaaatc tctgccttc cgcgcgctt cgcgcattcc ggctccgggc 300
cagaattgcc cgggtgttca caattttgcc tctccacgaa aaaacgcttc actttgtatt 360
ttggtcgggt ttccgatcct tcacgcaacc attctcattc aaagtaccca ctgtgtcgaa 420
cgtatcattg attttcaggt ccaaatattt tcttttcttc tgatcagc ctctgtcatt 480
ataagtcggg ggggaagtcca cctcgtggaa agcggggggg gtttagtaca gccgggtggc 540
gccttgagac tctcttgtgc agcctcggga ttcacctttt caaactacga tatgcattgg 600
gtgagggcagg ccacggcga ggggtcggag tgggtgagtg ctaatggcac cgcaggtgat 660
acttattatc caggttagcgt caaaggccgt ttcacgatta gtcgcgaaaa cgctaagaat 720
tctctctatt tgcagatgaa ttcactacga gcaggtgaca cagcagctca ctattgtcgc 780
agggacgttg atattgttgc gacctcacc gattacttcg attactgggg tcaaggcact 840
ttggttaacg tttctagtgc gagtactaag ggtccgagtg tatttccctt agcccttccc 900
tctaaaagca cttcaggcgg aacagcagcc cttggctgtt tgggtgaaaga ctacttccc 960
gaactgttaa ctgtgatcgt gaattctggc gcgctcacta gtggagttca cacttttcca 1020
gccgtgttac agtcatcggg tctttatttc ctgtcctccg tcgtgacggt cccgtcgagt 1080
tcactaggtg cgcagacata catttgtaac gtcaatcaca agccatcgaa cacaaagggt 1140
gacaagcgtg ttgaaccocaa ttcgatcgat aagacccaca cctgccctcc ctgccagct 1200
cccgagcttc tcggcggacc ttccggtttc ctctttccgc cgaagcccaa agatactttg 1260
atgatttcac ggcgcgcaga agtgacatgc gtgggtgttg acgtaagcca tgaggatccg 1320
gaagtgaatg ttaactggta cgtcgatgga gttgaggtcc ataacgccaa gaccaagccc 1380
cgcgaggaa aatataatc cacatatcgt gtgggtgcgg tacttacggt ccttcatcag 1440
gactgggtta atgaaaaaga atacaaatgc aaggtgtcca ataaagcgt ccccgctccc 1500
atcgaaaaga cgtacgcga ggcgaagggg caaccacgtg aaccgcaagt atacacgtc 1560
ccgcctcccc gagaagaaat gacaaaaaac caagtctccc tgacctgttt gggttaagg 1620
ttttaccctc cggacatcgc ggtagagtg ggtccaaacg gacaacctga gaataactac 1680
aaaaacgact ctcagtcctt ggaactcgga ggtactcttc tctgtactc caaattgacg 1740
gttgacaaat cccgatggca acaaggcaac gtttttagct gctcggtcat gcatgaagct 1800
ttgcacaacc actacacaca aaaatcgttg tcgctgtcgc caggaaaaga gggccgaggg 1860
tcgctcttga cgtgcggtga cgtcgaaaga aaccggggac ccatggattt ccaagtacaa 1920
atcttcagtt ttctacttat ctccggccagt gtaataatta gtaggggcga cattgtcatg 1980
acacaaactc ctctatcttc acctgtgacc cttggacaac ccgcctctat cagctgccgt 2040
tcctcccaat cgtcggtcca ttccgatgga aacacgtact tgtcttggat ccaacagcgc 2100
cctgggcagc ccccgcggtt cttgatttat aaaataagta acagatttag cggcggttccg 2160
gaccgggtta gtggttcggg cgcaggtacc gatttcaccc ttaagatctc ccgcgtagag 2220
gccgaggatg taggagtgta ttactgtatg caagcaactc aatttcccca ccaatggacg 2280
ttcgggtcaag ggactaaagt tgaattaag cgtacggttg ctgcgcctac cgtctttatt 2340
tttcgcgcac ctgacgaaca gtcacaaagc ggaacggctt ctgctgtctg tctcctgaac 2400
aatttttacc cgcgagaggg taaggttcag tggagggtcg ataagcgtt gcagtcgggc 2460
aactcgaggg aatctgtgac ggaacaggac agcaaggact ccacatacag cttatcgtcg 2520
acattgacac tgtcaaaagc ggactacgag aaacacaagg tgtatgcctg cgaagttacc 2580
catcagggat tgtcatcacc agtgaccaaa tcatttaatc gaggtgaatg ctaagtaaca 2640
atttacttta agactagcgc aatttttttc ttacattaga tgaacataaa acgacgatga 2700
ggatcactgt tagggaaacg atgaactacg ctgaggactt cctattatct gtcctttgat 2760
ggcgtggttg tcttttggtt tgcggccgag cgcacacgac aaggcctgta ccaaaaacga 2820
agttgataat cagataatgt agttgtaaca agaaactaaa gcattatttt agcagtagca 2880
cggatcagga tctaaaatga accgatccgg accagggcac aaataaaaac aaaggctcag 2940
tcgaaagact gggcctttcg ttttatctgt tgtttgtcgg tgaacgctct ctactagagt 3000
cacactggct caccttcggg tgggcttttc tgcgtttata tagagggaaa tgtgcgcgga 3060
accctatttt gtttattttt ctaaatatgt atccgctcat gagacaataa 3120
ccctgataaa tgcttcaata atattgaaaa aggaagagta tgagtattca acatttccgt 3180

```

-continued

```

gtgcgccctta ttcctctttt tgcggcattt tgccttccctg tttttgctca cccagaaacg 3240
ctgggtgaaag taaaagatgc tgaagatcag ttgggtgcac gagtgggtta catcgaaactg 3300
gatctcaaca gcggtgaagat ccttgagagt ttctgccccg aagaacgttt tccaatgatg 3360
agcaactttta aagttctgct atgtggcgcg gtattatccc gtattgacgc cgggcaagag 3420
caactcggtc gcccatataca ctattctcag aatgacttgg ttgagtactc accagtcaca 3480
gaaaagcatc ttacgggatg catgacagta agagaattat gcagtgtctg cataaccatg 3540
agtgataaca ctgcggccaa ctacttctg acaacgatcg gaggaccgaa ggagctaacc 3600
gcttttttgc acaacatggg ggatcatgta actcgccttg atcgttggga accggagctg 3660
aatgaagcca taccaaacga cgaagctgac accacgatgc ctgtagcaat ggcaacaacg 3720
ttgcgcaaac tattaactgg cgaactactt actctagctt cccggcaaca attaatagac 3780
tggatggagg cggataaagt tgaggacca ctctgcgct cgcccttcc ggctggctgg 3840
tttattgctg ataaatctgg agccggtgag cgtgggtcac gcggtatcat tgcagcactg 3900
gggcccagatg gtaagccctc ccgtatcgta gttatctaca cgacggggag tcaggcaact 3960
atggatgaac gaaatagaca gatcgctgag ataggtgcct cactgattaa gcattggtaa 4020
ctgtcagacc aagtttactc atataactt tagattgatt taaaacttca ttttttaatt 4080
aaaaggatct aggtgaagat cctttttgat aatctcatga ccaaaatccc ttaacgtgag 4140
ttttcgcttc actgagcgtc agaccccgta gaaaagatca aaggatcttc ttgagatcct 4200
ttttttctgc gcgtaactctg ctgcttgcaa acaaaaaaac cgctaccagc ggtggtttgt 4260
ttgcgggatc aagagctacc aactcttttt ccgaaggtaa ctggcttcag cagagcgcat 4320
ataccacaata ctgttcttct agtgtagccg tagttaggcc accactcaa gaactctgta 4380
gcaccgccta catacctgc tctgctaact ctgttaccag tggctgtctg cagtggcgat 4440
aagtcgtgtc ttaccgggtt ggactcaaga cgatagttac cggataaggc gcagcggctg 4500
ggctgaacgg ggggttctg caccagccc agcttggagc gaacgacctc caccgaactg 4560
agatacctac agcgtgagct atgagaaagc gccacgcttc ccgaagggag aaaggcggac 4620
aggtatccgg taagcggcag ggtcggaaca ggagagcgca cgaggagct tccaggggga 4680
aacgcctggt atctttatag tctgtcggg tttcgccacc tctgacttga gcgtcgattt 4740
ttgtgatgct cgtcaggggg gcggagccta tggaaaaacg ccagcaacgc ggcttttta 4800
cggttctctg ccttttctg ccttttctg cactgttct tctctgcgtt atccctgat 4860
tctgtggata accgtattac cgccagagaa tataaaaaagc cagattatta atccggctt 4920
tttattattt gacgcgtttt gagattcaaa attcaactct tttggacctg gataagacac 4980
aagagcgaca tctgacatg aacgcgtaaa acagcaaatc ctggttgaa acgtatcctt 5040
ttgggggctt ccgctacgac gctcgtctca gctgggctt ccttactata cacagcgcgc 5100
atatctcacg gttgccagat gtaacgatgg ccaagttagc cagtgcctt ccggtgctca 5160
ccgcgcgcga cgtcgcggga gcggtcgagt tctggaccga ccggtcggg ttctccggg 5220
acttcgtgga ggaagacttc gccggtgtgg tccgggacga cgtgacctg ttcatcagcg 5280
cggtcaggga ccaggtgggt ccggacaaca ccctggcctg ggtgtgggtg ccggcctg 5340
acgagctgta cgcagagtgg tcggaggtcg tgtccacgaa cttccgggac gctccgggc 5400
cggccatgac cggatcgctg gagcagccgt gggggcgagg gttcgccctg cgcgacccgc 5460
ccggcaactg cgtgcacttc gtggcgagg agcaggactg aacctcctt aaaaatttaa 5520
ttttcattag ttccagctac tccgctttg tttcacagtc aggaataaca ctagctctc 5580
ttccaccatg atgccaatct cgcctattca ttggtataaa aagttcaaca tccaaagcta 5640
gaacttttgg aaagagaaag aatatccgaa tagggcacgg cgtgccgtat tgttgagtg 5700
gactagcaga aagtggaggaa ggcacaggat gagttttctc gaaagacg 5748

```

```

SEQ ID NO: 46      moltype = DNA length = 5748
FEATURE            Location/Qualifiers
misc_feature        1..5748
                    note = Vector D
source              1..5748
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 46
catatgcgga agtgactgta aacgagaagt gcacgaagcc ttttctgtg acgtcacaaa 60
ccgaacagcc ctacgtggcg tgcgactttc cggcactttg gatatcgttg ccctatatgt 120
attgggtgta tacatgtcgt attccaacta cgacagaaaa cagtcagtta cgggtattcg 180
aacttcgggc acctgcagcg aggagttttg tgtccgtcgc acatcctccg tgetttcggc 240
acctacgtag gcggcaaaact tctgccttc cgcccgctt gcggcattcc ggctccgggc 300
cagaattgcc cgggtgttca caattttgct tcttcacgaa aaaacgttct actttgtatt 360
ttggtcgggt ttccgatact tccagcaacc attctcattc aaagtcacca cttgtgcgaa 420
cgtatcatgg attccaagt acaaatcttc agttttctac ttatctcggc cagtgttaata 480
attagtaggg gcgacattgt catgacacaa actcctctat cctcacctgt gacctttgga 540
caaccgcgct ctatcagctg ccgttctctc caatcgctgg tccattccga tggaaacacg 600
tacttgtctt ggctccaaca gcgccttggg cagcccccgc ggctcttgat ttataaaata 660
agtaacagat ttacggcgct tccggaccgg ttcagtgggt cgggcgcagg taccgatttc 720
acccttaaga tctccgcgt agaggccgag gatgtaggag tgtattactg tatgcaagca 780
actcaatttc cccaccaatg gacgttccgt caagggacta aagttgaaat taagcgtacg 840
gttgtgcgc catcgtctt tatttttccg ccactctgac aacagctcaa aagcggaaacg 900
gcttctgtcg tctgtctctt gaacaatttt tacccgcgag aggtcaaggt tcagtggaaag 960
gtcgataaat cgttgcatgc gggcaactcg caggaaactg tgacggaaca ggacagcaag 1020
gactccacat acagcttatt gtcgacattg acactgtcaa aagcggacta cgagaaacac 1080
aaggtgtatg cctgcgaagt taccatcag ggattgtcat caccagtgc caaatcattt 1140
aatcgaggtg aatcgaggg ccgagggctg ctcttgacgt gcggtgacgt cgaagaaaac 1200
ccgggaccca tggattttca ggtccaaata ttttcttttc ttctgatatc agcctctgtc 1260
attataagtc ggggggaagt ccacctctg gaaagcgggg ggggtttagt acagccgggt 1320
ggcgccttga gactctctt tgacgcctcg ggattcacct tttcaacta cgatatgcat 1380
tgggtgaggg agggccaccg caagggtctg gagtgggtga gtgctaatg caccgcaggt 1440

```

-continued

gatacttatt	atccaggtag	cgtcaaaggc	cgtttcacga	ttagtcgcga	aaacgctaag	1500
aattctctct	atttgcagat	gaattccacta	cgagcaggtg	acacagcagt	ctactattgt	1560
gcgaggggacg	ttgatattgt	tgcgaccatc	accgattact	tgcattactg	gggtcaaggc	1620
actttggtaa	cagtttctag	tgcgagtact	aagggtccga	gtgtatttcc	tttagccct	1680
tccctctaaa	gcacttcagg	cggaaacagca	gcccttggct	gtttggtgaa	agactacttc	1740
ccggaacctg	taactgtgag	ctggaattct	ggcgcgctca	ctagtggagt	tcacactttt	1800
ccagccgtgt	tacagtcac	gggtctttat	tccctgtcct	ccgtcgtgac	cgtcccgtcg	1860
agttcactag	gtacgcagac	atacatttgt	aacgtcaatc	acaagccatc	gaacacaaag	1920
gtggacaagc	gtgttgaaac	caagagctgc	gataagacct	acacctgccc	tccctgccca	1980
gctcccagac	ttctcggcgg	accttcggtt	tccctcttcc	cgcgaagcc	caaagatact	2040
ttgatgattt	cacggagccc	agaagtgaca	tgcgtgggtg	ttgacgtaa	ccatgaggat	2100
ccggaagtga	aatttaactg	gtacgtcgat	ggagttgagg	tccataacgc	caagaccaag	2160
ccccgcgagg	aacaatataa	ttccacatat	cgtgtgggtg	cggtaacttac	ggctcctcat	2220
caggactggc	taaatggaaa	agaatacaaa	tgcagggtgt	ccaataaagc	gctccccgct	2280
cccatcgaaa	agcagatcag	caaggcgaa	gggcaaccac	gtgaaccgca	agtatacacg	2340
ctcccgccct	ccgagaaga	aatgacaaaa	aaccaagtct	ccctgacctg	tttgggttaa	2400
ggattttacc	cttcggacat	cgcggtagag	tgggagtgca	acggacaacc	tgagaataac	2460
tacaaaacga	ctcctccagt	cctggactcg	gacggatcct	tcttcttgta	ctccaaattg	2520
acggttgaca	aatcccgatg	gcaacaagcg	aacgttttta	gctgctcggt	catgcatgaa	2580
gctttgcaca	accactacac	acaaaaatcg	ttgtcgctgt	cgcagggaaa	ataagtaaca	2640
atttacttta	agactagcgc	aatttttttc	ttacattaga	tgaacataaa	acgacgatga	2700
ggatcactgt	tagggaaaac	atgaactacg	ctgaggactt	cctattattc	gtcctttgat	2760
ggcgtgggtg	ttctttgggt	tgcggccgag	cgcacagacc	aaggcctgta	ccaaaaacga	2820
agttgataat	cagatatttg	agttgtaaca	agaaactaaa	gcattatttt	agcagtagca	2880
cggatcagga	tctaaaatga	accgatccgg	accaggcatc	aaataaaaac	aaaggctcag	2940
tcgaaagact	gggcctttcg	ttttatctgt	tgtttgtcgg	tgaacgctct	ctactagagt	3000
cacactggct	caccttcggg	tgggcctttc	tgcgtttata	tagagggaaa	tgtgcgcgga	3060
acccctattt	gtttattttt	tcaaatatgt	atccgctcat	gagacaataa		3120
ccctgataaa	gtcttcaata	atatgaaaa	aggaagagta	tgagtattca	acatttccgt	3180
gtcgccttta	ttcccttttt	tgcggcattt	tgccttctct	tttttgctca	cccagaaaacg	3240
ctggtgaaa	taaaagatgc	tgaagatcag	ttgggtgcac	gagtggtgta	catcgaaactg	3300
gatctcaaca	ggcgtaagat	ccttgagagt	tttcgccccg	aagaacgttt	tccaatgatg	3360
agcactttta	aagtctctct	atgtggcgcg	gtattatccc	gtattgacgc	cgggcaagag	3420
caactcggtc	gccgcataca	ctattctcag	aatgacttgg	ttgagtactc	accagtcaca	3480
gaaaagcatc	ttacggatgg	catgacagta	agagaattat	gcagtgtctg	cataaccatg	3540
agtgataaca	ctcgggccaa	cttacttctg	acaacgatcg	gaggaccgaa	ggagctaacc	3600
gcttttttgc	acaaactggg	ggatcatgta	actcgccttg	atcgttggga	accggagctg	3660
aatgaagcca	taccaaaccg	cgaagctgac	accacgatgc	ctgtagcaat	ggcaacacacg	3720
ttgcgcgaac	tattaactgg	cgaactactt	actctagctt	cccggcaaca	attaatagac	3780
tgatggagg	cggataaagt	tgcaggacca	cttctgcgct	cggcccttcc	ggctggctgg	3840
tttattgctg	ataaatctgg	agccggtgag	cgtgggtcac	gcggtatcat	tgacgactg	3900
gggcagatg	gtaagccctc	ccgtatcgta	gttatctaca	cgcggggag	tcaggcaact	3960
atggatgaac	gaaatagaca	gatcgtcgag	ataggtgcct	cactgattaa	gcattggtaa	4020
ctgtcagacc	aagtttactc	atatatactt	tagattgatt	taaaaactca	tttttaattt	4080
aaaaggatct	aggtgaagat	cttttttgat	aatctcatga	ccaaaatccc	ttaacgtgag	4140
ttttcgttcc	actgagcgtc	agaccccgta	gaaaagatca	aaggatcttc	ttgagatcct	4200
ttttttctgc	gcgtaactct	ctgcttgcaa	acaaaaaaac	cgtaccagc	gggtggttgt	4260
ttgcgggatc	aagagctacc	actctttttt	ccgaaggtaa	ctggcttcag	cagagcgag	4320
ataccaaaata	ctgttcttct	agcttagcgc	tagttaggcc	accacttcaa	gaactctgta	4380
gcaccgccta	catacctcgc	tctgctaatt	ctgttaccag	tggtctgctg	cagtggcgat	4440
aagtcgtgtc	ttaccgggtt	ggaactcaaga	cgatagttac	cggataaagg	gcagcgggtc	4500
ggctgaacgg	gggttctcgt	cacacagccc	agcttggagc	gaacgacctc	caccgaactg	4560
agatacctac	agcgtgagct	atgagaaaac	gccacgcttc	ccgaaggagg	aaaggcggac	4620
aggtatccgg	taagcggcag	ggtcggaaca	ggagagcgca	cgaaggagct	tccaggggga	4680
aacgcctggt	atctttatag	tccgttcggg	tttcgccacc	tctgacttga	gcgtcgattt	4740
ttgtgatgct	cgtcaggggg	gctggagccta	tggaaaaacg	ccagcaacgc	ggccttttta	4800
cggttcctgg	ccttttctg	gccttttgct	cacatgttct	tctctcggtt	atccccgat	4860
tctgtggata	accgtattac	cgcagagaaa	tataaaaaag	cagattatta	atccgctttt	4920
tttattattt	gacgcgtttt	gagattcaaa	attcaatctg	tttggacctg	gataagacac	4980
aagagcgaca	tcctgacatg	aacgccgtaa	acagcaaatc	ctggttgaa	acgtatcctt	5040
ttggggggcct	ccgctacgac	gctcgctcca	gctggggctt	ccttactata	cacagcgcgc	5100
atatttccag	gttgccagat	gtcaagatgg	ccaagttgac	cagtgcctgt	ccggtgctca	5160
ccgcgcgcga	cgtcgccgga	gctgtcgagt	tctggaccga	ccggtcgggg	ttctcccggg	5220
acttcgtgga	ggacgacttc	gcccgtgtgg	tccgggacga	cgtgacctg	ttcatcagcg	5280
cggtccagga	ccaggtgggt	ccggacaaca	ccttgccctg	ggtgtgggtg	cggggcctgg	5340
acgagctgta	cgccgagtg	tccgaggtcg	gttccacgaa	cttcggggac	gcctccgggc	5400
cggccatgac	cgagatcggc	gagcagccgt	gggggcggga	gttcgcccgt	cgcgacccgg	5460
ccggcaactg	cgtgcacttc	gtggccgagg	agcaggactg	aaccttctct	aaaaatttaa	5520
ttttcattag	ttgcagtcac	tccgctttgg	tttcacagtc	aggaataaca	ctagctcgct	5580
ttcccatggt	atgccaatct	cgcctattca	tgggtataaa	aagttcaaca	tccaaagcta	5640
gaacttttgg	aaagagaaa	aatatccgaa	tagggcacgg	cgtgccttat	tgttggagtg	5700
gactagcaga	aagtgaggaa	ggcacaggat	gagttttctc	gaaagacg		5748

SEQ ID NO: 47 moltype = DNA length = 6570
 FEATURE Location/Qualifiers
 misc_feature 1..6570

-continued

```

note = Vector Pro1D
source      1..6570
            mol_type = other DNA
            organism = synthetic construct

SEQUENCE: 47
tttttgctgcg tacttcacca cagcttttatt gaatagggac gagcttttat taatcatgtg 60
ctgctgtctct ttttgcatca ggattgtcac gtagccactt tggactgtaa ttaactttga 120
cttcttttato taacaaaagcg gcgagatcaa ggtcctttctc agtcaaaccc tgcagcttgt 180
gtgtcccaaat ttttacctcg atatctcggg aactagtaag atggaagtcg ggggtggtgac 240
tttcgcgttc agctatggcg cccatctcgt tgatgcattt taaagcagac tgaaaattgc 300
gtgccgtgaaa tttacgagat atatgcaaaag gttctgattc actcgagcaa agattccaca 360
ggatcccccct tatcccaagt ttgtccactg cactttcaaa tgtcaataga aaagatgcgt 420
ccagagttaga gcaaggagcg caatgtttcg gtccttgccg ggagcctgaa gccatcata 480
atactgtgtaa aatgagagcc ggcaattttc gaagactatg taagtctccg aacaggaagg 540
aagcgcaagc aaaccacaat tcgaactcca gataaacttg ctgttagcac aatcaacatg 600
accatgtaca agcgtctgtc gcttgtgagc ccgtcaatat gattgattga gggcagatcc 660
ttagcacttt aaccctgtaa acattttttg tttggctctc atcaacggca gctgtagcga 720
atctatagcc tgcggggcag tataacggtt atgacaaaa atgaagtga tataagtcca 780
taaatgtaac ccttttttct gctatataat tttttttgaa cgatgaacac taaacacaca 840
tagctatctg acatcgaaaca taaatgtaag agtcattcca aataagtaga gatgcaatgt 900
ggacatcatg ggatgatgtg tggatttacg atccctcaa ctgtaaaact attcttaggc 960
acaatttacc gagcgaaaca aaataactaa gaaacttagc atattttcga aagcattgta 1020
attttcaacc ttttgtgaaat tttttctata ttttcggaag tgtgacagtg aattccaaat 1080
cttgagtctg ctttgtgtac tttgacatga caactttcgg acacgctcac tttttgacga 1140
ccacacgttc atcgttgtct gcgctccggc acagactaaa gttcaggaat tgcctgtcac 1200
ctacgcagca tcaccactct ctgctctcta ccgtgattca aggcctgaatc atggatttcc 1260
aagtacaaat cttcagtttt ctacttatct cggccagtgt aataattagt aggggcgaca 1320
ttgtcatgac acaaaactct ctatctcac ctgtgacct tggacaaccc gctctatca 1380
gctgccttcc tcccacatcg ctgggtccatt ccgatggaaa cactgacttg tcttggtctc 1440
aacagcgccc tgggcagccc ccgcggtctc tgatttataa aataagtaac agatttagcg 1500
gcgttcggga ccggttcagt ggttcgggcg caggtaccga tttcacccct aagatctccc 1560
gcgtagaggg cgaggatgta ggagtgtatt actgtatgca agcaactcaa tttccccacc 1620
aatggacgtt cggccaaggg actaaagtgt aaattaagcg tacgggtgct gcgccatccg 1680
tctttatttt tccgccatct gaacgaacagc tcaaaagcgg aacggcttct gtcgtctgtc 1740
tctgaacaaa tttttaccgg cgagaggcta aggttcagtg gaaggctgat aatgcgttgc 1800
agtcggggca ctcgcaggaa tctgtgacgg aacaggacag caaggactcc acatacagct 1860
tatcgtcgac attgacatcg tcaaaagcgg actacgagaa acacaagggt tatgctgcg 1920
aagttacca tcagggatgt tcacaccag tgaccaaaac atttaatcga ggtgaatgcg 1980
agggccgagg gtcgctcttg acgtgcggtg acgtcgaaga aaacccggga cccatggatt 2040
ttcagggtcca atatttttct tttctctga tatcagctc tgcattata agtcgggggg 2100
aagtcacact cgtggaagc gggggggggt tagtacagcc ggggtggcgc ttgagactct 2160
cttgtgcagc ctcgggatcc acottttcaa actacgatat gcattgggtg aggcaggcca 2220
ccggcaaggg ctggagatgg gtgagtgcga atggcaccgc aggtgatact tattatccag 2280
gtagcgtcaa aggcggtttc acgattagtc gcgaaaacgc taagaattct ctctatttgc 2340
agatgaattc actacgagca ggtgacacag cagtctacta ttgtgcgagg gacgttgata 2400
ttgttgcgac catcacgcat tacttcgatt actggggtca aggcactttg gtaacagttt 2460
ctagtgcgag tactaagggt ccgagtgtat ttcctttagc ccttctctct aaaagcactt 2520
caggcggaac agcagccctt ggtgttttgg tgaagacta cttcccgaa cctgtaactg 2580
tgagctggaa ttctggcgcg ctcaactagt gaggttcacac ttttcagcc gtgttacagt 2640
catcgggtct ttattccctg tctcctcgct tgaccgtccc gtcgagttca ctaggtaacg 2700
agacatacat ttgtaacgtc aatcacaaag catcgaaac aaaggtggag aagcgtgtgt 2760
aaaccaagag ctgcgataag acccaacct gccctccctg ccagctccc gagcttctcg 2820
gcggaacctc ggttttctct tttccgccga agcccaaga tactttgatg attttcacgga 2880
cgccagaagt gacatgcgtg gttgttgacg taagccatga ggatccggaa gtgaaattta 2940
actggtacgt cgatggagtt gaggtccata acgccaagac caagccccc gaggaacaat 3000
ataattccac atatcgtgtg gtgtcgggtac ttacgggtct tcatcaggac tggctaaatg 3060
gaaaagaata caaatgcaag gtgtccaata aagcgctccc cgctcccat gaaaagacga 3120
tcagcaagcg gaaggggcga ccacgtgaac cgcaagtata cagctcccg cctcccgag 3180
aagaaatgac aaaaaaccaa gtctccctga cctgttttgt taaaggattt tacccttcgg 3240
acatcgcggt agagtgggag tccaacggac aacctgagaa taactacaaa acgactcctc 3300
cagtccgtga ctcggacgga tcttcttct tgtactcaca attgacggtt gacaaatccc 3360
gatggcaaca aggcaacggt tttagctgct cgggtcatgca tgaagctttg cacaaccact 3420
acacacaaaa atcgttgtcg ctgtcgccag gaaaataaaa ccaatttac tttaaagacta 3480
gcgcaatttt ttctttacat tagatgaaca taaaacgacg atgaggatca ctgttaggga 3540
aacgatgaac tacgtgaggt acttcttatt attcgtcctt tgatggcggtg gttgttctt 3600
ggtttgctgc cgagcgcaac gaccaaggcc tgtacaaaaa acgaagtga taatcagata 3660
tttgagtgtt aacaagaaac taagcatta ttttagcagt agcagcaatc aggatctaaa 3720
atgaacgcat ccggaccagg catcaataaa aacgaaaggc tcagtcgaaa gactgggcct 3780
ttcgttttat ctgttgtttg tcgggtgaacg ctctctacta gagtccactt ggctcacctt 3840
cgggtgggccc tttctcggtt tatatagagg gaaatgtgcg cggaacccct atttgtttat 3900
ttttctaaat acattcaaat atgtatccgc tcatgagaca ataacctga taaatgcttc 3960
aataatattg aaaaaggaag agtatgagta ttcaacattt ccgtgtcgcc cttattccct 4020
tttttgctgc attttgcctt cctgtttttg ctacccaga aacgctgggt aaagtaaaag 4080
atgctgaaga tcagttgggt gcacgagtggt gttacatcga actggatctc aacagcggt 4140
agatccttga gagttttcgc ccgaagaac gttttccaat gatgagcact tttaaagttc 4200
tgctatgtgg cgcggtatta tcccgattg acgcggggca agagcaactc ggtcgccga 4260

```

-continued

tacactat	tcagaatg	ttggttg	actcacc	cacagaaa	catcttac	4320
atggcatg	agtaagag	ttatgcag	ctgccata	catgagtg	aacactgc	4380
ccaactt	tctgaca	atcggagg	cgaaggag	aaccgcgt	ttgcaca	4440
tgggggat	tgtaact	cttgatc	gggaacc	gctgaatg	gccatacc	4500
acgacgag	tgacacc	atgcctg	caatggc	aacgttg	aaactatta	4560
ctggcgaa	acttact	gcttccc	aaacaatt	agactgg	gaggcgga	4620
aaagtgc	accactt	cgctcgg	ttccggc	ctggttt	gctgataa	4680
ctggagcc	tgagcgt	tcacgcg	tcattgc	actgggg	gatggtaa	4740
cctcccgt	cgtagtt	tacacga	ggagtc	aactatg	gaacgaat	4800
gacagat	tgagatg	gcctcact	ttaagc	gtaactg	gaccaagt	4860
actcatat	actttag	gatttaaa	ttcatttt	atttaaa	atctaggt	4920
agatcctt	tgataat	atgaccaa	tccttaac	tgagtttc	ttccactg	4980
cgtcagac	cgtagaaa	atcaagg	cttcttga	tccttttt	ctgcgcgt	5040
tctgctgt	gcaacaaa	aaaccac	taccagcg	ggtttgtt	ccggatca	5100
agctacca	tccttttc	aaggtaac	gcttcag	agcgagat	ccaaatac	5160
ttcttct	gtagccgt	ttaggcc	acttcaag	ctctgtag	ccgcctac	5220
acctcgct	gctaact	ttaccagt	ctgctgc	tgccgata	tcgtgtct	5280
ccgggttg	ctcaagac	tagttacc	ataaggcg	gcggtcgg	tgaaacgg	5340
gttcgtgc	acagccc	ttggagcg	cgacctac	cgaactga	tacctacg	5400
gtgagcta	agaaaag	acgcttc	aagggaga	ggcggac	tatccgg	5460
cgcgagg	cggaacag	gagcgac	gggagctt	aggggaaa	gcctggta	5520
tttatagt	tgctgggt	cgcacct	gaactgag	tcgatttt	tgatgctc	5580
cagggggg	gagcctat	aaaaacg	gcaacgcg	ctttttac	ttcctggc	5640
tttgctgc	ttttgctc	atgttctt	ctcggtat	ccctgatt	gtggataa	5700
gtattacc	cagagaat	aaaaagc	attatta	cggttttt	attatttg	5760
gcgttttg	attcaaat	caatctgt	ggacctg	aagacaca	agcgacat	5820
tgacatga	gccgtaaa	gcaaatc	gttgaac	tatccttt	ggggcctc	5880
gctacgac	tcgctcc	tggggctt	ttactata	cagcgcg	atttcagg	5940
tgccagat	caagatg	aaattgac	gtgccgtt	ggtgctac	gcgcgcg	6000
tcgcccga	ggctgagt	tggaacc	ggctcgg	ctcccggg	ttcgtggg	6060
acgaactc	cggtgtgt	cgggacg	tgacctgt	catcagcg	gtccagg	6120
agggtgtg	ggacaaac	ctggcctg	tgtgggtg	cggcctgg	gagctgt	6180
ccgagtg	ggaggctg	tcacgaac	tcgggagc	ctccggcg	gccatgac	6240
agatcggc	gcagccgt	ggcgggag	tcgcccgt	cgacccgg	ggcaactg	6300
tgcaactc	ggcgagg	caggactg	ccttcctt	aaatttaa	ttcattagt	6360
gcagtcact	cgctttgt	tcacagtc	gaataaac	agctcgtt	caccatgg	6420
gccaatct	cgtattcat	gtgtataa	gttcaac	caaagcta	acttttgg	6480
agagaaag	tatccga	gggcacg	tgccgtat	ttggagt	ctagcaga	6540
gtgagga	cacaggat	gttttctc				6570

SEQ ID NO: 48 moltype = DNA length = 6449
 FEATURE Location/Qualifiers
 misc_feature 1..6449
 note = Vector Pro2D
 source 1..6449
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 48

tttttcgg	tgaggact	ctaggagc	cataagcc	agactatg	ctgctgg	60
actctaaa	cggaacg	ttacgggt	gtgtttct	accattgt	tgatgg	120
cgctgagg	gtttctgc	gttgtagc	agattcaa	ctgcatct	gtataa	180
cgcaatgc	ctacagca	tcacatca	ctcgcgat	tggtcggt	caagacc	240
aaatttgc	gccttcag	aaggaata	ctgttggt	atatccag	cttggag	300
ggcaacac	caatgctg	ctgtatat	agtcagac	cacaccac	accatact	360
ccaaataa	gtatagcg	ttatcgaa	agctcgg	aaagaatt	tcctaaac	420
gtcggaaa	ttccaaa	gacaaa	aagtaag	agtcctct	gtgcttga	480
ggaaagct	agagggat	atttaaat	acgcgaaa	aaggagct	taagatgg	540
agagagag	accgcagt	tgtttaca	gtgaagct	aaaatgg	gaacaa	600
tgtaacta	tggtcact	taggtgg	ggctcct	gccaaat	cccaaac	660
accctagc	caattttt	actaatta	ctgtgggt	tcctatgc	cagggtgg	720
tcatttct	aagaacgt	tcgctccc	aggcgggt	gctccatt	tcattgat	780
cacgcata	atattcct	gactacc	tcataata	ccggtaac	aaacaat	840
aggctgat	tatgggtg	tttaacaa	cccaaac	tttttaac	gttactat	900
gcgattca	acatcaca	aatgacc	ttaaacg	agagtg	ctcatccg	960
aaagtagg	cgcatgtt	ctgagaga	cgcttc	gatctcat	gttgca	1020
gtctcaca	tgaacgtt	ataagcaa	acagcaa	tttgtgat	cagcta	1080
tgaccag	gcttctag	tggaacaa	ctctcatc	tcacag	tggaattc	1140
agtacaaa	ttcagttt	tacttatc	ggccagt	ataattag	ggggcg	1200
tgatcatg	caaaactc	tatctc	tgtagcc	ggacaacc	cctctat	1260
ctgcggtt	tcaccaat	tggtccat	cgatgga	acgtact	cttggctc	1320
acagcgcc	gggcagcc	cgcggctt	gattataa	ataagtaa	gatttag	1380
cgttccgg	cggttcag	gttcgggc	aggta	ttcacc	agatctcc	1440
cgtagagg	gaggatgt	gagtgat	ctgtatg	gcaactca	ttcccc	1500
atggacgt	ggtcaagg	ctaaag	aattaag	acggttg	cgccatcc	1560
ctttattt	cgcctatc	acgaac	caaaag	acggttc	tcgtctgt	1620
cctgaaca	ttttacc	gagagg	ggttcag	aaggtc	atgcgttg	1680

-continued

gtcgggcaac	tgcgaggaat	ctgtgacgga	acaggacagc	aaggactcca	catcacagctt	1740
atcgctcgaca	ttgacactgt	caaaagcggg	ctacgagaaa	cacaagggtg	atgcctgcga	1800
agttaccocat	cagggattgt	catcaccagt	gaccaaataca	tttaatcgag	gtgaatgcga	1860
gggcccgggg	tcgctcttga	cgtgcgggtga	cgtcgaagaa	aaccocggac	ccatggattt	1920
tcaggtccaa	atattttctt	ttcttctgat	atcagcctct	gtcattataa	gtcgggggga	1980
agtccaccto	gtggaagcgg	gggggggttt	agtacagccg	gggtggcgct	tgagactctc	2040
ttgtgcagcc	tcgggattca	ccttttcaaa	ctacgatatg	cattgggtga	ggcaggccac	2100
cggcaagggg	ctggagtggt	tgagtgcata	tggcaccgca	gggtatactt	attatccagg	2160
tagcgtcaaa	ggcgttttca	cgattagtgc	cgaaaacgct	aagaattctc	tctatttgca	2220
gatgaattca	ctacgagcag	gtgacacagc	agtctactat	tgtgcgaggg	acgttgatat	2280
tgttgcgacc	atcaccgatt	acttcgatta	ctgggggtcaa	ggcactttgg	taacagtttc	2340
tagtgcgagt	actaagggtc	cgagtgtatt	tccttttagcc	ccttcctcta	aaagcacttc	2400
aggcggaaaca	gcagcccttg	gctgtttggt	gaaagactac	ttcccggaac	ctgtaactgt	2460
gagctggaat	ctcggcgccg	tcactagtgg	agttcacact	tttccagccg	tgttacagtc	2520
atcgggtctt	tattccctgt	cctccgtcgt	gaccgtcccg	tcgagttcac	taggtacgca	2580
gacatacatt	tgtaacgtca	atcacaaagg	atcgaaacaca	aaggtggaca	agcgtgttga	2640
acccaagagc	tcgataaaga	cccacacctg	ccctccctgc	ccagctcccg	agcttctcgg	2700
cggaccttcg	gttttctctt	ttccgcccga	gcccacaaat	actttgatga	tttccaggac	2760
gcccagaagt	acatgctgtg	ttgttgacgt	aagccatgag	gatccggaag	tgaattttaa	2820
ctgggtacgtc	gatggagtgg	aggtccataa	cgccaaagacc	aagcccccgg	aggaacaata	2880
taattccaca	tatcgtgtgg	tgtcgttact	tacggctcct	catcaggagt	ggctaataatg	2940
aaaagaatac	aaatgcaagg	gtcccaataa	agcgctcccc	gctcccatcg	aaaagacgat	3000
cagcaaggcg	aagggggcaac	cacgtgaacc	gcaagtatac	acgctcccg	cctcccgaga	3060
agaaatgaca	aaaaaccaag	tctccctgac	ctgttttggtt	aaaggatttt	acccttcgga	3120
catcgcggtg	gagtgagggt	ccaacggaca	acctgagaat	aactacaaaa	cgactcctcc	3180
agtcctggac	tcggacggat	ccttcttctt	gtactccaaa	ttgacgggtg	acaaatcccc	3240
atggcaacaa	ggcaacgttt	ttagctgctc	ggctcatgat	gaagctttgc	acaaccacta	3300
cacacaaaaa	tcgttgtcgc	tgtcgccagg	aaaaataaac	ccaattttat	ttaagactag	3360
cgcaattttt	ttcttatcat	agatgaacat	aaaacgacga	tgaggatcac	tgttagggaa	3420
acgatgaact	acgctgagga	cttccctatta	ttcgtccttt	gatggcgctg	ttgttctttg	3480
gtttgcggcc	gagcgcaacg	accaaggcct	gtacacaaaa	cgaagttgat	aatcagatat	3540
ttgagttgta	acaagaaact	aaagcattat	tttagcagta	gcacgaatca	ggatctaaaa	3600
tgaacgcctc	cggaccaggc	atcaaaataa	acgaaaggct	cagtcgaaag	actgggcctt	3660
tcgtttttat	tgttgtttgt	cgttgaaacg	tctctactag	agtcacactg	gctcaccttc	3720
gggtgggcct	ttctgcgttt	atatagaggg	aaatgtgcgc	ggaaccccta	tttgtttatt	3780
ttcttaataa	catcaaaata	tgtatccgct	catgagacaa	taaccctgat	aaatgcttca	3840
ataatattga	aaaaggaaga	gtatgagtat	tcaacatttc	cgtgtcgccc	ttattccctt	3900
ttttgcggca	ttttgccttc	ctgtttttgc	tcacccagaa	acgctgggtg	aagtaaaaga	3960
tgctgaagat	cagttgggtg	cacgagtggt	ttacatcgaa	ctggatctca	acagcggtaa	4020
gactcttgag	agttttcgcc	ccgaagaacg	ttttccaatg	atgagcactt	ttaaagtctt	4080
gctatgtggc	gcgggtattt	cocgtattga	cgccgggcaa	gagcaactcg	gtcgcgcgat	4140
acactattct	cagaatgact	tggttgagta	ctcaccagtc	acagaaaagc	atcttacgga	4200
tggcatgaca	gtaagagatc	tatgcagtgc	tgccataacc	atgagtgata	acactgcggc	4260
caacttactt	ctgacaaacg	tcggaggacc	gaaggagcta	accgcttttt	tgacacacat	4320
gggggatcat	gtaactcgcc	tgtatcggtg	ggaacccgag	ctgaatgaag	ccataccaaa	4380
cgacgagcgt	gacaccagca	tgctgttagc	aatggcaaca	acgttgcgca	aactattaac	4440
tggcgaaact	cttactctag	cttcccgcca	acaattaata	gactggatgg	aggcgggata	4500
agttgcagga	ccacttctgc	gctcggccct	tcggcctggc	tggtttattg	ctgataaatc	4560
tgagcccggt	gagcgtgggt	cacgcggtat	cattgcagca	ctggggccag	atggtaagcc	4620
ctcccgatc	gtagttatct	acacgacggg	gagtcaggca	actatggatg	aacgaataag	4680
acagatcgct	gagatagggt	cctcactgat	taagcattgg	taactgtcag	accaagttta	4740
ctcatatata	ctttagattg	atttaaaact	tcatttttaa	tttaaaagga	tctaggtgaa	4800
gatccttttt	gataatctca	tgacaaaaat	cccttaacgt	gagttttcgt	tcactgagc	4860
gtcagacccc	gtagaaaaga	tcaaaaggat	ttcttgagat	ccttttttcc	tgcgcgtaat	4920
ctgctgcttg	caaacacaaa	aaccaccgct	accagcgggt	gtttgtttgc	cggatcaaga	4980
gctaccaact	ctttttccga	aggtaactgg	cttcagcaga	gcgcagatag	caaatactgt	5040
tcttctagt	tagccgtagt	taggccacca	cttcaagaac	tctgtagcac	cgcctacata	5100
cctcgtcttg	ctaactcctg	taccagtggc	tgctgccagt	ggcgataaag	cgtgtcttac	5160
cgggttgga	tcaagacgat	agttaccgga	taaggcgcag	cggtcgggct	gaacgggggg	5220
ttcgtgcaca	cagcccagct	tggagcgaac	gacctacacc	gaactgagat	acctacagcg	5280
tgagctatga	gaaagcgcca	cgttcccga	agggagaaaag	gcggacaggt	atccggtaa	5340
cggcagggtc	ggaacaggag	agcgcacgag	ggagcttcca	gggggaaacg	cctgggtatc	5400
ttatagtcct	gtcgggtttc	gccacctctg	acttgagcgt	cgatttttgt	gatgctcgtc	5460
agggggggcg	agcctatgga	aaaacgccag	caacgcggcc	tttttacggt	tcctggcctt	5520
ttgctggcct	tttgtccaca	gtttctttcc	tgcgttatcc	cctgattctg	tggaataacc	5580
tattaccgcc	agagaatata	aaaagccaga	ttattaatcc	ggctttttta	ttatttgacg	5640
cgttttgaga	ttcaaaatc	aatctgtttg	gacctggata	agacacaaga	gcgacatcct	5700
gacatgaacg	cogtaaacag	caaatcctgg	ttgaacacgt	atccttttgg	gggcctcccg	5760
ctacgacgct	cgctccagct	ggggcttctt	tactatacac	agcgcgcata	tttcacgggt	5820
gccagatgct	aagatggcca	agttgaccag	tgccgttccg	gtgctacccg	cgcgcgacgt	5880
cgcgggagcg	gtcgagttct	ggaccgaccg	gctcgggttc	tcccgggact	tcgtggagga	5940
cgaacttcgc	ggtgtggttc	gggacgacgt	gacctgttcc	atcagcgccg	tccaggacca	6000
ggtggtgccg	gacaacaccc	tggcctgggt	gtgggtgcgc	ggcctggacg	agctgtacgc	6060
cgagtggtcg	gaggtcgtgt	ccacgaactt	ccgggacgcc	tccggggccg	ccatgaccga	6120
gatcggcgag	cagccgtggg	ggcgggagtt	cgccctgcgc	gacccggccg	gcaactgcgt	6180
gcacttcgtg	gccgaggagc	aggactgaac	cttccttaaa	aatttaattt	tcattagttg	6240

-continued

cagtcactcc	gctttggttt	cacagtcagg	aataacacta	gctcgtcttc	accatggatg	6300
ccaatctcgc	ctattcatgg	tgtataaaaag	ttcaacatcc	aaagctagaa	cttttgga	6360
gagaaagaat	atccgaatag	ggcacggcgt	gccgtattgt	tggagtggac	tagcagaaa	6420
tgaggaaggc	acaggatgag	ttttctcga				6449

```

SEQ ID NO: 49      moltype = DNA length = 5859
FEATURE
misc_feature      Location/Qualifiers
                  1..5859
                  note = Vector Pro3D
source            1..5859
                  mol_type = other DNA
                  organism = synthetic construct

```

```

SEQUENCE: 49
ttttaatta tcaattgatg ttctgtctct aggaatttat agtcaagctt ccttttgcta 60
ctaccctgct cctcaatcag cgtcagggaa gcaagatcct cctcggctag ggtcttagca 120
tggttcttgg cctacgctag ctaggaagggt accataaaaa cgagagctat taatggctca 180
gcaattatcc gatacatggt taacagaaaag acagaaaaaca cgaaaaaagt tgttgacagt 240
gaatgctcca gatgtcacia gaaaggcgtt ggggcgaatc gattgggcct ttggaggcat 300
tagtaaaaaa gagaaagctt tttaatcttt gtcgacatgt actctatac aatattgcta 360
tagaattgaa ggacagcttg aaatgtctat cgacatagtg actccgactt aagggtcattc 420
actattttta cgcgtgacgg cggaccctaaa acccgaccgg cagttcctgc cccaaacata 480
atacaaatat tagctccaaa attgactctc ttgaaccga cgaccctaat ggttctacca 540
tggaattcca agtacaaatc ttccagtttc tacttatctc ggccagtgta ataattagta 600
ggggcgacat tgtcatgaca caaactcctc tatcctcacc tgtgaccctt ggacaaccgg 660
cctctatcag ctgcggttcc tcccaatcgc tgggtccattc cgatggaac acgtacttgt 720
cttggttcca acagcgccct gggcagcccc cgcggtcttt gatttataaa ataagtaaca 780
gatttagcgg cgttccggac cggttcagtg gttcggggcg aggtaccgat ttaccctta 840
agatctcccg cgtagagggc gaggatgtag gagtgtatta ctgtatgcaa gcaactcaat 900
tccccacca atggacgttc ggtcaaggga ctaaaagtga aattaagcgt acgggtgctg 960
cgccatccgt ctttattttt ccgccatctg acgaacagct caaaagcgga acgggtctctg 1020
tcgtctgtct cctgaacaaat tttaccgcc gagagggttaa ggttcagtgg aaggctcgata 1080
atcggttgca gtcgggcaac tcgcaggaat ctgtgacgga acaggacagc aaggactcca 1140
catcacagct atcgtcgaca ttgacactgt caaaagcgga ctacgagaaa cacaagggtg 1200
atgctctcga agttaccatc cagggtattgt catcaccagt gaccaaatca tttaatcgag 1260
gtgaatcgca gggccgaggg tcgctcttga cgtgcggtga cgtcgaagaa aacccgggac 1320
ccatggattt tcagggtcaa atattttctt ttctctgat atcagctctt gtcattataa 1380
gtcgggggga agtccacctc gtggaaaagc ggggggggtt agtacagccg ggtggcgcc 1440
tgagactctc ttgtgcagcc tcgggattca ccttttcaa ctacgatatg cattgggtga 1500
ggcagggcac cggcaagggt ctggagtggtg tgagtgtctaa tggcaccgca ggtgatactt 1560
attatccagg tagcgtcaaa ggcggtttca cgattagtgc gaaaacgctt aagaattctc 1620
tctatttgca taggaattca ctacgagcag gtgacacagc agtctactat tgtgcgaggg 1680
acgttgatag ttgtgcgacc atcaccgatt acttcgatta ctggggtcaa ggcaactttg 1740
taacagtttc tagtgcgagt actaagggtc cgagtgtatt tcttttagcc ccttctctca 1800
aaagcacttc aggcgggaac gcagcccttg gctgttttgt gaaagactac ttcccggaac 1860
ctgtaactgt gagctggaat tctggcgccg tcaactagtg agttccactt ttccagccg 1920
tggtacagtc atcgggtctt tattcctctg cctcgtctgt gaccgtcccg tcgagttcac 1980
taggtacgca gacatacatt tgtaacgtca atcacaagcc atcgaacaca aagggtggaca 2040
agcgtgttga acccaagagc tgcgataaga cccacacctg cctcctctgc ccagctcccg 2100
agcttctcgg cggacctctg ttttctctct ttccgcccga gcccaagat actttgatga 2160
tttcacggac gccagaagtg acatgcgtgg ttgttgacgt aagccatgag gatccggaag 2220
tgaaatttaa ctggtacgtc gatggagttg aggtccataa cgccaagacc aagcccccg 2280
aggaacaata taattcaaca tatcgtgtgg tgtcgtgact tacggtcctt catcaggact 2340
ggctaaatgg aaaagaatac aaatgcaagg tgtccaataa agcgtctccc gctcccactg 2400
aaaagacgat cagcaaggcg aaggggcaac cacgtgaacc gcaagtatac acgctcccg 2460
cctcccgaga agaaatgaca aaaaaccaag tctcctgac ctgtttggtt aaaggatttt 2520
acccttcgga catcgcggtg gagtgggagt ccaacggaca acctgagaat aactacaaaa 2580
cgactcctcc agtccctggc tcggacggat ccttctctt gtactccaaa ttgacgggtg 2640
acaaatcccg atggcaacaa ggcaacgttt tttagctgctc ggtcatgcat gaagctttgc 2700
acaaccacta cacacaaaaa tcgttgtctc tgtcgccagg aaaataaaac ccaatttact 2760
ttaagactag cgcaattttt ttcttacatt agatgaacat aaaaacgaga tgaggatcac 2820
tggttaggaa acgatgaact acgtgagga cttcctatta ttctccttt gatggcggtg 2880
ttgttctttg gtttgcggcc gagcgcaacg accaaggcct gtacaaaaa cgaagttgat 2940
aatcagatat ttgagttgta acaagaaact aaagcattat tttagcagta gcacgaatca 3000
ggatctaaaa tgaacgcac cggaccaggc atcaaaaaa acgaaaggct cagtcgaaag 3060
actgggcctt tcgtttttat cgtttttgtt cgttgaaacg tctctactag agtcacactg 3120
gctcaccttc ggggtgggct ttctgctgtt atatagaggg aaatgtgcgc ggaaccccta 3180
tttgtttatt tttctaata cattcaataa tgtatccgct catgagacaa taacctgat 3240
aaatgcttca ataatttga aaaaggaaga gtatgagat tcaacatttc cgtgtcgccc 3300
ttattccctt ttttgcggca ttttgccttc ctgtttttgc tccccagaa acgctgggtg 3360
aagtaaaaga tgctgaagat cagttgggtg cacgagtggt ttacatcgaa ctggatctca 3420
acagcggtaa gatccttgag gtttttcgcc ccgaagaacg ttttccaatg atgagcaatt 3480
ttaaggttct gctatgtggc gcggtattat cccgtattga cgccgggcaa gagcaactcg 3540
gtcgccgcat acactattct cagaatgact tgggtgagta ctaccagtc acagaaaagc 3600
atcttacgga tggcatgaca gtaagagaat tatgcagtc tgccataacc atgagtgata 3660
acactggcgc caacttact ctgacaacga tcggaggacc gaaggagcta accgttttt 3720
tgacaacat gggggatcat gtaactcgcc ttgatcgttg ggaacggag ctgaatgaag 3780

```

-continued

```

ccataccaaa cgacgagcgt gacaccacga tgcctgtagc aatggcaaca acgttgcgca 3840
aactattaac tggcgaaacta cttactctag cttcccgcca acaattaata gactggatgg 3900
aggcggataa agttgcagga ccactctctgc gctcggccct tccggtctggc tggtttattg 3960
ctgataaatc tggagccggt gagcgtgggt cagcggtgat cattgcagca ctggggccag 4020
atggtaagcc ctcccgatc gtatgtatct acacgacggg gagtcaggca actatggatg 4080
aacgaaatag acagatcgct gagatagggt cctcactgat taagcattgg taactgtcag 4140
accaagttta ctcatatata ctttagattg atttaaaact tcatttttaa tttaaaagga 4200
tctaggtgaa gatccttttt gataatctca tgacaaaaat cccttaacgt gagttttcgt 4260
tccactgagc gtcagacccc gttagaaaaga tcaaaaggatc ttcttgagat cctttttttc 4320
tgcgcgtaat ctgctgcttg caaacaacaaa aaccaccgct accagcggtg gttgttttgc 4380
cggatcaaga gctaccaact ctttttccga aggttaactgg cttcagcaga gcgcagatac 4440
caaatactgt tcttctagtg tagccgtagt taggccacca cttcaagaac tctgtagcac 4500
cgctacata cctcgctctg tcaatcctgt taccagtggc tgctgccagt ggcgataagt 4560
cgtgtcttac cgttttggtg tcaagacgat agttaccgga taaggcgag cggtcgggct 4620
gaacgggggg ttctgtgcaca cagcccagct tggagcgaaac gacctacacc gaactgagat 4680
acctacagcg tgagctatga gaaagcgcca cgcttccga agggagaaag gcggacaggt 4740
atccgggtaag cggcagggtc ggaacaggag agcgcacgag ggagcttcca gggggaaacg 4800
cctgggtatc ttatagtcct gtcgggtttc gccacctctg acttgagcgt cgatttttgt 4860
gatgctcgtc agggggggcg agcctatgga aaaacggcag caacgcggcc tttttacggt 4920
tcctggcctt ttgctggcct tttgtccaca tgttctttcc tgcgttatcc cctgattctg 4980
tggataaacg tattaccgcc agagaatata aaaaaccaga ttattaatcc ggctttttta 5040
ttatttgacg cgttttgaga ttcaaaattc aatctgtttg gacctggata agacacaaga 5100
gcgacatcct gacatgaacg ccgtaaacag caaatcctgg ttgaacacgt atccttttgg 5160
gggctctccc ctacgacgct cgctccagct ggggcttccct tactatacac agcgcgcata 5220
tttcacgggt gccagatgtc aagatggcca agttgaccag tgccgttccg gtgctcaacc 5280
cgcgcgacgt cgcgcgagcg gtcgagttct ggaccgacgg gctcgggttc tccggggact 5340
tcgtggagga cgacttcgcc ggtgtggtcc gggacgacgt gacctgttc atcagcgcg 5400
tcaggagcca ggtggtgcc gacaaacccc tggcctgggt gtgggtgcgc ggctggacg 5460
agctgtacgc cgagtgttcg gaggtcgtgt ccacgaactt ccgggacgcc tccggggact 5520
ccatgaccga gatcggcgag cagccgtggg ggcgggagtt cgccctgcgc gacccggcg 5580
gcaactgcgt gcaactcgtg gccgaggagc aggaactgaac cttccttaa aatttaattt 5640
tcattagtgt cagtcactcc gctttgggtt cacagtcagg aataacacta gctcgtcttc 5700
accatggatg ccaatctcgc ctattcatgg tgtataaaag ttcaacatcc aaagctagaa 5760
cttttgaaa gagaaagaat atccgaatag ggcacggcgt gccgtattgt tggagtggac 5820
tagcagaaaag tgaggaaggg acaggatgag ttttctcga 5859

```

```

SEQ ID NO: 50      moltype = DNA length = 6419
FEATURE           Location/Qualifiers
misc_feature       1..6419
                   note = Vector pro4D
source            1..6419
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 50
ttttaaatgt tatactccag cagtcgttta ataccggaga cggtagacgc aagagcagct 60
cacactcttt ggcacagact gtaaatgcct ctagtgttga cgagtttgcg tgcacgagag 120
aaattttcga gagaacacag gaaggacccc agtaactttg ggctcctttg agcttgcaac 180
gtctttcgtt tcgaaaacca aatccaaatc cctttcgaca caggggcttt ttctcgcgca 240
tgtgataaaa agcaaaacaa acccgagatg catccaaggt cgtcactgtc agaatacgg 300
gactctagaa ggcaatgcgg aactcccagc cggccgcata ctattgacga tgaatcatct 360
tgcttttcat tactataggt tgttgccgc ctctcgattt attttcagat gatctgttta 420
aaatgttctc cattcggtga attttcttgc agaccatgtc ccaagtccca cactaccaca 480
gcagatgatg gaattaaggt caccgtcaaa aggttagacag agtcgcgtta gaaacctatg 540
tagataaagg gtcaaacgga atgttgcgga tccccaggg ctatagacag gactacaaaa 600
gagatctgcg atcaaaattt tgttgcacag cacatatacc gaaagataac ttgagttatc 660
cttacattgg gagtcccaaa aatcaatctt tcgagcgtct tcgtgccgca acaattagat 720
ggccagttgg aacaaaaaaa gtatcaaatg gtcgaaaata ccaagagaa acgctcacta 780
atagaattct gtcgctctct tatatttctt tgaacatgag cgttattagc aggtaaacag 840
ttgtttcgta aatctgacat taatagtga gtgtgggtcat aatgcgcgc tcctctgatg 900
tcgaaactat ggattcatgt aaaacgggtc tcggctcatg caaaggaaact tcgcttttcta 960
gtaaaatacc acacaaacaa atttctcac ctcttctcgt gaatctggca tttccatcaa 1020
caccttttca tagctttgtg gatggattga gttgcgctcg ctgtcaccat ttgcgccgaa 1080
ttccttgccc ttacagtaa tggatttcca agtacaatc ttcagtttcc tacttatctc 1140
ggccagtgta ataattagta gggggcagat tgtcatgaca caaactcctc tatectcacc 1200
tgtgaccttt ggacaacccc cctctatcag ctgcggttcc tcccaatcgc tggttccattc 1260
cgatggaaaac acgtacttgt cttggctcca acagcgccct gggcagcccc cgcggctctt 1320
gatttataaa ataagtaaca gatttagcgg cggtccggac cggttcagtg gttcggggcg 1380
aggtagccgat ttacccttta agatctccc gtagaggcc gagtagtgag gattgtatta 1440
ctgtatgcaa gcaactcaat tccccacca atggacgttc ggtcaaggga ctaaagttag 1500
aattaaagcg acggttgtct cgccatccgt ctttattttt ccgccatctg acgaacagct 1560
caaaagcgga acggtctctg tcgtctgtct cctgaacaat tttaccgcg gagaggctaa 1620
ggttcagtgg aaggtcgata atgcgttgca gtccggcaac tcgcaggaaat ctgtgacgga 1680
acaggacagc aaggactcca catcacgctt atcgtcgaca ttgacactgt caaaagcgga 1740
ctacgagaaa ccaaggtgtg atgcctgcga agttacccat cagggtattgt catcaccagt 1800
gaccaaatac tttaatcgag gtgaatgcga gggccgaggg tcgctcttga cgtgcgggtg 1860
cgtcgaagaa aaccggggac ccatggattt tcaggtccaa atattttctt ttcttctgat 1920

```

-continued

atcagcctct	gtcattataa	gtcgggggga	agtcacctc	gtggaagcg	gggggggtt	1980
agtacagcgc	ggtggcgect	tgagactctc	ttgtgcagcc	tcgggattca	ccttttcaaa	2040
ctacgatatg	cattgggtga	ggcaggccac	cggcaagggt	ctggagtggg	tgagtgtctaa	2100
tggcaccgca	ggtgatactt	attatccagg	tagcgtcaaa	ggcggtttca	cgattagtgc	2160
cgaaaacgct	aagaattctc	tctatttgca	gatgaattca	ctacgagcag	gtgacacagc	2220
agtctactat	tgtgcgaggg	acgttgatct	tgttgcgacc	atcccgatt	acttcgatta	2280
ctgggggtcaa	ggcactttgg	taacagtttc	tagtgcgagt	actaagggtc	cgagtgtatt	2340
tcctttagcc	ccttcctcta	aaagcaactc	aggcggaaca	gcagcccttg	gctgtttggt	2400
gaaagactac	ttcccggaac	ctgtaactgt	gagctggaat	tctggcgccg	tcactagtgg	2460
agttcacact	tttccagcgc	tgttacagtc	atcgggtctt	tattccctgt	cctccgtcgt	2520
gaaccgtccc	tcgagttcac	taggtacgca	gacatacatt	tgtaacgtca	atcacagcc	2580
atcgaaacaca	aagggtggaca	agcgtgttga	acccaagagc	tgcgataaga	cccacacctg	2640
ccctccctgc	ccagctcccg	agcttctcgg	cggaccttcg	gttttctctt	ttccgcccga	2700
gcccaaagat	acttttgatga	tttcacggac	gccagaagtg	acatgctgtg	ttgttgacgt	2760
aagccatgag	gatccggaag	tgaattttaa	ctggtagctc	gatggagtgg	aggtccataa	2820
cgccaagacc	aagccccgcg	aggaacaata	taattccaca	tatcgtgtgg	tgtcggtagt	2880
tacggtcctt	catcaggact	ggctaaatgg	aaaagaatac	aaatgcaagg	tgccaataaa	2940
agcgtcccc	gctcccatcg	aaaagacgat	cagcaaggcg	aaggggcaac	cacgtgaacc	3000
gcaagtatac	acgctcccg	cctcccgaga	agaaatgaca	aaaaaccaag	tctccctgac	3060
ctgttttggt	aaaggatttt	acccttcgga	catcgcggtg	gagtgagggt	ccaacggaca	3120
acctgagaat	aactacaaaa	cgcactcctc	agtcctggac	tcggacggat	ccttcttctt	3180
gtactccaaa	ttgacgggtg	acaaatcccc	atggcaacaa	ggcaacgttt	ttagctgctc	3240
ggatcatgcat	gaagctttgc	acaaccacta	cacacaaaaa	tcgttgtcgc	tgccgccagg	3300
aaaataaaac	ccaatttact	ttaagactag	cgcatttttt	ttcttacatt	agatgaacat	3360
aaaacgacga	tgaggatcac	tgttagggaa	acgatgaact	acgctgagga	cttctctatta	3420
ttcgtccttt	gatggcggtg	ttgttctttg	gtttgcggcc	gagcgcaacg	accaaggcct	3480
gtacaaaaaa	cgaagtgtat	aatcagatat	ttgagttgta	acaagaaact	aaagcattat	3540
tttagcagta	gcacgaatca	ggacttaaaa	tgaacgcctc	cggaccaggc	atcaaataaa	3600
acgaaaggct	cagtcgaaag	actgggccc	tcgtttttat	tggtgtttgc	cggtgaacgc	3660
tctctactag	agtcacactg	gctcaccttc	gggtgggccc	ttctgcgttt	atatagaggg	3720
aaatgtgcgc	ggaacctcta	tttgtttatt	tttctaaata	cattcaataa	tgtatccgct	3780
catgagacaa	taacctgat	aaatgcttca	ataatatgta	aaaaggaaga	gtatgagtat	3840
tcaacatttc	cgtgtcgccc	ttattccctt	ttttgcggca	ttttgccttc	ctgtttttgc	3900
tcaccagaaa	acgctgggtga	aagtaaaaaga	tgctgaagat	cagttgggtg	cacgagtggtg	3960
ttacatcgaa	ctggatctca	acagcggtta	gatccttgag	agttttcgcc	ccgaagaacg	4020
ttttccaatg	atgagcactt	ttaaagtctt	gctatgtggc	gcggtattat	cccgatttga	4080
cgccgggcaa	gagcaactcg	gtcgcgcgat	acactattct	cagaatgact	tggttgagta	4140
ctcacccagtc	acagaaaagc	atcttacgga	tggcatgaca	gtaagagaat	tatgcagtg	4200
tgccataacc	atgagtgata	acactgcggc	caacttactt	ctgacaaacg	tcggaggacc	4260
gaaggagcta	accgcttttt	tgcaacaacat	gggggatcat	gtaactcgcc	ttgatcgttg	4320
ggaaaccggag	ctgaatgaag	ccataccaaa	cgaagagcgt	gacaccacga	tgctgttagc	4380
aatggcaaca	acgttgcgca	aactattaac	tggcgaaacta	cttactctag	cctcccgcca	4440
acaatttaata	cgttggtatg	aggcggataa	agttgcaggga	ccacttctgc	gctcggccct	4500
tcgggctggc	tggttttatg	ctgataaaatc	tggagccggg	gagcgtgggt	cacgcgggtat	4560
catctgcagc	ctggggccag	atggtaagcc	ctcccgatc	gtagtattct	acacgacggg	4620
gagtcaggca	actatggatg	aacgaaatag	acagatcgct	gagatagggt	cctcactgat	4680
taagcattgg	taactgtcag	accaagttaa	ctcatatata	ctttagattg	atttaaaact	4740
tcatttttaa	tttaaaagga	tctaggtgaa	gatccttttt	gataatctca	tgacaaaaat	4800
cccttaacgt	gagttttcgt	tcactgagc	gtcagacccc	gtagaaaaga	tcaaggatc	4860
ttcttgagat	cctttttttc	tgcgcgtaat	ctgctgcttg	caaaacaaaa	aaccaccgct	4920
accagcggtg	gtttgtttgc	cggatcaaga	gctacaaact	ctttttccga	aggttaactgg	4980
cttcagcaga	gctcagatac	caataactgt	tcttctagt	tagcgtagt	tagggcaccga	5040
cttcaagaac	ctctgtagc	cgcctacata	cctcgtctct	ctaactctgt	taccagtggc	5100
tgctgccagt	ggcgataagt	cgtgtcttac	cgggttggac	tcaagacgat	agttaccgga	5160
taaggcgag	cggctggggg	gaacgggggg	ttcgtgcaca	cagcccagct	tggagcggaac	5220
gacctacacc	gaactgagat	acctacagcg	tgagctatga	gaaagcgcca	cgttcccgga	5280
agggagaaaag	gcggacaggt	atccggttaag	cggcagggtc	ggaacaggag	agcgcacgag	5340
ggagcttcca	gggggaacgc	cctggtagct	ttatagtcct	gtcgggtttc	gccacctctg	5400
acttgagcgt	cgattttttg	gatgctcgtc	agggggggcg	agcctatgga	aaaacgccag	5460
caacgcggcc	tttttaacgtt	tcttgccctt	ttgctggcct	tttgctcaca	tgcttcttcc	5520
tgctttatcc	cctgattctg	tggataaacg	tattaccgcc	agagaatata	aaaagccaga	5580
ttattaatcc	ggctttttta	ttatttgacg	cgttttgaga	ttcaaaatcc	aatctgtttg	5640
gacctggata	agacacaaaga	gcgacatcct	gacatgaacg	ccgtaaacag	caaatcctgg	5700
ttgaacacgt	atcctttttg	gggcctcccg	ctacgacgct	cgtccagct	ggggcttcc	5760
tactatacac	agcgcgcata	tttcacgggt	gccagatgtc	aagatggcca	agttgaccag	5820
tgccgttcgc	gtgctcacgc	cgcgcgacgt	cgccggagcg	gtcaggtctt	ggaccgacgc	5880
gctcgggttc	tcccgggact	tcgtggagga	cgaactccgc	gggtgtggtc	gggacgacgt	5940
gaacctgttc	atcagcgcg	tcaggacca	gggtgtgcgc	gacaacacc	tgccctgggt	6000
gtgggtgcgc	ggcctggagc	agctgtacgc	cgaagtgtcg	gaggtcgtgt	ccacgaactt	6060
ccgggacgcc	tccggggccg	ccatgaccga	gatcggcgag	cagccgtggg	ggcgggaggt	6120
cgcctcgcgc	gaccggccgc	gcactgcgt	gccagggagc	aggactgaac		6180
cttccttaaa	aatattaatt	tcatagttg	cagtcactcc	gctttggttt	cacagtcagg	6240
aataacacta	gctcgtcttc	accatggatg	ccaatctcgc	ctattcatgg	tgtataaaag	6300
ttcaacatcc	aaagctagaa	cttttggaag	gagaaagaa	atccgaatag	ggcacggcgt	6360
gcgctattgt	tggagtgagc	tagcagaaag	tgaggaaagg	acaggatgag	ttttctcga	6419

-continued

```

SEQ ID NO: 51      moltype = DNA length = 6281
FEATURE            Location/Qualifiers
misc_feature        1..6281
                    note = Vector pro5D
source              1..6281
                    mol_type = other DNA
                    organism = synthetic construct

SEQUENCE: 51
ttttaatcat aggcaatggt tggtagacaa taaatgtgac gtagttcatt atgtgtctgca 60
atactgaata cttgccagga gtgacaggat tctagctcct gaggagcgct tacagtggaa 120
ttacacacat gacttgtagc gacagccaac taactataat aagaagcaat gcggacttgg 180
taggatcctt tcatcaattc aaacattcac tatcaatcaa aaattacaaa ttttacttgc 240
ttgcagtcag catatatttg tcaacgcatt ggggtcttct atagtcaaat agaaatgtca 300
cgagaatagc tcccggggccc gtcaagttct taattctgta actgtgtcct gccactccat 360
ttgcacttga cgtatcactt gtgcgtctca acacagcatt acacaatggc actctacaga 420
gcattggtta atgtaaatta ttgaaaggca aaacatagcc cttcttgact gtgagtccca 480
ccatcgctac gagtcaacta catgtaaaat atctgtcctt ccatcgaggg cacggaattc 540
tgaaacgaaa acaatcttgt tgggctcggg agtccgttac gcaaaaacgc ttctatcaat 600
tcaattagga ttggtcttgg ttggagctg cccgacgcgc tcaaaaattg actgggtatg 660
ggtagacttt ctctctgtcg aaatatattg tacagtcatt ttggttcttg ggtatgctta 720
ttggagatga tcagttgatg cgcgacggat tgagcccaaa caatttacct catagagata 780
tgcaacttgg aaaaagaccg ctgcggtcat ctgcaaatcc cacaagcttt ttgctctcga 840
accttcacaa tcttaccoga gttgctatcg ttgctagcag cctttcccat tgattgccaa 900
cattttgatc ctgacaaccc atagattcgc tctgttaaac caacagcaac agcaagcaaa 960
tatggatttc caagtacaaa tcttcagttt tctacttacc tcggccagtg taataattag 1020
tagggggcgc attgtcatga cacaaactcc tctatcctca cctgtgaccc ttggacaacc 1080
cgctctatcc agctgcggtt cctcccaatc gctggtccat tccgatggaa acacgtactt 1140
gtcttggtgc caacagcgcc ctgggcagcc cccgcggtcc ttgatttata aaataagtaa 1200
cagattttag ggcgttccgg accggttcag tgggtccggc gcagggtacc atttcacctt 1260
taagatctcc cgcgtagagg ccgaggatgt aggagtgat tactgtatgc aagcaactca 1320
atttccccac caatggacgt tcggtcaagg gactaaagt gaaattaaag gtacgggtgc 1380
tgccgccatc gtctttatct tccgccatc tgacgaacag ctcaaaagcg gaacggcttc 1440
tgtctgtctg ctctgaaca atttttacc cgcgagaggc aagggtcagt ggaaggctga 1500
taatgcgttg cactcgggca actcgcagga atctgtgacg gaacaggaca gcaaggactc 1560
cacatacagc ttatcgtcga cattgacact gtcaaaaagcg gactacgaga aacacaagggt 1620
gtatgcctgc gaagttaacc atcagggaat gtcatcacca gtgacaaaat catttaactc 1680
agggtgaatg gagggccgag ggtcgctctt gacgtgcggt gacgtcgaag aaaaccggcg 1740
accatggat tttcaggtcc aaatatcttc tttttctctg atatcagcct ctgtcattat 1800
aagtccgggg gaagtccacc tctgtgaaag cggggggggg ttagtacagc cgggtggcgc 1860
cttgagactc tctgtgcag cctcgggatt cactttttca aactacgata tgcattgggt 1920
gaggcaggcc accggcaagg gtctggagtg ggtgagtgct aatggcaccc cagggtgatac 1980
ttattatcca ggtagcgtca aaggccgttt cagcattagt cgcgaaaacg ctaagaattc 2040
tctctatttg cagatgaatt cactacgagc aggtgacaca gcagtctact attgtgcgag 2100
ggacgttgat attgttgca ccatcaccca ttactctgat tactggggtc aaggcacttt 2160
ggtaacagtt tctagtgcga gtactaaggg tccgagtgtt tttcctttag ccccttctc 2220
taaaagcact tcaggcggaa cagcagccct tggctgtttg gtgaaagact acttcccgga 2280
acctgtaact gtgagctgga attctggcgc gctcactagt ggagttcaca cttttccagc 2340
cgtgttacag tcatcgggtc tttatccct gtccctcgctc gtgacggtec cgtcgagttc 2400
actaggtacg cagacataca ttgtaacgt caatcacaa ccatcgaaca caaagggtgga 2460
caagcgtgtt gaacccaaga gctgcgataa gaccacaccc tgccctccct gccacgtcc 2520
cgagcttctc ggccgacatt cggttttctc ctttccgccc aagcccaag atactttgat 2580
gatttcacag accccagaag tgacatgcgt ggttgttgac gtaagccatg aggtacgga 2640
agtgaatttt aactggtacg tcgatggagt tgaggtccat aacgccaaag ccaagcccg 2700
cgaggaacaa tataattcca catatcgtgt ggtgtcggta cttacggtec ttcatcagga 2760
ctggctaaat ggaaaagaat acaaatgcaa ggtgtccaat aaagcgctcc ccgctcccat 2820
cgaaaagacg atcagcaagg cgaaggggca accacgtgaa ccgcaagtat acacgtccc 2880
gccctcccgga gaagaaatga caaaaacca agtctccctg acctgtttgg ttaaggatt 2940
ttacccttgc gacatcgcg tagagtggga gtccaacgga caacctgaga ataactacaa 3000
aacgactcct ccagtcctgg actcggacgg atccttcttc ttgtactcca aattgacggt 3060
tgacaaatcc cgtatggcaac aaggcaacgt ttttagctgc tcggtcatgc atgaagcttt 3120
gcacaaccac tacacacaaa aatcgttgc gctgtcgcca ggaaaataaa acccaattta 3180
ctttaagact agcgcgaatt ttttcttaca ttagatgaac ataaaacgac gatgaggatc 3240
actgttaggg aaacgatgaa ctacgctgag gacttccat tatctgctc ttgatggcgt 3300
ggtgttctct tgggtttcgg ccgagcgcaa cgaccaaggg ctgtaccaaa aacgaagttg 3360
ataatcagat atttgagttg taacaagaaa ctaaacgatt attttagcag tagcacgaat 3420
caggatctaa aatgaacgca tccggaccag gcatacaata aaacgaaagg ctcatcgaa 3480
agactggggc tttcgtttta tctgtgtgtt gtcggtgaac gctctctact agagtccac 3540
tggctcacct tcgggtgggc ctttctgcgt ttatatagag ggaatgtgc gcggaacccc 3600
tatttgttta tttttctaaa tacattcaaa tatgtatccg ctcatgagac aataacctcg 3660
ataaatgctt caataatatt gaaaaaggaa gagtatgagt attcaacatt tccgtgtcgc 3720
ccttatctcc ttttttgcgg cttttgcct tctgttttt gctcaccag aaacgctgggt 3780
gaaagtaaaa gatgctgaag atcagttggg tgcacgagtg ggttacatcg aactggatct 3840
caacagcggt aagatccttg agagttttcg ccccgaaaga cgttttccaa tgatgagcac 3900
ttttaagatt ctgctatgtg gcgcggtatt atcccgatt gacgcggggc aagagcaact 3960
cggtcgcccg atacactatt ctcaaatga cttggttgag tactcaccag tcacagaaaa 4020
gcatcttacg gatggcatga cagtaagaga attatgcagt gctgccataa ccatgagtga 4080

```

-continued

```

taacactgcy gccacttac tctgacacac gatcgaggga ccgaaggagc taaccgcttt 4140
tttgacacac atgggggagc atgtaactcg ccttgatcgt tgggaacggg agctgaatga 4200
agccatacca aacgacgagc gtgacaccac gatgcctgta gcaatggcaa caacgttgcy 4260
caaaactatta aatggcgaaac tactactct agcttcccgg caacaattaa tagactggat 4320
ggaggcggat aaagttagcag gaccacttct gcgctcgcc cttccggctg gctggtttat 4380
tgctgataaa tctggagccg gtgagcgtgg gtcacgcgg atcattgcag cactggggcc 4440
agatggtaag cctcccgta tctagttat ctacacgacg gggagtcagg caactatgga 4500
tgaacgaaat agacagatcg ctgagatagg tgctcactg attaagcatt ggtaactgtc 4560
agaccaagtt tactcatata tacttttagat tgatttaaaa cttcattttt aatttaaaag 4620
gatctagggt aagatccttt ttgataatct catgacaaa atccctaac gtgagttttc 4680
gttccactga gcgtcagacc ccgtagaaaa gatcaaaagg tcttcttgag atcctttttt 4740
tctgcgcgta atctgctgct tgcaaacaaa aaaaccaccg ctaccagcgg tggtttggtt 4800
gocggatcaa gagctaccaa ctttttttcc gaaggttaact ggcttcagca gagcgagat 4860
accaataact gttcttctag tgtagccgta gttaggccac cacttcaaga actctgtagc 4920
accgcctaca tactcgcctc tgctaactct gttaccagtg gctgctgcca gtggcgataa 4980
gtcgtgtctt accgggttgg actcaagacg atagttaccg gataaggcgc agcggtcggg 5040
ctgaacgggg ggttcgtgca cacagcccag ctggagcga acgacctaca ccgaactgag 5100
atacctacag cgtgagctat gagaaagcgc cacgcttccc gaaggagaa aggcgacag 5160
gtatccggtg agcggcaggg tcggaacagg agagcgcagc agggagcttc cagggggaaa 5220
cgctcggtat ctttatagtc ctgtcgggtt tcgccacctc tgacttgagc gtcgattttt 5280
gtgatgctcg tcaggggggc ggagcctatg gaaaaacgcc agcaacgcgg cctttttacg 5340
gttctggcc ttttctggc cttttgctca catgttctt cctgcgttat cccctgattc 5400
tttgataaac cgtattaccg ccagagaata taaaaagcca gattattaat ccggcttttt 5460
tattatttga cgcgttttga gattcaaaat tcaatctgtt tggacctgga taagacacaa 5520
gagcgacatc ctgacatgaa gcgcgtaaac agcaaatcct ggttgaacac gtatcctttt 5580
gggggctccc cgctacgacg ctgcctccag ctggggcttc cttactatac acagcgcgca 5640
tatttcacgg ttgcccagatg tcaagatggc caagttgacc agtgccgttc cgggtgctac 5700
cgcgcgacg gtcgcggagc cgttcgagtt ctggaccgac cggctcgggt tctcccgga 5760
cttctgtggg gacgacttcg ccgggtgtgt ccgggacgac gtgacctgt tcatcagcgc 5820
ggtccaggac caggtgtgtc cggacaacac cctggcctgg gtgtgggtgc gcggcctgga 5880
cgagctgtac gccagtggt aggaggtcgt gtcccgaac ttcggggagc cctccgggcc 5940
ggccatgacc gagatcgccg agcagccgtg ggggcccggg ttcgccctgc gcgaccggcc 6000
cggcaactgc gtgcacttcg tggccgagga gcaggactga accttcccta aaaatttaat 6060
tttctattgt tgcagtcaac cgcctttgtt ttcacagtca ggaataacac tagctcgtct 6120
tcaccatgga tgccaatctc ccctattcat ggtgtataaa agttcaacat ccaaagctag 6180
aacttttgga aagagaaaga atatccgaat agggcacggc gtgcgctatt gttggagtg 6240
actagcagaa agtgaggaag gcacaggatg agttttctcg a 6281

```

```

SEQ ID NO: 52      moltype = DNA length = 6196
FEATURE
misc_feature       1..6196
                    note = Vector Pro6D
source              1..6196
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 52
ttttgttgaa actacattgt tgtcggatga ctgcgcgact gcgatacgat agattggtga 60
tcgggagcgc ggaatgaaa gacagccgcg tgcgattgca gcttggcgtg ctctccgtgt 120
ggtagactag tgaagcaacg aggaagcggc caaagagcct gtgaaagggt ggcactgaat 180
caaaggaata gctaatgtga ttccacttc aagaagaaaa gatacaactg agcgattgtc 240
actatacttt tcaagttgga gagcagttgt cactggattt tgtctcccta ctagaaccga 300
cggtcgaaga tctgactggg gcgatgtgca cgagaaaaat atcataccgg ttggtagtct 360
tttgaaatgt ctctttggat ttgaaagcga atcgagattt ttatcacagt caattttagt 420
ttgtttcggg tagatagaac tgtaaacttg aacaattgta aataatcatt gagacataag 480
ttccgtcgga aatactttaa gcagtttgtg tgcaaacatg tacaacatt tacttactta 540
tatctactgt cgcattggaga ttttgcgttc ttgtctttta gttttgttc ccgcttatgc 600
ttcttagttc aagttgtgaa acagtgccat acttcttcag caggatacta acataaatcg 660
attttctttc ccccttacag ttaataggtc aaatggaagt agtcatcatt ttcattggcat 720
cgcgctggct accgagacat ttcaggaaac ggttgaagta ataaaacttg aaaaccggaa 780
taaaactgct tggcatcttg gcgagtttcg caagcactta catctcacia cagcaatact 840
cgtttctccc gttgtgtaga acaagcccta gagaatatgg atttccaaat acaaatcttc 900
agttttctac ttatctcggc cagtgttaata attagtaggg gcgacattgt catgacacaa 960
actcctctat cctcactgtg gacccttgga caaccgcct ctatcagctg ccgttctctc 1020
caatcgctgg tccattccga tggaaacacg tacttgtctt ggctccaaca gcgcctggg 1080
cagccccgcg ggtctctgat ttataaaata agtaacagat ttagcggcgt tccggaccgg 1140
ttcagtggtt cgggcgcagg taccgatttc acccttaaga tctcccgctg agaggccgag 1200
gatgtaggag tgtattactg tatgcaagca actcaatttc cccaccaatg gacgttcggt 1260
caagggacta aagttgaaat taagcgtacg gttgctgcgc catcgtctt tatttttccg 1320
ccatctgacg aacagctcaa aagcggaaac gcttctgtcg tctgtctcct gaacaatttt 1380
taccgcgagc aggttaaggt tcaagtgaag gtgcataatg cgttgacgac gggcaactcg 1440
caggaaatctg tgacggaaca ggacagcaag gactccacat acagcttatt gtcgacattg 1500
acactgtcaa aagcggacta cgagaaacac aaggtgtatg cctgcgaagt taccatcag 1560
ggattgtcat caccagtgac caaatcattt aatcgaggtg aatgcgaggg ccgagggtcg 1620
ctcttgacgt gcggtgacgt cgaagaaaaa ccgggaccca tggattttca ggtccaaata 1680
ttttcttttc ttctgatata agcctctgtc attataagtc ggggggaagt ccacctctg 1740
gaaagcgggg ggggtttagt acagccgggt ggcgccttga gactctcttg tgcagcctcg 1800

```

-continued

ggattcacct	tttcaaaacta	cgatatgcat	tgggtgaggg	aggccaccgg	caaggggtctg	1860
gagtggtgga	gtgctaattg	cacgcaggt	gatacttatt	atccaggtag	cgtcaaaggc	1920
cgtttcacga	ttagtcgcga	aaacgctaag	aattctctct	atttgcagat	gaattcacta	1980
cgagcaggtg	acacagcagt	ctactattgt	gcgagggacg	ttgatattgt	tgcgaccatc	2040
ccgattact	tcgattactg	gggtcaaggc	actttggtaa	cagtttctag	tgcgagtact	2100
aaggggtccga	gtgtatttcc	tttagccct	tcctctaaaa	gcacttcagg	cggaaacgca	2160
gcccttggtc	gtttggtgaa	agactacttc	ccggaacctg	taactgtgag	ctggaattct	2220
ggcgcgtca	ctagtggagt	tcacactttt	ccagccgtgt	tacagtcac	gggtctttat	2280
tcctgtctct	ccgtcgtgac	cgtcccgctg	agttcactag	gtacgcagac	atacatttgt	2340
aacgtcaatc	acaagccatc	gaacacaaag	gtggacaagc	gtgtgaacc	caagagctgc	2400
gataagagcc	acacctgccc	tccttgccca	gctcccgagc	ttctcgccgg	accttcggtt	2460
ttcctcttct	cgccgaagcc	caaagatact	ttgatgattt	cacggacgcc	agaagtgaca	2520
tgcgtggttg	ttgacgtaag	ccatgaggat	ccggaagtga	aatttaactg	gtacgtcgat	2580
ggagttagag	tcgataacgt	caagaccaag	ccccgcgagg	aacaatataa	ttccacatat	2640
cgtgtggtgt	cggtacttac	ggctcttcat	caggactggc	taaatggaaa	agaatacaaa	2700
tgcgaaggtg	ccaataaagc	gctccccgct	cccactgaaa	agacgatcag	caaggcgaa	2760
gggcaaccac	gtgaaccgca	agtatacacg	ctcccgcctc	cccagagaag	aatgacaaaa	2820
aaccaagtct	ccctgacctg	tttggttaaa	ggattttacc	cttcggacat	cgcggtagag	2880
tgggagtcca	ccggacaacc	tgagaataac	tacaaaacga	ctcctccagt	cctggactcg	2940
gacggatcct	ttctcttgta	ctccaaattg	acggttgaca	aatcccgatg	gcaacaaggc	3000
aacgttttta	gctgctcggt	catgcatgaa	gctttgcaca	accactacac	acaaaaatcg	3060
ttgtcgctgt	cgccaggaaa	ataaaaccca	atttacttta	agactagcgc	aatttttttc	3120
ttacattaga	tgaacataaa	acgacgatga	ggatcactgt	tagggaaaacg	atgaactacg	3180
ctgaggactt	cctattattc	gtcctttgat	ggcgtggttg	ttctttggtt	tgcggccgag	3240
cgcaacgacc	aaggcctgta	ccaaaacga	agttgataat	cagatatttg	agttgtaaca	3300
agaaactaaa	gcattatttt	agcagtagca	cgaatcagga	tctaaaatga	acgcatcccg	3360
accaggcatc	aaataaaaag	aaaggctcag	tcgaaagact	gggcctttcg	ttttatctgt	3420
tgtttgtcgg	tgaacgctct	ctactagagt	cacactggct	caccttcggg	tgggcctttc	3480
tgcgtttata	tgaagggaaa	tgtgcgcgga	acccctattt	gtttattttt	ctaaatcacat	3540
tcaaatatgt	atccgctcat	gagacaataa	ccctgataaa	tgcttcaata	atattgaaaa	3600
aggaagagta	ttagtattca	acatttccgt	gtcgccctta	ttcctttttt	tgcggcattt	3660
tgccttctcg	tttttgctca	cccagaaaacg	ctggtgaaag	taaaagatgc	tgaagatcag	3720
ttgggtgcac	gagtggtgta	catcgaactg	gatctcaaca	gcggtaaagt	ccttgagagt	3780
tttcgcccgc	aagaacgttt	tccaatgatg	agcactttta	aagttctgct	atgtggcgcg	3840
gtattatccc	gtattgacgc	cgggcgaagag	caactcggtc	gccgcataca	ctattctcag	3900
aatgacttgg	ttgagtactc	accagtcaca	gaaaagcatc	ttacggatgg	catgacagta	3960
agagaattat	gcagtatctc	cataaccatg	agtgataaca	ctgcggccaa	cttacttctg	4020
acaacgatcg	gagggacgaa	ggagctaaac	gcttttttgc	acaacatggg	ggatcatgta	4080
actcgccctg	atcggtggga	accggagctg	aatgaagcca	tacaaaacga	cgaagcgtgac	4140
accacgatgc	ctgtagcaat	ggcaacaacg	ttgcgcgaac	tattaaactg	cgaactctt	4200
actctagctt	cccggcaaca	attaatagac	tggatggagg	cggataaagt	tgcaggacca	4260
cttctgcgct	cgccctctcc	ggctggctgg	tttattgctg	ataaatctgg	agccgggtgag	4320
cgtgggtcac	cggttatcat	tgagcactg	gggcagatg	gtaagccctc	ccgtatcgta	4380
gttatctaca	cgacggggag	tcaggcaact	atggatgaac	gaaatagaca	gatcgctgag	4440
ataggtgcct	cactgattaa	gcattggtaa	ctgtcagacc	aagtttactc	atatatactt	4500
tagattgatt	tataacttca	tttttaattt	aaaaggatct	aggtgaagat	cctttttgat	4560
aatctcatga	ccaaaatccc	ttaacgtgag	ttttcggtcc	actgagcgtc	agaccccgta	4620
gaaaagatca	aaggatcttc	tttagatcct	ttttttctgc	gcgtaactcg	ctgcttgcaa	4680
acaaaaaaac	caccgctacc	agcgggtggt	gttttgcggg	atcaagagct	accaactcct	4740
tttcgaagg	taactggctt	cagcagagcg	cagataccaa	atactgttct	tctagtgtag	4800
ccgtagttag	gccaccactt	caagaactct	gtagcaccgc	ctacatacct	cgtctgtcta	4860
atcctgttac	cagtgtctgc	tgccagtggc	gataagtcgt	gtcttaccgg	gttggaactc	4920
agacgatagt	taccggataa	ggcgcagcgg	tcgggctgaa	cggggggttc	gtgcacacag	4980
cccagcttgg	agcgaacgac	ctacaccgaa	ctgagatacc	tacagcgtga	gctatgagaa	5040
agcgcacgc	ttcccgaagg	gagaaaaggc	gacaggtatc	cggtaagcgg	cagggtcgga	5100
acaggagagc	gcacgaggga	gcttccaggg	ggaaacgcct	ggatatctta	tagtctctgc	5160
gggtttcgcc	acctctgact	tgagcgtcga	tttttgatgt	gctcgtcagg	ggggcgaggc	5220
ctatggaaaa	acgcccacaa	cgcggccttt	ttacgggttc	tggccttttg	ctggcctttt	5280
gctcacatgt	ttttctctgc	gttatccctt	gattctgtgg	ataaccgtat	taccgccaga	5340
gaatataaaa	agccagatta	ttaatccggc	ttttttatta	tttgacgcgt	tttgagattc	5400
aaaaattcaat	ctgtttggag	ctggataaga	cacaagagcg	acatccgtac	atgaacgcgc	5460
taaacagcaa	atcctggttg	aaacagtatc	cttttggggg	cctcccgtca	cgcgctcgc	5520
tcacagctgg	gcttccctac	tatacacagc	gcgcataatt	cacggttgcc	agatgtcaag	5580
atggccaagt	tgaccagtgc	cgttccgggtg	ctcaccgcgc	gcgacgtcgc	cggagcgggc	5640
gagttctgga	ccgaccggct	cgggttctcc	cgggacttcg	tggaggacga	cttcgcgggt	5700
gtggtccggg	acgacgtgac	cctgttctac	agcgcgggtc	aggaccaggt	ggtgccggac	5760
aacaccctgg	cctgggtgtg	gggtgcgcgc	ctggacgagc	tgtacgcgga	gtggtcggag	5820
ctcgtgtcca	ggaacttccg	ggacgcctcc	gggcgggcca	tgaccgagat	cggcgagcag	5880
ccgtgggggc	gggagttcgc	cctgcgcgac	ccggccggca	actcgtgca	cttcgtggcc	5940
gaggagcagg	actgaacctt	ccttaaaaat	ttaattttca	ttagttgcag	tactccgct	6000
ttggtttcac	atcagggaat	aacactagct	cgtcttcacc	atggatgcca	atctcgcta	6060
ttcatggtgt	ataaaaagtc	aacatccaaa	gctagaactt	tggaaaagag	aaagaatatc	6120
cgaatagggc	acggcgtgcc	gtattgttgg	agtggaactag	cagaaagtga	ggaaggcaca	6180
ggatgagttt	tctcga					6196

-continued

FEATURE	Location/Qualifiers
misc_feature	1..6277
	note = Vector Pro7D
source	1..6277
	mol_type = other DNA
	organism = synthetic construct

SEQUENCE: 53

ttttttacca	ccaagcgtc	ttctatctac	cagtcaacac	taatgataat	gcacgtatat	60
gaatgacata	tttgtttttg	gactaaagat	tgacttgatc	tgtatcatta	aaacttgtag	120
gcggcggtgg	attgagacac	aggetgtatt	tcgccgcaaa	aagcatgaag	tctcgttttg	180
aaggggcggg	atcgctctac	acctgtactc	cgctattgtt	gtgcacacta	ggatgtgaaa	240
gcttgtcgaa	aggagtcatt	tcaggactgg	acagagaaac	aagcaaacct	aacatgacga	300
tgtctgactg	tttggattct	gaagggtgat	tacgttgctt	ggcttacatt	aattcatggt	360
tttatttact	gtcagtagac	ataaatagcg	atttttcggt	gaactaattt	ttttcaagt	420
gtttcgttga	aagcagtagg	atcggacgca	gagcatttac	gggttcacat	gagggttgaa	480
aggacgagcg	ttcccttgcg	tctgacacaa	atcatagatg	tttcacctgg	cgcgacgatc	540
ttcctgacgg	aacgctaaat	atggccacta	gattgccaac	ccatatgtat	ttcaatagtg	600
gaaaagcccg	tcaagaccta	ggcccaagta	tatatctcga	aggacatgtc	cgatatatct	660
ctacaaagga	tgagcggagt	cacactcggg	cgtagcgtct	ttcaaacact	tgctctcttg	720
gtttcatttt	tatcggctat	ttggctcattg	agcaaaaagg	ccctctggta	gcgtttgtag	780
gcatcgctac	ggcaaatgcc	tgttcagcac	ggtcacgagc	cattcgctct	tggtcgttgg	840
gccacttttc	tcggatttgt	gaagagtttc	agcatcagtc	gtcaaaaatat	caaattatat	900
tcagaggtca	acagcgcttc	tgccaaattg	caactctatc	gaaaaaacag	aacgaacatg	960
gatttccaag	tacaaatctt	cagttttcta	cttatctcgg	ccagtgtaat	aatttagtagg	1020
ggcgacattg	tcatgacaca	aactcctcta	tcctcacctg	tgacccttgg	acaaccggcc	1080
tctatcagct	gccgtttctc	ccaatcgctg	gtccattccg	atggaaacac	gtactgtgct	1140
tggtcccaac	agcccgcttg	gcagcccccg	cggctcttga	ttataaaaat	aagtaacaga	1200
tttagcggcg	ttccggacgg	gttcagtggt	tcgggcgcag	gtaccgattt	cacccttaag	1260
atctcccgcg	tagaggcgga	ggatgtagga	gtgtattact	gtatgcaagc	aactcaattt	1320
ccccaccaat	ggacgttcgg	tcaagggaact	aaagttgaaa	ttaagcgtac	gggtgctgcg	1380
ccatccgtct	ttatttttcc	gccactctga	gaacagctca	aaagcggaa	ggcttctgtc	1440
gtctgtctcc	tgaacaattt	ttaccgcgga	gaggctaagg	ttcagtgga	ggtcgataat	1500
gcgttgacgt	cgggcaactc	gcaggaatct	gtgacggaa	aggacagcaa	ggactccaca	1560
tacagcttat	cttcgacatt	gacactgtca	aaagcggact	acgagaaaca	caagggtgat	1620
gcctgcgaag	ttaccocatca	gggatgtgca	tcaccagtga	ccaaatcatt	taatcgaggt	1680
gaatgcgagg	gccgagggtc	gctcttgacg	tcgggtgacg	tcgaagaaaa	cccgggaccc	1740
atggattttc	aggtccaaat	attttctttt	cttctgatat	cagcctctgt	cattataagt	1800
cggggggaag	tccacctcgt	ggaaagcggg	gggggtttag	tacagccggg	tgccgccttg	1860
agactctctt	gtgcagcctc	gggatccacc	ttttcaaac	acgatatgca	ttgggtgagg	1920
caggccacat	gcaagggtct	ggagtgggtg	agtgtctaag	gcaccgcagg	tgatacttat	1980
tatccaggta	gcgtccaaag	ccgtttcacg	attagtccgc	aaaaacgtaa	gaattctctc	2040
tatttgcaga	tgaattcaat	acgagcaggt	gacacagcag	tctactattg	tcgaggggac	2100
gttgatattg	tttcgacatt	caccgattac	ttcgattact	gggttcaagg	cacttttgta	2160
acagttttta	gtgcgagtac	taagggtccg	agtgtatttc	cttttagccc	ttctctataa	2220
agcacttcag	gcggaacagc	agcccttggt	gtttgttgta	aagactactt	cccggaaact	2280
gttaactgtg	cgttgaattc	tgccgcgctc	actagtggag	ttcacacttt	tccagccgtg	2340
ttacagtcac	gggttcttta	ttccctgtcc	tcctcgtga	ccgtcccgct	gagttcacta	2400
ggtagcgaga	catacatttg	tacagtcac	cacaagccat	cgaacacaaa	gggtgacaag	2460
cgtgtttga	ccaagagcgt	cgataagacc	cacacctgcc	ctccctgccc	agctcccag	2520
cttctcggcg	gaccttcggt	tttctctttt	ccgcggaagc	ccaaagatac	tttgatgatt	2580
tcacggagcg	cagaagtgac	atcgctgggt	gttgacgtaa	gccatgagga	tcgggaagtg	2640
aaatttaact	ggtacgtcga	tgagattgag	ttccataacg	ccaagaccaa	gccccgcg	2700
gaacaatata	attccacata	tcgtgtgggt	tcggtaacta	cggtccctca	tcaggactgg	2760
ctaaatggaa	aagaatacaa	atgcaagggt	ttcaataaag	cgctcccgcc	ttccatcgaa	2820
aagacgatca	gcaaggcgaa	ggggcaacca	cgtgaaccgc	aagtatacac	gtccccgcc	2880
tcccagaga	aaatgacaaa	aaaccaagtc	ttccctgacct	gtttggttaa	aggattttac	2940
ccttcggaca	tcgcggtaga	gtgggagtc	aacggacaac	ctgagaataa	ctacaaaacg	3000
actcctccag	tcctcgactc	ggacggatcc	ttcttcttgt	actccaaatt	gacgggtgac	3060
aaatcccgat	ggcaacaagg	caacgttttt	agctgctcgg	tcatgcatga	agctttgcac	3120
aaccactaca	cacaaaaatc	gttgtcgctg	tcgccaggaa	aataaaaccc	aatttacttt	3180
aagactagcg	caattttttt	cttacattag	atgaacataa	aacgacgatg	aggatcactg	3240
ttagggaaac	gatgaactac	gctgaggact	tcctattatt	cgtcctttga	tgccgtgggt	3300
gttctttggt	ttgcggccga	gcgcaacgac	caaggcctgt	acaaaaaacg	aagttgataa	3360
tcagatattt	gagttgtaac	aagaaactaa	agcattattt	tagcagtagc	acgaatcagg	3420
atctaaaatg	aacgcattccg	gaccaggcat	caataaaaac	gaaaggctca	gtcgaaagac	3480
tgggcctttc	gttttatctg	ttgtttgtcg	gtgaacgcct	tctactagag	tcacactggc	3540
tcaccttcgg	gtgggcccct	ctgcgtttat	atagagggaa	atgtgcgcgg	aacccctatt	3600
tggttatttt	tctaaataca	ttcaaatatg	tatccgctca	tgagacaata	acctgataa	3660
atgcttcaat	aatattgaaa	aagggaagag	atgagtatcc	aacatttccg	tgctgcctct	3720
attccctttt	ttgcggcatt	ttgccttcct	gtttttgtct	accagaaaac	gctggtgaaa	3780
gtaaaagatg	ctgaagatca	gttgggtgca	cgagtgggtt	acatcgaaat	ggatctcaac	3840
agcggtaaga	tccttgagag	ttttcgcccc	gaagaacggt	ttccaatgat	gagcactttt	3900
aaagttctgc	tatgtggcgc	ggtattatcc	cgtattgacg	ccgggcaaga	gcaactcggt	3960
cgccgcatac	actattctca	gaatgacttg	gttgagtact	caccagtcac	agaaaagcat	4020
cttacggatg	gcacgacagt	aagagaatta	tcgagtgctg	ccataaccat	gagtgataac	4080
actgcggcca	acttacttct	gacaacgatc	ggaggaccga	aggagctaac	cgcttttttg	4140

-continued

```

cacaacatgg gggatcatgt aactcgcctt gatcgttggg aaccggagct gaatgaagcc 4200
ataccaaacg acgagcgtga caccacgatg cctgtagcaa tggcaacaac gttgcgcaaa 4260
ctattaactg gcgaactact tactctagct tcccggaac aattaataga ctggatggag 4320
gcggaataag ttgcaggacc acttctgcgc tcggcccttc cggtcggtcg gtttattgct 4380
gataaatctg gagcoggtga gcgtgggtca cgcggtatca ttgcagcact gggggccagat 4440
ggtaagccct cccgtatctg agttatctac acgacgggga gtcaggcaac tatggatgaa 4500
cgaaatagac agatcgctga gatagggtgc tcactgatta agcatttgga actgtcagac 4560
caagtttact catatatact tttagattgat ttaaaacttc atttttaatt taaaaggatc 4620
taggtgaaga tcctttttga taatctcatg accaaaatcc cttaacgtga gttttcgctc 4680
cactgagcgt cagaccocgt agaaaagatc aaaggatctt cttgagatcc ttttttctg 4740
cgcgtaatct gctgcttgca aacaaaaaaa ccaccgctac cagcggtggt ttgtttgccc 4800
gatcaagagc taccaactct ttttccgaag gtaactggct tcagcagagc gcagatacca 4860
aatactgttc ttctagtgtg gccgtagtta ggccaccact tcaagaactc tgtagcaccg 4920
cctacatacc tcgctctgct aatcctgtta ccagtggtcg ctgccagtgg cgataagtcg 4980
tgtcttaccg ggttggtactc aagacgatag ttaccggata aggcgcagcg gtcgggtctg 5040
acgggggggt cgtgcacaca gcccgacttg gagcgaacga cctacaccga actgagatac 5100
ctacagcgtg agctatgaga aagcgccacg cttccggaag ggagaaaagg ggacagggtat 5160
cgggtaagcg gcagggtcgg aacaggagag cgcacgaggg agcttccagg gggaaaacgc 5220
tggatctctt atagtctgtg cgggtttcgc cactctgac ttgagcgctg atttttgtga 5280
tgtctgctag gggggcgagg cctatggaaa aacgccagca acgcgccctt ttacagggtc 5340
ctggcctttt gctggccttt tgctcacatg ttcttctctg cggttatccc tgattctgtg 5400
gataaccgta ttaccgccag agaataaaa aagccagatt attaatccgg cttttttatt 5460
atgtgacgct ttttgagatt caaaattcaa tctgtttgga cctggataag acacaagagc 5520
gacatcctga catgaacgcc gtaaacagca aatcctggtt gaacacgtat ccttttgggg 5580
gcctcccgct acgacgctcg ctcagctgg ggcttctta ctatacacag cgcgcattat 5640
tcacgggtgc cagatgtcaa gatggccaag ttgaccagtg ccgttccggt gctcaccgcg 5700
cgcgacgtcg ccggagcggt cgaagtctgg accgaccggc tcgggttctc ccgggacttc 5760
gtggaggacg acttcgcggg tgtggtccgg gacgacgtga ccctggtcat cagcgcggtc 5820
caggaccagg tgggtgcggg caacaccctg gcctgggtgt ggggtgcggg cctggacgag 5880
ctgtacgccc agtgggtcgg ggtcgtgtcc acgaaactcc gggacgctc ccggccggcc 5940
atgaccgaga tcggcgagca gccgtggggg cgggagttcg ccctgcgcga cccggccggc 6000
aactgcgtgc acttcgtggc caggagcag gactgaacct tccttaaaaa tttaattttc 6060
attagttgca gtcactccgc ttgggtttca cagtcaggaa taacactagc tcgtcttcac 6120
catggatgcc aatctgcctt attcattggt tataaaaagt caacatccaa agctagaact 6180
tttggaaga gaaagaatat ccgaatatgg cagggcgtgc cgtattgttg gagtggacta 6240
gcagaaagtg aggaaggcac aggatgagtt ttctcga 6277

```

```

SEQ ID NO: 54      moltype = DNA length = 6282
FEATURE            Location/Qualifiers
misc_feature       1..6282
                    note = Vector Pro7D+ Kozak
source             1..6282
                    mol_type = other DNA
                    organism = synthetic construct

```

```

SEQUENCE: 54
ttttttacca caagcgctc ttctatctac cagtcaacac taatgataat gcacgtatat 60
gaatgacata tttgtttttg gactaaagat tgacttgatc tgtatcatta aaacttgtag 120
gcggcggtggc attgagacac aggcgtgtatt tcgcgcgaaa aagcatgaag tctcgttttg 180
aagggggcggg atgctctact acctgtactc cgtcattggt gtgcacacta ggatgtgaaa 240
gcttgtcgaa aggagtcagt tcaggactgg acagagaaac aagcaaacct aacatgacga 300
tgtctgactg tttggattct gaaggggtgat tacgttgcct ggcttacatt aattcatggt 360
tttatttact gtcagttagc ataaatagcg atttttcggt gaactaattt ttttcaaagt 420
gtttcgttga aagcagtagg atcggacgca gagcatttac ggggttcacat gagggttgaa 480
aggacgagcg ttcccttgcg tctgacacaa atcatagatg ttccacctgg cgcgacgatc 540
ttcctgacgg aacgctaaaat atggccacta gattgccaac ccatatgtat ttcaatagtg 600
gaaaagcccg tcaagaccta ggcccaagta tatattcgga aggcacatgct cgatatatct 660
ctacaaagga tgagcggagt cacactcggg cgtagcgtct ttcaaacact tgctctcttg 720
gtttcatttt tatcggctcat ttggtcattg agcaaaaagg ccctctggga cgtttttag 780
gcacgcgtac ggcaaatgcc tgttcagcac ggtcacgagc cattcgtctt tgttcgttgg 840
gccacttttc tcggatttgg gaagagtttc agcatcagtc gtcaaaaatat caaattatat 900
tcagaggtca acagcgcttc tgccaaattg caactctatc gaaaaaacag aacgaaccca 960
ccatgggattt ccaagtacaa atcttcagtt ttctacttat ctcggccagt gtaataatta 1020
gtagggggcga cattgtcatg acacaaactc ctctatctcc acctgtgacc cttggacaac 1080
ccgctcttat cagctgcggt tcttcccaat cgttgggtcca ttccgatgga aacacgtact 1140
tgtcttgctt caacacgcgc cctgggcagc ccccgcggtc cttgatttat aaaataagta 1200
acagatttag cggcgttccg gaccggttca gtgggtccgg cgcaggtacc gatttcaccc 1260
ttaagatctc ccggtagag gccgaggatg taggagtgta ttactgtatg caagcaactc 1320
aatttcccca ccaatggacg ttccgtcaag ggaactaaagt tgaaattaa cgtacgggtg 1380
ctgcgccate cgtctttatt tttccgccat ctgacgaaca gctcaaaagc ggaacggctt 1440
ctgtcgtctg tctcctgaac aatttttacc cgcgagaggg taaggttcag tggaaaggctg 1500
ataatgcgtt gcagtggggc aactcgcagg aatctgtgac ggaacaggac agcaaggact 1560
ccacatacac cttatcgtcg acattgacac tgtcaaaagc ggactacgag aaacacaagg 1620
tgtatgcctg cgaagtacc catcagggat tgtcatcacc agtgaccaa tcatttaatc 1680
gaggtgaatg cgagggccga ggttcgctct tgacgtgcgg tgacgtcgaa gaaaaccggg 1740
gacctatgga ttttcagggt caaatatttt cttttctctc gatatacgcc tctgtcatta 1800
taagtccggg ggaagtccac ctctgtgaaa gcggggggggg tttagtacag ccgggtggcg 1860

```

-continued

ccttgagact	ctcttgtgca	gctcgggat	tcaccttttc	aaactacgat	atgcattggg	1920
tgaggcaggc	caccggcaag	gggtctggagt	gggtgagtg	taatggcacc	gcagggtgata	1980
cttattatcc	aggtagcgtc	aaaggccgtt	tcacgattag	tcgcgaaaac	gctaagaatt	2040
ctctctat	gcagatgaat	tcactacgag	cagggtgacac	agcagtcctac	tattgtgcga	2100
gggacgttga	tattgttgcg	accatcacgc	attacttcga	ttactggggt	caaggcactt	2160
tggtaacagt	ttctagtcgc	agtactaagg	gtccgagtg	atttccttta	gccccctcct	2220
ctaaaagcac	ttcaggcgga	acagcagccc	ttggctgttt	ggtgaaagac	tacttcccgg	2280
aacctgtaac	tgtgagctgg	aattctggcg	cgctcactag	tggagttcac	acttttccag	2340
ccgtgttaca	gtcatcggtt	ctttattccc	tgctcctcgt	cgtagccgtc	ccgtcgagtt	2400
cactaggtac	gcagacatac	atttctaacg	tcaatcacaa	gccatcgaac	acaaaggtgg	2460
acaagcgtgt	tgaacccaag	agctcgata	agacccacac	ctgccccccc	tgcccagctc	2520
ccgagcttct	cgggcgacct	tcgggttttc	ctttcccgcc	gaagcccaaa	gatactttga	2580
tgatttcacg	gacgccagaa	gtgacatgcg	tggtgtgtga	cgtaagccat	gaggatccgg	2640
aagtgaattt	taactgtgtg	gtcgatggag	ttgaggtcca	taacgccaag	accaagcccc	2700
gcgaggaaca	atataattcc	acatatcggt	tggtgtcggt	acttacggtc	cttcatcagg	2760
actggctaaa	tggaaaagaa	tacaaatgca	agggtgtcaa	taaagcgctc	cccgtcccca	2820
tcgaaaagac	gatcagcaag	gcgaaggggc	aaccacgtga	accgcaagta	tacacgctcc	2880
cgccctcccg	agaagaatg	acaaaaaacc	aagtctccct	gacctgtttg	gttaaaggat	2940
tttacccttc	ggacatcgcg	gtagagtggg	agtcacaacg	acaacctgag	aataactaca	3000
aaacgactcc	tcagctcctg	gactcgagcg	gatccttctt	cttgtactcc	aaattgacgg	3060
ttgacaaatc	ccgatggcaa	caaggcaacg	tttttagctg	ctcggtcatg	catgaagctt	3120
tgcaacaaca	ctacacacaa	aaatcgttgt	cgctgtcgcc	aggaaaataa	aacctaat	3180
cccttaagac	tagcgcaatt	tttttcttac	attagatgaa	cataaaacga	cgatgaggat	3240
cactgttagg	gaaacgatga	actacgctga	ggacttccta	ttattcgctc	tttgatggcg	3300
tggtgtttct	ttggtttgcg	gccgagcgca	acgaccaagg	cctgtaccaa	aaacgaagtt	3360
gataatcaga	tatttgagtt	gtacaagaa	actaaagcat	tatttttagca	gtagcacgaa	3420
tcaggatcta	aaatgaacgc	atccggacca	ggcatcaaat	aaaacgaaag	gctcagtcga	3480
aaagctgggc	ctttcgtttt	atctgtttgt	tgctcggtgaa	cgctctctac	tagagtcaca	3540
ctggctcacc	ttcgggtggg	cttttctcg	tttatataga	gggaaatgtg	cgcggaaccc	3600
ctatttgttt	atttttctaa	atacatctaa	atatgtatcc	gctcatgaga	caataaacct	3660
gataaatgct	tcaataat	tgaaaaagga	agagtatgag	tattcaacat	ttcgtgtcgc	3720
cccttattcc	cttttttgcg	gcattttg	ttcctgtttt	tgctcaccga	gaaacgctgg	3780
tgaaagtaaa	agatgctgaa	gatcagttgg	gtgcacgagt	gggttacatc	gaactggatc	3840
tcaacagcgg	taagatcctt	gagagttttc	gccccgaaga	acgttttcca	atgatgagca	3900
cttttaaaagt	ttctgctatgt	ggcgcggtat	tatcccgat	tgacgcgggg	caagagcaac	3960
tcggctcgcc	catcacctat	tccagaatg	acttggttga	gtactcacca	gtcacagaaa	4020
agcatcttac	ggatggcatg	acagtaagag	aattatgcag	tgctgccata	acctagatg	4080
ataaactgc	ggccaactta	cttctgacaa	cgatcgagg	accgaaggag	ctaaccgctt	4140
ttttgcacaa	catgggggat	catgtaactc	gccttgatcg	ttgggaaccc	gagctgaatg	4200
aaagcatacc	aaacgacgag	cgtagaccca	cgatgcctgt	agcaatggca	acaacgttgc	4260
gcaaaactatt	aaactggcgaa	ctacttactc	tagcttcccg	gcaacaatta	atagactgga	4320
tgaggggcga	taaagtgtga	ggaccacttc	tgccgtcgcc	ccttcgggct	ggctggttta	4380
ttgtgtgata	atctgagcgt	ggtgagcgtg	ggtagcgcgg	tatcattgca	gcactggggc	4440
cagatggtaa	gcccctccgt	atcgtagtta	tctacacgac	ggggagtcag	gcaactatgg	4500
atgaacgaaa	tagacagatc	gctgagatag	gtgcctcact	gattaagcat	tggttaactgt	4560
cagaccaagt	ttactcatat	atactttaga	ttgattttaa	acttcatttt	taatttaaaa	4620
ggatctaggt	gaagatcctt	tttgataatc	tcatgaccaa	aatcccttaa	cgtgagtttt	4680
ggttccactg	agcgtcagac	cccgtagaaa	agatcaaaag	atcttcttga	gatccttttt	4740
ttctgcgcgt	aatctgtctg	ttgcaaacaa	aaaaaccacc	gctaccagcg	gtggtttgtt	4800
tgccggatca	agagctacca	actctttttc	cgaaggtaac	tggtctcagc	agagcgcaga	4860
taccaaaatc	tgttcttcta	gtgtagccgt	agttaggcca	ccacttcaag	aactctgtag	4920
caaccgctac	atacctcgct	ctgctaatac	gtttaccagt	ggctgctgcc	agtgccgata	4980
agtcgtgtct	taccgggttg	gactcaagac	gatagtacc	ggataaggcg	cagcggtcgg	5040
gctgaacggg	gggttcgtgc	acacagccca	gcttgagcgc	aacgacctac	accgaactga	5100
gataacctaca	gcgtgagcta	tgagaaaagc	ccacgcttcc	cgaagggaga	aaggcggaca	5160
ggtatccggt	aagcggcagg	gtcggaaacg	gagagcgcac	gaggggagctt	ccagggggaa	5220
acgcctggta	tctttatagt	cctgtcggtt	ttcgccacct	ctgacttgag	cgctcgat	5280
tggtgatgct	gtcagggggg	cgagccctat	ggaaaaacgc	cagcaacgcg	gcctttttac	5340
ggttcctggc	cttttctggt	ctttttgtct	acatgttctt	tcctgcgtta	tcccctgatt	5400
ctgtggataa	ccgtattacc	gccagagaa	ataaaaaagc	agattattaa	tccggttttt	5460
ttattatttg	acgcgttttg	agattcaaaa	ttcaatctgt	ttggacctgg	ataagacaca	5520
agagcgacat	cctgacatga	acgccgtaaa	cagcaaatcc	tggttgaaca	cgtatccttt	5580
tgggggccct	ccgtacgac	gctcgctcca	gctggggctt	ccttactata	cacagcgccg	5640
atatttcacg	ggtgccagat	gtcaagatgg	ccaagttgac	cagtgcctgt	ccggtgctca	5700
ccgcgcgcga	cgtcgccgga	cggtcgaggt	tctggaccga	ccggtcgggg	ttctcccgcc	5760
acttcgtgga	ggacgacttc	gccggtgtgg	tccgggacga	cgtgacctgt	ttcatcagcg	5820
cggtccagga	ccaggtgggt	ccggacaaca	ccctggcctg	ggtgtgggtg	cgcgccctgg	5880
acgagctgta	cgccgaggtg	tcggaggtcg	tgctccacga	cttcggggac	gcctccgggc	5940
cgcccatgac	cgagatcgcc	gagcagccgt	ggggggcgga	gttcgcctgt	cgcgaccctg	6000
ccggcaactg	cgtgcacttc	gtggccgagg	agcaggactg	aaccttctct	aaaaatttaa	6060
ttttcattag	tttcagctac	tcgcgttttg	tttcacagtc	aggaataaca	ctagctgctc	6120
ttcaccatgg	atgccaatct	cgccatttca	tggtgtataa	aagttcaaca	tccaaagcta	6180
gaacttttgg	aaagagaaag	aatatccgaa	tagggcacgg	cgtgccgtat	tgttgagtg	6240
gactagcaga	aagtgaggaa	ggcacaggat	gagttttctc	ga		6282

-continued

FEATURE	Location/Qualifiers
misc_feature	1..5716
	note = Vector NVB12
source	1..5716
	mol_type = other DNA
	organism = synthetic construct

SEQUENCE: 55

```

gagattcaaa attcaatctg tttggacctg gataagacac aagagcgaca tcccgacatg 60
aacgccgtaa acagcaaatc ctgggtgaac acgtatcctt ttgggggcct cccgctacga 120
cgctcgctcc agctgggggt tcttactat acacagcgcg catatttcac ggttgccaga 180
tgtcaagatg gccaaagtga ccagtgccgt tccggtgctc accgcgcgcg acgtcgcccg 240
agcggtcgag ttctggaccg accggctcgg gttctcccgg gacttcgtgg aggacgactt 300
cgccgggtgt gtcggggacg acgtgaccct gttcatcagc gcggtccagg accaggtgggt 360
gcccgaacaac accctggcct ggggtgtggg gcgcggcctg gacgagctgt acgccgagtg 420
gtcggaggtc gtgtccacga acttccggga cgcctccggg ccggccatga ccgagatcgg 480
cgagcagccg tggggggcgg agttcgccct gcgcgaccg gccggcaact gcggtgcactt 540
cgtggccgag gagcaggact gaaccttct taaaaattta attttcatta gttgcagtca 600
ctccgctttg gtttcacagt caggaataac actagctcgt cttcaccatg gatgccaatc 660
tcgctatttc atggtgtata aaagttcaac atccaaagct agaacttttg gaaagagaaa 720
gaatatccga atagggcacg gcgtgcccga ttggtggagt ggactagcag aaagtgcagg 780
aggcacagga tgagttttct cgaagacgcg atatcggaag gtgactgtaa acgagaagtg 840
cacgaagcct ttctgtgtga cgtcacaaac cgaacagccc tacgtggcgt gcgactttcc 900
ggcacttttg atactgttc cctatatgta ttgggtgtat acatgtcgtg ttccaaactac 960
gacagaaaac agtcagttac ggggtattcg acttccggca cctgcagcga ggagttttgt 1020
gtccgtcgca catctccgtt gcttccggca cctacgtagg cggcaaaact cctgccttcc 1080
gcccgccttg cggcattccg gctccggggc agaattgccc ggggtgtcac aattttgcct 1140
cctcacgaaa aaacgtttcta ctttgtattt tggtcgggtt tcgcatcctt ccagcaacca 1200
ttctcattca aagtcaccac ttgtgcgaac gatctatgga cttcaaagtg cagatcataa 1260
gcttttttgt aatctctgct tccgtaatta tgtcacgtgg ccaaattgtc ttgtgcgaat 1320
cccccgcat tttacttgca tccccgggtg agaaggtgac catgacatgc agagcatcgt 1380
catcggtgtc atacattcac tggttccaac agaagcccg tagttcgccg aaacctgga 1440
tttatgctac ctctaattctt gccagtgggg tacccggttc gttttcagga tccgggtcgg 1500
gaaccagcta ttctctaact atctcacgcg ttgaagcaga ggacgcggct acgtactatt 1560
gccagcagtg gactagtaat ccaccacgt tccgtggagg tacgaaactc gagatcaagc 1620
gaacagtcgc tgccccaaagc gtttttattt tcccgccttc cgtatgagcag ttgaaatccg 1680
gcaccgcgtc cgttgtttgc cttctcaata atttttacc aagggaagcg aaggttcaat 1740
ggaaagtcca caatgccttg caatctggaa actcccaaga atccgtgact gaacaggata 1800
gcaaggactc tacctacagt ctgagcagta cactcacgct gagtaaacgc gattacgaga 1860
aacataaggt ctacgcctgt gaagtcaacg accaaggctt atcctcgccg gtcacgaaat 1920
cgtttaaccg aggagaatgt gagggccgag ggtcgctctt gacgtgctgt gacgtcgaag 1980
aaaacccggg acccatgggg tggctactga tattgttgtt cctgtgtggc gtaggcgacac 2040
gagtatgttc tcaagtacag cttcagcaac cttgagccga actagtgaa cagggggcca 2100
gcgttaaat gtcttgcaag gctcagggtt acacatttac cagctataat atgcattggg 2160
tgaagcaaac acccgggcgg ggctgggaat ggattgggtc aatttatccc ggaacgggag 2220
atcgtctcta caaccagaaa tttaagggaag aagctacctt aacagccgat aaatcctcta 2280
gcacggccta ttgcagctg tcttcttga caagcgagga ctcgcgcgta tattattgtg 2340
cacgtagcac gtattacggc ggagactggt acttcaacgt ttggggggca gggacaacag 2400
ttacagtatc agcggcgagt accaaaggct ccagcgtatt cccgttggcg cccagctcta 2460
aatccacgag cggaggttac gctgcacttg gatgtctcgt gaaggattat ttctcgaac 2520
ctgtaactgt cagttggaat agcggagcac taacttcagg agtgcatacc ttcccgccg 2580
ttctccagtc ctcgggcctc tatagtctat cctcggtagt gactgtaccc tcatcctccc 2640
ttggtacgca aaagtcacatt tgcaatgtca accacaaacc gtccaatcag aaagtgcaga 2700
agaaagtga acctaagagt tgcgacaaaa cccacacctg tccaccatgc cgggtccag 2760
agttgtcggg ttggtccgtcc gtgttctctt ttcgcgcgaa accaaaggat acactgatga 2820
tctctaggac tccagagggt acttgcgtcg tctgtgatgt cagtcatgag gaccccgagg 2880
tgaagttcaa ttggtacgta gacggcgttg aggttcacaa cgccaaaact aaaccacgag 2940
aggagcaata caattcgacc tacccgggtc tatcggttct caccgtgtta caccaagact 3000
ggcttaatgg caagggaatc aaatgtaagg tatcaaataa ggcttgccg gcccccacg 3060
aaaagactat tagtaaaagt aagggccagc caagggaacc acaagtgtac acattacctc 3120
catcgcgta cgaattgact aagaatcaag tgtcattgac ctgcctgggtc aaaggattct 3180
acccgtcaga tatcgcggtg gaatgggaat cgaatggaca gccgggagaa aactacaaaa 3240
ccaccccccc ggttctcgat tccgacggct cgttctttct ctactcgaag ttgaccgtcg 3300
ataaatcgcg ctggcagcaa ggcaacgtgt ttctgtgctc ggtcatgcac gaagctctgc 3360
acaaccacta cagcagaaag tctgtatcgc ttctgccttg caaataaaa ccaatttact 3420
ttaagactag cgcatttttt ttcttacatt agatgaacat aaaacgacga tgaggatcac 3480
tggttaggaa acgatgaact acgctgagga cttcctatta ttctgtctt gatggcggtg 3540
ttgttctttg gtttgcgggc gagcgcgaacg accaaggcct gtacaaaaaa cgaagttgat 3600
aatcagatat ttgagttgta acaagaaact aaagcattat tttagcagta gcacgaatca 3660
ggatctaaaa tgaacgcacg cggaccaggc atcaaataaa acgaaaggct cagtcgaaa 3720
actgggcctt tctgtttatc tgttgtttgt cgggtgaacgc tctctactag agtcacactg 3780
ggtcaccttc ggggtggcct ttctgcgttt atatagaggg aaatgtgcgc ggaaccccta 3840
tttgtttatt ttcttaataa cattcaataa tgtatccgct catgagacaa taacctgat 3900
aaatgcttca ataatttga aaaaggaaga gtatgagtat tcaacatttc cgtgtcgccc 3960
ttattccctt ttgtgcggca ttttgccttc ctgtttttgc tcaccagaa acgctgggtg 4020
aagtaaaaag tctgtaagat cagtagtggg ttacatcgaa ctggatctca 4080
acagcggtaa gatccttgag agttttcgcc ccgaagaacg ttttccaatg atgagcactt 4140

```

-continued

ttaaagtctt	gctatgtggc	gcggtattat	cccgatttga	cgccgggcaa	gagcaactcg	4200
gtcgccgcat	acactattct	cagaatgact	tggttgagta	ctcaccagtc	acagaaaagc	4260
atcttacgga	tggcatgaca	gtaagagaat	tatgcagtcg	tgccataacc	atgagtgaata	4320
acactgccgc	caacttactt	ctgacaacga	tcggaggacc	gaaggagcta	accgcttttt	4380
tgcaacaacat	gggggatcat	gtgaactgcc	ttgatcgttg	ggaaccggag	ctgaatgaag	4440
ccataccaaa	cgacgagcgt	gacaccacga	tgccctgtagc	aatggcaaca	acgttgcgca	4500
aactattaac	tggcgaaacta	cttactctag	cttcccgcca	acaattaata	gactggatgg	4560
agggcggaata	agttgcagga	ccactctctg	gctcgccctt	tccggctggc	tggtttattg	4620
ctgataaato	tggagccggt	gagcgtgggt	cacgcgggat	cattgcagca	ctggggccag	4680
atggtaagcc	ctcccgatc	gtagtattct	acacgacggg	gagtcaggca	actatggatg	4740
aacgaaatag	acagatcgct	gagatagggt	cctcactgat	taagcattgg	taactgtcag	4800
accaagttta	ctcatatata	ctttagattg	atttaaaact	tcatttttaa	tttaaaagga	4860
tctaggtgaa	gatccttttt	gataatctca	tgacccaaat	cccttaacgt	gagttttcgt	4920
tccactgagc	gtcagacccc	gtagaaaaga	tcaaaggatc	ttcttgagat	cctttttttc	4980
tgccgctaata	ctgctgtctg	caacacaaaa	aaccaccgct	accagcgggt	gtttgtttgc	5040
cggatcaaga	gctaccaact	ctttttccga	aggttaactgg	cttcagcaga	gcgcagatac	5100
caaatactgt	tctctatgtg	tagccctagt	taggccacca	cttcaagaac	tctgtagcac	5160
cgctacata	cctcgctctg	ctaactcctgt	taccagtggc	tgctgccagt	ggcgataaagt	5220
cgtgtcttac	cggtgtggag	tcaagacgat	agttaccgga	taaggccgag	cggtcgggct	5280
gaacggggggg	tctgtgcaca	cagcccagct	tggagcgaa	gacctacacc	gaactgagat	5340
acctacacgc	tgagctatga	gaaagcgcca	cgtctcccga	agggagaaa	gcggacaggt	5400
atccggtaag	cgccaggggt	ggaacaggag	agcgcacgag	ggagcttcca	gggggaaacg	5460
cctgggtatc	ttatagtcct	gtcgggtttc	gccacctctg	acttgagcgt	cgatttttgt	5520
gatgctcgtc	agggggcgcg	agcctatgga	aaaacgccag	caacgcggcc	tttttacggt	5580
tcctggcctt	ttgtggcctt	ttgtctcaca	tgcttcttcc	tcggttatcc	cctgattctg	5640
tgataaacgc	tattaccgcc	agagaatata	aaaagccaga	ttattaatcc	ggctttttta	5700
ttatttgacg	cgtttt					5716

SEQ ID NO: 56 moltype = DNA length = 5686
FEATURE Location/Qualifiers
misc_feature 1..5686
 note = NVB13
source 1..5686
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 56

gagattcaaa	attcaatctg	tttggacctg	gataagacac	aagagcgaca	tcttgacatg	60
aacgcgcgta	acagcaaatc	ctgggtgaac	acgtatcctt	ttgggggccc	cccgctacga	120
cgctcgctcc	agctgggggt	ctcttactat	acacagcgcg	catatttcac	ggttgccaga	180
tgtaagatg	gccaagtgtg	ccagtgcctg	tccgggtgctc	accgcgcgcg	acgtcgcgcg	240
agcgggtcag	ttctgtgacc	accggctcgg	gttctcccg	gacttcgtgg	aggacgactt	300
cgccgggtgtg	gtccgggacg	acgtgaccct	gttcatcagc	gcggtccagg	accaggtggt	360
gcgggacaac	accctggcct	gggtgtgggt	gcgcggccgtg	gacgagctgt	acgcgagtg	420
gtcggaggtc	gtgtccacga	acttccggga	cgccctccggg	ccggccatga	ccgagatcgg	480
cgagcagccg	tggggggcgg	agttcgccct	gcgcgaccgc	gcgggcaact	gcgtgcactt	540
cgtggccgag	gagcaggact	gaaccttctt	taaaaattta	attttcatta	gttgacgtca	600
ctccgctttg	gtttccagct	caggaataac	actagctcgt	cttcaccatg	gatgccaatc	660
tcgctatttc	atggtgtata	aaagttcaac	atccaaagct	agaacttttg	gaaagagaaa	720
gaatatccga	atagggcacg	gcgtgcccga	ttgttgaggt	ggactagcag	aaagtggagg	780
aggcacagga	tgagttttct	cgattttcat	atgcgggaagt	gactgtaaac	gagaagtgtca	840
cgaagccttt	tctgtgtgac	tcacaaaccg	aacagcccta	cgtggcgctg	gactttccgg	900
caactttgga	atcgttgccc	tatatgtatt	gggtgtatag	atgtcgtatt	ccaactacga	960
cagaaaaacg	tcagttacgg	gtattcgaac	ttccggcacc	tgacgcgagg	agttttgtgt	1020
cgctcgacac	tcctccgtgc	ttccggcacc	tacgtaggcg	gcaaaacttc	tgcttccgcg	1080
ccgccttgcc	gcattccggc	tccgggcccag	aattgcgccg	gtgttcacaa	ttttgctctc	1140
tcacgaaaaa	agcttctact	ttgtattttg	gtcgggtttc	ggatccttcc	agcaaccatt	1200
ctcattcaaa	gtcaccactt	gtgcgaacga	tctatggact	tccaagtcca	gatcataaag	1260
tttttgttaa	tctctgcttc	cgtaattatg	tcacgtggcc	aaattgtctt	gtcgcgaatc	1320
cccgcgattt	tatctgcac	cccggttgag	aaggtgacca	tgacatgcag	agcatcgtca	1380
tcggtgtcat	acattcactg	gttccaacag	aagcccggtg	gttcgcgcga	accctggatt	1440
tatgctacct	ctaactctgc	cagtggggta	ccggttcggt	tttcaggatc	gggctcgga	1500
accagctatt	ctctaactat	ctcacgcgtt	gaagcagagg	acgcggctac	gtactattgc	1560
cagcagtgga	ctagtaatcc	accacggttc	ggtggaggta	cgaaactcga	gatcaagcga	1620
acagtcgctg	ccccaaagcgt	ttttattttc	ccgccttccg	atgagcagtt	gaaatccggc	1680
accgcgtccg	ttgttttgc	tttcaataat	ttttacccaa	gggaagcgaa	ggttcaatgg	1740
aaagtccgac	atgccttgca	atctggaaac	tcccagaagt	ccgtgactga	acaggatagc	1800
aaggactcta	cctacagctc	gagcagtaca	ctcacgctga	gtaaaagcga	ttacgagaaa	1860
cataaggtct	acgctgtgga	agtaacgcac	caaggcttat	cctcgccggg	cacgaaatcg	1920
tttaaccgag	gagaatgtga	gggcccagg	tcgctcttga	cgtgcgggtg	cgtcgaagaa	1980
aaccggggac	ccatgggggtg	gtcactgata	ttgtgtgtcc	ttgtggcggt	ggcgacacga	2040
gtattgtctc	aagtacagct	tcagcaacct	ggagccgaac	tagtgaagcc	aggggcccagc	2100
gttaaaatgt	cttgcaaggc	ctcagggtac	acatttacca	gctataatat	gcattgggtg	2160
aagcaaacac	ccggggcggg	cctggaatgg	attggtgcaa	tttatcccg	aaacggagat	2220
acgtcttaca	accagaaatt	taagggaaaa	gctaccttaa	cagccgataa	atcctctagc	2280
acggcctata	tgcagctgtc	gtccttgaca	agcagggact	ccgcgctata	ttattgtgca	2340
cgtagcacgt	attacggcgg	agactgggtac	ttcaacggtt	ggggggcgag	gacaacagtt	2400

-continued

acagtatcag	cggcgagtag	caaagggtccc	agcgtattcc	cgttggcgcc	cagctctaaa	2460
tccacgagcg	gaggtagggc	tgcacttgga	tgtctcgtga	aggattatctt	tctggaacct	2520
gtaactgtca	gttggaatag	cggagcacta	acttcaggag	tgcatacctt	tccggcggtt	2580
ctccagtcct	cggcctctta	tagtctatcc	tcggtagtga	ctgtaccctc	atcctccctt	2640
ggtagcgaaa	cgtacatttg	caatgtcaac	cacaaaccgt	ccaatacgaa	agtcgacaag	2700
aaagttgaac	ctaagagttg	cgacaaaacc	cacacctgtc	caccatgccc	ggctccagag	2760
ttgctgggtg	gtccgtccgt	gttccctctt	ccgcccgaac	caaaggatag	actgatgac	2820
tctaggactc	cagaggttac	ttgcgtcgtc	gttgatgtca	gtcatgagga	ccccaggtg	2880
aagttcaatt	ggtagcgtga	cggcgttgag	gttcacaacg	ccaaaactaa	accacgagag	2940
gagcaataca	attcgaccta	ccgggtcgta	tcggttctca	ccgtgttaca	ccaagactgg	3000
cttaaatggca	aggaatacaa	atgtaaggta	tcaataaagg	ccttgccggc	ccccatcgaa	3060
aagactatta	gtaagctaa	gggccagcca	agggaaccac	aagtgtacac	attacctcca	3120
tcgctgacg	aattgactaa	gaatcaagt	tcattgacct	gcctgggtcaa	aggattctac	3180
cgtcagata	tcgcggtgga	atgggaatcg	aatggacagc	cggagaacaa	ctacaaaacc	3240
acccccccgg	ttctcgattc	ggacggctcg	ttctttctct	actcgaagtt	gaccgtcgat	3300
aaatcgcgct	ggcagcaagg	caacgtgttt	tcgtgctcgg	tcatgcacga	agctctgcac	3360
aaccactaca	cgcagaagtc	gctatcgctt	tcgcctggca	aataaaaacca	ccttccttaa	3420
aaatttaatt	ttcattagtt	gcagtcactc	cgctttggtt	tcacagtcag	gaataacact	3480
agctcgtctt	caccatggat	gccaatctcg	cctattcatg	gtgtataaaa	gttcaacatc	3540
caaaagctaga	acttttggaa	agagaagaa	tatccgaata	gggcacggcg	tgcggtattg	3600
ttggagtgga	ctagcagaaa	gtgaggaagg	cacaggatga	gttttctcga	gagcccgagg	3660
atcaataaaa	acgaagggtc	cagtcgaaag	actgggcctt	tcgttttatc	tgtgttttgt	3720
cggtagaacg	ttctctactg	agtcacactg	gctcaccttc	gggtgggctt	ttctgcgttt	3780
atatagaggg	aatgtgctcg	ggaaccccta	ttgttttatt	tttctaataa	cattcaaat	3840
tgtatccgct	ctagagacaa	taacctgat	aaatgcttca	ataatattga	aaaaggaaga	3900
gtagtagtat	tcaacatttc	cgtgtcgccc	ttattccctt	ttttgcggca	ttttgccttc	3960
ctgtttttgc	tcacccagaa	acgctgggtg	aagtaaaaga	tgtctgaagat	cagttgggtg	4020
cacgagtggt	ttacatcgaa	ctggatctca	acagcggtaa	gatccttgag	agttttcgcc	4080
cgaagaagcg	ttttccaatg	atgagcactt	ttaaagttct	gctatgtggc	gcggtattat	4140
cccgtattga	cgccgggcaa	gagcaactcg	gtccgcgcac	acactattct	cagaatgact	4200
tgggtgagta	ctcacagtc	acagaaaagc	atcttacgga	tggcatgaca	gtaagagaat	4260
tatgagtcg	tgcataaac	atgagtata	acaactggcg	caacttactt	ctgacaaaca	4320
tcggaggacc	gaaggagcta	accgcttttt	tgcacaacat	gggggatcat	gtaactcgcc	4380
ttgatcgttg	ggaacggag	ctgaatgaag	ccataccaaa	cgacgagcgt	gacaccacga	4440
tgcctgtagc	aatggcaaca	acgttgcgca	aactattaac	tggcgaaact	cttactctag	4500
cttcccggca	acaattaata	gactggatgg	aggcgggata	agttgcagga	ccactctcgc	4560
gctcggccct	tcgggctggc	tggtttattg	ctgataaatc	tggagccggg	gagcgtgggt	4620
cacgcgggat	cattgcagca	ctggggccag	atggtaagcc	ctcccgtatc	gtagtattct	4680
acacgacggg	gagtcaggca	actatggatg	aacgaaatag	acagatcgct	gagatagggt	4740
cctcaactgat	taactgtcag	accaaagtta	ctcatatata	ctttagattg		4800
atttaaaact	tcatttttaa	tttaaaagga	tctaggtgaa	gatccctttt	gataatctca	4860
tgaccaaact	cccttaacgt	gagttttcgt	tccactgagc	gtcagacccc	gtagaaaaga	4920
tcaaggatc	ttcttgagat	cccttttttc	tgcgcgtaat	ctgctgcttg	caaacaaaaa	4980
aaccaccgct	accagcgggt	gtttgtttgc	cggatcaaga	gctaccaact	ctttttccga	5040
aggtaactgg	cttcagcaga	gcgcagatac	caaataactg	tcttctagtg	tagccgtagt	5100
taggccacca	cttcaagaac	tctgtagcac	cgccctacata	cctcgctctg	ctaatcctgt	5160
taccagtggt	tgctgccagt	ggcgataagt	cgtgtcttac	cgggttggac	tcaagacgat	5220
agttaccgga	taaggcgtag	cgttcgggct	gaacgggggg	ttcgtgcaca	cagcccagct	5280
tgagcgcaac	gacctacacg	acctacagcg	tgaactatga	gaaagcgcca		5340
cgtctcccga	aggagagaa	gcggacaggt	atccggtaag	cggcagggtc	ggaacaggag	5400
agcgcacgag	ggagctctca	gggggaaacg	cctggtatct	ttatagtcct	gtcgggtttc	5460
gccacctctg	acttgagcgt	cgttttttgt	gatgctcgtc	agggggcgcg	agcctatgga	5520
aaaacgcgag	caacgcggcc	tttttacggt	tcttgccctt	ttgtggtcct	ttgtctcaca	5580
tggtctttcc	tgcgttatcc	cctgattctg	tggataaccg	tattaccgcc	agagaatata	5640
aaaagccaga	ttattaatcc	ggctttttta	ttatttgacg	cgtttt		5686

SEQ ID NO: 57 moltype = DNA length = 5716
 FEATURE Location/Qualifiers
 misc_feature 1..5716
 note = NVB14
 source 1..5716
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 57

gagattcaaa	attcaatctg	tttggacctg	gataagacac	aagagcgaca	tcttgacatg	60
aacgcgcgt	aaacagcaaat	ctgggtgaac	acgtatcctt	ttgggggcct	cccgtcacga	120
cgctcgctcc	agctgggggt	tccttactat	acacagcgcg	catatttcac	ggttgccaga	180
tgtaaatgag	gccaaagtga	ccagtgccgt	tcgggtgctc	accgcgcgcg	acgtcgccgg	240
agcggtcgag	ttctggaccg	accggctcgg	gttctcccgg	gaactcgtgg	aggacgactt	300
gcgcgggtgt	gtccgggacg	acgtgacctt	gttcatcagc	gcggtcagg	accaggtggt	360
gcgggacaa	acctggcctt	gggtgtgggt	gcgcggcgct	gacgagctgt	acgcgcagtg	420
gtcggaggtc	gtgtccacga	acttcgggga	cgctccggg	ccggccatga	ccgagatcgg	480
cgagcagccg	tggggcgggg	agttcgccct	gcgcgacccg	gcgggcaact	gcgtgcactt	540
cgtggccgag	gagcaggact	gaaccttctt	taaaaaatta	attttcatta	gttgacgtca	600
ctccgctttg	gtttcacagt	cagggaataac	actagctcgt	cttaccatg	gatgccaatc	660
tcgcctattc	atggtgtata	aaagttcaac	atccaaagct	agaacttttg	gaaagagaaa	720

-continued

gaatatccga	atagggcacg	gcgtgccgta	ttgttggagt	ggactagcag	aaagtgagga	780
aggcacagga	tgagttttct	cgaaagacgc	atatgcggaa	gtgactgtaa	acgagaagtg	840
cacgaagcct	tttcttgtga	cgtcacaaac	cgaacagccc	tacgtggcgt	gcgactttcc	900
ggcactttgg	atategttgc	cctatatgta	ttgggtgtat	acatgtcgtg	ttccaactac	960
gacagaaaac	agtcagttac	gggtatttca	acttcoggca	cctgcagcga	ggagttttgt	1020
gtccgtcgca	catcctccgt	gctttcggca	cctacgtagg	cggaacactt	cctgccttcc	1080
gcccccttg	cggcattccg	gctccggggc	agaattgccc	gggtgttcac	aattttgcct	1140
cctcacgaaa	aaacgtctta	ccttgtattt	tggtcgggtt	tcggatcctt	ccagcaacca	1200
ttctcattca	aagtcaccac	ttgtgcgaac	gatctatgga	catgcgagtc	ccggctcaac	1260
ttctcggatt	actactgttg	tggttgtcgg	gtgcccgtcg	ccaaattgtc	ttgtcgaat	1320
ccccgcgat	tttatctgca	tccccgggtg	agaaggtgac	catgacatgc	agagcatcgt	1380
catcgggtgc	atacattcac	tggttccaac	agaagcccgg	tagttcgcgg	aaacctgga	1440
tttatgctac	ctctaattctt	gccagtgggg	taccggttcg	gttttcagga	tcgggctcgg	1500
gaaccagcta	ttctctaact	atctcacgcg	ttgaagcaga	ggacgcggct	acgtactatt	1560
gccagcagtg	gactagtaat	cccccacagt	tcggtgaggg	tacgaaactc	gagatcaagc	1620
gaacagtcgc	tgccccaaagc	gtttttattt	tcccccttc	cgatgagcag	ttgaaatccg	1680
gcaccgcgtc	cgttgtttgc	cttctcaata	atttttaccc	aaggggaagc	aaggttcaat	1740
ggaaagtcga	caatgccttg	caatctggaa	actcccaaga	atccgtgact	gaacaggata	1800
gcaaggagtc	tactacacgt	ctgagcagta	cactcacgct	gagtaaagcg	gattacgaga	1860
aacataaggt	ctacgcctgt	gaagtaacgc	accaaaggctt	atcctcgccg	gtcacgaaat	1920
cgtttaaccg	aggagaatgt	gagggccgag	ggtcgctctt	gacgtgcggg	gacgtcgaag	1980
aaaaaccggg	accatggag	tttgggctgt	catgggtatt	cctcgttgcc	ctgttccgag	2040
gtgtgcagtg	ccaagtacag	cttcagcaac	ctggagccga	actagtgaag	ccagggggcca	2100
gcgttaaaat	gtcttgcaag	gcctcagggt	acacatttac	cagctataat	atgcattggg	2160
tgaagcaaac	accggggcgg	ggcctggaat	ggattggtgc	aatttatccc	ggaaacggag	2220
atacgtctta	caaccagaaa	tttaaggga	aagctacctt	aacagccgat	aaatcctcta	2280
gcacggccta	tatgcagctg	tcgtccttga	caagcgagga	ctccgcctga	tattattgtg	2340
cacgtagcac	gtattacggc	ggagactggg	acttcaacgt	ttggggggca	gggacaacag	2400
ttacagtatc	agcggcgagt	accaaaggtc	ccagcgtatt	cccgttgccg	cccagctcta	2460
aatccacgag	cggaggtagc	gctgcacttg	gatgtctcgt	gaaggattat	tttctcgaac	2520
ctgtaactgt	cagttggaat	agcggagcac	taacttcagg	agtgcatacc	tttccggcgg	2580
ttctccagtc	ctccggcctc	tatagtctat	cctcggtagt	gactgtaccc	tcacccctcc	2640
ttggttacga	aacgtacatt	tgcaatgtca	accacaaaac	gtccaatacg	aaagtcgaca	2700
agaaagttag	acctaaagct	tgcgacaaa	cccacacctg	tccaccatgc	ccggctccag	2760
agttgctggg	tggtccgtcc	gtgttccctc	ttccgccgaa	accaaaggat	acactgatga	2820
ttcttaggac	tccagagggt	acttgcgtcg	tcgttgatgt	cagtcgatgag	gaccccgagg	2880
tgaagtccaa	ttggtacgta	gacggcggtg	aggttcacaa	cgccaaaact	aaaccacgag	2940
aggagcaata	caatttcgac	taccgggtcg	tatcgggtct	caccgtgtta	caccaagact	3000
ggcttaattg	caaggaatac	aaatgtaagg	tatcaataaa	ggccttgccg	gccccactcg	3060
aaaagactat	tgaataagct	aagggccagc	caaggggaac	acaagtgtac	acattacctc	3120
catcgcgtga	cgaattgact	aagaatcaag	tgctattgac	ctgcctggtc	aaaggattct	3180
accgcgtcga	tatcggcgtg	gaatgggaat	cgaatggaca	gccggagaa	aaactacaaa	3240
ccaccccccc	gggtctcgat	tcggacggct	cggtctttct	ctactcgaag	ttgacccgtc	3300
ataaatcgcg	ctggcagcaa	ggcaacgtgt	tttctgtctc	ggctcatgcac	gaagctctgc	3360
acaaccacta	cacgcagaag	tcgctatcgc	tttcgcctgg	caataaaaa	ccaatttact	3420
ttaaagactag	cgcatttttt	ttcttaccat	agatgaacat	aaaacgacga	tgaggatcac	3480
tgtaggggaa	acgatgaact	acgctgagga	cttctctatta	ttcgtccttt	gatggcgtgg	3540
ttgttctttg	gtttgcggcg	gagcgcgaac	accaaaggct	gtacccaaaa	cgaagtgtat	3600
aatcagatat	ttgagttgta	acaagaaact	aaagcattat	tttagcagta	gcacgaatca	3660
ggatctaaaa	tgaacgcatac	cggaccaggc	atcaataaaa	acgaaaggct	cagtcgaaag	3720
actgggcctt	tcgttttatc	tgttgtttgt	cgggtgaacgc	tctctactag	agtcacactg	3780
gctcaccttc	gggtgggcct	ttctgcgttt	atatagaggg	aaatgtgcgc	ggaaccccta	3840
tttgtttatt	tttctaaata	cattcaaaata	tgtatccgct	catgagacaa	taaccctgat	3900
aaatgcttca	ataatattga	aaaaggaaga	gtatgagtat	tcaacatttc	cgtgtcgccc	3960
ttattccctt	ttttgcggca	ttttgccttc	ctgtttttgc	tcaccacaga	acgctgggtga	4020
aagtaaaaaga	tgctgaagat	cagttgggtg	cacgagtggg	ttacatcgaa	ctggatctca	4080
acagcggtaa	gatccttgag	agttttcgcc	cgaagaacgc	ttttccaatg	atgagcactt	4140
ttaaagtctc	gctatgtggc	cgggtattat	cccgtattga	cgccggggcaa	gagcaactcg	4200
gtcgcgcgat	acactattct	cagaatgact	tggttgagta	ctcaccagtc	acagaaaagc	4260
atcttacgga	tgccatgaca	gtaagagaat	tatgcagtgc	tgccataaac	atgagtgata	4320
acactgcggc	caacttactt	ctgacaacga	tcggaggacc	gaaggagcta	accgcttttt	4380
tgcaacaacat	gggggatcat	gtaactcgcc	ttgatcgttg	ggaaccggag	ctgaatgaag	4440
ccataccaaa	cgacgagcgt	gacaccacga	tgccgtgtagc	aatggcaaca	acgttgcgca	4500
aaactattaac	tggcgaaacta	cttactctag	cttcccgcca	acaattaata	gactggatgg	4560
agggcggtaa	agttgcaggga	ccacttctgc	gctcggccct	tcggcgtggc	tggtttattg	4620
ctgataaaatc	tggagccgggt	gagcgtgggt	cacgcgggtat	cattgcagca	ctggggccag	4680
atggtaagcc	ctcccgctac	gtagttatct	acacgacggg	gagtcaggca	actatggatg	4740
aacgaaatag	acagatcgct	gagatagggt	cctcactgat	taagcatttg	taactgtcag	4800
accaagttta	ctcatatata	ctttagattg	atttaaaaat	tcatttttta	tttaaaaagg	4860
tctaggtgaa	gatccttttt	gataatctca	tgaccaaaa	cccttaacgt	gagttttcgt	4920
ttccactgagc	gtcagacccc	gtagaaaaa	tcaaaggatc	ttcttgagat	cctttttttc	4980
tgccgcgtaat	ctgctgcttg	caaaacaaaa	aaacacccgt	accagcgggt	gtttgttttc	5040
cggatcaagga	gctaccaact	ctttttccga	aggtaactgg	cttcagcaga	gcgcagatac	5100
caaatactgt	tcttctagt	tagccgtagt	taggccacca	cttcaagaac	tctgtagcac	5160
cgctacata	cctcgctctg	ctaatcctgt	taccagtggc	tgctgcccag	ggcgataagt	5220
cgtgtcttac	cgggttggac	tcaagacgat	agttaccgga	taaggcgcag	cggtcgggct	5280

-continued

gaacgggggg	ttcgtgcaca	cagcccagct	tggagcgaac	gacctacacc	gaactgagat	5340
acctacagcg	tgagctatga	gaaagcgcca	cgcttoccga	agggagaaaag	gcggaacaggt	5400
atccggtaag	cggcagggtc	ggaacaggag	agcgccacgag	ggagcttcca	gggggaaacg	5460
cctggatatc	ttatagtcct	gtcgggttcc	gccacctctg	acttgagcgt	cgatttttgt	5520
gatgctcgtc	aggggggcgg	agcctatgga	aaaacgccag	caacgcggcc	tttttacggt	5580
tectggcctt	ttgtctggct	ttgtctcaca	tggtctttcc	tgcgttatcc	cctgattctg	5640
tgataaccg	tattaccgcc	agagaatata	aaaagccaga	ttattaatcc	ggctttttta	5700
ttatttgacg	cgtttt					5716

SEQ ID NO: 58 moltype = DNA length = 5705
FEATURE Location/Qualifiers
misc_feature 1..5705
 note = Vector NVB16
source 1..5705
 mol_type = other DNA
 organism = synthetic construct

SEQUENCE: 58

gagattcaaa	attcaatctg	tttggacctg	gataagacac	aagagcgaca	tcttgacatg	60
aacgcggtaa	acagcaaatc	ctggttgaac	acgtatcctt	ttgggggcct	cccgtacga	120
cgctcgctcc	agctggggct	tccttactat	acacagcgcg	catatttcac	ggttgccaga	180
tgtaagatg	gccaaagtga	ccagtgcctg	tccggtgctc	accgcgcgcg	acgtgcgccg	240
agcggtcgag	ttctggaccg	accggctcgg	gttctcccgg	gacttcgtgg	aggacgactt	300
cgccgggtg	gtccgggacg	acgtgacctt	gttcatcagc	gcggtccagg	accaggtggt	360
gcccagacaac	accttgccct	gggtgtgggt	gcgcggccctg	gacgagctgt	acgccgagtg	420
gtcggaggtc	gtgtccacga	acttcgggga	cgctccggg	ccggccatga	ccgagatcgg	480
cgagcagccg	tgggggcggg	agtctgcctt	gcgcgaccgg	gcccggcaact	gcgtgcactt	540
cgtaggcgag	gagcaggact	gaaccttcct	taaaaaatta	attttcatta	gttgacgtca	600
ctccgctttg	gtttccacgt	caggaataac	actagctcgt	cttcaccatg	gatgccaatc	660
tcgcctattc	atggtgtata	aaagtccaac	atccaaagct	agaacttttg	gaaagagaaa	720
gaatatccga	atagggcacg	gcgtgcctga	ttgttggagt	ggactagcag	aaagtgagga	780
aggcacagga	tgagttttct	cgattttcat	atgcggaagt	gactgtaaac	gagaagtga	840
cgaagccttt	tctgtgacg	tcacaaaccg	aacagcccta	cgtaggcgtg	gactttccgg	900
cactttggat	atcgtttccc	tatatgtatt	gggtgtatac	atgtcgtatt	ccaactacga	960
cagaaaacag	tcagtacagg	gtattcgaac	ttccggcacc	tgacgcgagg	agttttgtgt	1020
ccgtgcgaca	tcctccgtgc	tttcggcacc	tacgtaggcg	gcaaaactcc	tgcttccgc	1080
ccgccttgcg	gcattccggc	tcggggccag	aattgcccgg	gtgttcacaa	ttttgcctcc	1140
tcacgaaaaa	acgttctact	ttgtattttg	gtcgggttcc	ggatccttcc	agcaaccatt	1200
ctcattcaaa	gtcaccactt	gtgcgaacgt	atcatgaccc	gactcacggt	cctagccctt	1260
ttggctggac	tggtggcgtc	gtcccgcgca	caaatttgtct	tgtagcaatc	ccccgcgatt	1320
ttatctgcat	ccccgggtga	gaaggtgacc	atgacatgca	gagcatcgct	atcgggtgca	1380
tacattcact	ggttccaaac	gaagcccggt	agttccgcga	aaacctggat	ttatgtacc	1440
tcataatctg	ccagtggggg	accggttcgg	ttttcaggat	cggtgctcgg	aaccagctat	1500
ttcttaacta	tcagacgctg	tgaagcagag	gacgcggcta	cgtactattg	ccagcagtg	1560
actagtaatc	caccacaggt	cggtggaggt	acgaaactcg	agatcaagcg	aacagtcgct	1620
gccccaaagc	tttttatttt	ccgcgcttcc	gatgagcagt	tgaaatccgg	caccgcgctc	1680
gtgtgtttgc	ttctcaataa	tttttaccac	agggaaagcg	aggttcaatg	gaaagtcgac	1740
aatgccttgc	aatctggaaa	ctcccaagaa	tcctgtgactg	aacaggatag	caaggactct	1800
acctacagtc	tgagcagtag	actcacgctg	agtaaaagcg	attacgagaa	acataaggtc	1860
tacgcctgtg	aagtaaacgc	ccaaggctta	tcctcgcggg	tcacgaaatc	gtttaaccga	1920
ggagaatgtg	agggccgagg	gtcgtctctg	acgtgcgggtg	acgtcgaaga	aaacccgagg	1980
cccatgaccc	gactcacggg	cctagccctt	ttggttgga	tggtggcgct	gtcccgcgca	2040
caagtacagc	ttcagcaaac	tgagccgaa	ctagtgaagc	caggggcccag	cgttaaaaatg	2100
ttctgcaagg	cctcagggtg	acacattacc	agctataata	tgcatgtggg	gaagcaaaac	2160
cccgggcggg	gcctggaatg	gatttggtga	atttatcccg	gaaacggaga	tacgtcttac	2220
aaaccagaaat	ttaagggaaa	agctaccctta	acagccgata	aatcctctag	cacggcctat	2280
atgcagctgt	cgtccttgac	aagcgaggac	tcgcgcgtat	attattgtgc	acgtagcacg	2340
tattacggcg	gagactggga	cttcaacgtt	tgggggggcag	ggacaacagt	tacagtatca	2400
gcggcgagta	ccaaaggtcc	cagcgtattc	ccgttggcgc	ccagctctaa	atccacgagc	2460
ggaggtacgg	ctgcacttgg	atgtctcgtg	aaggattatt	ttcctgaacc	tgtaactgtc	2520
agttggaata	cgggagcact	aacttcagga	gtgcatacct	ttccggcggt	tctccagctc	2580
tcgcgctctc	atagctctatc	ctcggtagtg	actgtaccct	catcctccct	tggtacgcaa	2640
acgtacattt	gcaatgtcaa	ccacaaaccg	ttcaatacga	aagtcgacaa	gaaagttgaa	2700
cctaagagtt	gcgacaaaac	ccacacctgt	ccacctgcc	cggtccagaa	gttgctgggt	2760
ggtccgtccg	tggtctctct	tcgcgcgaaa	ccaaaggata	cactgatgat	ctctaggact	2820
ccagaggtta	cttgcgtcgt	cggtgatgtc	agtcagaggg	accccgaggt	gaagttcaat	2880
tggtacgtag	acggcggtga	ggttcacaac	gccaaaacta	aaccacgaga	ggagcaatac	2940
aattcgacct	accgggtcgt	atcgggttct	accgtgttac	accaagactg	gcttaatggc	3000
aaggaaataca	aagtgaaggt	atcaataaag	gccttgcggg	ccccatcga	aaagactatt	3060
agtaaagcta	agggccagcc	aagggaacca	caagtgtaca	cattacctcc	atcgcgtgac	3120
gaattgacta	agaatcaagt	gtcattgacc	tgcttggtca	aaggattcta	cccgtcagat	3180
aatccgggtg	aatgggaatc	gaatggacag	ccggagaaca	actacaaaac	cacccccccg	3240
gttctcgatt	cgaacggctc	gttcttttct	tactcgaagt	tgacctcgca	taaatcgccg	3300
tggaagcaag	gcaacgtgtt	ttcgtgctcg	gtcatgcacg	aagctctgca	caaccactac	3360
acgcagaagt	cgctatcgct	ttcgcctggc	aaataaaacc	caatttactt	taagactagc	3420
gcaatttttt	tcttacatta	gatgaacata	aaacgacgat	gaggatcact	gttagggaaa	3480
cgatgaacta	cgctgaggac	ttcctattat	tcgtcctttg	atggcgtggt	tggtcttttg	3540

-continued

```

tttgcgccg agcgcaacga ccaaggcctg taccaaaaac gaagttgata atcagatatt 3600
tgagttgtaa caagaaacta aagcattatt ttagcagtag cacgaatcag gatctaaaat 3660
gaacgcatcc ggaccaggca tcaataaaaa cgaaaggctc agtcgaaaga ctgggccttt 3720
cgttttatct gttgtttgtc ggtgaacgct ctctactaga gtcacactgg ctcaccttcg 3780
ggtagggcctt tctgcgttta tatagaggga aatgtgcgcg gaacccttat ttgtttattt 3840
ttctaaatcc attcaaatat gtatccgctc atgagacaat aacctgata aatgcttcaa 3900
taatatggaa aaaggaagag tatagattt caacatttcc gtgtgcctct tatccctttt 3960
tttgcgccat tttgoccttc tgtttttgct caccagaaa cgctgggtgaa agtaaaagat 4020
gctgaagato agtgggtgac acgagtggtg tacatcgaa tggatctcaa cagcggtaag 4080
atccttgaga gtttccgccc cgaagaacgt tttccaatga tgagcacttt taaagtctcg 4140
ctatgtggcg cggtattatc ccgtattgac gccgggcaag agcaactcgg tcgccgcata 4200
cactattctc agaatgactt ggttgagtac tcaccagtca cagaaaagca tcttacggat 4260
ggcatgacag taagagaatt atgcagtgtc gccataacca tgagtataaa cactgcggcc 4320
aacttacttc tgacaacgat cggaggaccg aaggagctaa ccgctttttt gcacaacatg 4380
ggggatcatg taactcgccc tgatcggttg gaaccggagc tgaatgaagc cataccaaac 4440
gacgagcgtg taaccacgat gccgttagca atggcaacaa cgttgcgcaa actattaact 4500
ggcgaactac ttactctagc ttcccggcaa caattaatag actggatgga ggccgataaa 4560
gttgaggac cacttctgcg ctccgcccct ccggctggtt ggtttatgct tgataaatct 4620
ggagccggtg agcgtgggtc acgcggtatc attgcagcac tggggccaga tggtaagccc 4680
tccgtatcgt tagttatcta acgacggggg agtcaggcaa ctatggatga acgaaataga 4740
cagatcgctg agtaggtgct ctactgatt aagcattggt aactgtcaga ccaagtttac 4800
tcataatata tttagattga tttaaaactt catttttaat ttaaaaggat ctagggtgaag 4860
atcctttttg ataattctcat gacaaaaatc ccttaacgtg agttttctgt cactgagcg 4920
tcagaccctg tagaaaagat caaaggatct tcttgagatc cttttttctt gcgcgtaac 4980
tgctgcttgc aaacaaaaaa accaccgcta ccagcgggtg tttgtttgcc ggatcaagag 5040
ctaccaactc tttttccgaa ggttaactggc ttacgcagag cgcagataac aaatactgtt 5100
cttctagtgt agccgtagtt aggccaccac ttcaagaact ctgtagcacc gcctacatac 5160
ctcgctctgc taatcctgtt accagtggct gctgccagtg gcgataagtc gtgtcttacc 5220
gggttggaat caacagcata gttaccggat aaggcgagc ggtcgggtg aacgggggggt 5280
tcgtgcacac agcccgagct ggagcgaacg acctacaccg aactgagata cctacagcgt 5340
gagctatgag aaagcgccac gcttccgaa gggagaaagg cggacaggtt tccggtaagc 5400
ggcaggggtc gaacaggaga gcgcacgagg gagcttcag ggggaaacgc ctggatctct 5460
tatagtctcg tcgggtttcg ccacctctga cttgagcgtc gattttgtgt atgctcgtca 5520
ggggggcgga gccatggaaa aaacgccagc aacgcggcct tttacgggtt cctggccttt 5580
tgctggcctt ttgtctacat gttctttctt gcgttatccc ctgattctgt ggataaccgt 5640
attaccgcca gagaatataa aaagccagat tattaatccg gcttttttat tatttgacgc 5700
gtttt 5705

```

```

SEQ ID NO: 59      moltype = DNA length = 500
FEATURE           Location/Qualifiers
misc_feature       1..500
                   note = Termination sequence of Phatr3_J37038 (500 pb after
                   STOP codon)
source             1..500
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 59
atttcagatt aagttttgaa caatttgctt ttattatggt gtgcaggcac gattgaaact 60
gtgaacgggt tgggttggtat gctgctcac tgtcaattta cggtgggtta tcagggaatcg 120
gatcaggatt ggcaaaaggct aaacctatcc ttgcccacct ctaagattct aatcatcaat 180
caacatttgc tttggagatt tggctgctgg aatcagccaa actgatttgg gtacgagcca 240
ttgccagacc agaaatagcg acccatataa gctctctgcc agcaaatgga aggtatctct 300
tcttaccatc ttctttcccc gagcggtagc ggatacgaag accgtatcaa atatatatac 360
ttacatttca ataaaggcgt ccagtgcaga taatgggata ctgcgtatct tgtctcacga 420
tgtagtgcgt gcaatagtta gaaaaaatgt gcaagtgtac cataccttgc aatttttttg 480
ggaccgaggg acttttttgg 500

```

```

SEQ ID NO: 60      moltype = DNA length = 500
FEATURE           Location/Qualifiers
misc_feature       1..500
                   note = Termination sequence of Phatr3_Jdraft1443 (500 pb
                   after STOP codon)
source             1..500
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 60
gtagatggta caaacaatat gtatagtagc tcttttctgt gaaagttgta ttctctctct 60
cctttctcac agaaggacga cttgatcttc catgcgagaa aatgggctat gcactgggtg 120
ttcgatggca tctccatccc gtgagcctct ccacttttct ttagacaata aactcatcac 180
acaaattagt atgacaagct gctctttaa atcaatctca ttgatcagtc cgtattggat 240
ttgggaatta tcttggcata ccggtagttc tagacaaaga aaggcaagga ctatacttac 300
ttaaagtcac gtaaaagatc agtcagaacg gggatcgagc cttgatagag agacaatctt 360
cgccagctat aaaatatgta ctatccgatg tgtgtccctt tgcctttctt gaaatgactc 420
ttttgtccga atgtgcctct gactatttagc aagaagctcc caaggggttaa tctaccattt 480
tcaagctcga aaaaattgga 500

```

-continued

```

SEQ ID NO: 61      moltype = DNA length = 175
FEATURE           Location/Qualifiers
misc_feature       1..175
                   note = Termination sequence of Phatr3_J35102 (175 pb after
                   STOP codon)
source            1..175
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 61
cgcttattat attgcagatc atggttggtc atggttcggag catgggctgc tgtaacaaca 60
accctgaaac atgatcacat tctgcgtgac gtcgaaaaaa gagagcgagg ctggtgaaat 120
gtatgccctt ttgttacagc gtagtattac aaaaaaaaaa gaaaatttac tgttta 175

```

```

SEQ ID NO: 62      moltype = DNA length = 500
FEATURE           Location/Qualifiers
misc_feature       1..500
                   note = Termination sequence of Phatr3_J50252 (500 pb after
                   codon STOP)
source            1..500
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 62
atggaattcg atcatggaaa atggaaccta aagtgtgacc tgatgtagat gatttgcgaa 60
taactctaaa tcggcaaaga gcttacaatt aatcttttaa tacgtcgtct gtcaagatat 120
aagtgaatat gcttgatata ccttaaatgt gacgagggtca aacgacgtga ttgggggata 180
gcgctccgtg gttcaattcc atggcacgtt aattttgaca gtcgtcaatc aaaactcgct 240
attagaccaa gataggaaat ctatgctttg tcgtaacgac caatagctgt ggtgctgac 300
tacgaaccaa aaaaagctgt tagacctgga attgctcgct ggctatgcaa gaaaattgag 360
cttgcatttt acaaatggga aacggaacgg ctactcccat gggaaatcgg cgacaatccc 420
gatcctctca caaatgact cggaatccat gaattgtaaa accttaagct ataaagactg 480
attcctgatt taatgaatta 500

```

```

SEQ ID NO: 63      moltype = DNA length = 229
FEATURE           Location/Qualifiers
misc_feature       1..229
                   note = Termination sequence of Phatr3_J34085 (229 pb after
                   STOP codon)
source            1..229
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 63
actcctttgg ctctttcaac agtatgtaac gaccatcatc tgatggtaaa cttggaaatt 60
attgaattag tctaattata ttgcatatc gttcagtgtg agtcagttgt gccatagttc 120
attcaacggc gtgtacgtgc gtcaaggagg tcgtgaaatt taggtttctc gaaaactccg 180
aagcaagatg aaggtgcaaa cgtcgatgga atacactgtc aacaatcgc 229

```

```

SEQ ID NO: 64      moltype = DNA length = 114
FEATURE           Location/Qualifiers
misc_feature       1..114
                   note = Termination sequence of Phatr3_J48164 (114 pb after
                   STOP codon)
source            1..114
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 64
tcggaacata gtgctggtct tgttcgggtca cagcagcata taggtaaacc ttaaaagtgt 60
cggaatttat acagtgtaga ctgtctaatt agtacaagag gcttagtacc atat 114

```

```

SEQ ID NO: 65      moltype = DNA length = 500
FEATURE           Location/Qualifiers
misc_feature       1..500
                   note = Termination sequence of Phatr3_EG02422 (500 pb after
                   STOP codon)
source            1..500
                   mol_type = other DNA
                   organism = synthetic construct

```

```

SEQUENCE: 65
atggagcgag gcaattcttc ctccatttta ttggtgctgt tatattctaa aagtatttgt 60
tacatttcag ttgcatcgta tcgaggtaga attgtggaga gatgtgtttt gatttgtaga 120
atgaccggcc ttgcagttaa ctccaagatg caacacagcc tatgcgagat gtcgacgcac 180
atccagggtg ggcaaatata tatcgcaaga tgacagacgg acagaatctc ggcgaaagac 240
gttcgttcac tttttcgaag tactctccag cgtcatgcgg ctctgtgtaa acgaaaggaa 300
ttggcaatct attcgtcatg aaattaccgc tgcgcctaag atgccttggg ctctttcttt 360
cgttcctccc ctcgtttcca tgggtcacc cgtacggtgt caaaacttcg ctccgtcttc 420
caatttgttc gaaaatttgt cctcgacttc agtccgttac acttcggggg gatcggatag 480
ttttgctctc agaatccgcg 500

```

-continued

```

SEQ ID NO: 66      moltype = DNA length = 148
FEATURE           Location/Qualifiers
misc_feature       1..148
                   note = Termination sequence of Phatr3_J42538 (148 pb after
                   STOP codon)
source            1..148
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 66
ttgtgggagcg tctcattacg ggtcagcgag ggattatatt aggtagtcga tgttttgcat 60
ataaagaata cggcgagacg aagccatttt ttgcttaca acaatacctt cagaaaccta 120
gagatgcttg cagtcaggat tgactacc 148

SEQ ID NO: 67      moltype = DNA length = 351
FEATURE           Location/Qualifiers
misc_feature       1..351
                   note = Termination sequence of Phatr3_J48356 (351 pb after
                   STOP codon)
source            1..351
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 67
ttgctcacag tcaacgatga atccttcacg atgagaggct cacgttagcg ttgcgggggt 60
caagattgga cgatgaggtg gttgatccga acggaagggt gacgagattg gccctggact 120
ccgtcgggtca aagaaaaaca attgcccatc gcatagtaga cacaaattca tagtatttca 180
aagaagaatc aactgtctat atattgtagg attgactttg tcgcatttaa caccacaaaa 240
tatgttataa tgtcttaaa aacggtaaa agcaggcaat aaggtagttc ctatttcaaa 300
atgggtactcg aggtgtagcg atcaagcttt tgtcttcacg tcaaaagtag a 351

SEQ ID NO: 68      moltype = DNA length = 20
FEATURE           Location/Qualifiers
misc_feature       1..20
                   note = Primer TRA-F
source            1..20
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 68
cgataatgcg ttgcagtcgg 20

SEQ ID NO: 69      moltype = DNA length = 20
FEATURE           Location/Qualifiers
misc_feature       1..20
                   note = Primer TRA-R
source            1..20
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 69
tcggaccctt agtactcgca 20

SEQ ID NO: 70      moltype = DNA length = 20
FEATURE           Location/Qualifiers
misc_feature       1..20
                   note = Primer RTX-F
source            1..20
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 70
atctcacgcg ttgaagcaga 20

SEQ ID NO: 71      moltype = DNA length = 20
FEATURE           Location/Qualifiers
misc_feature       1..20
                   note = Primer RTX-R
source            1..20
                   mol_type = other DNA
                   organism = synthetic construct

SEQUENCE: 71
ccctggcttc actagttcgg 20

SEQ ID NO: 72      moltype = DNA length = 21
FEATURE           Location/Qualifiers
misc_feature       1..21
                   note = Primer HC trastuzumab - F
source            1..21
                   mol_type = other DNA

```

-continued

```

                                organism = synthetic construct
SEQUENCE: 72
ctgtcattat aagtcggggg g                                     21

SEQ ID NO: 73          moltype = DNA  length = 21
FEATURE               Location/Qualifiers
misc_feature           1..21
                        note = Primer HC trastuzumab - R
source                 1..21
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 73
tcgcgactaa tcgtgaaacg g                                     21

SEQ ID NO: 74          moltype = DNA  length = 21
FEATURE               Location/Qualifiers
misc_feature           1..21
                        note = Primer LC trastuzumab - F
source                 1..21
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 74
aaactcctct atcctcacct g                                     21

SEQ ID NO: 75          moltype = DNA  length = 20
FEATURE               Location/Qualifiers
misc_feature           1..20
                        note = Primer LC trastuzumab - R
source                 1..20
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 75
caactttagt cccttgaccg                                       20

SEQ ID NO: 76          moltype = DNA  length = 22
FEATURE               Location/Qualifiers
misc_feature           1..22
                        note = Single primer trastuzumab - F
source                 1..22
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 76
cagcaaggac tccacataca gc                                    22

SEQ ID NO: 77          moltype = DNA  length = 21
FEATURE               Location/Qualifiers
misc_feature           1..21
                        note = Single primer trastuzumab - R
source                 1..21
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 77
tcgcgactaa tcgtgaaacg g                                     21

SEQ ID NO: 78          moltype = DNA  length = 66
FEATURE               Location/Qualifiers
misc_feature           1..66
                        note = Primer 1 TU from pPhaNR-eGFP
source                 1..66
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 78
gctctagata gttggaatgg tacgtaccaa ctccataagg atccatgggtg agcaagggcg 60
aggagc                                                         66

SEQ ID NO: 79          moltype = DNA  length = 41
FEATURE               Location/Qualifiers
misc_feature           1..41
                        note = Primer 2 TU from pPhaNR-eGFP
source                 1..41
                        mol_type = other DNA
                        organism = synthetic construct
SEQUENCE: 79
acatgcatgc tccggatgcg ttcatttttag atcctgatcc g             41

SEQ ID NO: 80          moltype = DNA  length = 20

```

-continued

FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Primer eGFP - F	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 80		
atcatggccg acaagcagaa		20
SEQ ID NO: 81	moltype = DNA length = 20	
FEATURE	Location/Qualifiers	
misc_feature	1..20	
	note = Primer eGFP - R	
source	1..20	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 81		
gactgggtgc tcaggtagt		20
SEQ ID NO: 82	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Signal peptide sequence putative PDI Phaeodactylum	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 82		
MKFTAAITLA FAASVSA		17
SEQ ID NO: 83	moltype = DNA length = 499	
FEATURE	Location/Qualifiers	
misc_feature	1..499	
	note = Sequence of Promotor HASP1	
source	1..499	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 83		
catacagtga atgtaacttt cgaattgaca gtattagtag tcgtattgac agtgaggcac		60
gcccctcaat gtgcgagggt gaaaatatac cagcatgaca atgaatcttg gagattcttt		120
tgctgtcatc aagattcacc gccaaatctt caggaacctc tcacgtccac aggcgatgtt		180
aattcttgag tcgtcaaac aaagtccgtg cctacctgta gaagttgaca gcgagcaatt		240
gtatgcaaac ttctgacttt gttataataa cattaaaggt aattaagtat cttcaattag		300
gcattttgtc actgtcagtc cgttccgaca atataggtag atttggaaag aatcttttct		360
atgctgtctg gaactctgta cacctttgag gccgtagatt ctgtccgacg aagcgataat		420
tattgcaaaa tacatggact cattattttg attcgatttc tttttggtat ccgactcgaa		480
aagatccatc acggcgagc		499
SEQ ID NO: 84	moltype = DNA length = 600	
FEATURE	Location/Qualifiers	
misc_feature	1..600	
	note = Sequence of Promotor 40SRPS8	
source	1..600	
	mol_type = other DNA	
	organism = synthetic construct	
SEQUENCE: 84		
ccctgcgata gaccttttcc aaactcacgc agtccaagaa aacaaagggg tgagaagtat		60
acgcaccttt cggttttcggc ataattctta aactcttgtg gtcactttct tgtgaagaag		120
ctaggggcac tcgttttccc tcagagcctg caaacacaaa attcctgcag tcaattgtcc		180
caacactcgg caaaccgat gcgcaagcaa cgatgcgcag aaggccgtgg atggatggcg		240
actcgcgata tggcttcttg ggtcgccagt gtggtacgtc cggcgatagt caatacgcga		300
attcggacga ctggcatctc taggaggagg attccttctt ttatgacatg tttattttat		360
atacattgat gctttccgac agtcggaagt aataaatgaa tttatttcaa gactacctat		420
actcctttga cttgttcgac taatcttacc gcttactaaa atctcgaaat cacgcttgac		480
ctctcgcacg caaatttttg ctgctggacg ctacgcactc ggcccaattc ttctcggtcc		540
tcgtcgtcgc aattgctggt gcgttgatct tgcaccgaag gaatcagaga atagaatacc		600
SEQ ID NO: 85	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
REGION	1..18	
	note = modified signal peptide B from human serum albumin	
source	1..18	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 85		
KWVTFISLLF LFSSSSRA		18

SEQ ID NO: 86	moltype = DNA length = 57	
FEATURE	Location/Qualifiers	
misc_feature	1..57 note = Codon-optimized DNA sequence coding for signal peptide ''E''	
source	1..57 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 86		
atgacccgac tcacggtcct	agcccttttg gctggactgt tggcgctcgtc ccgcgcga	57
SEQ ID NO: 87	moltype = DNA length = 57	
FEATURE	Location/Qualifiers	
misc_feature	1..57 note = Codon-optimized DNA sequence coding for signal peptide ''H5''	
source	1..57 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 87		
atggactgga cctggcgatt	cttgtttgtt gtcgccgcgg ctacgggagt gcaatcg	57
SEQ ID NO: 88	moltype = DNA length = 57	
FEATURE	Location/Qualifiers	
misc_feature	1..57 note = Codon-optimized DNA sequence coding for signal peptide ''H7''	
source	1..57 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 88		
atggagtttg ggctgtcatg	ggtattcttc gttgccctgt tccgaggtgt gcagtgcc	57
SEQ ID NO: 89	moltype = DNA length = 66	
FEATURE	Location/Qualifiers	
misc_feature	1..66 note = Codon-optimized DNA sequence coding for signal peptide ''L1''	
source	1..66 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 89		
atggacatgc gagtcccgcc	tcaacttctc ggattactac tgttgtgggt gtcgggtgcc	60
cgctgc		66
SEQ ID NO: 90	moltype = DNA length = 54	
FEATURE	Location/Qualifiers	
misc_feature	1..54 note = Codon-optimized DNA sequence coding for signal peptide of the human BIP	
source	1..54 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 90		
atgaagctat ccttggtcgc	agccatgctg cttctcttgt cggctgcgcg agcc	54
SEQ ID NO: 91	moltype = DNA length = 51	
FEATURE	Location/Qualifiers	
misc_feature	1..51 note = Codon-optimized DNA sequence coding for signal peptide of the human PDI	
source	1..51 mol_type = other DNA organism = synthetic construct	
SEQUENCE: 91		
atqctacgac qtqccettgct	ctqccctacq qtaqctgcc tgaqccgcgc a	51

1. A single polynucleotide vector expressing a therapeutic monoclonal antibody or a functional fragment or derivative thereof, said vector sequence comprising:

- a) nucleotide sequences encoding, in the same or in two different Open Reading Frame(s):
 - the light chain of said therapeutic antibody, fragment or derivative thereof, preceded at its N-terminal end by a first heterologous secretion signal peptide,
 - the heavy chain of said therapeutic antibody, fragment or derivative thereof, preceded at its N-terminal end by a second heterologous secretion signal peptide,
 said light and heavy chain being separated by a nucleotide sequence encoding a self-cleaving linker,
- b) at least one promoter operatively linked with the sequences encoding said light and heavy chains, said promoter being able to drive the expression of said light and heavy chains in the microalgae *Phaeodactylum tricornutum*,
- c) at least one translation termination sequence, and
- d) optionally, at least one detectable marker.

2. The polynucleotide vector of claim 1, wherein said heterologous secretion signal peptide is chosen among: E of SEQ ID NO:1, H5 of SEQ ID NO:2, H7 of SEQ ID NO:3, B of SEQ ID NO:85 or L1 of SEQ ID NO:4.

3. The polynucleotide vector of claim 1 wherein said first and second heterologous secretion signal peptides are different, preferably L1 and H7 respectively.

4. The polynucleotide vector of claim 1 wherein said first and second heterologous secretion signal peptides are identical, and preferably the signal peptide B of SEQ ID NO:85 or the signal peptide E of SEQ ID NO:1.

5. The polynucleotide vector of claim 1 wherein said first and second heterologous secretion signal peptides are followed by a sequence encoding a cleavage site.

6. The polynucleotide vector of claim 1 wherein said linker is chosen in the group consisting of: E2A (SEQ ID NO:6), F2A (SEQ ID NO:7 or SEQ ID NO:12), T2A (SEQ ID NO:9), P2A (SEQ ID NO:8), IRES (SEQ ID NO:16 or 17), preferably in the group consisting of: T2A and F2A.

7. The polynucleotide vector of claim 1 wherein said promoter is chosen in the group consisting of: the promoter of the Nitrate Reductase gene from *Phaeodactylum tricornutum* (SEQ ID NO:26), the promoter of the FcpA gene from *Phaeodactylum tricornutum* (SEQ ID NO: 27), the promoter of the FcpB gene from *Phaeodactylum tricornutum* (SEQ ID NO:28), the promoter of the V-ATPase-C gene (SEQ ID NO:25), the promoter of the histone H4-1B gene (SEQ ID NO:24), the promoter of the Phatr3_J37038 gene (SEQ ID NO:29), the promoter of the Phatr3_Jdraft1443 gene (SEQ ID NO:30), the promoter of the Phatr3_J35102 gene (SEQ ID NO:31), the promoter of the Phatr3_J50252 gene (SEQ ID NO:32), the promoter of the Phatr3_J34085 gene (SEQ ID NO:33), the promoter of the Phatr3_J48164 gene (SEQ ID NO:34), the promoter of the Phatr3_EG02422

gene (SEQ ID NO:35), the promoter of the Phatr3_J42538 gene (SEQ ID NO:36), and the promoter of the Phatr3_J48356 gene (SEQ ID NO:37).

8. The polynucleotide vector of claim 1 wherein said translation termination sequence is chosen among: the termination sequence of the Nitrate Reductase gene (SEQ ID NO:20) or the termination sequence of the FcpA gene of *Phaeodactylum tricornutum* (SEQ ID NO:19), the termination sequence of the V-ATPase-C gene (SEQ ID NO:22), the termination sequence of the γ -tubulin gene (SEQ ID NO:X), the termination sequence of the Phatr3_J34085 gene (SEQ ID NO:63) and the termination sequence of the histone H4 gene (SEQ ID NO:21).

9. The polynucleotide vector of claim 1 wherein said promoter is the promoter of the Nitrate Reductase gene of SEQ ID NO:26 or the promoter of the V-ATPase-C gene of SEQ ID NO:25 or the promoter of the Phatr3_J34085 gene of SEQ ID NO:33 and said translation termination sequence is the termination sequence of the FcpA gene of *Phaeodactylum tricornutum* of SEQ ID NO:19.

10. The polynucleotide vector of claim 1 containing, in this order:

- the promoter of V-ATPase-C of SEQ ID NO:25 or the promoter of the Phatr3_J34085 gene of SEQ ID NO:33,
- the sequence encoding the secretion peptide signal E of SEQ ID NO:1,
- the sequence encoding the light chain of said therapeutic antibody, fragment or derivative thereof,
- the sequence encoding the linker T2A of SEQ ID NO:15,
- the sequence encoding the secretion peptide signal E of SEQ ID NO:1,
- the sequence encoding the heavy chain of said therapeutic antibody, fragment or derivative thereof,
- the termination sequence of the FcpA gene of *Phaeodactylum tricornutum* of SEQ ID NO:19.

11. A transformed cell of *Phaeodactylum tricornutum* comprising the polynucleotide vector of claim 1, said cell being preferably from the strain Pt1, Pt3, Pt4, Pt5 or Pt7.

12. A method of using the transformed cell of claim 11, comprising producing and secreting a therapeutic antibody, a functional fragment or derivative thereof.

13. An in vitro method for producing a therapeutic antibody, a functional fragment or derivative thereof, said method comprising:

- (i) culturing the transformed cell of claim 11 in appropriate culture conditions,
- (ii) harvesting the extracellular medium of said culture, and
- (iii) purifying the therapeutic antibody, fragment or derivative thereof, which is secreted in said extracellular medium.

14. The method of claim 13, wherein said appropriate culture conditions involve the use of a culture medium containing a nitrate concentration comprised between 0.1 g/L and 1 g/L and low salt concentration comprised between 5 to 30%.

15. The polynucleotide of claim 1 wherein said therapeutic antibody is a chimeric, humanized, or human anti-viral, auto-immune or anticancer antibody or any variant of this, preferably chosen from the group consisting of: rituximab, trastuzumab, adalimumab, bevacizumab, infliximab, cetuximab, motavizumab, palivizumab, alemtuzumab, dinutux-

imab, naxitamab, but also comprising for instance, benralizumab, catumaxomab, daratumumab, elotuzumab, epratuzumab, farletuzumab, galiximab, gentuzumab ozogamicin, ibritumomab tiuxetan, lumiliximab, necitumumab, nimotuzumab, ocrelizumab, ofatumumab, oregovomab, pertuzumab, tositumomab, zalutumumab, and zanolimumab, more preferably Trastuzumab or Rituximab, dinutuximab and naxitamab or derivatives.

* * * * *